# Local laws for the occupation time of on/off processes with heavy-tailed, finite-mean periods: bulk, big jumps, boundary layers and aging

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**Abstract**

Let $A_N$ be the number of time slots spent in the on state during the first $N$ slots of a discrete-time on/off process whose on and off periods are independent and have regularly varying tails with indices in $(1, 2)$. Equivalently, $A_N$ counts the right steps of a lattice Lévy walk. The limit law of $A_N$ at the scale $N^{1/\alpha}$ is stable and classical. We describe $\mathbb{P}(A_N = k)$ at every point. In the bulk it is a stable density, uniformly over the whole range, under regular variation and aperiodicity alone. On each side of the bulk and up to the endpoint one kernel gives the law under local regular variation. It adds a complete giant period and the period in progress at time $N$, and it contains a renewal boundary layer at each end and an atom at each endpoint. This is a lattice form of the infinite density of the Lévy walk. We find the exact effect of an arbitrary initial period on each piece, and the effect of aging, when the process is observed from time $\theta N$ on. The bulk does not age. The atom, the layer and the big-jump zone do, and we give their aged forms in closed form. For the stationary start and for every aged start, the most likely value of $A_N$ is an endpoint for all large $N$ when the index is below the golden ratio and is interior when it is above. For fresh and fair starts, aperiodic periods, comparable scales, a second-order smoothness condition and no ties in the boundary profiles, the law has, for all large $N$, one more mode than the two boundary profiles have local maxima. Hence it has 3 modes for strictly log-convex period laws. The results give the exposure law of an experiment that switches arm at the regenerations of a slotted queue with heavy-tailed service. We prove that the cycle law of the queue satisfies the smoothness conditions, so the mode count applies to it.

**Keywords:** on/off process; alternating renewal process; occupation time; Lévy walk; local limit theorem; stable law; big-jump principle; infinite density; aging; renewal theorem; number of modes; switchback experiment; busy period.
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# 1 Introduction and main results

An on/off process is a two-state process in discrete time whose on periods and off periods alternate and have independent lengths. Let the on periods have the law of $X_+$ and the off periods the law of $X_-$, two laws on $\{1, 2, \dots\}$ with regularly varying tails of indices $\alpha_+$ and $\alpha_-$ in $(1, 2)$, so that a period has a finite mean and an infinite variance. Let $A_N$ be the number of time slots spent in the on state among the first $N$. The law of $A_N$ in distribution is classical. Under a finite variance $A_N$ is Gaussian at the scale $\sqrt{N}$ [65, 66, 41], under infinite means $A_N/N$ has the generalized arcsine law of Lamperti [44], and in the regime of this paper $A_N - pN$ is stable at the scale $N^{1/\alpha}$, where $p$ is the long-run fraction of on time [65, 31]. These are statements about distribution functions. We describe the mass function $P_N(k) = \mathbb{P}(A_N = k)$ at every point $k$ of its range $\{0, \dots, N\}$. The bulk is a stable density, uniformly over the whole range (Theorem A). On each side of the bulk and up to the endpoint the law is given by one kernel, which adds a complete giant period and the period in progress at time $N$, and which contains a renewal boundary layer at each end and an atom at each endpoint (Theorem B). An arbitrary initial period enters each of these pieces through a sharp criterion (Theorem C), and observation from a later time ages the atom, the layer and the big-jump zone but not the bulk (Theorem D). The shape follows. The endpoint is the global mode above or below the golden ratio (Theorem E), an interior minimum sits at the stable scale of the opposite periods (Theorem F), and the law has exactly $M_+ + M_- + 1$ modes for all large $N$, where $M_\pm$ counts the local maxima of the two boundary profiles (Theorem G). Section 7 reads these theorems as the exposure law of an experiment that switches treatment at the regenerations of a queue with heavy-tailed service.

The same process is a lattice Lévy walk. Read an on slot as a step to the right and an off slot as a step to the left. The walk moves in runs of alternating direction, its position after $N$ steps is $2A_N - N$, and $P_N$ is the law of the walk on its natural lattice. Lévy walks are a standard model of superdiffusion [78], and the same walk describes heat transport in one-dimensional lattices [19, 71]. In continuous time the propagator of a Lévy walk is a stable bulk between two ballistic peaks at the fronts, formed by the walkers still in their first flight [78, Section II.A.2 and Fig. 2] (see also [77, 18]), and the law between the bulk and the fronts is the infinite density of Rebenshtok, Denisov, Hänggi and Barkai [58, 59]. On the lattice the peaks are the endpoint atoms and the boundary layers, and the infinite density is Theorem B read at $k = vN$. The picture is known. What we add is its exact form at a fixed time, its dependence on the start, and an exact count of its peaks. We keep the vocabulary of the walk in the body of the paper. A period is a run, an on slot is a plus step, an off slot is a minus step, and the two ends of the range are the plus end $k = N$ and the minus end $k = 0$.

## 1.1 The four pieces of the law

Figure 1 shows $P_N$ for $N = 5000$ on a logarithmic scale. Four pieces make up the law.

- The bulk. Near $pN$ the law is a stable density at the scale $B_N$ of the larger of the two run-length fluctuations. Here $B_N$ grows like $N^{1/\alpha_*}$ up to a slowly varying factor, with $\alpha_*$ the smaller index.
- The big-jump zones. A count $k$ with $k - pN$ much larger than $B_N$ needs one plus run of macroscopic length, a giant run, and the law of that run gives $P_N(k)$ its size, which is of order $T_+(N)$ on the ballistic range, with $T_+$ the tail of a plus run. The giant run is either a complete run in the interior or the run in progress at time $N$, and $P_N(k)$ is the sum of the two contributions.
- The boundary layers. At $k = N - j$ with $j$ fixed, the giant run is the one in progress

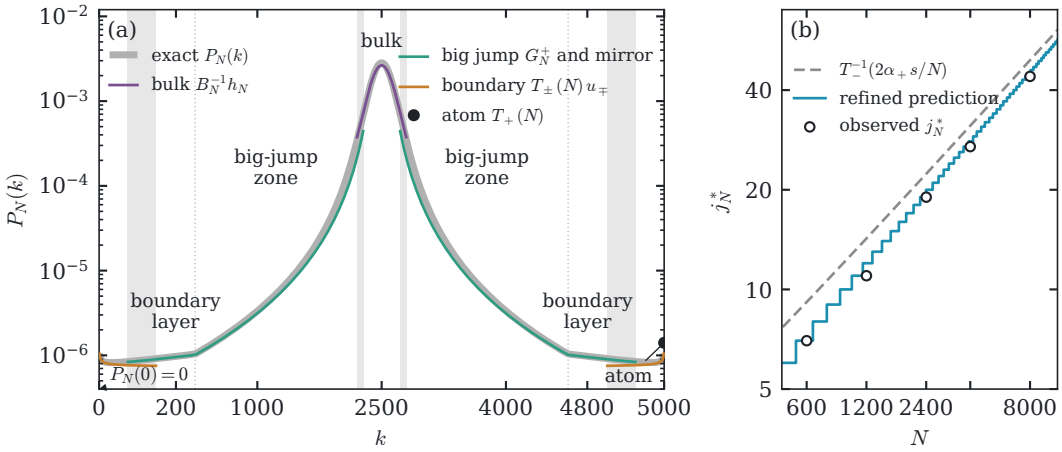



Figure 1: (a) The law $P_N(k)$ for $N = 5000$, run lengths with $f(n) \propto n^{-5/2}$ for both signs and a fresh plus start, on a logarithmic scale, with the four pieces of Theorems A and B. The two ends of the horizontal axis are stretched. (b) The position $j_N^*$ of the interior minimum near the plus end against $N$, on logarithmic axes, with the law of Theorem F and the refined prediction with $C(j)$ from Theorem 6.1. Section 9 describes the computations.

at time $N$, and the only question is where it started. It starts right after the $j$-th minus step exactly when a minus run ends there, so the layer is $T_+(N)$ times the renewal mass function $u_-(j)$ of the minus runs.

- The atoms. The point $k = N$ is the event that the first run covers all $N$ steps, and its mass is the tail of the first run.

The first run of the walk is its initial run. It may be an ordinary run, a run seen from a typical time, the run in progress at the time the observation starts, or anything else, and it enters each of the four pieces through its own criterion.

### 1.2 Main results

*Hypotheses.* We state them in Section 2.2. They are regular variation of the two tails with indices in $(1, 2)$, written (RV), the local form $f_\sigma(n) \sim \alpha_\sigma T_\sigma(n)/n$ of the mass functions, written (LR), and aperiodicity, written (SPAN). Sections 6.4 and 6.5 also use bounds on the first and second differences of the mass functions, written (LD) and (LD2), and Section 6.5 a comparison of the two tails, written (EQ). The run laws may differ between the two signs, so the two stable scales $b_{+,N}$ and $b_{-,N}$ may differ, and their ratio may oscillate. Every theorem below except Theorems 4.14 and G holds for every relation between the two scales. We write $s$ for the sum of the two mean run lengths and $p$ for the fraction of the plus mean.

*The bulk.* Theorem A says that for a fresh start, under (RV) and (SPAN),

$$\sup_{k\in\mathbb{Z}} \bigl| B_N P_N(k) - h_N\bigl((k - pN)/B_N\bigr)\bigr| \to 0, \qquad \sup_k P_N(k) \le \frac{C}{B_N},$$

where $h_N$ is the density of $(1-p)\theta_{+,N}Z_+ - p\,\theta_{-,N}Z_-$, with $Z_\pm$ independent totally skewed stable variables. The supremum runs over the whole range of the count. On a window $|k - pN| \le KB_N$ the local limit also follows from a strong renewal theorem of Berger [5, Theorem 3.3], and what is new is the uniformity over every point under (RV) and (SPAN) alone. Outside the bulk the theorem says only that $P_N(k) = o(1/B_N)$, and the next theorem says what $P_N(k)$ is there.

*Big jumps and the boundary layer.* Put $L_k = (k - pN)/(1-p)$, the length of the plus run that carries a typical configuration to the count $k$. Theorem B says that for a fresh plus start, under (RV) and (SPAN) for both signs and (LR) for the plus sign,

$$P_N(k) = \Big[\gamma(N-k)\, f_+(\lfloor L_k \rfloor) + T_+(\lfloor L_k \rfloor)\, u_-(N-k)\Big](1+o(1))$$

uniformly on $pN + K\lambda_N B_N \le k \le N$, in the sense that the error tends to zero as $N \to \infty$ and then $K \to \infty$. Here $\gamma(j)$ is an explicit finite sum with $\gamma(j) \sim js/\mu_-^2$, and $\lambda_N = 1 + \log(B_N/b_{+,N})$, which equals $1$ when the plus scale is the larger one. The first term is a complete giant run and the second the run in progress at time $N$. At $k = N - j$ with $j$ fixed the theorem reduces to $P_N(N-j)/T_+(N) \to u_-(j)$, which needs only (RV) for the plus sign. As $j$ grows, $u_-(j) \to 1/\mu_-$ and the layer joins the big-jump zone continuously. Read at $k = vN$ the theorem gives the infinite density

$$\frac{P_N(\lfloor vN \rfloor)}{T_+(N)} \to \frac{1}{\mu_-}\Big[\alpha_+(1-\ell)\ell^{-\alpha_+-1} + \ell^{-\alpha_+}\Big], \qquad \ell = \frac{v-p}{1-p},$$

where the second term is the run in progress at time $N$ (Corollary 4.9). (LR) suffices for the theorem, and Example 4.12 shows that (RV) alone does not, since the kernel reads $f_+$ at a single point. This parallels the optimality of the local hypothesis for sums [6, Eq. (2.4)]. When the plus scale is the larger one, the bulk profile up to a slowly growing multiple of $B_N$ and the kernel beyond it give the whole plus side in a single limit, while no fixed switching point does (Remark 4.10). When the plus scale is the smaller one, the window $KB_N \le k - pN \le K\lambda_N B_N$ lies between the two theorems, and in general we have no relative error there. For a fresh start with $b_{-,N}/b_{+,N} \to \infty$, Theorem 4.14 narrows it. The kernel takes over at $pN + (1+\delta)\xi_N B_N$ and not at $pN + (1-\delta)\xi_N B_N$, where $\xi_N = o(\lambda_N)$ comes from a Cramér transform of the minus runs, and before it the law is a moderate deviation of the minus runs, described on the logarithmic scale. For sums of independent variables with a heavy left tail of index in $(1,2)$ and a lighter right tail, the same crossover at a logarithmic multiple of the stable scale is due to Rozovskiĭ [62] for the tail and to Denisov, Dieker and Shneer [16, Section 9.2] for local probabilities. What Theorem 4.14 adds is the count at a fixed time, where the number of runs is random and the big jump may be made by the run in progress at time $N$.

*The initial run.* Wang, Schulz, Deng and Barkai [73] left "the effect of initial preparation on the rare fluctuations" for future work. Their question covers all the observables of their paper and also the range $0 < \alpha < 1$. Theorem C answers it for the occupation count with $1 < \alpha < 2$, for every law of the initial run and on every piece of the law. It rests on the first-run decomposition of Proposition 2.4. Write $\nu_\sigma(n)$ for the probability that the first run has sign $\sigma$ and length $n$, and $Q_\sigma(n)$ for the probability that it has sign $\sigma$ and length at least $n$. On every macroscopic interior of the range of the count the bulk approximation holds with no condition on the initial run. It holds uniformly over $1 \le k < N$ exactly when $B_n(\nu_+(n) + \nu_-(n)) \to 0$, which we call the bulk criterion. The initial run is invisible in the boundary layer at the plus end exactly when $\nu_+(n) = o(T_+(n))$. On the big-jump zone up to the end the same condition is sharp, while on its interior only window averages of $\nu_+$ matter. The conditions are nested, and when one fails the theorem gives the exact correction. The stationary initial run, the run in progress at a typical time of a two-sided stationary process, satisfies the bulk criterion and fails the other two, and Corollary 5.4 gives its corrections in closed form. Corollary 5.5 shows that no trimming of the range $\{0, \dots, N\}$ by $o(N)$ points removes the dependence on the initial run for every initial run at once.

*Aging.* Let the process start fresh at time $0$ and count the plus steps among the steps $t_a + 1, \dots, t_a + N$, with $t_a/N \to \theta$. This is the aged occupation time of the physics

literature, whose law in distribution for $0 < \alpha < 1$ is due to Akimoto, Sera, Yamato and Yano [1]. For $1 < \alpha < 2$ the run in progress at time $t_a$ is an initial run whose law depends on $N$. Theorem C is stated for such families, under a tightness condition that costs nothing, and Theorem D is its corollary for the aged law. The bulk does not age, for every $t_a(N)$. The atom becomes $NT_+(N)(1-(1+\theta)^{1-\alpha_+})/(s(\alpha_+-1))$. On the whole big-jump zone up to the end the kernel gains the term $s^{-1}[T_+(L_k)-T_+(L_k+t_a)](1+\mu_+u_-(N-k))$, the probability that the run in progress at time $t_a$ is the giant run, so the boundary layer becomes

$$\frac{P_N(N-j)}{T_+(N)} \to p\,u_-(j) + \frac{1}{s} + \frac{1}{s}\big(1-(1+\theta)^{-\alpha_+}\big)\big(1+\mu_+u_-(j)\big)$$

and the interior of the zone gains $[T_+(L_k)-T_+(L_k+t_a)]/\mu_-$. At $\theta = 0$ these are the fresh forms with the atom replaced by $t_aT_+(N)/s$, and at $\theta = \infty$ they are the stationary forms of Corollary 5.4. The third term of the layer is the complete initial run, which the fresh start never sees.

*The global mode and the golden ratio.* Theorem E compares the endpoint atoms with the bulk for an initial run that satisfies the bulk criterion. An endpoint is the global mode when its atom exceeds $H_N/B_N$ by a fixed factor, with $H_N$ the height of the profile, and the bulk wins when both atoms are smaller by a fixed factor. For the stationary initial run with equal indices this is the golden ratio threshold. The endpoints win for $\alpha < \varphi = (1+\sqrt{5})/2$, the bulk wins for $\alpha > \varphi$, and at $\alpha = \varphi$ with pure power tails the bulk wins by the explicit factor $1/\mathcal{C}(\varphi) = 1.288\ldots$, with $\mathcal{C}$ as in (6.1). The threshold is the exponent identity $\alpha - 1 = 1/\alpha$, and Corollary 5.7 shows that it survives aging for every $\theta \in (0,\infty]$. For a fresh start the mode is always interior.

*The antimode and the extrema near the ends.* Theorem F locates the interior minimum between the bulk and the plus end for a fresh plus start. If $u_-(j) > 1/\mu_-$ for every $j \geq 1$, which holds for every strictly log-convex law of the minus runs, the minimum sits at $N - j_N^*$ with $T_-(j_N^*) \sim 2\alpha_+ s/N$, so $j_N^*$ grows like the stable scale of the minus runs. It is asymptotic to the minimizer of $u_-(j) + C(j)/N$, with $C(j)$ the exact second-order coefficient of Theorem 6.1. When $u_-$ dips below its limit the minimum stays at bounded distance from the end. Section 6.4 compares neighbouring values. A difference form of the renewal theorem, $u_-(j) - u_-(j-1) \sim -T_-(j)/\mu_-^2$ under (RV) and (SPAN), makes the end profile eventually monotone whenever the initial run has $\nu_-(n) = (c+o(1))T_-(n)$ with $c \neq 1/\mu_-$ (Lemma 6.6). Each local maximum of the profile at an end is copied by $P_N$. So when $\nu_\sigma = o(T_\sigma)$ for both signs, $P_N$ has at least one mode more than the two profiles together (Theorem 6.8), and some prepared starts of a log-convex law have at least 4 modes (Corollary 6.9). Under (LD), up to distance $N^{1-\eta}$ from each end the local extrema of $P_N$ are those of the profile, followed by an odd number of extrema in a window of width $2\eta' a_-(N)$ around the interior minimum, for every $\eta' > 0$, and $P_N$ is strictly monotone in between (Theorem 6.12).

*The exact number of modes.* Theorem G assumes (LR), (SPAN) and the second-difference bound (LD2) for both signs, and (EQ), which says that the two tails are comparable. For a fresh or fair start with no ties in the end profiles, the first differences of $P_N$ change sign in a fixed order for all large $N$, and $P_N$ has exactly $M_+ + M_- + 1$ modes, where $M_\pm$ is the number of local maxima of the end profile at the end of sign $\pm$. So $P_N$ has exactly 3 modes for all large $N$, a central peak and two peaks at the ends, if and only if each end profile has a single local maximum (Corollary 6.13). This holds for all strictly log-convex run laws (Corollary 6.15). The criterion is not monotonicity of the renewal mass function, and Lemma 6.14 says when the two agree. The picture of 3 peaks fails for every aperiodic law without runs of length 1. Then the fair start has at least 5 modes, and for the law $f(n) \propto n^{-5/2}$ on $n \geq 2$ it has at least 15 (Example 6.17). The proof needs three steps beyond Theorem 6.12. A second-order form of Theorem 6.11

gives a single antimode. A difference form of the local big-jump theorem for sums of $m$ runs, uniform in $m$ (Lemma 6.18), makes $P_N$ monotone on the big-jump zone, where $m$ is of order $N$. A local limit theorem for the differences of $P_N$ in the bulk, together with the bell shape of stable densities, gives a single bulk mode. Given equal indices, the comparison (EQ) is needed only at the scale $N$ (Corollary 6.16). Unequal scales and a bound for every $N$ remain open (Remark 6.19 and Section 8).

*Exposure in switchback experiments.* A switchback experiment alternates two treatments over time, and its exposure is the number of slots spent under one of them. If the system is a slotted queue and the experiment switches at the regenerations, the exposure is $A_N$ with the cycle laws of the two arms as run laws. Proposition 7.1 shows that the cycle law of a slotted queue with service of locally regularly varying tail satisfies (LR), (SPAN), (LD) and (LD2), so that every theorem of the paper applies to it. The local tail of the cycle is known [42, 37, 17], and the difference bounds are new. Corollaries 7.2 to 7.4 give the exposure imbalance at the scale $N^{1/\alpha}$, the probability of a severe imbalance, the probability that the experiment never switches, the factor of order $N$ by which a start at an arbitrary time inflates it, and the golden ratio rule for the most likely exposure. The designs in the literature assume geometric mixing and so a Gaussian exposure at the scale $\sqrt{N}$. In the language of statistics, the theorems of this paper are the exact sampling law of the treated share $A_N/N$, including its dependence on when the experiment starts.

### 1.3 Related work

*On/off processes.* The occupation time of an alternating renewal process is a classical object. Takács [65, 66] and Kesten [41] proved its limit laws in distribution, Lamperti [44] the generalized arcsine law for infinite means, with later refinements by Mitov and Yanev [50], and Starreveld, Bekker and Mandjes [64] a central limit theorem and a large deviation principle under exponential moments, with dependent sojourns. The heavy-tailed on/off source is the standard model of long-range dependence in teletraffic. Heath, Resnick and Samorodnitsky [32] derived the slowly decaying covariance of the stationary indicator, Taqqu, Willinger and Sherman [67] the self-similar limit of many superposed sources, and Mikosch, Resnick, Rootzén and Stegeman [49] the stable and fractional Brownian limits according to the order of the limits. These works concern the stationary indicator, its covariance, and aggregates of many sources. The object here is one source at a fixed horizon, through its mass function at every point. Every statement in the sources above is a statement in distribution, while every statement here concerns single points.

*Infinite densities and big jumps.* Rebenshtok, Denisov, Hänggi and Barkai [58, 59] derived the infinite density of a Lévy walk from the Montroll and Weiss equation, formally and in continuous time. For the two-state velocity model their density [58, Eqs. (8) and (10)] has the shape $|v|^{-1-\alpha}[\alpha-(\alpha-1)|v|]$. Schulz and Barkai [63, Eqs. (25) and (26)] found the infinite density of the occupation time of a site for an alternating process with two sojourn laws and matched it to the stable bulk, and Wang, Schulz, Deng and Barkai [73, Eq. (83)] found it for one law. A complete giant run alone would give the linear term $-\alpha|v|$, and the run in progress adds $+|v|$, so these formulas already contain the run in progress. Vezzani, Barkai and Burioni [69, Eqs. (12) and (13)] (see also [70]) derived the tail from a single big jump as the sum of exactly these two terms. So the shape of the infinite density and its two-term mechanism are known formally. What Theorem B and Corollary 4.9 add is a proof on the lattice at a fixed time, with the exact kernels $\gamma$ and $\kappa$ at finite distance from the end, uniformly up to the endpoint, together with the boundary layer and the role of the local hypothesis. In the physics model each flight chooses its direction independently, and consecutive flights in the same direction merge into one run. This gives the alternating walk with run laws thinned geometrically

with parameter $\frac{1}{2}$.

*Aging and the initial condition.* The paper [73] works in continuous time through a double Laplace transform, with one law for both states and a start of fair random sign. It finds a boundary layer of tail times renewal density for the backward recurrence time when $1 < \alpha < 2$ [73, Eq. (52)] and an atom with a boundary layer for the occupation time when $0 < \alpha < 1$ [73, Section VII.A]. Its authors write that "the effect of initial preparation on the rare fluctuations is left for a future work" [73, Section VIII]. Wang, Vezzani, Burioni and Barkai [74] then compared the ordinary and the equilibrated start formally for a biased continuous-time random walk. They found that the rare fluctuations remember the start, through an added term $\xi^{-\alpha}$ in the infinite density and an atom at the origin [74, Eqs. (28), (46) and (47)]. This is the structure of Corollary 5.4(a) and (b), where the stationary initial run adds the complete initial run to the kernel. Theorem C treats an arbitrary initial run at every point, with a sharp criterion in each zone for when the start is invisible and the exact correction when it is not. Akimoto, Sera, Yamato and Yano [1] proved the aged arcsine law in distribution for $0 < \alpha < 1$, and Akimoto [4] computed the moments of the occupation time for the ordinary, the equilibrium and the aged start and showed that they differ for $1 < \alpha < 2$. Theorem D is the local aging law for $1 < \alpha < 2$. Wang, Höll and Barkai [72] treat the infinite-mean case. The golden ratio appears in Wang, Li and Hänggi [71] as the exponent at which the side peaks and the central peak decay at the same rate, with the edge exponent of Cipriani, Denisov and Politi [13], and Lee-Dadswell, Nickel and Gray [46] found the conduction exponent $(3 - \sqrt{5})/2$ by a hydrodynamic argument. What Theorem E adds is the comparison of single points as a theorem and the constant $\mathcal{C}(\varphi)$.

*Local limit theorems and renewal theory.* Doney [21] proved a bivariate local limit theorem in the stable domain, and Berger [5] proved multivariate local large deviations and strong renewal theorems. His Theorem 3.3 applied to the walk of run lengths, together with the fresh-plus identity of Proposition 2.3, gives the window $|k - pN| \le KB_N$ of Theorem A, and Section 3 explains what the rest of the theorem adds and why it needs new work. Melbourne, Pène and Terhesiu [51] prove mixing local limit theorems with multivariate stable limits on Young towers under the invariant measure. For the walk they give one point at the stable scale for the stationary initial run, with the first and the last run restricted to fixed lengths. In the i.i.d. renewal setting Theorems A and C remove these restrictions. We use the local large deviation bound of Caravenna and Doney [10] (see also [52]) for sums of a fixed number of variables. The local one-big-jump theorems for sums of independent variables are due to Denisov, Dieker and Shneer [16] and Berger, Birkner and Yuan [7], and the one under exactly the local condition (LR) is [6, Theorem 2.4]. That theorem gives the probability that a fixed number of plus runs adds up to a given large value. It does not give Theorem B directly. The number of runs before step $N$ is random and tied to the run lengths, the big jump may be made by the run in progress at time $N$, which no statement about complete sums sees, and the kernel must hold uniformly up to the endpoint, where only boundedly many minus steps remain. The proof of Theorem B does not use these theorems. The second-order coefficient rests on the second-order renewal theorem of Frenk [28], see also Rogozin [60] and Doney [23], and the monotonicity of $u_-$ on de Bruijn and Erdős [14]. Lemma 6.18 is a difference form of the local big-jump theorems of Doney [20], of Denisov, Dieker and Shneer and of Berger, which give $f^{*m}(n)$ up to a factor $1 + o(1)$ and so say nothing about its differences. For a fixed number of summands the difference form is within reach of the expansions of Barbe and McCormick [3], and what is new is the uniformity in $m$. The bulk mode rests on Kwaśnicki's theorem that stable densities are bell-shaped [43]. The infinite-mean case $0 < \alpha < 1$, where the local theory rests on strong renewal theorems [22, 10] and the bulk is the Lamperti law, is the subject of the companion paper [57], which uses the

exact identities of Section 2.

*The walk.* The walk is the directionally reinforced random walk of Mauldin, Monticino and von Weizsäcker [48], with limit laws by Horváth and Shao [33], and the persistent random walk of Cénac, Le Ny, de Loynes and Offret [11, 12]. For geometric run lengths its crossings and a related walk with a Landau profile are treated in [55, 56]. Magdziarz, Scheffler, Straka and Żebrowski [47] prove limit theorems for Lévy walks in continuous space and time, and Theorem A is the local form of the marginal of these limits at one time. Godrèche and Luck [31] give the transform of the occupation time, which is the generating function of the fresh-plus identity.

### 1.4 Structure of the paper

Everything rests on the exact identities of Section 2, which write $P_N(k)$ through products of renewal kernels of the two signs and reduce every initial run to the two fresh laws. Section 3 proves Theorem A and Section 4 Theorem B, with the infinite density, the examples and the window theorem. Section 5 proves Theorem C, its corollaries for the stationary start, and Theorem D. Section 6 proves Theorems E and F and the results on the extrema near the ends, and states Theorem G. Section 7 treats the queue. Section 8 lists the open problems and Section 9 describes the computations. Appendix A collects classical inputs, and Appendix B proves Theorem G.

## 2 Setting, hypotheses and exact identities

### 2.1 Runs, initial run and the occupation count

The walk takes $N$ steps and each step has a sign, $+$ or $-$. A plus step moves right and a minus step moves left. The signs come in maximal runs of equal sign, and consecutive runs have opposite signs. All runs after the first are independent. A run of sign $\sigma \in \{+,-\}$ has a length distributed as $X_\sigma$, a random variable on $\{1,2,\dots\}$ with mass function $f_\sigma(n) = \mathbb{P}(X_\sigma = n)$ and survival function

$$T_\sigma(x) = \mathbb{P}(X_\sigma \geq x), \qquad x \in \mathbb{R}.$$

We extend $f_\sigma$ by zero to nonpositive integers, and we write $\bar\sigma$ for the sign opposite to $\sigma$.

We call the first run the initial run. It has a sign $\Sigma$ and a length $D$, and the pair $(\Sigma, D)$ is independent of all later runs. We put

$$\nu_\sigma(d) = \mathbb{P}(\Sigma = \sigma,\, D = d), \qquad Q_\sigma(N) = \mathbb{P}(\Sigma = \sigma,\, D \geq N),$$

and we assume that $D$ is finite almost surely. A fresh start of sign $\sigma_0$ has $\nu_{\sigma_0} = f_{\sigma_0}$ and $\nu_{\bar\sigma_0} = 0$, so its initial run is an ordinary run. The stationary initial run has $\nu_\sigma = T_\sigma/s$ with $s$ as in (2.1) below. It is the law of the current run seen from a typical time of a two-sided stationary sequence of runs, and it is the equilibrated start of the physics literature.

Let $A_N$ be the number of plus steps among $1,\dots,N$, and

$$P_N(k) = \mathbb{P}(A_N = k), \qquad 0 \leq k \leq N.$$

The walk is at $2A_N - N$ after $N$ steps. For a fresh start of sign $\sigma$ we write $F_N^\sigma(k)$ for $P_N(k)$, with $F_N^\sigma(k) = 0$ for $k \notin \{0,\dots,N\}$. We also need the law of the count of the starting sign,

$$\Pi_m^+(b) = F_m^+(b), \qquad \Pi_m^-(b) = F_m^-(m-b),$$

so $\Pi_m^\sigma(b)$ is the probability that a fresh walk of sign $\sigma$ has exactly $b$ steps of its own sign among its first $m$ steps.

### 2.2 Hypotheses

The hypotheses carry fixed names, and every statement says which it uses.

(RV) For each sign, $T_\sigma$ is regularly varying at infinity with index $-\alpha_\sigma$, where $1 < \alpha_\sigma < 2$. We write $\alpha_* = \min(\alpha_+, \alpha_-)$.

(LR) For each sign there is $\alpha_\sigma \in (1,2)$ with $f_\sigma(n) \sim \alpha_\sigma T_\sigma(n)/n$ as $n \to \infty$.

(SPAN) For each sign, the support of $X_\sigma$ is not contained in any arithmetic progression $b + h\mathbb{Z}$ with $h \geq 2$.

(LD) For each sign, $|f_\sigma(n) - f_\sigma(n+1)| = O(f_\sigma(n)/n)$ as $n \to \infty$.

(LD2) For each sign, $|f_\sigma(n+1) - 2f_\sigma(n) + f_\sigma(n-1)| = O(f_\sigma(n)/n^2)$ as $n \to \infty$.

(EQ) There are $0 < c \leq C$ with $c \leq T_+(n)/T_-(n) \leq C$ for all large $n$.

A hypothesis assumed "for the plus sign" is assumed for that sign only. Under (LR) we have $T_\sigma(n+1)/T_\sigma(n) = 1-(\alpha_\sigma+o(1))/n$, so the representation theorem [9, Theorem 1.3.1] shows that (LR) implies (RV) with the same index, and (LR) implies (SPAN), since $f_\sigma(n) > 0$ for all large $n$. (LR) says that the mass function is regularly varying, not only the tail. (SPAN) is the aperiodicity needed for local limit theorems. (LD) is used only in Sections 6.4 and 6.5, and (LD2) and (EQ) only in Section 6.5 and Appendix B. (LD) and (LD2) hold for instance when $f_\sigma(n) \propto n^{-\alpha_\sigma - 1}$ for large $n$. Under (LR), (LD2) implies (LD) (Lemma B.1). (EQ) says that the two scales are comparable. It forces $\alpha_+ = \alpha_-$. Under (RV) the runs have finite means, and we put

$$\mu_\sigma = \mathbb{E}X_\sigma, \qquad s = \mu_+ + \mu_-, \qquad p = \mu_+/s, \tag{2.1}$$

so $p$ is the long-run fraction of plus steps and $A_N/N \to p$ in probability.

*Constants.* Unindexed $C$ and $c$ are positive constants that depend only on the laws of the runs and may change from one use to the next. A constant with a numerical index, such as $C_3$ or $c_2$, or named after a source, such as the constant $C_{\mathrm{LD}}$ of the local large deviation bound in Lemma A.1(iii), which is not tied to (LD), keeps the value it receives where it is introduced to the end of the paper, and no index is used twice. A constant indexed by a parameter, such as $C_\eta$, also depends on it and is local to its statement or proof.

### 2.3 Scales and the moving profile

Under (RV) choose for each sign an increasing function $a_\sigma : (0,\infty) \to (0,\infty)$ with $t\,T_\sigma(a_\sigma(t)) \to 1$ as $t \to \infty$. So $a_\sigma$ is regularly varying with index $1/\alpha_\sigma$, and $a_\sigma(m)$ is the scale of the stable fluctuation of a sum of $m$ runs of sign $\sigma$. Put

$$b_{\sigma,N} = a_\sigma(N/s), \qquad B_N = \max(b_{+,N}, b_{-,N}), \qquad \theta_{\sigma,N} = b_{\sigma,N}/B_N \in (0,1], \tag{2.2}$$

and $x_{N,k} = (k - pN)/B_N$. The walk makes about $N/s$ runs of each sign in $N$ steps, so $b_{\sigma,N}$ is the fluctuation scale of the total length of the runs of sign $\sigma$, and $B_N$ is the larger of the two. When $\alpha_+ \neq \alpha_-$ one ratio $\theta_{\sigma,N}$ tends to zero, and when the indices are equal the ratios may oscillate, so we keep them in the statements. We collect the facts about regular variation that recur.

**Lemma 2.1** (regular variation)**.** *Let $f$ be regularly varying with index $\varrho$.*

*(i) (Uniform convergence.) The supremum of $|f(x)/f(y) - 1|$ over $x, y \geq x_0$ with $|x/y - 1| \leq r$ tends to zero as $x_0 \to \infty$ and $r \to 0$ [9, Theorems 1.2.1 and 1.5.2].*

*(ii) (Potter's bounds [9, Theorem 1.5.6].) For every $\delta > 0$ there is $x_\delta$ with $f(y)/f(x) \le 2\max\{(y/x)^{\varrho+\delta}, (y/x)^{\varrho-\delta}\}$ for $x, y \ge x_\delta$. Under (RV), for every $c \ge 1$ there is $C_c$ with $T_\sigma(x/c) \le C_c T_\sigma(x)$ for $x \ge c$.*

*(iii) [9, Theorem 1.5.3] If $\varrho > 0$ and $f$ is bounded above and away from zero on compact subsets of $[1,\infty)$, then $\sup_{1\le y\le x} f(y) \le C_f f(x)$ for $x \ge 1$.*

*(iv) (Karamata [9, Propositions 1.5.8 and 1.5.10].) Under (RV) there is $C_{\mathrm{K}}$ such that, for both signs and all $w \ge 1$,*

$$\sum_{n<w} n\, T_\sigma(n) \le C_{\mathrm{K}}\, w^2 T_\sigma(w), \qquad \sum_{n\ge w} T_\sigma(n) \le C_{\mathrm{K}}\, w T_\sigma(w), \tag{2.3}$$

*and $\sum_{n\ge w} T_\sigma(n) \sim w T_\sigma(w)/(\alpha_\sigma - 1)$ as $w \to \infty$.*

*(v) Under (RV), $B_N$ is nondecreasing and regularly varying with index $1/\alpha_*$. Moreover $B_N T_\sigma(N) \to 0$, $N T_\sigma(b_{\sigma,N}) \to s$ and $B_N \log N = o(N)$.*

*(vi) Under (RV), for both signs, all large $N$ and all $L \ge b_{\sigma,N}$,*

$$\frac{s}{4}\Big(\frac{L}{b_{\sigma,N}}\Big)^{-\alpha_\sigma - 1} \le N T_\sigma(L) \le 4s\Big(\frac{L}{b_{\sigma,N}}\Big)^{-(\alpha_\sigma+1)/2}. \tag{2.4}$$

The proof is routine and is in Appendix A.1.

Let $K_\alpha = -\Gamma(1-\alpha) > 0$ for $1 < \alpha < 2$, and let $Z_+$ and $Z_-$ be independent with

$$\mathbb{E}e^{itZ_\sigma} = \exp\big(K_{\alpha_\sigma}(-it)^{\alpha_\sigma}\big),$$

with the principal branch. These are the totally skewed stable laws that arise as limits of centered sums of the runs, and we write $g_\sigma$ for the density of $Z_\sigma$. For $\theta = (\theta_+, \theta_-) \in [0,1]^2$ let $h_\theta$ be the density of

$$W_\theta = (1-p)\,\theta_+ Z_+ - p\,\theta_- Z_-, \tag{2.5}$$

let $\Theta$ be the compact set of $\theta \in [0,1]^2$ with $\max(\theta_+, \theta_-) = 1$, and put $\theta_N = (\theta_{+,N}, \theta_{-,N}) \in \Theta$. The moving profile is $h_N = h_{\theta_N}$. It is a stable density when $\alpha_+ = \alpha_-$, and otherwise a convolution of two stable densities with different indices. We write $H_N = \sup_x h_N(x)$ for its height. For each $N$ let $\tau = \tau_N$ be a sign with $b_{\tau,N} = B_N$, taking $\tau = +$ if the two scales are equal. We call $\tau$ the dominant sign and $\bar\tau$ the other sign, and we put $\varsigma_+ = 1$ and $\varsigma_- = -1$.

**Lemma 2.2** (the profile)**.** *Assume (RV).*

*(a) There are $C_1, C_2$ with $\sup_N H_N \le C_1$ and $|h_N(x) - h_N(y)| \le C_2|x-y|$ for all $N$ and $x, y$.*

*(b) $\varepsilon_h(R) = \sup_N \sup_{|x|\ge R} h_N(x)$ and $\varepsilon'_h(R) = \sup_N \int_{|x|>R} h_N(x)\,dx$ tend to zero as $R \to \infty$.*

*(c) The map $\theta \mapsto h_\theta$ is continuous from $\Theta$ into the bounded continuous functions with the uniform norm. So $h_N \to h_\theta$ uniformly along every subsequence with $\theta_N \to \theta$, and $\|h_{N'} - h_N\|_\infty \to 0$ whenever $N'/N \to 1$.*

*(d) $\inf_N H_N > 0$, and $\inf_N \min_{|x|\le K} h_N(x) > 0$ for every $K \ge 0$.*

*(e) For every $N$ and $x \in \mathbb{R}$,*

$$h_N(x) = \frac{s}{\mu_{\bar\tau}} \int_{\mathbb{R}} g_{\bar\tau}(y)\, g_\tau\Big(\varsigma_\tau \frac{sx}{\mu_{\bar\tau}} + \frac{\mu_\tau}{\mu_{\bar\tau}}\theta_{\bar\tau,N}\, y\Big) dy. \tag{2.6}$$

The proof is in Appendix A.1.

### 2.4 Renewal kernels

For each sign let $u_\sigma$ be the renewal mass function of the runs of sign $\sigma$,

$$u_\sigma(0) = 1, \qquad u_\sigma(j) = \sum_{m \geq 1} f_\sigma^{*m}(j) \quad (j \geq 1),$$

the probability that some partial sum of independent copies of $X_\sigma$ equals $j$. We also need

$$v_\sigma(j) = \sum_{m \geq 1} m\, f_\sigma^{*m}(j),$$

and the iterated kernels $u_m^\sigma = f_\sigma^{*m}$ for $m \geq 0$, with $u_0^\sigma = \delta_0$, and, for $m \geq 1$,

$$q_m^\sigma(k) = \sum_{r<k} u_{m-1}^\sigma(r)\, T_\sigma(k-r) \quad (k \geq 1), \qquad q_m^\sigma(k) = 0 \quad (k \leq 0).$$

This is the convolution of $u_{m-1}^\sigma$ with $T_\sigma$ restricted to positive arguments of $T_\sigma$. The restriction matters, since $T_\sigma = 1$ at nonpositive arguments. Let $X_1^\sigma, X_2^\sigma, \dots$ be independent copies of $X_\sigma$ and $S_m^\sigma = X_1^\sigma + \cdots + X_m^\sigma$, with $S_0^\sigma = 0$. Then $S_m^\sigma$ has the law $u_m^\sigma$, and since $X_m^\sigma$ is independent of $S_{m-1}^\sigma$, for $m, k \geq 1$

$$\mathbb{P}\big(S_{m-1}^\sigma < k \leq S_m^\sigma\big) = \sum_{r<k} u_{m-1}^\sigma(r)\, T_\sigma(k-r) = q_m^\sigma(k). \tag{2.7}$$

So $q_m^\sigma(k)$ is the probability that the $m$-th run covers level $k$ in the time of its sign. Every run has length at least 1, so $S_m^\sigma$ is strictly increasing in $m$, and for each $k \geq 1$ exactly one $m \geq 1$ has $S_{m-1}^\sigma < k \leq S_m^\sigma$. Hence, for $k \geq 1$,

$$\sum_{m \geq 1} u_m^\sigma(k) = u_\sigma(k), \qquad \sum_{m \geq 1} q_m^\sigma(k) = 1, \qquad \sum_{m \geq 1} q_{m+1}^\sigma(k) = 1 - T_\sigma(k), \tag{2.8}$$

where the third sum is the second minus its term $q_1^\sigma = T_\sigma$. Summing the middle expression of (2.7) over $m$ gives the covering identity

$$\sum_{m=1}^{k} T_\sigma(m)\, u_\sigma(k-m) = 1 \qquad (k \geq 1). \tag{2.9}$$

Under (RV) and (SPAN) for the sign $\sigma$ the renewal theorem holds,

$$u_\sigma(j) \to 1/\mu_\sigma \qquad (j \to \infty), \tag{2.10}$$

and we prove it in local form in Lemma 3.1 (see also [26, Section XIII.3]).

The boundary kernel at the plus end records where a plus run can begin. For a start of sign $\sigma_0$ let

$$\kappa_+^{(\sigma_0)}(j) = \mathbb{P}\big(\text{a plus run starts immediately after the } j\text{-th minus step}\big),$$

where for $j = 0$ the event is that a plus run starts at step 1. Count time in minus steps only. In both cases the ends of the minus runs form a renewal process with interarrival law $X_-$, started at 0. A plus run starts right after the $j$-th minus step exactly when $j$ is a renewal epoch, with one exception. For a minus start the epoch $j = 0$ is followed by the first minus run. Hence

$$\kappa_+^{(+)}(j) = u_-(j) \quad (j \geq 0), \qquad \kappa_+^{(-)}(j) = u_-(j)\mathbf{1}_{\{j \geq 1\}}. \tag{2.11}$$

The kernel at the minus end is defined in the same way with the signs exchanged. Every statement below is made at the plus end and holds at the minus end with the signs exchanged.

We refine $\kappa_+^{(\sigma_0)}(i)$ by the number of plus steps before the run. For $i \geq 0$ and $x \geq 0$ let $\rho_i^{(\sigma_0)}(x)$ be the probability that a plus run starts immediately after the $i$-th minus step (at step 1 if $i = 0$) and that exactly $x$ plus steps precede it. At most one $x$ is possible for each realization, so

$$\sum_{x \geq 0} \rho_i^{(\sigma_0)}(x) = \kappa_+^{(\sigma_0)}(i). \tag{2.12}$$

A plus run that starts after the $i$-th minus step is preceded by $m$ complete minus runs with $S_m^- = i$, and by $m$ plus runs for a plus start and $m-1$ plus runs for a minus start. The two families of runs are independent, so

$$\rho_i^{(+)}(x) = \sum_{m \geq 0} u_m^-(i)\, u_m^+(x), \qquad \rho_i^{(-)}(x) = \sum_{m \geq 1} u_m^-(i)\, u_{m-1}^+(x). \tag{2.13}$$

### 2.5 Exact identities

The following identities hold for every $N$ and need no hypothesis.

**Proposition 2.3** (the fresh-plus identity)**.** *For $k, j \geq 1$ with $k + j = N$,*

$$F_N^+(k) = \sum_{m \geq 1} \big[u_m^-(j)\, q_{m+1}^+(k) + u_m^+(k)\, q_m^-(j)\big],$$

*and $F_N^+(N) = T_+(N)$, $F_N^+(0) = 0$. Exchanging the signs gives the fresh-minus identity $F_N^-(k) = \sum_m [u_m^+(k) q_{m+1}^-(j) + u_m^-(j) q_m^+(k)]$ for $k, j \geq 1$, with $F_N^-(0) = T_-(N)$ and $F_N^-(N) = 0$.*

*Proof.* We split according to the run that contains step $N$. For a fresh plus start the run lengths $X_1^+, X_2^+, \dots$ and $X_1^-, X_2^-, \dots$ are independent, and the runs alternate starting with a plus run. So the $i$-th plus run fills the steps in $(S_{i-1}^+ + S_{i-1}^-,\, S_i^+ + S_{i-1}^-]$, the $i$-th minus run fills those in $(S_i^+ + S_{i-1}^-,\, S_i^+ + S_i^-]$, and these intervals partition $\{1, 2, \dots\}$. Let $k + j = N$ with $0 \leq k \leq N$.

If step $N$ lies in the $i$-th plus run, the first $N$ steps contain exactly the $S_{i-1}^-$ minus steps of the first $i-1$ minus runs, so $A_N = k$ means $S_{i-1}^- = j$, and then step $N = k + j$ lies in that run exactly when $S_{i-1}^+ < k \leq S_i^+$. The two events concern independent sequences, so by (2.7) this case has probability $u_{i-1}^-(j)\, q_i^+(k)$. If step $N$ lies in the $i$-th minus run, then $A_N = S_i^+$, so $A_N = k$ means $S_i^+ = k$, and step $N$ lies in that run exactly when $S_{i-1}^- < j \leq S_i^-$. This case has probability $u_i^+(k)\, q_i^-(j)$.

The cases are disjoint. For $k, j \geq 1$ the plus case with $i = 1$ vanishes because $u_0^-(j) = 0$, and $m = i - 1$ in the plus case and $m = i$ in the minus case give the formula. For $k = N$ only the plus case with $i = 1$ survives, with probability $q_1^+(N) = T_+(N)$. For $k = 0$ the plus case is impossible because step $N$ is a plus step, and the minus case is impossible because $S_i^+ \geq 1$. The fresh minus start is the same with the signs exchanged. □

**Proposition 2.4** (the first-run decomposition)**.** *For every initial run and every $0 \leq k \leq N$,*

$$P_N(k) = Q_+(N)\mathbf{1}_{\{k=N\}} + Q_-(N)\mathbf{1}_{\{k=0\}} + \sum_{d<N} \nu_+(d)\, F_{N-d}^-(k-d) + \sum_{d<N} \nu_-(d)\, F_{N-d}^+(k).$$

*In particular $P_N(N) = Q_+(N)$ and $P_N(0) = Q_-(N)$.*

*Proof.* We condition on the first run. If $\Sigma = +$ and $D \geq N$, then $A_N = N$, and this has probability $Q_+(N)$. If $\Sigma = +$ and $D = d < N$, the first $d$ steps are plus steps and steps $d+1, \dots, N$ are the first $N - d$ steps of a fresh walk of sign $-$, independent of $(\Sigma, D)$. So $A_N = d + A'$, where $A'$ has the law $F_{N-d}^-$. The case $\Sigma = -$ is the same with $A_N = 0$

when $D \geq N$ and $A_N$ of law $F^+_{N-d}$ when $D = d < N$. This gives the formula. For $d < N$, Proposition 2.3 gives $F^-_{N-d}(N-d) = 0$ and $F^+_{N-d}(0) = 0$, and $F^+_{N-d}(N) = F^-_{N-d}(-d) = 0$ trivially, so only the atoms survive at $k = N$ and $k = 0$. □

**Lemma 2.5** (the Green identity)**.** *For each sign and every $b \geq 1$,*

$$\sum_{m \geq 1} \Pi^\sigma_m(b) = 1 + \mu_{\bar\sigma}\, u_\sigma(b).$$

*Proof.* Take a fresh walk of sign $\sigma$ continued forever, with runs of sign $\sigma$ of lengths $X^\sigma_1, X^\sigma_2, \dots$ and runs of sign $\bar\sigma$ of lengths $X^{\bar\sigma}_1, X^{\bar\sigma}_2, \dots$, all independent, the $i$-th run of sign $\bar\sigma$ coming right after the $i$-th run of sign $\sigma$. Let $C_m$ be the number of steps of sign $\sigma$ among the first $m$. By Tonelli's theorem $\sum_m \Pi^\sigma_m(b) = \mathbb{E}\,\#\{m \geq 1 : C_m = b\}$. Let $t_b < t_{b+1}$ be the $b$-th and $(b+1)$-th steps of sign $\sigma$. Then $C_m = b$ exactly for $t_b \leq m < t_{b+1}$. If step $t_b$ is not the last step of its run, then $t_{b+1} = t_b + 1$. If it is, it ends the $i$-th run of sign $\sigma$ for the unique $i$ with $S^\sigma_i = b$, and then $t_{b+1} - t_b = 1 + X^{\bar\sigma}_i$. Hence $\#\{m : C_m = b\} = 1 + \sum_{i \geq 1} \mathbf{1}_{\{S^\sigma_i = b\}} X^{\bar\sigma}_i$. Since $X^{\bar\sigma}_i$ is independent of $S^\sigma_i$, taking expectations and using (2.8) gives $1 + \mu_{\bar\sigma} u_\sigma(b)$. □

The next identity isolates the final run, which is the run that contains step $N$.

**Proposition 2.6** (the final-run identity)**.** *For a fresh plus start, $N \geq 1$ and $0 \leq j < N$,*

$$\begin{aligned} P_N(N-j) = &\sum_{r \geq 0} u^-_r(j)\, \mathbb{E}\big[T_+(N-j-S^+_r);\, S^+_r \leq N-j-1\big] \\ &+ \sum_{m=1}^{j} T_-(m) \sum_{r \geq 0} u^-_r(j-m)\, u^+_{r+1}(N-j). \end{aligned} \tag{2.14}$$

*The first sum is the event that the final run is a plus run, and the second the event that it is a minus run of length at least $m$ that starts at step $N-m+1$. Only $r \leq j$ contributes, and for $j = 0$ the identity reads $P_N(N) = T_+(N)$.*

*Proof.* We use the notation of the proof of Proposition 2.3. The $(r+1)$-th plus run starts at step $S^+_r + S^-_r + 1$ with $S^+_r$ plus steps before it. It is the final run and $A_N = N - j$ exactly when the steps before it contain $j$ minus steps, that is $S^-_r = j$, and its length satisfies $X^+_{r+1} \geq N - j - S^+_r \geq 1$. These conditions involve independent variables, which gives the term $r$ of the first sum. The $(r+1)$-th minus run starts at step $S^+_{r+1} + S^-_r + 1$ with $S^+_{r+1}$ plus steps before it. If it is the final run and starts at step $N-m+1$, then $A_N = N - j$ means $S^+_{r+1} = N - j$, hence $S^-_r = N - m - (N-j) = j - m$, which needs $1 \leq m \leq j$, and its length satisfies $X^-_{r+1} \geq m$. This gives the second sum. Exactly one run is final, so the cases are disjoint. Finally $u^-_r(i) = 0$ for $r > i$ and $u^-_r(0) = \mathbf{1}_{\{r=0\}}$. □

By (2.13) the first sum of (2.14) equals $\sum_{x \leq N-j-1} \rho^{(+)}_j(x) T_+(N-j-x)$. Section 9 describes how these identities compute the exact law.

**Lemma 2.7** (the counting lemma)**.** *Fix $j \geq 0$ and $N > j$. On the event $\{A_N = N-j\}$ at most $j+1$ plus runs meet the first $N$ steps (at most $j$ if the first run is a minus run), and at least one of them has at least $(N-j)/(j+1)$ steps among the first $N$. For a fresh start of either sign and every $L \geq 1$, the probability that $A_N = N-j$ and two plus runs that meet the first $N$ steps have length at least $L$ is at most $(j+1)^2 T_+(L)^2$.*

*Proof.* On $\{A_N = N-j\}$ the first $N$ steps contain $j$ minus steps. Let $r$ be the number of plus runs that meet them. Between two consecutive such runs lies a minus run inside the first $N$ steps, so $r - 1 \leq j$, and if the first run is a minus run it adds one more minus step, so $r \leq j$. The $N-j$ plus steps are shared among $r \leq j+1$ runs, so one of them has at least $(N-j)/(j+1)$ steps. For a fresh start the plus runs have independent lengths

$X_1^+, X_2^+, \dots$, and on the event of the lemma there are $1 \le a < b \le j+1$ with $X_a^+ \ge L$ and $X_b^+ \ge L$. The union bound gives at most $\binom{j+1}{2} T_+(L)^2 \le (j+1)^2 T_+(L)^2$, and counting only the steps among the first $N$ makes the event smaller. □

## 3 The bulk

The bulk is the window $|k-pN| = O(B_N)$, where the law of $A_N$ is stable. The theorem of this section is a statement about every point of the range $\{0, \dots, N\}$ for a fresh start, with one error term and one bound that are uniform in $k$.

**Theorem A** (the bulk, fresh start)**.** *Assume (RV) and (SPAN). For each starting sign $\sigma_0$,*

*(a)* $\sup_{k\in\mathbb{Z}} \bigl| B_N F_N^{\sigma_0}(k) - h_N(x_{N,k}) \bigr| \to 0$.

*(b)* $\sup_{N\ge 1} \sup_k B_N F_N^{\sigma_0}(k) < \infty$.

*(c) for every $K > 0$,* $\displaystyle\sup_{|k-pN|\le KB_N} \Bigl| \frac{B_N F_N^{\sigma_0}(k)}{h_N(x_{N,k})} - 1 \Bigr| \to 0$.

*The convergence covers every relation between the two scales $b_{+,N}$ and $b_{-,N}$, including ratios that oscillate.*

*What is new.* Part (c) on a fixed window $|k-pN| \le KB_N$ also follows from Berger's strong renewal theorem [5, Theorem 3.3] for the walk of run lengths and the fresh-plus identity. The content is in (a) and (b). They control every point, including the remote points with $|k-pN| \gg B_N$, by one error term and one bound $C/B_N$, under (RV) and (SPAN) alone.

The fresh identity writes $F_N^{\sigma_0}(k)$ as a sum over the run index $m$ of products of a kernel of each sign, and away from the bulk most of the sum comes from indices at which one factor is a local large deviation. Local bounds for the walk of run lengths away from its mean direction [5, Theorems 4.1 to 4.3] assume a local condition on the run law, which (RV) does not give. We bound only the factor of the dominant sign, by $C/B_N$, and sum the other factor, whose total mass is at most one. The two scales may differ and their ratio may oscillate, and Lemma 3.1 resolves the index of the covering run on the scale of its own sign. The classical facts on sums of runs (Lemma A.1) and the kernel bounds ((K1) to (K4), Lemmas A.2 and A.3) are in Appendix A.

### 3.1 The index of the covering run

The next lemma is the renewal theorem in local form, with the index of the run that covers a level resolved on its natural scale. For $j \ge 1$ and $K > 0$ let

$$I_j^\sigma(K) = \bigl\{ m \ge 1 :\ |j - m\mu_\sigma| \le K\, a_\sigma(j/\mu_\sigma) \bigr\}, \qquad y_m^\sigma(j) = \frac{j - m\mu_\sigma}{a_\sigma(j/\mu_\sigma)}.$$

We call $I_j^\sigma(K)$ the index window. For a modulus $\omega$ (a function with $\omega(r) \to 0$ as $r \to 0$) let $\mathcal{F}(\omega)$ be the set of functions $\psi$ on $\mathbb{R}$ with $|\psi| \le 1$ and $|\psi(y) - \psi(y')| \le \omega(|y-y'|)$.

**Lemma 3.1** (the index of the covering run)**.** *Assume (RV) and (SPAN), and let* $\mathsf{k}$ *be a kernel of sign $\sigma$.*

*(a) For every $K \ge 2$,*

$$\limsup_{j\to\infty} \sum_{m\notin I_j^\sigma(K)} \mathsf{k}_m(j) \le \eta_\sigma(K) = \mathbb{P}(Z_\sigma \ge K-1) + e^{C_-(\sigma)+1-K/2}.$$

*(b) For every modulus $\omega$,*

$$\sup_{\psi\in\mathcal{F}(\omega)} \Bigl| \sum_{m\ge 1} \mathsf{k}_m(j)\, \psi\bigl(y_m^\sigma(j)\bigr) - \frac{c_{\mathsf{k}}}{\mu_\sigma} \int_{\mathbb{R}} g_\sigma(y)\psi(y)\, dy \Bigr| \to 0 \qquad (j\to\infty).$$

*In particular* $\sum_m u^\sigma_m(j) \to 1/\mu_\sigma$*, which is* (2.10)*.*

*Proof.* Write $\mu = \mu_\sigma$, $a = a_\sigma$, $J = J_\sigma(j)$ and $m_\pm = (j \pm Ka(j/\mu))/\mu$. Note that $a(x)/x \to 0$.

(a) Write $I = I^\sigma_j(K)$. By (K3), $\sum_{m\notin I} u_m(j)$ and $\sum_{m\notin I} q_m(j)$ are at most $\mathbb{P}(J \notin I)$, and $\sum_{m\notin I} q_{m+1}(j) \le \mathbb{P}(J-1 \notin I)$. All three are at most $\mathbb{P}(J \le m_- + 1) + \mathbb{P}(J > m_+)$. A count of runs below its typical value forces the sum of the runs above its mean, a right deviation, and a count above its typical value forces the sum below its mean, a left deviation.

*Too few runs.* Let $m' = \lfloor m_- + 1 \rfloor$. If $J \le m'$, then $S_{m'} \ge j$. Now $j - m'\mu \ge Ka(j/\mu) - \mu$, and $a(m') \le a(j/\mu)$ for large $j$. So

$$\mathbb{P}(J \le m_- + 1) \le \mathbb{P}\Big(\frac{S_{m'} - m'\mu}{a(m')} \ge K - \frac{\mu}{a(j/\mu)}\Big) \le \mathbb{P}(Z_{m'} \ge K - 1)$$

for large $j$. Since $m' \to \infty$, Lemma A.1(i) gives $\limsup_j \mathbb{P}(Z_{m'} \ge K-1) \le \mathbb{P}(Z_\sigma \ge K-1)$.

*Too many runs.* Let $m'' = \lfloor m_+ \rfloor$. If $J > m_+$, then $S_{m''} < j$, and $m''\mu - j \ge Ka(j/\mu) - \mu$. The local bound Lemma A.1(iii) does not apply to this left deviation, and Lemma A.1(iv) gives

$$\mathbb{P}(J > m_+) \le \exp\Big(C_- - \frac{Ka(j/\mu) - \mu}{a(m'')}\Big) \le e^{C_- + 1 - K/2}$$

for large $j$, since $m'' \le 2j/\mu$, $a(2j/\mu) \le 2a(j/\mu)$ and $\mu/a(j/\mu) \le 1$ eventually.

(b) Fix $K \ge 2$. The terms with $m \notin I^\sigma_j(K)$ contribute at most $\eta_\sigma(K) + o(1)$ by (a), and $\int_{|y|>K} g_\sigma \le \mathbb{P}(|Z_\sigma| > K)$. For $m \in I^\sigma_j(K)$ we have $m\mu/j \to 1$ uniformly, so $a(m)/a(j/\mu) \to 1$ uniformly. With $y_m = y^\sigma_m(j) \in [-K, K]$ the argument $(j - m\mu)/a(m) = y_m\, a(j/\mu)/a(m)$ differs from $y_m$ by at most $K|a(j/\mu)/a(m) - 1|$. Lemma A.2 and the uniform continuity of $g_\sigma$ give, uniformly in $m \in I^\sigma_j(K)$,

$$\mathsf{k}_m(j) = \frac{c_\mathsf{k}\, g_\sigma(y_m) + o(1)}{a(j/\mu)}.$$

The points $y_m$ form an arithmetic progression with step $\Delta_j = \mu/a(j/\mu) \to 0$ in $[-K, K]$, with at most $2K/\Delta_j + 1$ points, so the $o(1)$ terms add up to $o(1)$. What remains is $c_\mathsf{k}/\mu$ times a Riemann sum of $g_\sigma\psi$ over $[-K, K]$ with mesh $\Delta_j$, whose distance from $\int_{-K}^{K} g_\sigma\psi$ is at most $2K(\omega_{g_\sigma}(\Delta_j) + \|g_\sigma\|_\infty \omega(\Delta_j)) + 2\Delta_j\|g_\sigma\|_\infty$, uniformly in $\psi \in \mathcal{F}(\omega)$. Hence the $\limsup$ of the supremum in (b) is at most $\eta_\sigma(K) + (c_\mathsf{k}/\mu)\mathbb{P}(|Z_\sigma| > K)$, and we let $K \to \infty$. With $\psi \equiv 1$ and $\mathsf{k} = u^\sigma$ we get (2.10). □

### 3.2 Proof of Theorem A

We write $F^{\sigma_0}_N(k)$ through the fresh identity as a sum over the index $m$ of products of a kernel of sign $+$ and a kernel of sign $-$. In each product we resolve the factor of the dominant sign $\tau$ by its local limit and sum the factor of the other sign through Lemma 3.1. The points near the ends, the indices far from the index window of the other sign, the remote points and the central points are treated in turn, and no assumption on the ratio of the two scales enters.

*Proof of Theorem A(a).* Throughout we fix

$$\varepsilon = \min\Big\{\frac{p}{2},\ \frac{1-p}{2},\ \frac{1}{16\mu_+},\ \frac{1}{16\mu_-}\Big\},$$

and we write $\mu_{\max} = \max_\sigma \mu_\sigma$, $\mu_{\min} = \min_\sigma \mu_\sigma$, $\eta(K) = \max_\sigma \eta_\sigma(K)$ and $\delta(n) = \sup_{m\ge n} \max_{\sigma,\mathsf{k}} \delta_\mathsf{k}(m)$, the maximum over the six kernels. Then $\eta(K) \to 0$ as $K \to \infty$ and $\delta(n) \to 0$ as $n \to \infty$. A constant called $C_i$ or $c_i$ depends only on the laws of the runs and on $\varepsilon$, never on $N$, $k$, $K$ or $R$.

**Step 1. The fresh identity as a sum of products.** We write $\mathcal{E}_N(k) = |B_N F_N^{\sigma_0}(k) - h_N(x_{N,k})|$ for $k \in \mathbb{Z}$. Let $1 \le k \le N-1$ and write $k_+ = k$ and $k_- = N-k$. By Proposition 2.3,

$$F_N^{\sigma_0}(k) = \sum_{i=1}^{2} \sum_{m \ge 1} \mathsf{k}_m^{+,i}(k_+)\, \mathsf{k}_m^{-,i}(k_-), \tag{3.1}$$

where $(\mathsf{k}^{+,1}, \mathsf{k}^{-,1}, \mathsf{k}^{+,2}, \mathsf{k}^{-,2})$ is $(q_{m+1}^+, u_m^-, u_m^+, q_m^-)$ for $\sigma_0 = +$ and $(u_m^+, q_{m+1}^-, q_m^+, u_m^-)$ for $\sigma_0 = -$. Each is a kernel of the sign it carries, and in both cases

$$c_{\mathsf{k}^{+,1}} c_{\mathsf{k}^{-,1}} + c_{\mathsf{k}^{+,2}} c_{\mathsf{k}^{-,2}} = \mu_+ + \mu_- = s. \tag{3.2}$$

The endpoint atoms are $T_\sigma(N)$ or $0$, and $B_N T_\sigma(N) \to 0$ by Lemma 2.1(v). The profile at $x_{N,k}$ for $k \le 0$ or $k \ge N$ is at most $\varepsilon_h(\min(p, 1-p)N/B_N)$, which tends to zero by Lemma 2.2(b). So it remains to treat $1 \le k \le N-1$.

We also need a uniform bound for one factor. Let $N_1$ be so large that for $N \ge N_1$ and both signs $C_5(\sigma,\varepsilon) b_{\sigma,N} e_{\sigma,N} \le 1$ and $a_\sigma(N/s) \le (8/\varepsilon) a_\sigma(\varepsilon N/(8\mu_\sigma))$. The second inequality holds eventually because the ratio tends to $(8\mu_\sigma/(\varepsilon s))^{1/\alpha_\sigma} < 8/\varepsilon$. Then for $N \ge N_1$, $\varepsilon N \le j \le N$, all $m \ge 1$ and every kernel $\mathsf{k}$ of sign $\sigma$,

$$\mathsf{k}_m(j) \le \frac{C_6}{b_{\sigma,N}}, \qquad C_6 = 1 + 8C_3/\varepsilon. \tag{3.3}$$

Indeed, for $8m\mu_\sigma \le \varepsilon N$ this is (A.1) with $\beta = \varepsilon$, and for larger $m$ Lemma A.2 gives $\mathsf{k}_m(j) \le C_3/a_\sigma(\varepsilon N/(8\mu_\sigma)) \le 8C_3/(\varepsilon b_{\sigma,N})$. For the dominant sign the bound is $C_6/B_N$. For the other sign it may be much larger, and we never use it there. In every product of (3.1) we bound the factor of the dominant sign and sum the factor of the other sign.

**Step 2. Small counts and the ends of the range.** Let $k_- < \varepsilon N$. By (K2) every nonzero product in (3.1) has $m \le k_- < \varepsilon N$. The plus factor sits at $k_+ > (1-\varepsilon)N \ge N/2$, and $8m\mu_+ \le N/2$, so Lemma A.3 with $\beta = 1/2$ bounds it by $C_5(+, \frac{1}{2}) e_{+,N}$. The minus factors add up to at most $1$ by (K1). Hence $B_N F_N^{\sigma_0}(k) \le 2C_5(+,\frac{1}{2}) B_N e_{+,N} \to 0$ uniformly in these $k$, and the case $k_+ < \varepsilon N$ is the same with the signs exchanged. In both cases $|k - pN| \ge \varepsilon N$, so the profile is at most $\varepsilon_h(\varepsilon N/B_N) \to 0$. So

$$\sup_{\min(k, N-k) < \varepsilon N} \mathcal{E}_N(k) \to 0. \tag{3.4}$$

From now on $\varepsilon N \le k_+, k_- \le N$, and (3.3) applies to both factors.

**Step 3. Indices far from the index window of the other sign.** Fix $K \ge 2$ and write $I = I_{k_{\bar\tau}}^{\bar\tau}(K)$. By (3.3) for the dominant factor and Lemma 3.1(a) for the other factor at the level $k_{\bar\tau} \ge \varepsilon N$,

$$\limsup_{N\to\infty} \sup_{\varepsilon N \le k \le (1-\varepsilon)N} B_N \sum_{i=1}^{2} \sum_{m \notin I} \mathsf{k}_m^{+,i}(k_+) \mathsf{k}_m^{-,i}(k_-) \le 2C_6\, \eta(K). \tag{3.5}$$

For $m \in I$ we record three facts. Let $N \ge N_2(K)$, where $N_2(K) \ge N_1$ is so large that $K a_\sigma(N/\mu_\sigma) \le \varepsilon N/2$ for both signs. Then every $m \in I$ satisfies

$$\frac{\varepsilon N}{2\mu_{\bar\tau}} \le m \le \frac{2N}{\mu_{\bar\tau}}, \tag{3.6}$$

so Lemma A.2 applies to the dominant factor with the error $\delta(\lceil \varepsilon N/(2\mu_{\max})\rceil) \to 0$. Next, the definitions of $p$ and $s$ give the exact identity

$$k_\tau - m\mu_\tau = \varsigma_\tau \frac{s(k - pN)}{\mu_{\bar\tau}} + \frac{\mu_\tau}{\mu_{\bar\tau}} \big(k_{\bar\tau} - m\mu_{\bar\tau}\big). \tag{3.7}$$

For $\tau = +$ the right side is $(sk - \mu_+ N + \mu_+ k_-)/\mu_- - m\mu_+ = k - m\mu_+$, and $\tau = -$ is the same computation. Finally, by Lemma 2.1(ii) there is $N_3 \geq N_2(K)$ such that for $N \geq N_3$

$$a_\sigma(N/\mu_\sigma) \leq c_1\, b_{\sigma,N}, \qquad a_\sigma(2N/\mu_{\min}) \leq c_2 b_{\sigma,N}, \qquad B_N \leq c_3\, a_\tau(\varepsilon N/(2\mu_{\max})), \tag{3.8}$$

with $c_1 = s/\mu_{\min}$, $c_2 = 2s/\mu_{\min}$ and $c_3 = 2/\varepsilon$, because the ratios tend to $(s/\mu_\sigma)^{1/\alpha_\sigma}$, $(2s/\mu_{\min})^{1/\alpha_\sigma}$ and $(2\mu_{\max}/(\varepsilon s))^{1/\alpha_\tau}$.

**Step 4. Remote points.** Let $R \geq 4\mu_{\max}/\mu_{\min}$ and put $K = K(R) = sR/(2c_1\mu_{\max}) = \mu_{\min}R/(2\mu_{\max}) \geq 2$. Let $N \geq N_3$ with $N_3$ taken for this $K$, and let $\varepsilon N \leq k \leq (1-\varepsilon)N$ and $|k - pN| > RB_N$. For $m \in I$ we have $|k_{\bar\tau} - m\mu_{\bar\tau}| \leq Ka_{\bar\tau}(N/\mu_{\bar\tau}) \leq c_1 KB_N$ by (3.8), so (3.7) gives

$$|k_\tau - m\mu_\tau| \geq \frac{sRB_N}{\mu_{\bar\tau}} - \frac{\mu_\tau}{\mu_{\bar\tau}} c_1 KB_N \geq \frac{sRB_N}{2\mu_{\bar\tau}}.$$

By (3.6) and (3.8), $a_\tau(m) \leq c_2 B_N$, so the scaled argument $z = (k_\tau - m\mu_\tau)/a_\tau(m)$ of the dominant factor has modulus at least $c_4 R$ with $c_4 = s/(2c_2\mu_{\max})$. Lemma A.2 and (3.8) give

$$B_N \mathsf{k}_m^{\tau,i}(k_\tau) \leq \frac{B_N}{a_\tau(m)}\big(\mu_{\max}\bar g_\tau(c_4R) + \delta(m)\big) \leq c_3\big(\mu_{\max}\bar g_\tau(c_4R) + \delta(\lceil \varepsilon N/(2\mu_{\max})\rceil)\big).$$

The factors of sign $\bar\tau$ add up to at most 1 for each $i$ by (K1). Together with (3.5) and Lemma 2.2(b),

$$\limsup_{N\to\infty} \sup_{\substack{\varepsilon N \leq k \leq (1-\varepsilon)N \\ |k-pN| > RB_N}} \mathcal{E}_N(k) \leq e(R) = 2c_3\mu_{\max} \max_\sigma \bar g_\sigma(c_4R) + 2C_6\eta(K(R)) + \varepsilon_h(R), \tag{3.9}$$

and $e(R) \to 0$ as $R \to \infty$.

**Step 5. The central sum.** Fix $R \geq 1$ and $K \geq 2$, and let $|k - pN| \leq RB_N$, so $x = x_{N,k} \in [-R, R]$. For large $N$ these points satisfy $\varepsilon N \leq k \leq (1-\varepsilon)N$, since $B_N = o(N)$. Let $m \in I$ and $y_m = y_m^{\bar\tau}(k_{\bar\tau}) \in [-K, K]$. Since $k_+/\mu_+ = N/s + xB_N/\mu_+$ and $k_-/\mu_- = N/s - xB_N/\mu_-$, we have $k_{\bar\tau}/\mu_{\bar\tau} = (N/s)(1+o(1))$ and $m = (N/s)(1+o(1))$, uniformly in $|x| \leq R$ and $m \in I$. Lemma 2.1(i), applied to $a_+$ and to $a_-$ separately, gives

$$a_{\bar\tau}(k_{\bar\tau}/\mu_{\bar\tau}) = b_{\bar\tau,N}(1 + o(1)), \qquad a_\tau(m) = B_N(1 + o(1)),$$

uniformly in $|x| \leq R$ and $m \in I$, whichever sign is dominant. Since $b_{\bar\tau,N} = \theta_{\bar\tau,N} B_N$, the identity (3.7) gives for the argument of the dominant factor

$$z_m = \frac{k_\tau - m\mu_\tau}{a_\tau(m)} = \varsigma_\tau \frac{sx}{\mu_{\bar\tau}}\, r_1 + \frac{\mu_\tau}{\mu_{\bar\tau}} \theta_{\bar\tau,N}\, y_m\, r_2, \qquad r_1 = \frac{B_N}{a_\tau(m)}, \quad r_2 = \frac{a_{\bar\tau}(k_{\bar\tau}/\mu_{\bar\tau})}{b_{\bar\tau,N}} \frac{B_N}{a_\tau(m)},$$

where $r_1, r_2 \to 1$ uniformly. Since $|x| \leq R$, $|y_m| \leq K$ and $\theta_{\bar\tau,N} \leq 1$,

$$\sup_{|x|\leq R} \sup_{m\in I} \big|z_m - \zeta_{N,x}(y_m)\big| \to 0, \qquad \zeta_{N,x}(y) = \varsigma_\tau \frac{sx}{\mu_{\bar\tau}} + \frac{\mu_\tau}{\mu_{\bar\tau}} \theta_{\bar\tau,N}\, y.$$

Lemma A.2, $B_N/a_\tau(m) \to 1$ and the uniform continuity of $g_\tau$ give, uniformly in $|x| \leq R$ and $m \in I$,

$$B_N \mathsf{k}_m^{\tau,i}(k_\tau) = c_{\mathsf{k}^{\tau,i}}\, g_\tau\big(\zeta_{N,x}(y_m)\big) + o(1). \tag{3.10}$$

Let $\psi_{N,x} = g_\tau \circ \zeta_{N,x}/\|g_\tau\|_\infty$. The slope of $\zeta_{N,x}$ is at most $\mu_{\max}/\mu_{\min}$, so $\psi_{N,x} \in \mathcal{F}(\omega)$ for all $N$ and $x$, with $\omega(r) = \max_\sigma \omega_{g_\sigma}(\mu_{\max} r/\mu_{\min})/\|g_\sigma\|_\infty$. We multiply (3.10) by $\mathsf{k}_m^{\bar\tau,i}(k_{\bar\tau})$ and sum over $m \in I$. The $o(1)$ terms add up to $o(1)$ by (K1). Then we add back the indices

$m \notin I$ in the sum of $\mathsf{k}_m^{\bar\tau,i}\psi_{N,x}(y_m)$, at a cost of at most $\eta(K)+o(1)$ by Lemma 3.1(a), and apply Lemma 3.1(b) at the level $k_{\bar\tau} \ge \varepsilon N$. This gives, uniformly in $|x| \le R$,

$$B_N \sum_{m\in I} \mathsf{k}_m^{+,i}(k_+)\mathsf{k}_m^{-,i}(k_-) = \frac{c_{\mathsf{k}^{\tau,i}} c_{\mathsf{k}^{\bar\tau,i}}}{\mu_{\bar\tau}} \int_{\mathbb{R}} g_{\bar\tau}(y)\, g_\tau\big(\zeta_{N,x}(y)\big)dy + O\big(\eta(K)\big) + o(1),$$

with the constant $\mu_{\max}\max_\sigma \|g_\sigma\|_\infty$ in $O(\eta(K))$. We sum over $i$ and use (3.2). By Lemma 2.2(e) the integral with the weight $s/\mu_{\bar\tau}$ is exactly $h_N(x)$. Adding (3.5) for the indices outside $I$,

$$\limsup_{N\to\infty} \sup_{|k-pN|\le RB_N} \mathcal{E}_N(k) \le C_7\,\eta(K), \tag{3.11}$$

with $C_7 = 2C_6 + 2\mu_{\max}\max_\sigma \|g_\sigma\|_\infty$. Since $K$ is arbitrary, the left side is zero.

**Combination.** Let $R \ge 4\mu_{\max}/\mu_{\min}$. Every $k \in \mathbb{Z}$ is an endpoint or lies outside $\{0,\dots,N\}$ (Step 1), or within $\varepsilon N$ of an end (Step 2), or is a remote point with $|k-pN| > RB_N$ (Step 4), or a central point with $|k-pN| \le RB_N$ (Step 5). By (3.4), (3.9) and (3.11), $\limsup_N \sup_{k\in\mathbb{Z}} \mathcal{E}_N(k) \le e(R)$, and we let $R\to\infty$. No step assumed that $\theta_N$ converges, since every error term is a maximum over the two signs of quantities that do not involve $\theta_N$, and $\theta_N$ enters only through the slope of $\zeta_{N,x}$, which is at most $\mu_{\max}/\mu_{\min}$. □

*Proof of Theorem A(b) and (c).* For (b), part (a) and Lemma 2.2(a) give an $N_4$ with $B_N F_N^{\sigma_0}(k) \le C_1 + 1$ for all $N \ge N_4$ and all $k$, and $B_N F_N^{\sigma_0}(k) \le B_N \le B_{N_4}$ for $N < N_4$, since $B_N$ is nondecreasing. For (c), Lemma 2.2(d) gives $c_K > 0$ with $h_N \ge c_K$ on $[-K,K]$ for all $N$, and dividing the uniform error of (a) by $c_K$ gives (c). □

The proof used (SPAN) only through Lemma A.1(ii), and it did not use (LR).

## 4 Big jumps and the boundary layer

Outside the bulk $P_N(k) = o(1/B_N)$ at each point by Theorem A. How much smaller it is depends on the distance of $k$ from $pN$. On the ballistic range $k - pN \asymp N$ the values are of order $T_+(N)$, smaller than $1/B_N$ by a power of $N$. Near $k - pN = KB_N$ with $K$ fixed they are still of order $1/B_N$, with a factor of order $K^{-\alpha_+-1}$ when the plus scale is the larger one (Remark 4.10). This section describes these values and the boundary layer with one kernel.

Throughout we write $j = N - k$ for the number of minus steps. For $k > pN$ put

$$L_k = \frac{k-pN}{1-p}, \qquad \text{so that} \qquad N - L_k = \frac{j}{1-p} = \frac{js}{\mu_-}. \tag{4.1}$$

This is the length of the plus run that moves a typical configuration to the plus count $k$. For a start of sign $\sigma_0$ and $j \ge 1$ put

$$\gamma_+^{(\sigma_0)}(j) = \sum_{i=0}^{j-1} \kappa_+^{(\sigma_0)}(i)\big(1 + \mu_+ u_-(j-i)\big), \qquad \gamma_+^{(\sigma_0)}(0) = 0, \tag{4.2}$$

and define the big-jump kernel by

$$G_N^{(\sigma_0)}(k) = \gamma_+^{(\sigma_0)}(N-k)\, f_+(\lfloor L_k\rfloor) + T_+(\lfloor L_k\rfloor)\, \kappa_+^{(\sigma_0)}(N-k), \qquad pN < k \le N. \tag{4.3}$$

The first term is a complete plus run of length about $L_k$. The coefficient $\gamma_+^{(\sigma_0)}(j)$ counts its admissible positions. The run starts right after the $i$-th minus step, which has probability $\kappa_+^{(\sigma_0)}(i)$, and the fresh minus walk that follows it must hold the other $j-i$ minus steps. By the Green identity, Lemma 2.5, the expected number of lengths of that walk that do

so is $1+\mu_+u_-(j-i)$. The second term is a plus run in progress at time $N$, of length at least $L_k$. By (2.10) and (2.11), $\kappa_+^{(\sigma_0)}(i)\to 1/\mu_-$ and $u_-(i)\to 1/\mu_-$, so a Cesàro average gives

$$\gamma_+^{(\sigma_0)}(j)\sim \frac{js}{\mu_-^2}=\frac{N-L_k}{\mu_-}\qquad (j\to\infty). \tag{4.4}$$

At $k=N-j$ with $j$ fixed the first term is of order $T_+(N)/N$ and the second is $T_+(N)\kappa_+^{(\sigma_0)}(j)$ up to a factor $1+o(1)$, so the kernel reduces to the boundary layer. As $j\to\infty$ the coefficient of the second term tends to $1/\mu_-$, its value in the interior. So one kernel covers the big-jump zone and the boundary layer.

The zone starts at a multiple of $B_N$ when the plus scale is the larger one. Otherwise it starts a little later, and we put

$$\lambda_N=1+\log\frac{B_N}{b_{+,N}}=1+\log\frac{1}{\theta_{+,N}}\ \ge 1. \tag{4.5}$$

So $\lambda_N=1$ when $b_{+,N}=B_N$, for instance for all large $N$ when $\alpha_+<\alpha_-$, and $\lambda_N=O(\log N)$ in all cases.

**Theorem B** (big jumps and the boundary layer)**.** *Let the start be fresh with sign $\sigma_0$, and write $\kappa=\kappa_+^{(\sigma_0)}$ and $G_N=G_N^{(\sigma_0)}$.*

*(a) (The boundary layer.) Assume (RV) for the plus sign. For every fixed $j\ge 0$,*

$$\frac{P_N(N-j)}{T_+(N)}\to\kappa(j).$$

*(b) (The big-jump zone.) Assume (RV) and (SPAN) for both signs and (LR) for the plus sign. Then*

$$\lim_{K\to\infty}\limsup_{N\to\infty}\sup_{pN+K\lambda_NB_N\le k\le N}\Big|\frac{P_N(k)}{G_N(k)}-1\Big|=0,$$

*where $G_N(k)=0$ only if $P_N(k)=0$, and then the ratio is read as $1$. In particular, with the size function*

$$\widehat G_N(k)=\frac{N}{L_k}\,T_+(\lfloor L_k\rfloor), \tag{4.6}$$

*we have*

$$\lim_{K\to\infty}\limsup_{N\to\infty}\sup_{pN+K\lambda_NB_N\le k\le N}\frac{|P_N(k)-G_N(k)|}{\widehat G_N(k)}=0,$$

*and this additive form also holds for the kernel $\overline G_N=\overline G_N^{(\sigma_0)}$ obtained from* (4.3) *by replacing $\gamma_+^{(\sigma_0)}(N-k)$ with $(N-L_k)/\mu_-$.*

*The same holds at the minus end with the signs exchanged.*

The exact coefficient $\gamma_+^{(\sigma_0)}(j)$ matters for the ratio form at bounded $j$ with $\kappa(j)=0$. If $f_-(1)=0$, then for a plus start $P_N(N-1)=f_+(N-1)$ and $G_N(N-1)=f_+(\lfloor N-s/\mu_-\rfloor)$, whose ratio tends to one by (LR), while the limit coefficient $s/\mu_-^2$ would not. In the additive form the two coefficients agree.

**Remark 4.1** (the uniform boundary layer)**.** Under the hypotheses of (b), for every $\delta_N\to 0$,

$$\sup_{0\le j\le\delta_NN}\Big|\frac{P_N(N-j)}{T_+(N)}-\kappa(j)\Big|\to 0,$$

and $P_N(N-j)/(T_+(N)\kappa(j))\to 1$ uniformly over the same $j$ with $\kappa(j)>0$, a condition that excludes nothing but $j=0$ for a minus start when $f_-(1)>0$. This is the additive form of (b) read near the end. Let $0\le j\le\delta_NN$. Then $k-pN\ge(1-p)N/2$ for large $N$, and

$\lambda_N B_N = o(N)$ by Lemma 2.1(v), so these points lie in the zone of (b) for every $K$ once $N$ is large, and the additive form gives $\sup_{j\le\delta_N N} |P_N(N-j)-G_N(N-j)|/\widehat{G}_N(N-j) \to 0$. Here $L_k = N - js/\mu_- \ge N(1-\delta_N s/\mu_-)$, so $T_+(\lfloor L_k\rfloor)/T_+(N) \to 1$ and $\widehat{G}_N(N-j)/T_+(N) \to 1$ uniformly by Lemma 2.1(i), and by (LR) and $\gamma(j) \le (1+\mu_+)j$ from (4.2), $\gamma(j)f_+(\lfloor L_k\rfloor) \le C(\delta_N + 1/N)T_+(N)$. Hence $|P_N(N-j) - \kappa(j)T_+(N)| = o(T_+(N))$ uniformly, which is the first statement. The second follows because $\inf\{\kappa(j) : \kappa(j) > 0\} > 0$, since $\kappa(j) \to 1/\mu_-$ by (2.10) and (2.11).

The double limit in (b) is equivalent to the single limit $\sup |P_N(k)/G_N(k) - 1| \to 0$ over $pN + K_N\lambda_N B_N \le k \le N$ for every sequence $K_N \to \infty$. It cannot be replaced by a fixed $K$. Fix $K$ and let $k = pN + KB_N$. Theorem A(c) gives $B_N P_N(k) = h_N(x_{N,k}) + o(1)$. By Remark 4.10 the ratio of $h_N(K)$ to $B_N G_N(k)$ tends to one only as $K \to \infty$. When $\theta_{+,N} \to 0$ a factor in front of $B_N$ that tends to infinity is also needed (Remark 4.11). Theorem 4.14 in Section 4.5 finds the right factor. It is a quantity $\xi_N = o(\lambda_N)$ given by a Cramér transform of the minus runs, and the zone starts at $(1+\delta)\xi_N B_N$ and not at $(1-\delta)\xi_N B_N$. So the window $KB_N \le k - pN \le K\lambda_N B_N$ left by (b) shrinks to $K \le x_{N,k} \le (1+\delta)\xi_N$. On it Theorem 4.14 describes $P_N$ on the logarithmic scale, and the strip $(1\pm\delta)\xi_N$ around the start remains open. This window lies on the side of the sign with the smaller scale. When the minus scale is the smaller one it is the window at the minus end given by the same statements with the signs exchanged.

### 4.1 Giant runs

Every point mass in this section is decomposed according to the plus runs that occupy many steps. We use the weights $\rho_i^{(\sigma_0)}(x)$ of Section 2.4, with the representation (2.13), and for $i \ge 1$ and $x \ge 0$ we put

$$\beta_i(x) = \Pi^-_{i+x}(i),$$

the probability that a fresh minus walk on $i+x$ steps has exactly $i$ minus steps. Then

$$\sum_{x\ge0} \rho_i^{(\sigma_0)}(x) = \kappa_+^{(\sigma_0)}(i), \qquad \sum_{x\ge0} \beta_i(x) = 1 + \mu_+ u_-(i), \tag{4.7}$$

by (2.12) and by the Green identity, Lemma 2.5.

We need $\beta_i$ in terms of sums of runs. Take a fresh minus walk with minus runs $X_1^-, X_2^-, \dots$ and plus runs $X_1^+, X_2^+, \dots$ in the order $X_1^- X_1^+ X_2^- X_2^+ \cdots$. Let $J = J_-(i)$ be the index of the minus run that contains the $i$-th minus step, in the notation of (K3) in Appendix A.3, and put $\xi_i = X_J^+ \mathbf{1}_{\{S_J^- = i\}}$. The $i$-th minus step is step $i + S_{J-1}^+$. The next minus step follows it at once unless the run $X_J^-$ ends there, and then it follows after $X_J^+$ plus steps. Hence

$$\beta_i(x) = \mathbb{P}\big(S_{J-1}^+ \le x \le S_{J-1}^+ + \xi_i\big). \tag{4.8}$$

Here $J$ depends only on the minus runs, and $X_J^+$ is independent of $(J, S_{J-1}^+, X_1^-, \dots, X_J^-)$.

For an integer $g \ge 1$ let $\mathcal{G}_g$ be the number of plus runs that have at least $g$ steps among the first $N$ steps. We call these runs giant. A complete run counts with its length, and the final run counts with the number of its steps up to step $N$. Let $M_N$ be the largest number of steps that a plus run has among the first $N$ steps, so $\{M_N < g\} = \{\mathcal{G}_g = 0\}$.

**Lemma 4.2** (the giant identity)**.** *For every start of sign $\sigma_0$, every $N$, every $0 \le j < N$ with $k = N - j$, and every integer $g \ge 1$,*

$$\mathbb{E}\big[\mathcal{G}_g \mathbf{1}_{\{A_N = k\}}\big] = \mathrm{Cen}_g(k) + \mathrm{Int}_g(k), \tag{4.9}$$

*where, with* $\rho_i = \rho_i^{(\sigma_0)}$,

$$\mathrm{Cen}_g(k) = \sum_{x\ge 0,\ N-j-x\ge g} \rho_j(x)\, T_+(N-j-x),$$

$$\mathrm{Int}_g(k) = \sum_{i=0}^{j-1} \sum_{x,x'\ge 0,\ N-j-x-x'\ge g} \rho_i(x)\, \beta_{j-i}(x')\, f_+(N-j-x-x').$$

*Consequently*

$$P_N(k) = \mathbb{P}(A_N = k,\ M_N < g) + \mathrm{Cen}_g(k) + \mathrm{Int}_g(k) - \mathbb{E}\big[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = k\}}\big], \tag{4.10}$$

*and, with* $f_+^*(n) = \sup_{m\ge n} f_+(m)$,

$$\mathrm{Cen}_g(k) \le \kappa_+^{(\sigma_0)}(j)\, T_+(g), \qquad \mathrm{Int}_g(k) \le \gamma_+^{(\sigma_0)}(j)\, f_+^*(g). \tag{4.11}$$

*Proof.* By linearity, the left side of (4.9) is the sum over all plus runs of the probability that the run is giant and $A_N = k$. We split the runs into the final run and the complete runs.

Let the final run be a plus run that starts at step $t+1$. All $j$ minus steps precede it, so $t = j + x$, where $x$ is the number of plus steps before it. The run has $N - j - x$ steps up to step $N$, and its length is at least $N - j - x$. The length is independent of the runs before it. So this case contributes $\rho_j(x) T_+(N-j-x)$ for each $x$ with $N - j - x \ge g$, which is $\mathrm{Cen}_g(k)$.

Let a complete plus run start right after the $i$-th minus step, with $x$ plus steps before it, and let its length be $y$. It occupies steps $i+x+1$ to $i+x+y$, and $i+x+y \le N-1$. The runs after it form a fresh minus walk, independent of everything before. On $\{A_N = k\}$ this walk holds the remaining $j - i$ minus steps among its $N - i - x - y \ge 1$ steps. Since its first step is a minus step, $j - i \ge 1$. Write $N - i - x - y = (j - i) + x'$. Then $y = N - j - x - x'$, and this case contributes $\rho_i(x) f_+(y) \beta_{j-i}(x')$. A fresh plus start has $\rho_0 = \delta_0$, and then the first run has law $f_+$, as every other plus run. Summing over $i < j$, $x$ and $x'$ with $y \ge g$ gives $\mathrm{Int}_g(k)$.

For (4.10) note that $\mathbf{1}_{\{\mathcal{G}_g \ge 1\}} = \mathcal{G}_g - (\mathcal{G}_g - 1)^+$. For (4.11) bound $T_+$ by $T_+(g)$ and $f_+$ by $f_+^*(g)$ on the ranges of summation, and use (4.7). □

### 4.2 The boundary layer

We use two consequences of (RV) for the plus sign. First, $T_+(n+1)/T_+(n) \to 1$, so $f_+(n)/T_+(n) \to 0$ and

$$f_+^*(n) \le \sup_{m\ge n} \frac{f_+(m)}{T_+(m)}\, T_+(n) = o\big(T_+(n)\big). \tag{4.12}$$

Second, a sum of $m' \ge 1$ plus runs equal to $x$ has a summand of size at least $x/m'$. Conditioning on the other summands gives

$$\mathbb{P}(S_{m'}^+ = x) \le m'\, f_+^*(x/m'). \tag{4.13}$$

*Proof of Theorem B(a).* Fix $j \ge 0$. If $j = 0$ then $P_N(N)$ is $T_+(N)$ for a plus start and $0$ for a minus start, which is $\kappa(0) T_+(N)$. Let $j \ge 1$ and take $g = g'_N = \lceil (N-j)/(j+1) \rceil$ in Lemma 4.2. By Lemma 2.7, on $\{A_N = N - j\}$ some plus run has at least $g'_N$ steps, so the first term of (4.10) vanishes. Since $g'_N \ge N/(2j+2)$ for large $N$, Lemma 2.1(ii) gives $T_+(g'_N) \le C_j T_+(N)$.

The last term of (4.10). There are at most $j+1$ plus runs, so $(\mathcal{G}_g - 1)^+ \le (j+1)\mathbf{1}_{\{\mathcal{G}_g \ge 2\}}$. The second part of Lemma 2.7 gives

$$\mathbb{E}\big[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = N-j\}}\big] \le (j+1)^3\, T_+(g'_N)^2 = O\big(T_+(N)^2\big) = o\big(T_+(N)\big).$$

The term $\mathrm{Int}_g$. It contains the configurations of the second sum of (2.14), a complete giant plus run followed by a final minus run of length $m \le j$, which are the leading term of $P_N(N-j)$ when $\kappa(j) = 0$. By (4.11) and (4.12), $\mathrm{Int}_g(N-j) \le \gamma_+^{(\sigma_0)}(j) f_+^*(g'_N) = o(T_+(g'_N)) = o(T_+(N))$.

The term $\mathrm{Cen}_g$. Split the sum at $x = N/4$. For $x \le N/4$ and large $N$ we have $N - j - x \ge N/2 \ge g'_N$, and $T_+(N-j-x)/T_+(N)$ is at most $T_+(\lfloor N/2 \rfloor)/T_+(N)$, which is bounded by Lemma 2.1(ii). For each fixed $x$ the ratio tends to one. Since $\rho_j$ is a finite measure that does not depend on $N$, dominated convergence and (4.7) give

$$\frac{1}{T_+(N)} \sum_{x \le N/4} \rho_j(x)\, T_+(N-j-x) \to \sum_{x \ge 0} \rho_j(x) = \kappa(j).$$

For $x > N/4$ we use (2.13). The plus sums there have at most $j$ terms, and $\sum_m u_m^-(j) \le 1$. So (4.13) gives $\rho_j(x) \le j\, f_+^*(N/(4j))$. Since $\sum_x T_+(N-j-x) \le \sum_{n \ge 1} T_+(n) = \mu_+$,

$$\sum_{x > N/4} \rho_j(x)\, T_+(N-j-x) \le j\mu_+ f_+^*\big(N/(4j)\big) = o\big(T_+(N)\big),$$

by (4.12) and Lemma 2.1(ii). Adding the four terms gives $P_N(N-j) = \kappa(j) T_+(N) + o(T_+(N))$. □

At fixed $j$ the giant run is the run in progress at time $N$, and its law enters only through $T_+$, which is why (a) needs no local hypothesis.

### 4.3 The big-jump zone

In the zone we take $g = \lceil \eta L_k \rceil$ in Lemma 4.2 with a small fixed $\eta$. Then (4.10) has four terms, and we treat them in three lemmas. Lemma 4.4 bounds the contribution with no giant run, Lemma 4.5 the contribution with two, and Lemma 4.7 evaluates $\mathrm{Cen}_g$ and $\mathrm{Int}_g$. Note that $k \ge L_k$ for $pN < k \le N$, since $k - L_k = p(N-k)/(1-p)$.

We also need a bound for right deviations without big summands. It is the elementary exponential estimate behind the inequalities of Fuk and Nagaev [29] and [53, Theorems 1.1 and 1.2], and we include the short proof.

**Lemma 4.3** (truncated right deviations)**.** *Assume (RV) for the plus sign. For all $m \ge 1$, $w \ge 1$ and $z > 0$,*

$$\mathbb{P}\Big(S_m^+ - m\mu_+ \ge z,\ \max_{i \le m} X_i^+ < w\Big) \le e^{C_{\mathrm{K}}} \big(m\, T_+(w)\big)^{z/(2w)}.$$

*Proof.* If $mT_+(w) \ge 1$ there is nothing to prove. Otherwise let $\vartheta = \frac{1}{2}\log(1/(mT_+(w))) > 0$. For $0 \le v \le \vartheta$ we have $e^v \le 1 + v + \frac{1}{2}v^2 e^{\vartheta}$. We use this with $v = \vartheta X/w$ on $\{X < w\}$, where $X = X_+$. By summation by parts and (2.3), $\mathbb{E}[X^2 \mathbf{1}_{\{X<w\}}] \le 2\sum_{n<w} nT_+(n) \le 2C_{\mathrm{K}} w^2 T_+(w)$. With $1 + v \le e^v$ this gives $\mathbb{E}[e^{\vartheta X/w} \mathbf{1}_{\{X<w\}}] \le \exp(\vartheta\mu_+/w + C_{\mathrm{K}}\vartheta^2 e^{\vartheta} T_+(w))$. Markov's inequality and independence give the bound $\exp(-\vartheta z/w + C_{\mathrm{K}} m\vartheta^2 e^{\vartheta} T_+(w))$. Here $m\vartheta^2 e^{\vartheta} T_+(w) = \vartheta^2 e^{-\vartheta} \le 1$, and $e^{-\vartheta z/w} = (mT_+(w))^{z/(2w)}$. □

**Lemma 4.4** (no giant run)**.** *Assume (RV) and (SPAN). Let $0 < \eta \le 1/16$ and $r = 1/(16\eta)$, so $r \ge 1 > 1/\alpha_+$. There are constants $C, c, C_8 > 0$ such that, for every fresh start, all large $N$ and all $pN < k \le N$ with $L_k \ge C_8 B_N$,*

$$\mathbb{P}\big(A_N = k,\ M_N < \eta L_k\big) \le C\, \frac{(NT_+(L_k))^r}{b_{+,N}} + \frac{C}{b_{-,N}}\, e^{-cL_k/b_{-,N}} + C\, e^{-N^c}.$$

*Proof.* Write $L = L_k$, $j = N - k$ and $P_0$ for the probability on the left. If $j = 0$ then the whole walk is one plus run and $P_0 = 0$. Let $j \ge 1$, and take first a plus start. On $\{A_N = k\}$

step $N$ lies either in the $(m+1)$-th plus run, and then the first $m$ minus runs sum to $j$ and $S^+_m < k \le S^+_{m+1}$, or in the $m$-th minus run, and then $S^+_m = k$ and $S^-_{m-1} < j \le S^-_m$, as in the proof of Proposition 2.3. On $\{M_N < \eta L\}$ the plus runs that lie inside the first $N$ steps are shorter than $\eta L$, and in the first case the final run has $k - S^+_m < \eta L$ steps. The plus and minus runs are independent. Hence $P_0 \le E_1 + E_2$ with

$$E_1 = \sum_{m\ge 0} u^-_m(j)\, q^\eta_{m+1}, \qquad q^\eta_{m+1} = \sum_{1\le v<\eta L} \mathbb{P}\Big(S^+_m = k - v,\ \max_{i\le m} X^+_i < \eta L\Big) T_+(v),$$
$$E_2 = \sum_{m\ge 1} \mathbb{P}\Big(S^+_m = k,\ \max_{i\le m} X^+_i < \eta L\Big)\, q^-_m(j).$$

For a minus start the plus run that contains step $N$ is the $m$-th one, preceded by $m$ minus runs and $m-1$ plus runs, and the minus run that contains step $N$ is the $(m+1)$-th one, preceded by $m$ plus runs. So $E_1$ becomes $\sum_{m\ge1} u^-_m(j) q^\eta_m$, where $q^\eta_m$ is built on $\mathbb{P}(S^+_{m-1} = k - v, \dots)$, and $E_2$ becomes $\sum_{m\ge1} \mathbb{P}(S^+_m = k, \dots)\, q^-_{m+1}(j)$. We note below the two places where this changes the argument.

We split the indices. Put $d_1 = k - m\mu_+$ and $d_2 = m\mu_- - j$. Then $\mu_- d_1 + \mu_+ d_2 = \mu_- k - \mu_+ j = s(k - pN) = \mu_- L$. Let $\mathcal{A}$ be the set of $m$ with $d_1 \ge L/2$ and $\mathcal{B}$ its complement. For $m \in \mathcal{B}$ we get $d_2 \ge y_2 = \mu_- L/(2\mu_+)$. The kernel sums $\sum_m u^-_m(j)$, $\sum_m q^-_m(j)$, $\sum_m q^-_{m+1}(j)$, $\sum_m q^+_m(k)$ and $\sum_m u^+_m(k)$ are at most 1 by (2.8). So $E_1 + E_2$ is at most the sum of the suprema over $m \in \mathcal{A}$ of the plus factors and the suprema over $m \in \mathcal{B}$ of the minus factors.

*The indices in $\mathcal{A}$.* Let $m \in \mathcal{A}$ and let $v'$ be an integer with $v' - m\mu_+ \ge L/4$. Every $k - v$ in $q^\eta_{m+1}$ is such a $v'$, since $k - v - m\mu_+ \ge L/2 - \eta L$, and so is $k$. For a minus start the relevant sum $S^+_{m-1}$ has one term fewer, so its deviation $k - v - (m-1)\mu_+$ is larger by $\mu_+$, and the bound below applies with $m-1$ in place of $m$. For $m \le 1$ the probability $\mathbb{P}(S^+_m = v', \max_i X^+_i < \eta L)$ vanishes, because $v' \ge L/4 > \eta L$. For $m \ge 2$ split the indices into two halves of sizes $\lfloor m/2 \rfloor$ and $\lceil m/2 \rceil$. On $\{S^+_m = v'\}$ one half exceeds its mean by at least $L/8$, and then the other half takes one exact value. By Lemma 4.3 with $w = \eta L$ and $z = L/8$, and by Lemma A.1(ii),

$$\mathbb{P}\Big(S^+_m = v',\ \max_{i\le m} X^+_i < \eta L\Big) \le 2e^{C_{\mathrm{K}}} \big(mT_+(\eta L)\big)^r \min\Big(1, \frac{\bar u_+}{a_+(\lfloor m/2 \rfloor)}\Big).$$

The indices in $\mathcal{A}$ satisfy $m \le m_+ = k/\mu_+ \le N/\mu_+$, and $m_+ \ge N/s$ since $k > pN$. Since $r > 1/\alpha_+$, Lemma 2.1(ii) for $a_+$ shows that the right side is largest up to a constant at $m = m_+$. Also $a_+(m_+) \ge a_+(N/s) = b_{+,N}$ and $T_+(\eta L) \le C_\eta T_+(L)$. With $\sum_v T_+(v) \le 1 + \mu_+$ in $q^\eta_{m+1}$ the plus factors are at most $C(NT_+(L))^r / b_{+,N}$.

*The indices in $\mathcal{B}$.* Let $m \in \mathcal{B}$, so $x_m = m\mu_- - j \ge y_2$ and in particular $m \ge y_2/\mu_-$, which is large. On $\{S^-_m = j\}$ one half of the minus runs falls below its mean by at least $x_m/2$, and the other half takes one exact value. Lemma A.1(iv) for the half, with $a_-(\lceil m/2 \rceil) \le a_-(m)$, and Lemma A.1(ii) give

$$u^-_m(j) \le 2e^{C_-} e^{-x_m/(2a_-(m))} \min\Big(1, \frac{\bar u_-}{a_-(\lfloor m/2 \rfloor)}\Big).$$

Next, $q^-_m(j) = \sum_{v\ge1} T_-(v) u^-_{m-1}(j-v) \le \mu_- \sup_{v\ge1} u^-_{m-1}(j-v)$, and $(m-1)\mu_- - (j-v) \ge x_m - \mu_- \ge x_m/2$. So $q^-_m(j)$ obeys the same bound with $\mu_-$ in front, $x_m/4$ in the exponent and $\lfloor (m-1)/2 \rfloor$ in place of $\lfloor m/2 \rfloor$. For a minus start, $q^-_{m+1}(j) = \sum_{v\ge1} T_-(v) u^-_m(j-v) \le \mu_- \sup_{v\ge1} u^-_m(j-v)$ and $m\mu_- - (j-v) \ge x_m$, so it obeys the bound for $u^-_m(j)$ with $\mu_-$ in front. Let $m \le 2N/\mu_-$. Then $t = a_-(m) \le c_5 b_{-,N}$, and Lemma 2.1(ii) gives $a_-(m) \le c_6 a_-(\lfloor (m-1)/2 \rfloor)$. So all these bounds are at most $Ct^{-1}e^{-y_2/(4t)}$. The function $t \mapsto t^{-1}e^{-y_2/(4t)}$ increases on $(0, y_2/4]$. Choose $C_8 = 8c_5\mu_+/\mu_-$. Then $L \ge C_8 B_N$ gives

$c_5 b_{-,N} \le y_2/4$, and the bounds are at most $C b_{-,N}^{-1} e^{-cL/b_{-,N}}$. Let $m > 2N/\mu_-$. Then $x_m \ge m\mu_- - N \ge m\mu_-/2$, and Lemma 2.1(ii) gives $a_-(m) \le m^{1-2c}$ for large $m$ with $c = (1-1/\alpha_-)/4$. So the bounds are at most $Ce^{-cm^{2c}} \le Ce^{-N^c}$.

Adding the bounds for $\mathcal{A}$ and $\mathcal{B}$ proves the lemma. □

The second term of Lemma 4.4, from minus runs falling short of their mean, is the reason for $\lambda_N$.

**Lemma 4.5** (two giant runs)**.** *Assume (RV) and (LR) for the plus sign, and let $0 < \eta < 1$. There is a constant $C$ such that, for every fresh start, all $N$ and all $pN < k \le N$ with $\eta L_k \ge 1$ and $g = \lceil \eta L_k \rceil$,*

$$\mathbb{E}\big[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = k\}}\big] \le C\Big[(j+1)^2 \, \frac{T_+(L_k)^2}{L_k} + (j+1)\, T_+(L_k)^2\Big].$$

*Proof.* Since $(n-1)^+ \le n(n-1)/2$ for integers $n \ge 0$, it suffices to bound the expected number of pairs of giant runs on $\{A_N = k\}$. The second run of a pair is complete or final. As in the proof of Lemma 4.2, the walk between the two runs is a fresh minus walk until a plus run starts after $i_2 \ge 1$ of its minus steps, with $x_2$ plus steps before it, which has probability $\rho^{(-)}_{i_2}(x_2)$. For two complete runs of lengths $y_1, y_2$ the expected number of pairs is

$$\sum_{i_1+i_2+i_3=j} \; \sum_{x_1,x_2,x_3} \rho^{(\sigma_0)}_{i_1}(x_1)\, \rho^{(-)}_{i_2}(x_2)\, \beta_{i_3}(x_3) \sum_{y_1+y_2=N-j-x_1-x_2-x_3,\; y_1,y_2 \ge g} f_+(y_1) f_+(y_2),$$

with $i_2, i_3 \ge 1$. For a complete run followed by the final run the expected number is the same with $\beta_{i_3}(x_3)$ removed, $i_1 + i_2 = j$, and $T_+(y_2)$ in place of $f_+(y_2)$. By (LR), $f_+(n) \le C_{\mathrm{LR}} T_+(n)/n$ for all $n \ge 1$, so $f_+^*(g) \le C_{\mathrm{LR}} T_+(g)/g$. Let $y_1 + y_2 = y$ with both at least $g$, so $y/2 \ge g$. The larger of $y_1, y_2$ is at least $y/2$. So

$$\sum f_+(y_1) f_+(y_2) \le 2 f_+^*(g) T_+(g), \qquad \sum f_+(y_1) T_+(y_2) \le f_+^*(g) \sum_{n \ge g} T_+(n) + T_+(g)^2,$$

and the second sum is at most $(C_{\mathrm{K}} C_{\mathrm{LR}} + 1) T_+(g)^2$ by (2.3). The weights have total mass at most $(1+\mu_+)(j+1)^2$ in the first case and $j+1$ in the second, by (4.7). Finally $T_+(g) \le C_\eta T_+(L_k)$ and $g \ge \eta L_k$. □

The terms $\mathrm{Cen}_g$ and $\mathrm{Int}_g$ need the spread of the weights. A plus run that starts after the $i$-th minus step is preceded by about $\bar{x}_i = i\mu_+/\mu_-$ plus steps, and the fluctuation is of order $B_N$. For $z > 0$ put

$$\mathrm{sp}_i^\rho(z) = \sum_{x \,:\, |x - \bar{x}_i| > z} \rho_i^{(\sigma_0)}(x), \qquad \mathrm{sp}_i^\beta(z) = \sum_{x \,:\, |x - \bar{x}_i| > z} \beta_i(x).$$

**Lemma 4.6** (spread of the weights)**.** *Assume (RV). For every $\varepsilon > 0$ there are $Z$ and $N_0$ such that $\mathrm{sp}_i^\rho(ZB_N) \le \varepsilon$ and $\mathrm{sp}_i^\beta(ZB_N) \le \varepsilon$ for all $N \ge N_0$ and all $0 \le i \le N$.*

*Proof.* Let $\varpi_\sigma(z) = \sup_{m \ge 1} \mathbb{P}(|S^\sigma_m - m\mu_\sigma| > z\, a_\sigma(m))$. By Lemma A.1(i) the laws of $(S^\sigma_m - m\mu_\sigma)/a_\sigma(m)$ converge, so they are tight, and $\varpi_\sigma(z) \to 0$ as $z \to \infty$. By Lemma 2.1(ii) there is $c_7$ with $a_\sigma(m) \le c_7 B_N$ for $m \le 4N/\mu_-$ and large $N$.

The number of minus runs. By (K3), $\{S^-_m = i\} \subset \{J_-(i) = m\}$, so for $w > 0$

$$\sum_{m \,:\, |m\mu_- - i| > w} u^-_m(i) \le \mathbb{P}\big(|J_-(i)\mu_- - i| > w\big) \le \mathbb{P}\big(S^-_{m_1} \ge i\big) + \mathbb{P}\big(S^-_{m_2} < i\big),$$

with $m_1 = \lfloor (i-w)/\mu_- \rfloor$ (the first term is absent if $i \le w$) and $m_2 = \lfloor (i+w)/\mu_- \rfloor$. Indeed $J_-(i) > (i+w)/\mu_-$ gives $J_-(i) > m_2$, and then $S^-_{m_2} \le S^-_{J_-(i)-1} < i$. The first event is a

deviation of at least $w$ above the mean, and the second a deviation of at least $w-\mu_-$ below it. For $w \le N$ both indices are at most $3N/\mu_-$. So for $w = zB_N$ the right side is at most $2\varpi_-(z/(2c_7))$ for large $N$.

The bound for $\mathrm{sp}_i^{\rho}$. Take a plus start. On $\{S_m^- = i\}$ we have $m \le i \le N$ and $|S_m^+ - \bar{x}_i| \le |S_m^+ - m\mu_+| + (\mu_+/\mu_-)|m\mu_- - i|$. By (2.13),

$$\begin{aligned}\mathrm{sp}_i^{\rho}(ZB_N) &\le \sup_{m\le N} \mathbb{P}\big(|S_m^+ - m\mu_+| > ZB_N/2\big) + \sum_{m\,:\,|m\mu_- - i| > \mu_- ZB_N/(2\mu_+)} u_m^-(i) \\ &\le \varpi_+\Big(\frac{Z}{2c_7}\Big) + 2\varpi_-\Big(\frac{\mu_- Z}{4\mu_+ c_7}\Big).\end{aligned}$$

For a minus start (2.13) has $S_{m-1}^+$ in place of $S_m^+$, and $|S_{m-1}^+ - (m-1)\mu_+| > ZB_N/2 - \mu_+$ is needed for the first event. For large $N$ this is at least $ZB_N/4$, so the same bound holds with $Z/(4c_7)$ in place of $Z/(2c_7)$ in the first term.

The bound for $\mathrm{sp}_i^{\beta}$. Write $S = S_{J-1}^+$ in (4.8). Then $\mathrm{sp}_i^{\beta}(z)$ is the expected number of integers $x \in [S, S+\xi_i]$ with $|x - \bar{x}_i| > z$, which is at most $(1+\xi_i)\mathbf{1}_{\{|S-\bar{x}_i|>z/2\}} + \xi_i \mathbf{1}_{\{\xi_i > z/2\}}$. Since $X_J^+$ is independent of $(J,S)$ and $\xi_i \le X_J^+$,

$$\mathrm{sp}_i^{\beta}(z) \le (1+\mu_+)\mathbb{P}(|S-\bar{x}_i| > z/2) + \mathbb{E}\big[X_+ \mathbf{1}_{\{X_+ > z/2\}}\big].$$

The last term tends to zero as $z \to \infty$. For the first, $J$ is independent of the plus runs, and $\mathbb{P}(J > 3N/\mu_-) \le \mathbb{P}(S^-_{\lfloor 3N/\mu_-\rfloor} < N) \to 0$ uniformly in $i \le N$ by Lemma A.1(iv). On $\{J \le 3N/\mu_-\}$ we bound $|S - \bar{x}_i|$ as for $\mathrm{sp}_i^{\rho}$, with $J-1$ in place of $m$. The plus part is controlled by $\varpi_+$ at the index $J-1$, and the number of minus runs by $\varpi_-$ through the display above, since $J = J_-(i)$ and $|(J-1)\mu_- - i| \le |J\mu_- - i| + \mu_-$. Taking $z = ZB_N$ with $Z$ large and then $N$ large proves the lemma. □

**Lemma 4.7** (one giant run)**.** *Assume (RV) and (SPAN), let the start be fresh with sign $\sigma_0$, let $0 < \eta \le 1/2$, and put $g = \lceil \eta L_k \rceil$. Write $\kappa = \kappa_+^{(\sigma_0)}$ and $\gamma = \gamma_+^{(\sigma_0)}$. Then $\mathrm{Cen}_g(k) = 0$ when $\kappa(N-k) = 0$, $\mathrm{Int}_g(k) = 0$ when $\gamma(N-k) = 0$, and*

$$\lim_{K\to\infty} \limsup_{N\to\infty} \sup_{k\,:\,L_k \ge KB_N,\ \kappa(N-k)>0} \Big| \frac{\mathrm{Cen}_g(k)}{\kappa(N-k)\,T_+(\lfloor L_k \rfloor)} - 1 \Big| = 0.$$

*If in addition (LR) holds for the plus sign, then also*

$$\lim_{K\to\infty} \limsup_{N\to\infty} \sup_{k\,:\,L_k \ge KB_N,\ \gamma(N-k)>0} \Big| \frac{\mathrm{Int}_g(k)}{\gamma(N-k)\,f_+(\lfloor L_k \rfloor)} - 1 \Big| = 0.$$

*Proof.* The two vanishing statements follow from (4.11). By (2.10) and (2.11), $\kappa(i) \to 1/\mu_-$, and by (4.4), $\gamma(i)/i \to s/\mu_-^2$. Hence

$$\kappa_* = \inf\{\kappa(i) \mid \kappa(i) > 0\} > 0, \qquad \gamma_* = \inf\{\gamma(i)/(i+1) \mid \gamma(i) > 0\} > 0.$$

Write $L = L_k$, $\hat{L} = \lfloor L \rfloor$ and $j = N - k$. Note that $N - j - x = L - (x - \bar{x}_j)$ by (4.1). Fix $\varepsilon > 0$ and take $Z, N_0$ such that Lemma 4.6 holds with $Z/2$ in place of $Z$. Since $\mathrm{sp}_i^{\rho}(z)$ and $\mathrm{sp}_i^{\beta}(z)$ decrease in $z$, the same bounds hold at $ZB_N$.

The term $\mathrm{Cen}_g$. Let $\kappa(j) > 0$ and write $\mathrm{Cen}_g(k) - \kappa(j)T_+(\hat{L}) = \sum_x \rho_j(x)[T_+(N-j-x)\mathbf{1}_{\{N-j-x \ge g\}} - T_+(\hat{L})]$. If $|x - \bar{x}_j| \le ZB_N$, then $|N-j-x-\hat{L}| \le ZB_N + 1 \le (Z/K + 1/L)L$. Since $Z/K \le \eta/2$ for large $K$, the indicator is 1, and by Lemma 2.1(i) the bracket is at most $\omega_T(Z/K)T_+(\hat{L})$, where $\omega_T(\delta) \to 0$ as $\delta \to 0$ once $N$ is large. For the other $x$ the bracket is at most $T_+(g) + T_+(\hat{L}) \le C_\eta T_+(\hat{L})$, and their total weight is $\mathrm{sp}_j^{\rho}(ZB_N) \le \varepsilon \le (\varepsilon/\kappa_*)\kappa(j)$. Hence

$$\big|\mathrm{Cen}_g(k) - \kappa(j)T_+(\hat{L})\big| \le \big(\omega_T(Z/K) + C_\eta \varepsilon/\kappa_*\big)\kappa(j)T_+(\hat{L}).$$

The term $\mathrm{Int}_g$. Let $\gamma(j) > 0$ and define the measure $\chi_j$ by

$$\chi_j(z) = \sum_{i<j} \sum_{x+x'=z} \rho_i(x)\beta_{j-i}(x').$$

It has total mass $\gamma(j)$ by (4.7), and

$$\mathrm{Int}_g(k) = \sum_z \chi_j(z) f_+(N-j-z)\mathbf{1}_{\{N-j-z\ge g\}}.$$

If $|z-\bar{x}_j| > ZB_N$, then the split $(x,x')$ has $|x-\bar{x}_i| > ZB_N/2$ or $|x'-\bar{x}_{j-i}| > ZB_N/2$. So Lemma 4.6 and (4.7) bound the mass of these $z$ by

$$\sum_{i<j}\Big[(1+\mu_+)\mathrm{sp}_i^\rho(ZB_N/2) + \kappa(i)\mathrm{sp}_{j-i}^\beta(ZB_N/2)\Big] \le (2+\mu_+)\, j\,\varepsilon \le \frac{(2+\mu_+)\varepsilon}{\gamma_*}\,\gamma(j).$$

Under (LR), $x \mapsto T_+(x)/x$ is regularly varying and $f_+(n) \sim \alpha_+ T_+(n)/n$. So Lemma 2.1(i) gives $|f_+(n)/f_+(\hat{L})-1| \le \omega_f(Z/K)$ for $|n-\hat{L}| \le ZB_N+1$ and large $N$, and Lemma 2.1(ii) gives $f_+^*(g) \le C_\eta f_+(\hat{L})$. As for $\mathrm{Cen}_g$,

$$\big|\mathrm{Int}_g(k) - \gamma(j)f_+(\hat{L})\big| \le \Big(\omega_f(Z/K) + \frac{(C_\eta+1)(2+\mu_+)\varepsilon}{\gamma_*}\Big)\gamma(j)f_+(\hat{L}).$$

Let $N\to\infty$, then $K\to\infty$, then $\varepsilon\to 0$. □

*Proof of Theorem B(b).* Write $\kappa,\gamma$ as in Lemma 4.7, $L = L_k$, $\hat{L} = \lfloor L\rfloor$ and $j = N-k$. Note that $T_+(\hat{L}) \ge T_+(L)$. By (2.10) and (4.4) there is $J_0 \ge 1$ with $\kappa(j) \ge 1/(2\mu_-)$ and $\gamma(j) \ge js/(2\mu_-^2)$ for $j \ge J_0$. Fix

$$\eta = \min\Big\{\frac{1}{32},\ \frac{1}{2J_0+2}\Big\}, \qquad g = \lceil \eta L\rceil,$$

so that $r = 1/(16\eta) \ge 2$. Fix $\varepsilon > 0$. By (4.10),

$$\begin{aligned}\big|P_N(k) - G_N(k)\big| &\le \mathbb{P}(A_N = k,\ M_N < \eta L) + |\mathrm{Cen}_g(k) - \kappa(j)T_+(\hat{L})| \\ &\quad + |\mathrm{Int}_g(k) - \gamma(j)f_+(\hat{L})| + \mathbb{E}\big[(\mathcal{G}_g-1)^+\mathbf{1}_{\{A_N=k\}}\big].\end{aligned}$$

In the zone $k \ge pN + K\lambda_N B_N$ we have $L \ge k - pN \ge K\lambda_N B_N \ge KB_N$. By Lemma 4.7 the second and third terms are at most $\varepsilon[\kappa(j)T_+(\hat{L}) + \gamma(j)f_+(\hat{L})] = \varepsilon G_N(k)$ for $K \ge K_1(\varepsilon)$ and $N \ge N_1(K)$. It remains to compare the first and last terms with $G_N(k)$. We use (2.4) for the plus sign.

*Small $j$.* Let $j < J_0$. For large $N$ we have $\eta L \le \eta N \le (N-j)/(j+1)$, so Lemma 2.7 gives $\mathbb{P}(A_N = k, M_N < \eta L) = 0$. If $G_N(k) = 0$, then $\kappa(j) = \gamma(j) = 0$, so $\mathrm{Cen}_g = \mathrm{Int}_g = 0$ and $\mathcal{G}_g = 0$ almost surely on $\{A_N = k\}$, and by (4.10) $P_N(k) = 0$. If $G_N(k) > 0$, then $G_N(k) \ge \min(\kappa_* T_+(\hat{L}), \gamma_* f_+(\hat{L})) \ge cT_+(L)/L$ by (LR). Also $L \ge N/2$ for large $N$. By Lemma 4.5 the last term is at most $CJ_0^2 T_+(L)^2$, and its ratio to $G_N(k)$ is at most $CJ_0^2 LT_+(L)$, which tends to zero since $\alpha_+ > 1$.

*Large $j$.* Let $j \ge J_0$. By (LR), $f_+(\hat{L}) \ge \alpha_+ T_+(\hat{L})/(2L)$ for large $L$. Since $js/\mu_-^2 = (N-L)/\mu_-$ by (4.1),

$$G_N(k) \ge \frac{T_+(L)}{2\mu_-}\Big(\frac{\alpha_+(N-L)}{2L} + 1\Big) \ge c_8\,\frac{N}{L}\,T_+(L).$$

By Lemma 4.5, with $j+1 \le N$, the last term divided by $G_N(k)$ is at most $C(j+1)T_+(L) \le CNT_+(L) \le CK^{-(\alpha_++1)/2}$ by (2.4), since $L/b_{+,N} \ge L/B_N \ge K$. By Lemma 4.4, for $K \ge C_8$, the first term divided by $G_N(k)$ is at most $C$ times

$$\big(NT_+(L)\big)^{r-1}\frac{L}{b_{+,N}} + \frac{L}{b_{-,N}}\,e^{-cL/b_{-,N}}\,\frac{1}{NT_+(L)} + \frac{e^{-N^c}}{T_+(N)}.$$

The last part tends to zero. The first part is at most $C(L/b_{+,N})^{(1-\alpha_+)/2}$ by (2.4) and $r \geq 2$, and this is at most $CK^{(1-\alpha_+)/2}$. For the middle part put $x = L/b_{-,N}$ and $\varrho = b_{-,N}/b_{+,N}$, so $L/b_{+,N} = x\varrho$. By (2.4) the middle part is at most $Cx^{\alpha_++2}\varrho^{\alpha_++1}e^{-cx}$. If $\varrho \leq 1$, then $x \geq L/B_N \geq K$ and the bound is at most $C\sup_{x\geq K} x^{\alpha_++2}e^{-cx}$. If $\varrho > 1$, then $B_N = b_{-,N}$, $\lambda_N = 1 + \log\varrho$ and $x \geq K(1+\log\varrho)$. For $cK \geq 2(\alpha_+ + 1)$ this gives $\varrho^{\alpha_++1}e^{-cx/2} \leq e^{(\alpha_++1)\log\varrho - cK(1+\log\varrho)/2} \leq e^{-cK/2}$, and the bound is at most $Ce^{-cK/2}\sup_x x^{\alpha_++2}e^{-cx/2}$. In every case the first term is at most $\varepsilon G_N(k)$ for $K$ large and then $N$ large. This proves the ratio form of (b).

For the additive form note that $\gamma(j) \leq (1+\mu_+)j \leq (1+\mu_+)N$ by (4.2), $(N-L)/\mu_- \leq N/\mu_-$, $f_+(\hat{L}) \leq 2C_{\mathrm{LR}}T_+(\hat{L})/L$ by (LR), and $\kappa \leq 1 \leq N/L$. Hence

$$G_N(k) \leq C_{\mathrm{G}}\widehat{G}_N(k), \qquad \overline{G}_N(k) \leq C_{\mathrm{G}}\widehat{G}_N(k), \tag{4.14}$$

with $C_{\mathrm{G}}$ depending only on the laws, and the same holds at every horizon $n$ and for both starting signs whenever $L_k^{(n)} \geq 1$. The ratio form gives the additive form. Replacing $\gamma(j)$ by $js/\mu_-^2 = (N-L)/\mu_-$ changes $G_N(k)$ by $|\gamma(j) - js/\mu_-^2| f_+(\hat{L}) \leq 2C_{\mathrm{LR}}|\gamma(j) - js/\mu_-^2|\,T_+(\hat{L})/L$. By (4.4) the factor $|\gamma(j) - js/\mu_-^2|$ is at most $\varepsilon N$ for $j \geq J_\varepsilon$ and at most a constant $C_\varepsilon$ for $j < J_\varepsilon$, and in the second case $L \geq N/2$. In both cases the change is at most $(2C_{\mathrm{LR}}\varepsilon + 4C_{\mathrm{LR}}C_\varepsilon/N)\widehat{G}_N(k)$. This proves (b). □

**Corollary 4.8** (the lower bound)**.** *Under the hypotheses of Theorem B(b),*

$$\lim_{K\to\infty} \liminf_{N\to\infty} \inf_{pN+KB_N \leq k \leq N,\ G_N(k)>0} \frac{P_N(k)}{G_N(k)} \geq 1.$$

*That is, for every $\varepsilon > 0$ there is $K$ such that $P_N(k) \geq (1-\varepsilon)G_N(k)$ for all $pN + KB_N \leq k \leq N$ and all large $N$.*

*Proof.* The first term of (4.10) is nonnegative, and it is the only term whose bound in the proof of (b) used $\lambda_N$. So $P_N(k) \geq \mathrm{Cen}_g(k) + \mathrm{Int}_g(k) - \mathbb{E}[(\mathcal{G}_g - 1)^+ \mathbf{1}_{\{A_N = k\}}]$, with $J_0$, $\eta$ and $g$ as in that proof. For $pN + KB_N \leq k \leq N$ we have $L_k \geq KB_N$. Lemma 4.7 needs only this, and it gives $\mathrm{Cen}_g(k) + \mathrm{Int}_g(k) \geq (1-\varepsilon)G_N(k)$ for $K$ large and then $N$ large. The bounds for the last term in the proof of (b) also use only $L_k \geq KB_N$, and they make it at most $\varepsilon G_N(k)$. □

### 4.4 Consequences and an example

**Corollary 4.9** (the infinite density)**.** *Assume (RV) and (SPAN) for both signs and (LR) for the plus sign. Fix $v \in (p,1)$ and put $\ell = (v-p)/(1-p)$. For a fresh start of either sign,*

$$\frac{P_N(\lfloor vN \rfloor)}{T_+(N)} \longrightarrow I_+(v) = \frac{1}{\mu_-}\Big[\alpha_+(1-\ell)\ell^{-\alpha_+-1} + \ell^{-\alpha_+}\Big].$$

*The function $I_+$ is not integrable at $v \downarrow p$.*

*Proof.* Let $k = \lfloor vN \rfloor$. Then $k - pN \sim (v-p)N$, which exceeds $K\lambda_N B_N$ for every $K$ once $N$ is large, by Lemma 2.1(v). So Theorem B(b) gives $P_N(k) \sim G_N(k)$. Here $j = N - k \sim (1-v)N = (1-p)(1-\ell)N$ and $L_k \sim \ell N$. By (4.4), $\gamma_+^{(\sigma_0)}(j) \sim (1-\ell)N/\mu_-$. By (LR) and (RV), $f_+(\lfloor L_k \rfloor) \sim \alpha_+\ell^{-\alpha_+-1}T_+(N)/N$ and $T_+(\lfloor L_k \rfloor) \sim \ell^{-\alpha_+}T_+(N)$. By (2.10), $\kappa_+^{(\sigma_0)}(j) \to 1/\mu_-$. Adding the two terms of (4.3) gives $I_+(v)$. Near $v = p$ the first term is of order $(v-p)^{-\alpha_+-1}$, which is not integrable. □

Rebenshtok, Denisov, Hänggi and Barkai [58, Eqs. (8) and (10)] (see also [59, Eqs. (38) and (52)]) obtained for the two-state velocity model an infinite density proportional to $|v|^{-1-\alpha}[\alpha - (\alpha-1)|v|]$, formally and in continuous time. Since $\alpha(1-\ell)\ell^{-\alpha-1} + \ell^{-\alpha} =$

$\ell^{-\alpha-1}[\alpha-(\alpha-1)\ell]$, the function $I_+$ has this shape. The complete giant run alone gives $\ell^{-\alpha-1}[\alpha-\alpha\ell]$. The run in progress at time $N$ supplies the term $\ell^{-\alpha}$, which changes the coefficient of the linear term from $-\alpha$ to $-(\alpha-1)$. So their formula already contains the run in progress. The same shape appears for occupation times in [63, Eq. (25)] and [73, Eq. (83)].

Suppose that $b_{+,N} = B_N$ for all large $N$, so $\lambda_N = 1$. Then there is a sequence $K_N \to \infty$ with

$$\sup_{|k-pN|\le K_N B_N} \Big| \frac{B_N P_N(k)}{h_N(x_{N,k})} - 1 \Big| \to 0 \qquad \text{and} \qquad \sup_{pN+K_N B_N \le k \le N} \Big| \frac{P_N(k)}{G_N(k)} - 1 \Big| \to 0.$$

Indeed, let $\epsilon_N(K)$ be the first supremum with $K$ in place of $K_N$. By Theorem A(c), $\epsilon_N(m) \to 0$ for each integer $m$, so there are $N_1 < N_2 < \cdots$ with $\epsilon_N(m) \le 1/m$ for $N \ge N_m$, and $K_N = m$ for $N_m \le N < N_{m+1}$ gives the first limit. The second holds for every $K_N \to \infty$, by the single-limit form of Theorem B(b). So the bulk profile up to a slowly growing multiple of $B_N$ and the big-jump kernel beyond it give $P_N(k)$ at every point on the plus side with relative error $o(1)$ in a single limit. Schulz and Barkai matched the stable bulk and the infinite density by patching them along the common power law [63, Eq. (26)], formally. This statement is a rigorous form of that matching. The switching point is not explicit, because no statement here has a rate.

**Remark 4.10** (the double limit)**.** The bulk profile and the big-jump kernel are different expressions on their common ground. With a fixed switching point $K$, neither the bulk profile nor the big-jump kernel nor their sum has relative error $o(1)$ uniformly over the plus side. Assume $b_{+,N} = B_N$ and (LR) for the plus sign, and let $k = \lceil pN + KB_N \rceil$ with $K$ fixed. By Theorem A(c), $B_N P_N(k)/h_N(K) \to 1$, and $h_N(K)$ is bounded below by Lemma 2.2(d). On the other hand, $j = N - k \sim (1-p)N$ and $L_k \sim KB_N/(1-p)$, so by (4.4) and (LR)

$$B_N G_N(k) = \frac{\alpha_+}{s} \frac{N T_+(L_k)}{K} (1 + o(1)) = \alpha_+ (1-p)^{\alpha_+} K^{-\alpha_+ - 1} (1 + o(1)),$$

where the second step uses $NT_+(b_{+,N}) \to s$ and Lemma 2.1(i), and the second term of (4.3) is negligible because $B_N T_+(L_k) \to 0$. The right side is the leading term of the tail of $h_N$, but it is not $h_N(K)$. Put $r_N(K) = h_N(K)/(\alpha_+(1-p)^{\alpha_+}K^{-\alpha_+-1})$. Then

$$\frac{P_N(k)}{G_N(k)} = r_N(K) + o(1), \qquad \frac{P_N(k)}{B_N^{-1} h_N(K) + G_N(k)} = \frac{r_N(K)}{1 + r_N(K)} + o(1).$$

Since $h_N \le C_1$ by Lemma 2.2(a), $r_N(K)$ is bounded in $N$ for each $K$, so the second ratio stays below a constant smaller than one for large $N$. The first ratio need not converge when $\theta_{-,N}$ oscillates, and it may equal one at isolated values of $K$. In the example of Section 9, where $\theta_N = (1,1)$, $r_N(K)$ does not depend on $N$ and moves from $0.31$ at $K = 1$ to $1.68$ at $K = 4$, so it equals one at some $K$ in between. So no fixed $K$ makes the kernel accurate in general. Section 9 reports the check. By Theorem B(b) and Theorem A(c), $\lim_{K\to\infty} \limsup_{N\to\infty} |r_N(K) - 1| = 0$, and the ratio $P_N/G_N$ tends to one in this double limit, which is the form of Theorem B(b).

**Remark 4.11** (the start of the zone)**.** The zone of Theorem B(b) cannot start at $pN + KB_N$ with $K$ fixed when $\theta_{+,N} \to 0$. Some factor in front of $B_N$ that tends to infinity is needed. Assume (LR) for the plus sign and let $\theta_{+,N} \to 0$ along a subsequence. Fix $K$ and let $k = \lceil pN + KB_N \rceil$. By Theorem A(c) and Lemma 2.2(d), $B_N P_N(k)$ is bounded below. By (LR) and the bounds $\gamma(j) \le (1+\mu_+)N$ and $\kappa \le 1$,

$$B_N G_N(k) \le C\, N T_+(L_k) \frac{B_N}{L_k} + B_N T_+(L_k) \le \frac{C}{K} N T_+(L_k) + B_N T_+(B_N).$$

Since $L_k \geq KB_N = Kb_{+,N}/\theta_{+,N}$, the bound (2.4) gives $NT_+(L_k) \leq 4s\,(K/\theta_{+,N})^{-(\alpha_++1)/2}$, which tends to zero, and $B_NT_+(B_N) \to 0$. So $P_N(k)/G_N(k) \to \infty$ along the subsequence. The values there are carried by short minus runs, the second term of Lemma 4.4, and not by a giant plus run. The factor need not be as large as $\lambda_N$, which comes from the exponential left-tail bound of Lemma A.1(iv). Theorem 4.14(c) shows that the zone starts at $(1+\delta)\xi_NB_N$ for every $\delta > 0$, that it cannot start at $(1-\delta)\xi_NB_N$, and that $\xi_N = (\log 1/\theta_{+,N})^{(\alpha_- -1)/\alpha_- + o(1)} = o(\lambda_N)$. So the threshold is sharp up to the factor $1 \pm \delta$. The lower bound needs no factor at all (Corollary 4.8).

**Example 4.12** ((RV) without (LR))**.** Fix $1 < \alpha < 2$ and let $n_i = 2^{2^i}$ for $i \geq 1$ and $n_0 = 0$. Put $h(n) = 1/\log n_i$ for $n_{i-1} < n \leq n_i$, and

$$T(n) = \frac{n^{-\alpha}\big(1+h(n)\big)}{1+h(1)}, \qquad f(n) = T(n) - T(n+1) \qquad (n \geq 1).$$

Let both run laws be this law, so $p = 1/2$ and $L_k = 2k - N$. Then (RV) and (SPAN) hold and (LR) fails. Let $N_i = 2n_i$ and $k_i = 3n_i/2$. Then $k_i - pN_i = n_i/2$, so $k_i$ lies in the zone of Theorem B(b) for every $K$ once $i$ is large, and

$$\frac{P_{N_i}(k_i)}{G_{N_i}(k_i)} \to 0.$$

So the conclusion of Theorem B(b), whose kernel reads $f_+$ at a single point, fails under (RV) and (SPAN) alone, while (a) holds. For sums of a fixed number of variables the analogous optimality is due to Berger [6, Eq. (2.4)]. Under regular variation of the tail alone the local large deviation bound of his Theorem 2.3 cannot be improved, while his Theorem 2.4 gives the local asymptotics under the local condition.

*Verification.* Write $h_i = 1/\log n_i = 1/(2^i \log 2)$, so $h_{i+1} = h_i/2$ and $h$ is nonincreasing and tends to zero. Put $c = 1/(1+h(1))$. Then $T(1) = 1$, and $T$ is decreasing and tends to zero, so $f$ is a probability mass function with $f(n) > 0$ for all $n$. This gives (SPAN), and $T(n) \sim cn^{-\alpha}$ gives (RV). For $n_{i-1} < n < n_i$ the points $n, n+1$ lie in the same block, so $f(n) = c(1+h_i)(n^{-\alpha} - (n+1)^{-\alpha}) \leq 2c\alpha n^{-\alpha-1}$. At the block ends,

$$f(n_i) = c\Big[\big(n_i^{-\alpha} - (n_i+1)^{-\alpha}\big)(1+h_{i+1}) + n_i^{-\alpha}\,\frac{h_i}{2}\Big],$$

so $\frac{c}{2}h_in_i^{-\alpha} \leq f(n_i) \leq Ch_in_i^{-\alpha}$. Since $n_ih_i \to \infty$, $n_if(n_i)/T(n_i) \to \infty$ and (LR) fails.

Let $N = N_i$, $k = k_i$, $j = n_i/2$, so $L_k = n_i$. By (4.4), $\gamma(j) \geq n_i/(2\mu)$ for large $i$, where $\mu$ is the common mean. Hence

$$G_N(k) \geq \gamma(j)f(n_i) \geq \frac{c}{4\mu}\,h_in_i^{1-\alpha}.$$

For the upper bound take $\eta = 1/32$ and $g = \lceil n_i/32 \rceil$ in Lemma 4.2, and drop the last term of (4.10). Lemma 4.4 uses only (RV) and (SPAN). With $L_k = n_i$, $N = 2n_i$ and $r = 2$ it bounds the first term by $Cn_i^{2-2\alpha}$ plus terms that decay faster than every power of $n_i$, and $n_i^{2-2\alpha} = o(h_in_i^{1-\alpha})$. By (4.11), $\mathrm{Cen}_g(k) \leq T(g) \leq Cn_i^{-\alpha}$. For $\mathrm{Int}_g$ note that $n_{i-1} = \sqrt{n_i} < n_i/32$ and $n_{i+1} = n_i^2 > 2n_i$ for large $i$. So $n_i$ is the only block end in $[g, N]$, and $f(n) \leq Cn_i^{-\alpha-1}$ for the other $n$ there. With the measure $\chi_j$ of the proof of Lemma 4.7,

$$\mathrm{Int}_g(k) \leq Cn_i^{-\alpha-1}\gamma(j) + f(n_i)\sup_z \chi_j(z) \leq Cn_i^{-\alpha} + Ch_in_i^{-\alpha}\sup_z \chi_j(z).$$

It remains to show $\sup_z \chi_j(z) = o(j)$. Since $\chi_j = \sum_{i'<j} \rho_{i'} * \beta_{j-i'}$ and the two factors have masses at most 1 and $1+\mu$, $\sup_z(\rho_{i'} * \beta_{j-i'})(z) \leq \min\{(1+\mu)\sup_x \rho_{i'}(x),\ \sup_x \beta_{j-i'}(x)\}$.

One of $i'$ and $j-i'$ is at least $j/2$. So it suffices that $\sup_x \rho_n(x) \to 0$ and $\sup_x \beta_n(x) \to 0$ as $n \to \infty$. Let $m_1 = \lfloor n/(2\mu) \rfloor$. By (2.13) and Lemma A.1(ii), $\rho_n(x) \le \mathbb{P}(S_{m_1} \ge n) + \bar{u}/a(m_1-1)$, because a renewal at $n$ with at most $m_1$ runs forces $S_{m_1} \ge n$. Both terms tend to zero, by the law of large numbers and since $a(m) \to \infty$. By (4.8) and the independence of $X_J^+$, $\beta_n(x) \le \sum_{r\ge 0} T(r)\mathbb{P}(S_{J-1} = x-r)$ with $T(0)=1$, so $\beta_n(x) \le (1+\mu)[\mathbb{P}(J \le m_1) + \bar{u}/a(m_1-1)]$, and $\mathbb{P}(J \le m_1) \le \mathbb{P}(S_{m_1} \ge n)$, which tends to zero. Hence $P_N(k) = o(h_i n_i^{1-\alpha})$, and the ratio tends to zero. □

**Remark 4.13** (the hypothesis (LR))**.** The spike at $n_i$ enters $G_N(k_i)$ at full weight but enters $P_N(k_i)$ only through a window of width of order $B_N$ around $L_k$. The proof of Theorem B(b) uses (LR) to replace $f_+(n)$ by $f_+(\lfloor L_k \rfloor)$ for $|n - L_k| \le ZB_N$, to bound $f_+^*(\eta L_k)$ by $CT_+(L_k)/L_k$, and to bound $G_N$ below. The first use is the insensitivity of the step law to shifts on the scale of the fluctuations, which Denisov, Dieker and Shneer identify as the condition for local big-jump asymptotics of sums [16, Section 1 and condition (6) of Corollary 2.1]. Their local theorem for infinite variance [16, Theorem 9.1] assumes it in the form of intermediate regular variation of $x \mapsto \mathbb{P}(x < \xi \le x+T)$, which for an integer step $\xi$ and $T=1$ is the mass function. In our zone the windows have relative width at most $Z/K$, and under (RV) insensitivity at relative width $o(1)$ is equivalent to (LR), since $T_+(n) - T_+((1+\delta)n)$ is then both $\delta n f_+(n)(1+o_\delta(1))$ and $T_+(n)(\alpha_+\delta + O(\delta^2))$. We do not prove that (LR) is necessary for Theorem B(b).

### 4.5 The start of the zone

When $\theta_{+,N} \to 0$ the zone of Theorem B(b) starts at $K\lambda_N B_N$, and by Remark 4.11 it cannot start at $KB_N$. Here we locate its start up to a factor $1 \pm \delta$ and describe $P_N$ before it on the logarithmic scale. Below the start the values are carried by minus runs that fall short of their mean, and the cost of the shortfall is a Cramér transform of the minus runs.

For sums the crossover is known. Let $S_n$ be a sum of $n$ independent centred variables whose left tail is regularly varying with index $-\alpha$, $\alpha \in (1,2)$, and whose right tail is lighter, regularly varying with index $-\gamma$ for some $\gamma > \alpha$, and let $b_n$ be the stable scale. In this setting Rozovskiĭ [62, Corollary 2A] proved that the right tail of $S_n$ is $n$ times that of one step, uniformly for $x \ge t(c\log n)^{(\alpha-1)/\alpha} b_n$, for every $t > 1$ and an explicit constant $c$ that depends on the two indices. His asymptotic representation of the tail holds uniformly in $x > 0$, so it also covers the range below this boundary. Denisov, Dieker and Shneer [16, Proposition 9.2 and Theorem 9.2] proved the local form on the same range. The mechanism is the one here. Below the boundary many small steps on the heavy side cost less than one jump on the lighter side. In Theorem 4.14 the minus runs are the heavy side and the plus runs the lighter one. When the indices differ, $\log(1/\theta_{+,N})$ is of order $\log N$, and $\xi_N B_N$ has the order $(\log N)^{(\alpha_- -1)/\alpha_-} b_{-,N}$ of their boundary. So Theorem 4.14 is the analogue of these results for the count. What it adds is the count at a fixed time. The number of runs before step $N$ is random and tied to their lengths, and the big jump may be made by the run in progress at time $N$. The theorem also covers equal indices with a slowly varying ratio of the scales, where $\log(1/\theta_{+,N}) = o(\log N)$.

Throughout this subsection $b_{+,N} \le b_{-,N}$ for all large $N$, so $B_N = b_{-,N}$. We write $X = X_-$, $\alpha = \alpha_-$, $\mu = \mu_-$, $a = a_-$, $T = T_-$, $S_m = S_m^-$, $n_N = N/s$ and $x_{\min} = \min\{n : f_-(n) > 0\}$. For $\vartheta, v, d \ge 0$ and $m > 0$ put

$$\psi(\vartheta) = \log \mathbb{E} e^{-\vartheta(X-\mu)}, \qquad \psi^*(v) = \sup_{\vartheta \ge 0} \big(\vartheta v - \psi(\vartheta)\big), \qquad \mathcal{I}(m,d) = m\,\psi^*(d/m).$$

So $\mathcal{I}(m,d)$ is the Cramér cost of a shortfall $d$ below the mean for a sum of $m$ minus runs.

We put

$$\mathcal{I}_N(x) = \mathcal{I}\big(n_N,\, xB_N/p\big), \qquad \beta_- = \frac{\alpha_-}{\alpha_- - 1}, \qquad \varkappa_- = (\alpha_- - 1)\,\alpha_-^{-\beta_-}\,K_{\alpha_-}^{-1/(\alpha_- - 1)}. \quad (4.15)$$

If $A_N = pN + xB_N$ and the plus runs keep their typical length, then about $n_N$ minus runs hold the $N - A_N$ minus steps, and they fall short of their mean by $xB_N/p$. When $\theta_{+,N} \to 0$ let $\xi_N$ be the root of $\mathcal{I}_N(\xi_N) = \alpha_+ \log(1/\theta_{+,N})$. It exists for large $N$ by the proof of Theorem 4.14(c), and it is unique since $\mathcal{I}_N$ is strictly increasing. A function $\phi$ on $(0, v_0)$ is regularly varying at $0+$ with index $\varrho$ when $x \mapsto \phi(1/x)$ is regularly varying with index $-\varrho$, and Lemma 2.1(i) to (iii) then hold at $0+$.

**Theorem 4.14** (the start of the zone)**.** *Assume (RV) and (SPAN) for both signs and (LR) for the plus sign, let the start be fresh with either sign, let $G_N$ be its kernel* (4.3)*, and assume $b_{+,N} \le b_{-,N}$ for all large $N$. Write $x = x_{N,k}$.*

*(a) (Two-sided bound.) For every $\varepsilon > 0$ there is $K$ such that, for every sequence $X_N$ with $X_N B_N = o(N)$, for all large $N$ and all $k$ with $K \le x \le X_N$,*

$$\max\Big\{(1-\varepsilon)G_N(k),\ \frac{e^{-(1+\varepsilon)\mathcal{I}_N(x)}}{B_N}\Big\} \le P_N(k) \le (1+\varepsilon)G_N(k) + \frac{e^{-(1-\varepsilon)\mathcal{I}_N(x)}}{B_N}.$$

*(b) (The exponent.) For fixed $x > 0$, $\mathcal{I}_N(x) \to \varkappa_-(x/p)^{\beta_-}$. Let $X_N B_N = o(N)$. For every $\delta > 0$ there are $c, C > 0$ with $cx^{\beta_- - \delta} \le \mathcal{I}_N(x) \le Cx^{\beta_- + \delta}$ for $1 \le x \le X_N$ and large $N$, and $\mathcal{I}_N(cx)/\mathcal{I}_N(x) \to c^{\beta_-}$ uniformly in $1 \le x \le X_N$ for every fixed $c > 0$. If $T_-(n) \sim c_- n^{-\alpha_-}$, then $\mathcal{I}_N(x) = \varkappa_-(x/p)^{\beta_-}(1 + o(1))$ uniformly in $1 \le x \le X_N$.*

*(c) (The start.) Assume in addition $\theta_{+,N} \to 0$. Then $\xi_N \to \infty$, $\xi_N/\lambda_N \to 0$ and $\xi_N = (\log 1/\theta_{+,N})^{(\alpha_- - 1)/\alpha_- + o(1)}$. For every $\delta \in (0,1)$ and every $K_0 > 0$, with the ratio read as in Theorem B(b),*

$$\sup_{pN + (1+\delta)\xi_N B_N \le k \le N} \Big|\frac{P_N(k)}{G_N(k)} - 1\Big| \to 0,$$

$$\inf_{pN + K_0 B_N \le k \le pN + (1-\delta)\xi_N B_N} \frac{P_N(k)}{G_N(k)} \to \infty.$$

*If $T_-(n) \sim c_- n^{-\alpha_-}$, then $\xi_N \sim p\big(\alpha_+ \log(1/\theta_{+,N})/\varkappa_-\big)^{(\alpha_- - 1)/\alpha_-}$.*

*The same holds at the minus end with the signs exchanged.*

So the conclusion of Theorem B(b) holds from $(1+\delta)\xi_N B_N$ on, in a single limit. Below the start, on $K \le x \le (1-\delta)\xi_N$, the proof of (c) shows that the light term of (a) dominates, so $\log(B_N P_N(k)) = -(1 + O(\varepsilon))\mathcal{I}_N(x)$ there for $K \ge K(\varepsilon)$. The values are those of a moderate deviation of the minus runs. The strip $(1-\delta)\xi_N \le x \le (1+\delta)\xi_N$, where both terms of (a) matter, is not described with a relative error.

**Remark 4.15** (a conjecture)**.** Below the start the factors $e^{\pm\varepsilon\mathcal{I}_N(x)}$ in (a) are unbounded, so (a) gives no relative error there. Assume the hypotheses of (c) and $T_\sigma(n) = c_\sigma n^{-\alpha_\sigma}(1 + O(n^{-\varrho}))$ for both signs and some $\varrho > 0$, and let $\tau_N(x) = \alpha_+((1-p)\theta_{+,N})^{\alpha_+} x^{-\alpha_+ - 1}$ be the leading term of the right tail of $h_N$. We conjecture that

$$\lim_{K\to\infty} \limsup_{N\to\infty} \sup_{pN + KB_N \le k \le pN + N/K} \Big|\frac{B_N P_N(k)}{h_N(x) + B_N G_N(k) - \tau_N(x)} - 1\Big| = 0.$$

For sums the global form of such a statement is known. The representation of Rozovskiĭ [62] joins the moderate-deviation term and the big-jump term in one formula for the tail of $S_n$, uniformly in $x > 0$ (see also [61]). The conjecture asks for the local form

for the count. The first missing step is a local estimate of $\mathbb{P}(S_m^- = m\mu_- - d)$ with relative error $o(1)$ for $d$ up to $C(\log N)^{1/\beta_-}a_-(m)$, where $\mathcal{I}$ is of order $\log N$ and a second-order condition is needed.

We need three facts on the minus runs.

The proof of Theorem 4.14 rests on the Cramér transform $\psi^*$ of the minus runs, through a bound on the Laplace exponent of a minus run and tilted versions of the exact identities of Section 2.5, and it occupies 5 pages. It is in the arXiv version of this paper, at the end of its Section 4. The journal version omits it because the theorem concerns the one window on which the paper has no relative error, and no other result here depends on it.

## 5 The initial run and aging

The first run is the only part of the walk that the two fresh laws do not describe, and Proposition 2.4 says exactly how it enters. This section turns that identity into a criterion for each zone of the range of the count.

### 5.1 Statements

The law of the initial run may depend on $N$. Throughout this section $(\Sigma, D) = (\Sigma^{(N)}, D^{(N)})$ has a law $\nu = \nu^{(N)}$ that may vary with $N$, and we assume the tightness

$$\varepsilon_J = \sup_N \mathbb{P}\big(D^{(N)} > J\big) \to 0 \qquad (J \to \infty). \tag{5.1}$$

A fixed law satisfies (5.1) because $D$ is finite almost surely, and it is the case we have in mind. The general case costs nothing in the proofs, and Section 5.8 needs it. We write $\nu_\sigma$, $Q_\sigma$ and $\pi_\sigma = \mathbb{P}(\Sigma = \sigma)$ without the index $N$, and we put

$$\bar\nu_\sigma(n) = \sup_N \sup_{d \ge n} \nu_\sigma^{(N)}(d), \qquad \bar\nu(n) = \bar\nu_+(n) + \bar\nu_-(n), \tag{5.2}$$

the upper envelopes of the family. For a fixed law, $\bar\nu_+(n) = o(T_+(n))$ if and only if $\nu_+(n) = o(T_+(n))$, since $T_+$ is nonincreasing, and $B_n\bar\nu(n/2) \to 0$ if and only if $B_n\mathbb{P}(D = n) \to 0$, since $B_n \le CB_d$ for $d \ge n/2$ by Lemma 2.1(ii) and (v).

For a sign $\sigma$ and $b \in \mathbb{Z}$ define the directional defect

$$\mathcal{D}_N^\sigma(b) = B_N \sum_{1 \le m < N/2} \nu_\sigma(N-m)\,\Pi_m^{\bar\sigma}(b), \qquad \Delta_{\sigma,N} = \sup_{b \ge 1} \mathcal{D}_N^\sigma(b), \tag{5.3}$$

with $\Pi_m^{\bar\sigma}(b) = 0$ for $b \notin \{0, \dots, m\}$. Note that $\Pi_m^{\bar\sigma}(0) = 0$ for $m \ge 1$, since a fresh walk has at least one step of its own sign. So $\mathcal{D}_N^\sigma(b) = 0$ for $b \le 0$ and for $b \ge N/2$. Take a first run of sign $\sigma$ that ends within the last $N/2$ steps. Times $B_N$, its contribution at the point $k$ is $\mathcal{D}_N^+(N-k)$ for $\sigma = +$ and $\mathcal{D}_N^-(k)$ for $\sigma = -$. So the argument $b$ is the number of steps of the opposite sign. We say that a fixed initial run satisfies the *bulk criterion* if $B_n\mathbb{P}(D = n) \to 0$.

For the big-jump zone let $L_k$ be as in (4.1) and $\widehat{G}_N$ as in (4.6), and define for $pN < k \le N$ and $j \ge 0$

$$\kappa_+(j) = \pi_+ u_-(j)\mathbf{1}_{\{j \ge 1\}} + \sum_{d=1}^{j} \nu_-(d)\,u_-(j-d), \tag{5.4}$$

$$G_N^\nu(k) = \frac{N - L_k}{\mu_-}\, f_+(\lfloor L_k \rfloor) + T_+(\lfloor L_k \rfloor)\,\kappa_+(N-k), \tag{5.5}$$

$$\Psi_N(k) = \sum_{L_k/2 \le d < N} \nu_+(d)\,F_{N-d}^-(k-d). \tag{5.6}$$

The kernel $\kappa_+(j)$ is the probability that a plus run other than the first run starts immediately after the $j$-th minus step. For a fresh start of sign $\sigma_0$ it equals $\kappa_+^{(\sigma_0)}(j)$ for $j \geq 1$, by (2.11). The kernel $G_N^\nu$ uses the limit coefficient $(N - L_k)/\mu_-$. The term $\Psi_N(k)$ is the exact contribution of a first plus run that supplies at least half of the big jump. We also use two conditions on $\nu_+$. The first is the regularity condition

$$\begin{aligned} &\sup_{d \geq n/2} \nu_+(d) \leq C_\nu\big(\nu_+(n) + T_+(n)\big) \ \text{ for all large } n, \\ &\lim_{\delta \downarrow 0} \limsup_{n \to \infty} \sup_{|d-n| \leq \delta n} \frac{|\nu_+(d) - \nu_+(n)|}{\nu_+(n) + T_+(n)} = 0, \end{aligned} \tag{R$\nu$}$$

where for a family both bounds are required uniformly in $N$. It says that $\nu_+$ is continuous at the scale $n$ up to errors of the order of $T_+(n)$. Every regularly varying $\nu_+$ with a negative index satisfies (R$\nu$), by Lemma 2.1(i) and (ii), and so do $\nu_+ = 0$ and, under (LR), the fresh plus start $\nu_+ = f_+$. The second is a condition on the window averages

$$W_n = \frac{1}{B_n} \sum_{|d-n| \leq B_n} \nu_+(d),$$

written $W_n^{(N)}$ when the law depends on $N$.

**Theorem C** (the initial run)**.** *Let the law of the initial run $(\Sigma, D)$ depend on $N$ or not, subject to* (5.1)*.*

*(a) (The exact defect.) Assume (RV) and (SPAN) for both signs. Uniformly over $0 \leq k \leq N$,*

$$\begin{aligned} B_N P_N(k) = h_N(x_{N,k}) + B_N Q_+(N)\mathbf{1}_{\{k=N\}} + B_N Q_-(N)\mathbf{1}_{\{k=0\}} \\ + \mathcal{D}_N^+(N-k) + \mathcal{D}_N^-(k) + o(1). \end{aligned}$$

*(b) (The bulk criterion.) Under the hypotheses of (a),*

$$\begin{aligned} &\sup_{1 \leq k < N} \big|B_N P_N(k) - h_N(x_{N,k})\big| = \max\big(\Delta_{+,N}, \Delta_{-,N}\big) + o(1), \\ &\sup_{0 \leq k \leq N} \big|B_N P_N(k) - h_N(x_{N,k})\big| = \max\big(\Delta_{+,N}, \Delta_{-,N}, B_N Q_+(N), B_N Q_-(N)\big) + o(1). \end{aligned}$$

*The first error tends to zero if $B_N \bar{\nu}(N/2) \to 0$ and only if $B_N \mathbb{P}(D^{(N)} = N-1) \to 0$, and the second needs in addition $B_N Q_\sigma(N) \to 0$ for both signs. For a fixed law the first error tends to zero if and only if $B_n \mathbb{P}(D = n) \to 0$, and the second if and only if $B_n \mathbb{P}(D \geq n) \to 0$. For every fixed $\varepsilon > 0$ the approximation holds on $\varepsilon N \leq k \leq (1-\varepsilon)N$ with no condition on the initial run beyond* (5.1)*.*

*(c) (The big-jump zone.) Assume (RV) and (SPAN) for both signs and (LR) for the plus sign, and let $\lambda_N$ be as in* (4.5)*. Then*

$$\lim_{K \to \infty} \limsup_{N \to \infty} \sup_{pN + K\lambda_N B_N \leq k \leq N} \frac{\big|P_N(k) - Q_+(N)\mathbf{1}_{\{k=N\}} - G_N^\nu(k) - \Psi_N(k)\big|}{\widehat{G}_N(k)} = 0.$$

*The same holds with $\Psi_N$ removed if $\bar{\nu}_+(n) = o(T_+(n))$, and only if $\nu_+(N-1) = o(T_+(N))$. For a fixed law both conditions read $\nu_+(n) = o(T_+(n))$. If $\nu_+$ satisfies (R$\nu$), it holds with $\Psi_N(k)$ replaced by $\nu_+(\lfloor L_k \rfloor)(1 + \mu_+ u_-(N-k))\mathbf{1}_{\{k<N\}}$ and $\widehat{G}_N(k)$ replaced by $\widehat{G}_N(k) + \nu_+(\lfloor L_k \rfloor)$.*

*(d) (The boundary layer.) Assume (RV) for the plus sign. For every fixed $j \geq 0$,*

$$P_N(N-j) = Q_+(N)\mathbf{1}_{\{j=0\}} + T_+(N)\,\kappa_+(j) + B_N^{-1}\mathcal{D}_N^+(j) + o(T_+(N)),$$

*where $B_N^{-1}\mathcal{D}_N^+(j) = \sum_{1\leq m<N/2} \nu_+(N-m)\Pi_m^-(j)$ does not involve the scales. The defect term is $o(T_+(N))$ for every $j \geq 1$ if $\bar\nu_+(n) = o(T_+(n))$, and if it is $o(T_+(N))$ for one $j \geq 1$, then $\nu_+(N-m_0) = o(T_+(N))$ for some fixed $m_0 = m_0(j) \geq 1$. For a fixed law both conditions read $\nu_+(n) = o(T_+(n))$, and then $P_N(N-j)/T_+(N) \to \kappa_+(j)$ for every $j \geq 1$. If $\nu_+$ satisfies (R$\nu$), then $B_N^{-1}\mathcal{D}_N^+(j) = \nu_+(N)\big(1+\mu_+u_-(j)\big) + o\big(\nu_+(N) + T_+(N)\big)$ for every $j \geq 1$.*

*(e) (Interior of the big-jump zone.) Assume the hypotheses of (c). For $0 < \varepsilon < \frac{1}{2}$ let $Z_N^\varepsilon = \{k : \varepsilon N \leq L_k \leq (1-\varepsilon)N\}$, and consider the property*

$$\sup_{k\in Z_N^\varepsilon} \frac{|P_N(k) - G_N^\nu(k)|}{\widehat{G}_N(k)} \to 0 \quad (N\to\infty) \qquad \text{for every } \varepsilon \in (0,\tfrac{1}{2}). \tag{5.7}$$

*The condition $\bar\nu_+(n) = o(T_+(n))$ implies* (5.7)*. Conversely* (5.7) *implies the window condition $W_n^{(2n)} = o(T_+(n))$. If (LR) holds for the minus sign as well, then $\sup_{\varepsilon N/4\leq n\leq N} W_n^{(N)} = o(T_+(N))$ for every $\varepsilon \in (0,\frac{1}{2})$ implies* (5.7)*. For a fixed law the last two conditions read $W_n = o(T_+(n))$.*

*At the plus end the criteria are nested. Since $\nu_+ \leq Q_+$, the atom condition $\sup_N Q_+(n) = o(T_+(n))$ implies the big-jump condition $\bar\nu_+(n) = o(T_+(n))$. Since $B_nT_+(n) \to 0$, the big-jump condition implies $B_n\bar\nu_+(n/2) \to 0$, which is the plus half of the bulk criterion.*

The zone in (c) starts where the zone of Theorem B(b) starts, and by Remark 4.11 its factor $\lambda_N$ cannot be replaced by a constant, already for a fresh start. For fresh starts Theorem 4.14 gives the sharp factor. At $k = N$ we have $\Psi_N(N) = 0$, $G_N^\nu(N) = 0$ and $\mathcal{D}_N^+(0) = 0$, so (c) and (d) both reduce to the identity $P_N(N) = Q_+(N)$ of Proposition 2.4. Part (e) refines (c) on the interior of the zone, where (5.7) is the statement of (c) without $\Psi_N$. The regular case of (c) gives one formula on the whole zone up to the end, with the term $\nu_+(\lfloor L_k\rfloor)(1+\mu_+u_-(N-k))$ for the first run as the giant run, and the regular case of (d) is its value at $j = N-k$ fixed, since $\nu_+(\lfloor L_k\rfloor) = \nu_+(N)(1+o(1)) + o(T_+(N))$ there by (R$\nu$).

**Remark 5.1** (the prepared boundary kernel)**.** By (2.10) and dominated convergence, $\kappa_+(j) \to 1/\mu_-$, the fresh value. With generating functions $F$, $U = 1/(1-F)$ and $\widehat\nu_-$ of $f_-$, $u_-$ and $\nu_-$, the identity $\kappa_+ = u_-$ on $j \geq 1$ reads $\widehat\nu_-U = \pi_-(U-1)$, that is, $\nu_- = \pi_- f_-$. So the prepared kernel is the fresh one exactly when the first run, given that it is a minus run, has the law of $X_-$.

### 5.2 Tools

For a horizon $n$ and a point $k > pn$ we write $L_k^{(n)} = (k-pn)/(1-p)$ for the quantity (4.1) at the horizon $n$, and $\widehat{G}_n(k)$, $G_n^{(\sigma_0)}(k)$ and $\lambda_n$ for (4.6), (4.3) and (4.5) at the horizon $n$. Recall from Lemma 2.1(v) that $B_n$ is nondecreasing and regularly varying with index $1/\alpha_*$.

Fix $c \in (0,1)$. We have $\lambda_n = 1 + \log\max(1, r(n))$ with $r(n) = b_{-,n}/b_{+,n}$, which is regularly varying with index $\varrho = 1/\alpha_- - 1/\alpha_+$. By the uniform convergence theorem [9, Theorem 1.5.2], $r(n)/r(N) = (n/N)^\varrho(1+o(1))$ uniformly in $n \in [cN, N]$. If $\varrho \geq 0$, then $(n/N)^\varrho \leq 1$, so $r(n) \leq r(N)(1+o(1))$. If $\varrho < 0$, then $r(N) \to 0$, so $r(n) \leq c^\varrho r(N)(1+o(1)) < 1$ for large $N$ and $\lambda_n = \lambda_N = 1$. Since $x \mapsto \log\max(1,x)$ is nondecreasing and changes by at most $\log(1+\delta)$ when $x$ is multiplied by $1+\delta$, in both cases $\lambda_n \leq \lambda_N + o(1)$ uniformly in $n \in [cN, N]$. Since $\lambda_N \geq 1$,

$$\lambda_n \leq 2\lambda_N \qquad (cN \leq n \leq N) \tag{5.8}$$

once $N$ is large. By Lemma 2.1(v), $\lambda_N B_N = o(N)$. So for each $c > 0$ the points with $k - pN \geq cN$ lie in the zone of Theorem C(c) for every $K$ once $N$ is large.

**Lemma 5.2** (tightness of the count)**.** *Assume (RV) and (SPAN). For a fresh start of either sign,* $\sup_{n\geq 1} \mathbb{P}(|A_n - pn| > RB_n) \to 0$ *as* $R \to \infty$.

*Proof.* Fix the starting sign and put $\varepsilon_n = \sup_k |B_n F_n(k) - h_n(x_{n,k})|$, which tends to zero by Theorem A(a). Let $R \geq 1$ and let $k$ run over the integers with $|k - pn| \leq RB_n$. The points $x_{n,k}$ are spaced $1/B_n$ apart, and the intervals $[x_{n,k}, x_{n,k} + 1/B_n)$ cover $[-R + 1/B_n, R]$. On each of them $h_n$ exceeds $h_n(x_{n,k})$ by at most $C_2/B_n$, by Lemma 2.2(a). There are at most $2RB_n + 1$ such $k$. Hence, for $B_n \geq 1$,

$$\begin{aligned}\mathbb{P}(|A_n - pn| \leq RB_n) &\geq \sum_k \frac{h_n(x_{n,k}) - \varepsilon_n}{B_n} \geq \int_{-R+1/B_n}^{R} h_n - 3R\Big(\varepsilon_n + \frac{C_2}{B_n}\Big) \\ &\geq 1 - \varepsilon_h'(R-1) - 3R\Big(\varepsilon_n + \frac{C_2}{B_n}\Big).\end{aligned}$$

So $\limsup_n \mathbb{P}(|A_n - pn| > RB_n) \leq \varepsilon_h'(R-1)$, which tends to zero by Lemma 2.2(b). For each of the finitely many remaining $n$ the probability is zero once $RB_n > n$. □

The measure $m \mapsto \Pi_m^-(j)$ of the Green identity sits within $O(B_N)$ of $j/(1-p)$, the time at which a typical walk collects $j$ minus steps.

**Lemma 5.3** (the Green measure is concentrated)**.** *Assume (RV) and (SPAN). For every* $\varepsilon > 0$ *there are* $R$ *and* $N_0$ *such that for all* $N \geq N_0$ *and all* $1 \leq j \leq N$,

$$\sum_{m\geq 1:\ |m - j/(1-p)| > RB_N} \Pi_m^-(j) \leq \varepsilon.$$

*Proof.* Take a fresh minus walk continued forever, as in the proof of Lemma 2.5. Its minus runs $X_1^-, X_2^-, \dots$ and its plus runs $X_1^+, X_2^+, \dots$ are independent, and the $i$-th plus run comes right after the $i$-th minus run. Let $C_m$ be the number of minus steps among the first $m$, let $t_j$ be the step of the $j$-th minus step, and let $r$ be the index of the minus run that contains it. Put $\xi_j = X_r^+$ if $S_r^- = j$ and $\xi_j = 0$ otherwise. As in the proof of Lemma 2.5, $C_m = j$ exactly for $t_j \leq m \leq t_j + \xi_j$. The index $r$, the event $\{S_r^- = j\}$ and $t_j = j + X_1^+ + \cdots + X_{r-1}^+$ are functions of the minus runs and of $X_1^+, \dots, X_{r-1}^+$. So $X_r^+$ is independent of them and has the law of $X_+$.

Put $c = j/(1-p)$, $\mathcal{S} = \{m : |m - c| > RB_N\}$ and $\mathcal{V} = \{m : |m - c| \leq RB_N/2\}$. If $t_j \in \mathcal{V}$ and $\xi_j < RB_N/2$, then the steps $t_j, \dots, t_j + \xi_j$ all lie outside $\mathcal{S}$. Hence, by Tonelli's theorem,

$$\sum_{m\in\mathcal{S}} \Pi_m^-(j) = \mathbb{E}\,\#\{m \in \mathcal{S} : C_m = j\} \leq \mathbb{E}\big[(1 + X_r^+)\mathbf{1}_{\{t_j \notin \mathcal{V}\}}\big] + \mathbb{E}\big[(1 + X_r^+)\mathbf{1}_{\{X_r^+ \geq RB_N/2\}}\big].$$

By the independence just noted, the first term is $(1 + \mu_+)\mathbb{P}(t_j \notin \mathcal{V})$ and the second is $\mathbb{E}[(1 + X_+)\mathbf{1}_{\{X_+ \geq RB_N/2\}}]$, which tends to zero as $N \to \infty$ because $\mu_+ < \infty$.

It remains to bound $\mathbb{P}(t_j \notin \mathcal{V})$. Let $m_1 = \lfloor c + RB_N/2 \rfloor$ and $m_2 = \lceil c - RB_N/2 \rceil$. If $t_j \notin \mathcal{V}$, then $t_j > m_1$ or $t_j \leq m_2$. Now $\{t_j > m_1\} = \{C_{m_1} < j\} = \{A_{m_1} > m_1 - j\}$, and $m_1 - j - pm_1 = (1-p)m_1 - j \geq (1-p)(RB_N/2 - 1) \geq (1-p)RB_N/4$ for large $N$. In the same way $\{t_j \leq m_2\} = \{A_{m_2} \leq m_2 - j\}$ and $(1-p)m_2 - j \leq -(1-p)RB_N/4$, and this event is empty if $m_2 < 1$. For $N \geq N_0$ we have $RB_N \leq N$, so $m_1, m_2 \leq c_9 N$ with $c_9 = 2 + 1/(1-p)$, and $B_{m_i} \leq B_{c_9 N} \leq c_{10} B_N$ for some $c_{10}$ by Lemma 2.1(ii). Hence

$$\mathbb{P}(t_j \notin \mathcal{V}) \leq 2 \sup_{n\geq 1} \mathbb{P}\Big(|A_n - pn| > \frac{(1-p)R}{4c_{10}}\, B_n\Big),$$

which is small for large $R$ by Lemma 5.2. □

Finally we record the fresh big-jump asymptotics at a general horizon. For a horizon $n$ and a point $k \le n$ with $L = L_k^{(n)} \ge 1$ let $\overline{G}_n^{(\sigma_0)}(k) = (n-L)f_+(\lfloor L \rfloor)/\mu_- + T_+(\lfloor L \rfloor)\kappa_+^{(\sigma_0)}(n-k)$ be the kernel (4.3) at the horizon $n$ with the limit coefficient. Assume (RV) and (SPAN) for both signs and (LR) for the plus sign. By the last clause of Theorem B(b), applied at the horizon $n$, for every $\varepsilon > 0$ there are $K_0$ and $n_0$ such that

$$\bigl|F_n^{\sigma_0}(k) - \overline{G}_n^{(\sigma_0)}(k)\bigr| \le \varepsilon\, \widehat{G}_n(k) \qquad (n \ge n_0,\ pn + K_0\lambda_n B_n \le k \le n) \tag{5.9}$$

for both starting signs, and by (4.14) both $\overline{G}_n^{(\sigma_0)}(k)$ and $G_n^{(\sigma_0)}(k)$ are at most $C_{\mathrm{G}}\widehat{G}_n(k)$ whenever $L_k^{(n)} \ge 1$.

### 5.3 The bulk, parts (a) and (b)

*Proof of Theorem C(a).* By the definition of $\Pi_m^\sigma$ we have $F_{N-d}^-(k-d) = \Pi_{N-d}^-(N-k)$ and $F_{N-d}^+(k) = \Pi_{N-d}^+(k)$. Fix $J \ge 1$. By Proposition 2.4, for $N > 2J$,

$$B_N P_N(k) = B_N Q_+(N)\mathbf{1}_{\{k=N\}} + B_N Q_-(N)\mathbf{1}_{\{k=0\}} + E_1(k) + E_2(k) + E_3(k),$$

where $E_1$, $E_2$ and $E_3$ collect the first-run lengths $d \le J$, $J < d \le N/2$ and $N/2 < d < N$, each multiplied by $B_N$. In $E_3$ put $m = N - d$, so that $1 \le m < N/2$. By (5.3) this gives exactly $E_3(k) = \mathcal{D}_N^+(N-k) + \mathcal{D}_N^-(k)$.

For $E_2$ the horizon is $N - d \ge N/2$. By Theorem A(b) there is $C_{10}$ with $B_n F_n^{\sigma_0}(k) \le C_{10}$ for all $n$, $k$ and $\sigma_0$. By Lemma 2.1(v), $B_N/B_{N-d} \le B_N/B_{\lceil N/2 \rceil} \le C$. Hence $0 \le E_2(k) \le CC_{10}\varepsilon_J$ for all $k$, by (5.1).

For $E_1$ we claim that for each fixed $d \ge 1$

$$\sup_{k\in\mathbb{Z}}\bigl|B_N F_{N-d}^-(k-d) - h_N(x_{N,k})\bigr| \to 0, \qquad \sup_{k\in\mathbb{Z}}\bigl|B_N F_{N-d}^+(k) - h_N(x_{N,k})\bigr| \to 0. \tag{5.10}$$

We prove the first, and the second is the same with $pd$ in place of $-(1-p)d$ below. Put $n = N - d$. By Theorem A(a), $B_n F_n^-(k-d) = h_n(y_k) + o(1)$ uniformly in $k$, with $y_k = (k - d - pn)/B_n = (k - pN - (1-p)d)/B_n$. Also $B_N/B_n \to 1$ and $h_n$ is bounded. By Lemma 2.2(c), $\|h_n - h_N\|_\infty \to 0$. Fix $R$. If $|x_{N,k}| \le R$, then $|y_k - x_{N,k}| \le R|B_N/B_n - 1| + (1-p)d/B_n \to 0$, and the Lipschitz bound of Lemma 2.2(a) gives $|h_N(y_k) - h_N(x_{N,k})| \to 0$. If $|x_{N,k}| > R$, then $|y_k| \ge R/2$ for large $N$, and both values are at most $\varepsilon_h(R/2)$. Let $N \to \infty$ and then $R \to \infty$. This proves (5.10). Summing it over the finitely many $d \le J$ gives $\sup_k |E_1(k) - \mathbb{P}(D \le J)h_N(x_{N,k})| \to 0$.

Let $e_N(k)$ be the difference between $B_N P_N(k)$ and the right side of (a) without the term $o(1)$. Collecting the three parts, $\limsup_{N\to\infty} \sup_{0\le k\le N} |e_N(k)| \le (C_1 + CC_{10})\varepsilon_J$, with $C_1$ the bound of Lemma 2.2(a). This tends to zero as $J \to \infty$, by (5.1). □

*Proof of Theorem C(b).* The two defects are nonnegative. If $\mathcal{D}_N^+(N-k) \ne 0$, then some $m < N/2$ has $\Pi_m^-(N-k) > 0$, so $N - k \le m < N/2$ and $k > N/2$. In the same way $\mathcal{D}_N^-(k) \ne 0$ forces $k < N/2$. So the two defects have disjoint supports in $1 \le k < N$, and as $k$ runs over $1 \le k < N$ the arguments $N-k$ and $k$ run over all $b$ at which $\mathcal{D}_N^\sigma(b)$ can be positive. Hence $\sup_{1\le k<N}\bigl(\mathcal{D}_N^+(N-k) + \mathcal{D}_N^-(k)\bigr) = \max(\Delta_{+,N}, \Delta_{-,N})$. Part (a) and the inequality $|\sup|f+g| - \sup|f|| \le \sup|g|$ then give

$$\sup_{1\le k<N}\bigl|B_N P_N(k) - h_N(x_{N,k})\bigr| = \max(\Delta_{+,N}, \Delta_{-,N}) + o(1).$$

At $k = N$ part (a) gives $B_N P_N(N) = B_N Q_+(N) + h_N(x_{N,N}) + o(1)$, since $\mathcal{D}_N^+(0) = 0$ and $\mathcal{D}_N^-(N) = 0$. Here $x_{N,N} = (1-p)N/B_N \to \infty$, so $h_N(x_{N,N}) \to 0$ by Lemma 2.2(b). The same holds at $k = 0$ with $Q_-$. This proves the formula for the error over the whole range.

Now we compare the defects with the atoms. The term $m = 1$ of (5.3) at $b = 1$ is $B_N\nu_+(N-1)\Pi_1^-(1) = B_N\nu_+(N-1)$, since a fresh minus walk of one step has one minus step. So $\Delta_{+,N} \geq B_N\nu_+(N-1)$. In the other direction, $\nu_+(N-m) \leq \bar\nu_+(N/2)$ for $m < N/2$, and the Green identity, Lemma 2.5, together with $u_- \leq 1$ gives

$$\Delta_{+,N} \leq (1+\mu_+)\, B_N \sup_{d>N/2} \nu_+(d) \leq (1+\mu_+)\, B_N\, \bar\nu_+(N/2). \tag{5.11}$$

The same bounds hold for the minus sign. So $\max(\Delta_{+,N}, \Delta_{-,N}) \to 0$ if $B_N\bar\nu(N/2) \to 0$ and only if $B_N\mathbb{P}(D^{(N)} = N-1) \to 0$, and for a fixed law both conditions read $B_n\mathbb{P}(D = n) \to 0$, by the remark after (5.2) and $B_N/B_{N-1} \to 1$. This is the interior criterion. For the whole range we need in addition $B_N Q_\sigma(N) \to 0$ for both signs, and for a fixed law, since $\nu_\sigma \leq Q_\sigma$, the two conditions together read $B_n\mathbb{P}(D \geq n) \to 0$.

Finally let $\varepsilon N \leq k \leq (1-\varepsilon)N$. A nonzero term of $\mathcal{D}_N^+(N-k)$ has $m \geq N-k \geq \varepsilon N$, and Theorem A(b) gives $\Pi_m^-(N-k) \leq C_{10}/B_m \leq C_{10}/B_{\lceil \varepsilon N\rceil}$. Hence $\mathcal{D}_N^+(N-k) \leq C_{10}(B_N/B_{\lceil\varepsilon N\rceil})\, Q_+(\lceil N/2\rceil) \to 0$, since $Q_+(\lceil N/2\rceil) \leq \varepsilon_J$ once $N > 2J$. The minus defect is the same, and part (a) gives the approximation on this range. □

### 5.4 The boundary layer, part (d)

*Proof of Theorem C(d).* The proof uses only (RV) for the plus sign, through Theorem B(a) and the bound $T_+(\lceil N/2\rceil) \leq CT_+(N)$ of Lemma 2.1(ii). Fix $j \geq 0$ and let $N > 2j$. Proposition 2.4 at $k = N-j$ gives

$$P_N(N-j) = Q_+(N)\mathbf{1}_{\{j=0\}} + \sum_{d<N} \nu_-(d)\, F_{N-d}^+(N-j) + \sum_{d<N} \nu_+(d)\, F_{N-d}^-(N-d-j).$$

In the first sum the fresh plus walk on $N-d$ steps has $N-j$ plus steps, so $d \leq j$, and it then has $j-d$ minus steps. For each such $d$, Theorem B(a) at the horizon $N-d$ and (2.11) give $F_{N-d}^+(N-j)/T_+(N-d) \to u_-(j-d)$, and $T_+(N-d)/T_+(N) \to 1$ by (RV). The sum has at most $j$ terms. So the first sum equals $T_+(N)\big(\sum_{d=1}^j \nu_-(d)u_-(j-d) + o(1)\big)$.

Split the second sum at $J$ and at $N/2$, with $J$ fixed. For $d \leq J$, Theorem B(a) for a fresh minus start and (2.11) give $F_{N-d}^-(N-d-j)/T_+(N-d) \to u_-(j)\mathbf{1}_{\{j\geq 1\}}$, so this part equals $T_+(N)\big(\mathbb{P}(\Sigma = +, D \leq J)u_-(j)\mathbf{1}_{\{j\geq 1\}} + o(1)\big)$. For $J < d \leq N/2$ the horizon is $n = N-d \geq N/2$. The convergent sequence $F_n^-(n-j)/T_+(n)$ is bounded by some $C_j$ for $n \geq N/2$ and large $N$, and $T_+(n) \leq T_+(\lceil N/2\rceil) \leq CT_+(N)$. So this part is at most $CC_j\varepsilon_J T_+(N)$, by (5.1). For $N/2 < d < N$ put $m = N-d$. Then $F_{N-d}^-(N-d-j) = \Pi_m^-(j)$, and by (5.3) this part equals $B_N^{-1}\mathcal{D}_N^+(j)$ exactly. Collecting terms and using (5.4),

$$\limsup_{N\to\infty} \frac{\big|P_N(N-j) - Q_+(N)\mathbf{1}_{\{j=0\}} - T_+(N)\kappa_+(j) - B_N^{-1}\mathcal{D}_N^+(j)\big|}{T_+(N)} \leq (1+CC_j)\,\varepsilon_J,$$

and the right side tends to zero as $J \to \infty$. This is the expansion.

Let $j \geq 1$. If $\bar\nu_+(n) = o(T_+(n))$, then the Green identity, Lemma 2.5, and $u_- \leq 1$ give $B_N^{-1}\mathcal{D}_N^+(j) \leq (1+\mu_+)\bar\nu_+(N/2) = o(T_+(\lceil N/2\rceil)) = o(T_+(N))$. Conversely, suppose that the defect term is $o(T_+(N))$ for one $j \geq 1$. By Lemma 2.5 the weights $\Pi_m^-(j)$, $m \geq 1$, add up to $1 + \mu_+u_-(j) \geq 1$, so there is an $m_0 = m_0(j) \geq 1$ with $\Pi_{m_0}^-(j) > 0$, and it does not depend on $N$. (If $T_-(j) > 0$ one may take $m_0 = j$, with $\Pi_j^-(j) = T_-(j)$.) For $N > 2m_0$ the term $m = m_0$ of (5.3) gives $B_N^{-1}\mathcal{D}_N^+(j) \geq \nu_+(N-m_0)\Pi_{m_0}^-(j)$, so $\nu_+(N-m_0) = o(T_+(N))$. For a fixed law, $T_+(N)/T_+(N-m_0) \to 1$ gives $\nu_+(n) = o(T_+(n))$, and then the expansion gives $P_N(N-j)/T_+(N) \to \kappa_+(j)$.

Finally let $\nu_+$ satisfy (R$\nu$) and $j \geq 1$, and put $r_N = \nu_+(N) + T_+(N)$. Fix $M$. For $m \leq M$ the second part of (R$\nu$) gives $|\nu_+(N-m) - \nu_+(N)| \leq \epsilon_N r_N$ with $\epsilon_N \to 0$, and for

$M < m < N/2$ the first part gives $\nu_+(N-m) + \nu_+(N) \le (C_\nu + 1) r_N$. By Lemma 2.5 the weights $\Pi^-_m(j)$ have the finite sum $1 + \mu_+ u_-(j)$. Hence

$$\Big| B_N^{-1} \mathcal{D}_N^+(j) - \nu_+(N)\big(1 + \mu_+ u_-(j)\big) \Big| \\ \le \epsilon_N (1+\mu_+) r_N + (C_\nu + 1) r_N \sum_{m > M} \Pi^-_m(j) + \nu_+(N) \sum_{m \ge N/2} \Pi^-_m(j),$$

and letting $N \to \infty$, then $M \to \infty$, the right side is $o(r_N)$. □

### 5.5 The big-jump zone, parts (c) and (e)

*Proof of the decomposition in Theorem C(c).* Fix $\varepsilon \in (0, 1]$ and take $K_0$ and $n_0$ from (5.9). Let $K \ge 4K_0$ and $J \ge 1$. Let $pN + K\lambda_N B_N \le k \le N$, and write $L = L_k$ and $j = N - k$. Then $L \ge K\lambda_N B_N/(1-p) \ge K B_N/(1-p)$, since $\lambda_N \ge 1$, and $N - L = j/(1-p)$. Take $N$ so large that $\min(p, \frac{1}{2})N \ge n_0$, that $J < L/2$, and that (5.8) holds with $c = \min(p, \frac{1}{2})$. By Proposition 2.4,

$$P_N(k) = Q_+(N)\mathbf{1}_{\{k=N\}} + E_1 + E_2 + E_3 + E_4 + \Psi_N(k),$$

where $E_1$ and $E_2$ collect $\nu_-(d) F^+_{N-d}(k)$ over $d \le \min(J, j)$ and over $J < d \le j$, and $E_3$ and $E_4$ collect $\nu_+(d) F^-_{N-d}(k-d)$ over $d \le J$ and over $J < d < L/2$. The terms $\nu_-(d) F^+_{N-d}(k)$ with $d > j$ vanish, since then $k > N - d$.

*Step 1 (geometry).* Let $d \le j$ and $n = N - d$. Then $n \ge k > pN$, so $n \ge n_0$, and $L^{(n)}_k = L + pd/(1-p) \ge L$. Also $k - pn \ge k - pN \ge K\lambda_N B_N \ge (K/2)\lambda_n B_n$, by (5.8) and since $B$ is nondecreasing. So $k$ lies in the zone of (5.9) at the horizon $n$, and $\widehat{G}_n(k) \le (N/L) T_+(\lfloor L \rfloor) = \widehat{G}_N(k)$ because $T_+$ is nonincreasing. Let $d < L/2$ and $n = N - d \ge N/2$. Then $L^{(n)}_{k-d} = L - d \ge L/2$, and $(k-d) - pn = (1-p)(L-d) \ge K\lambda_N B_N/2 \ge (K/4)\lambda_n B_n$, again by (5.8), so $k - d$ lies in that zone at the horizon $n$. By Lemma 2.1(ii), $\widehat{G}_n(k-d) \le (2N/L) T_+(\lfloor L/2 \rfloor) \le C_* \widehat{G}_N(k)$ for large $N$.

*Step 2 (intermediate first-run lengths).* By Step 1 and (5.9), every fresh probability in $E_2$ and $E_4$ is at most $(C_{\mathrm{G}} + 1)$ times the corresponding $\widehat{G}_n$, since $\varepsilon \le 1$. Hence $E_2 + E_4 \le (C_{\mathrm{G}} + 1)(1 + C_*)\, \varepsilon_J\, \widehat{G}_N(k)$, by (5.1).

*Step 3 (short first-run lengths).* Let $d \le J$. By Step 1 and (5.9),

$$|F^+_{N-d}(k) - \overline{G}^{(+)}_{N-d}(k)| \le \varepsilon \widehat{G}_N(k), \qquad |F^-_{N-d}(k-d) - \overline{G}^{(-)}_{N-d}(k-d)| \le \varepsilon C_* \widehat{G}_N(k).$$

With $L' = L + pd/(1-p)$ and $(N-d) - L' = (j-d)/(1-p)$, and with (2.11), the kernels are

$$\overline{G}^{(+)}_{N-d}(k) = \frac{j-d}{(1-p)\mu_-} f_+(\lfloor L' \rfloor) + T_+(\lfloor L' \rfloor)\, u_-(j-d), \\ \overline{G}^{(-)}_{N-d}(k-d) = \frac{N-L}{\mu_-} f_+(\lfloor L - d \rfloor) + T_+(\lfloor L - d \rfloor)\, u_-(j) \mathbf{1}_{\{j \ge 1\}}.$$

By (LR) and (RV), $f_+(m+i)/f_+(m) \to 1$ and $T_+(m+i)/T_+(m) \to 1$ as $m \to \infty$, uniformly in $|i| \le 2J/(1-p)$. Let $r_N$ be the largest deviation from one of these ratios over $m \ge K B_N/(2(1-p))$. Then $r_N \to 0$. Note that $(j-d)/(1-p) = (N-L) - d/(1-p)$ and that $f_+(\lfloor L' \rfloor) \le 2C T_+(\lfloor L \rfloor)/L$ for large $N$, by (LR). Hence

$$\Big| \overline{G}^{(+)}_{N-d}(k) - \frac{N-L}{\mu_-} f_+(\lfloor L \rfloor) - T_+(\lfloor L \rfloor) u_-(j-d) \Big| \le C\Big( r_N + \frac{J}{L} \Big) \widehat{G}_N(k),$$

and $\overline{G}^{(-)}_{N-d}(k-d)$ differs from $(N-L) f_+(\lfloor L \rfloor)/\mu_- + T_+(\lfloor L \rfloor) u_-(j) \mathbf{1}_{\{j \ge 1\}}$ by at most $C r_N \widehat{G}_N(k)$. Here we used $(N-L) f_+(\lfloor L \rfloor) \le C \widehat{G}_N(k)$ and $T_+(\lfloor L \rfloor) \le \widehat{G}_N(k)$.

Now we sum over $d$. In $E_3$ the weights sum to $\mathbb{P}(\Sigma=+, D\le J)$, which differs from $\pi_+$ by at most $\varepsilon_J$. In $E_1$ the weights of the interior term sum to $\mathbb{P}(\Sigma=-, D\le \min(J,j))$. If $j\ge J$ this differs from $\pi_-$ by at most $\varepsilon_J$. If $j<J$, then $N-L<J/(1-p)$ and the whole interior term is at most $CJ\,T_+(\lfloor L\rfloor)/L\le C(J/L)\widehat G_N(k)$. The boundary terms in $E_1$ sum to $T_+(\lfloor L\rfloor)\sum_{d\le \min(J,j)}\nu_-(d)u_-(j-d)$, which differs from $T_+(\lfloor L\rfloor)\sum_{d\le j}\nu_-(d)u_-(j-d)$ by at most $\varepsilon_J\widehat G_N(k)$, since $u_-\le 1$. By (5.4) and (5.5),

$$|E_1+E_3-G_N^\nu(k)|\le C\Big(\varepsilon+r_N+\frac{J}{L}+\varepsilon_J\Big)\widehat G_N(k).$$

*Step 4 (conclusion).* Steps 2 and 3 and $J/L\le (1-p)J/(KB_N)$ give, for $K\ge 4K_0$ and large $N$,

$$\sup_{pN+K\lambda_N B_N\le k\le N}\frac{|P_N(k)-Q_+(N)\mathbf{1}_{\{k=N\}}-G_N^\nu(k)-\Psi_N(k)|}{\widehat G_N(k)} \le C\Big(\varepsilon+r_N+\frac{(1-p)J}{KB_N}+\varepsilon_J\Big).$$

Let $N\to\infty$, then $J\to\infty$. The bound becomes $C\varepsilon$ for every $K\ge 4K_0(\varepsilon)$, and $\varepsilon$ is arbitrary. □

*Proof of the criteria in Theorem C(c). The whole zone.* By the decomposition, the statement without $\Psi_N$ holds if and only if $\lim_K \limsup_N \sup_k \Psi_N(k)/\widehat G_N(k)=0$, the supremum being over the zone. Suppose $\bar\nu_+(n)=o(T_+(n))$. Every $d$ in (5.6) has $d\ge L_k/2$, and Lemma 2.5 with $u_-\le 1$ gives, with $j=N-k$,

$$\Psi_N(k)\le \sup_{d\ge L_k/2}\nu_+(d)\sum_{m\ge 1}\Pi_m^-(j)\le (1+\mu_+)\,\bar\nu_+(L_k/2), \tag{5.12}$$

because $F_{N-d}^-(k-d)=\Pi_{N-d}^-(j)$ and distinct $d$ give distinct $m=N-d$. The right side is $o(T_+(\lceil L_k/2\rceil))=o(T_+(\lfloor L_k\rfloor))$ by Lemma 2.1(ii), and $T_+(\lfloor L_k\rfloor)\le \widehat G_N(k)$. Conversely, for large $N$ the point $k=N-1$ lies in the zone for every $K$, because $\lambda_N B_N=o(N)$, and it has $L=N-1/(1-p)\ge N/2$. The term $d=N-1$ of (5.6) gives $\Psi_N(N-1)\ge \nu_+(N-1)F_1^-(0)=\nu_+(N-1)$, and $\widehat G_N(N-1)\le 2T_+(\lfloor N/2\rfloor)\le CT_+(N)$. So the criterion forces $\nu_+(N-1)=o(T_+(N))$, which for a fixed law is $\nu_+(n)=o(T_+(n))$.

*The regular case.* Let $\nu_+$ satisfy (R$\nu$) and fix $\varepsilon>0$. Take $R$ and $N_0$ from Lemma 5.3, and $\delta\in(0,\frac12]$ and $n_\delta$ such that the first part of (R$\nu$) holds for $n\ge n_\delta$ and $|\nu_+(d)-\nu_+(n)|\le \varepsilon(\nu_+(n)+T_+(n))$ for $|d-n|\le\delta n$ and $n\ge n_\delta$. Let $K\ge R(1-p)/\delta+2$ and let $k<N$ lie in the zone, so $j=N-k\ge 1$ and $L_k\ge KB_N/(1-p)$, and put $n=\lfloor L_k\rfloor$ and $r=\nu_+(n)+T_+(n)$. For large $N$ we have $n\ge n_\delta$. Let $\mathcal V=\{d\le N-1:|d-L_k|\le RB_N\}$. Since $m=N-d$ satisfies $|m-j/(1-p)|=|d-L_k|$, Lemma 5.3 gives $\sum_{d\le N-1,\,d\notin\mathcal V}\Pi_{N-d}^-(j)\le\varepsilon$, and Lemma 2.5 then gives $\sum_{d\in\mathcal V}\Pi_{N-d}^-(j)\in[1+\mu_+u_-(j)-\varepsilon,\,1+\mu_+u_-(j)]$. Every $d\ge L_k/2$ has $d\ge n/2$, so $\nu_+(d)\le C_\nu r$. Every $d\in\mathcal V$ has $d\ge L_k/2$ and $|d-n|\le RB_N+1\le\delta n$, because $n\ge KB_N/(1-p)-1$, so $|\nu_+(d)-\nu_+(n)|\le\varepsilon r$. Hence

$$\begin{aligned}\Big|\Psi_N(k)-\nu_+(n)\big(1+\mu_+u_-(j)\big)\Big| &\le \sum_{d\notin\mathcal V}\nu_+(d)\Pi_{N-d}^-(j)+\sum_{d\in\mathcal V}\big|\nu_+(d)-\nu_+(n)\big|\Pi_{N-d}^-(j)\\ &\quad+\nu_+(n)\,\varepsilon \;\le\; \big(C_\nu+2+\mu_+\big)\,\varepsilon\, r.\end{aligned}$$

Since $T_+(n)\le\widehat G_N(k)$, the decomposition gives the stated form, with the double limit taken as before. At $k=N$ both $\Psi_N$ and the replacement vanish. □

*Proof of Theorem C(e). Reduction.* Fix $\varepsilon\in(0,\frac12)$. For large $N$ the set $Z_N^\varepsilon$ lies in the zone of (c) for every $K$, because $\lambda_N B_N=o(N)$, and it contains no $k=N$. So the

decomposition of (c) gives $\sup_{k\in Z_N^\varepsilon}|P_N(k)-G_N^\nu(k)-\Psi_N(k)|/\widehat{G}_N(k)\to 0$, and (5.7) holds if and only if $\Psi_N=o(\widehat{G}_N)$ uniformly on $Z_N^\varepsilon$ for every $\varepsilon$. On $Z_N^\varepsilon$ we have $T_+(N)\le \widehat{G}_N(k)\le \varepsilon^{-1}T_+(\lfloor\varepsilon N\rfloor)\le C_\varepsilon T_+(N)$. So the condition reads $\sup_{k\in Z_N^\varepsilon}\Psi_N(k)=o(T_+(N))$. Sufficiency of $\bar\nu_+(n)=o(T_+(n))$ follows from (5.12).

For the necessity of the window condition, let $n$ be large, put $N=2n$ and choose $k$ with $|L_k-n|\le 1/(1-p)$, which is possible because $L_k$ moves in steps of $1/(1-p)$. Then $k\in Z_N^{1/4}$. For $|d-n|\le B_n$ put $m=N-d$. Then $m\in[N/4,3N/4]$, and $(m-j)-pm=(1-p)(L_k-d)$ has absolute value at most $2(1-p)B_n\le c_{11}B_m$, with $c_{11}$ depending only on the laws, by Lemma 2.1(ii). Theorem A(c) and Lemma 2.2(d) give $B_m\Pi_m^-(j)=B_mF_m^-(m-j)\ge c_{12}>0$ for large $n$. Also $d\ge L_k/2$ and $B_m\le CB_n$. Hence $\Psi_N(k)\ge (c_{12}/C)W_n^{(N)}$ with $N=2n$. If (5.7) holds, then $\Psi_N=o(T_+(N))$ on $Z_N^{1/4}$ by the reduction, so $W_n^{(2n)}=o(T_+(2n))=o(T_+(n))$.

*Sufficiency of the window condition under (LR) for both signs.* Let $k\in Z_N^\varepsilon$, with $j=N-k\ge(1-p)\varepsilon N$. A nonzero term of (5.6) has $m=N-d\ge j$, so $m\in[(1-p)\varepsilon N,N]$ and $d\in[\varepsilon N/2,N]$. Hence $B_m\ge c_\varepsilon B_N$, and Theorem A(b) gives $\Pi_m^-(j)\le C_\varepsilon/B_N$. Put $\overline{W}_N=\sup_{\varepsilon N/4\le n\le N}W_n^{(N)}$, which is $o(T_+(N))$ by hypothesis, and for a fixed law with $W_n=o(T_+(n))$ by (RV). Split the $d$ axis into the blocks $\mathcal{B}_i=\{d: iB_N\le d-L_k<(i+1)B_N\}$, $i\in\mathbb{Z}$. Each block meets $[\varepsilon N/2,N]$ in a set covered by at most $C_\varepsilon$ windows $[n-B_n,n+B_n]$ with $\varepsilon N/4\le n\le N$, so $\nu_+(\mathcal{B}_i\cap[\varepsilon N/2,N])\le C_\varepsilon B_N\overline{W}_N$. The blocks with $|i|\le R$ contribute at most $(2R+1)C_\varepsilon^2\overline{W}_N$ to $\Psi_N(k)$.

For the blocks with $|i|>R$ we do not use (5.9), since its zone, and its analogue at the minus end, carry logarithmic factors that are unbounded when $\alpha_+\ne\alpha_-$. We use a bound that needs only an excess of order $B_m$. Let $m$ be a large horizon and $k'>pm$ a point, and put $L=L_{k'}^{(m)}$ and $g=\lceil L/32\rceil$. We apply (4.10) at the horizon $m$ for a fresh minus start and drop its last term. We bound its first term by Lemma 4.4 with $\eta=1/32$ and $r=2$, and $\mathrm{Cen}_g$ and $\mathrm{Int}_g$ by (4.11), with $\kappa_+^{(-)}\le 1$ and $\gamma_+^{(-)}(m-k')\le(1+\mu_+)m$. By (LR) for the plus sign, $f_+^*(g)\le C_{\mathrm{LR}}T_+(g)/g$, and $T_+(g)\le CT_+(L)$ by Lemma 2.1(ii). Since $L\le m$, this gives

$$F_m^-(k')\le C\,\frac{(mT_+(L))^2}{b_{+,m}}+\frac{C}{b_{-,m}}\,e^{-cL/b_{-,m}}+Ce^{-m^c}+C\,\frac{m}{L}\,T_+(L)\qquad(L\ge C_9B_m),\tag{5.13}$$

with $C$ and $c$ depending only on the laws and $C_9\ge C_8$ the larger of the thresholds of Lemma 4.4 at the two ends. The minus-end forms of Lemmas 4.2 and 4.4, with (LR) for the minus sign, give the same bound for $k'<pm$ with the signs exchanged, with the same constants. There $L$ is replaced by $L^-=(pm-k')/p$, $T_+$ by $T_-$, and $b_{+,m}$ and $b_{-,m}$ by $b_{-,m}$ and $b_{+,m}$.

Now let $|i|>R$ and $d\in\mathcal{B}_i$, so that $m=N-d\in[(1-p)\varepsilon N,N]$ and $\Pi_m^-(j)=F_m^-(k')$ with $k'=m-j$. Then $k'-pm=(1-p)(L_k-d)$. If $d<L_k$, then $k'>pm$ and $L=L_k-d>(|i|-1)B_N$. If $d>L_k$, then $k'<pm$ and $L^-=(1-p)(d-L_k)/p\ge(1-p)|i|B_N/p$. Put $c_*=\min(1,(1-p)/p)$, so the excess is at least $y_i=c_*(|i|-1)B_N$ in both cases. Choose $R$ with $c_*R\ge\max(C_9,1)$. Then $y_i\ge C_9B_N\ge C_9B_m$, so (5.13) or its minus-end form applies, and $y_i\ge B_N\ge b_{\sigma,N}$ for both signs. Each term of (5.13) decreases in $L$. For these $m$ we have $mT_\sigma\le NT_\sigma$ and $c_\varepsilon b_{\sigma,N}\le b_{\sigma,m}\le b_{\sigma,N}$. We treat the blocks with $i<0$, and those with $i>0$ are the same with the signs exchanged. We multiply the four terms of (5.13) at $L=y_i$ by the block mass $C_\varepsilon B_N\overline{W}_N$, sum over $i<-R$, and leave out the common factor $C_\varepsilon\overline{W}_N$ below. Put $t=|i|-1\ge 1$.

The first term. By (2.4), $(NT_+(y_i))^2\le 16s^2(y_i/b_{+,N})^{-\alpha_+-1}$, so the term is at most $C(B_N/b_{+,N})^{-\alpha_+}t^{-\alpha_+-1}\le Ct^{-\alpha_+-1}$. These sum to a constant.

The second term. Put $x=B_N/b_{-,N}\ge 1$. The term is at most $Cxe^{-cc_*tx}$, and $\sum_{t\ge1}xe^{-cc_*tx}=x/(e^{cc_*x}-1)\le 1/(cc_*)$.

The third term. At most $N/B_N + 2$ blocks meet $[0, N]$, so these terms add up to at most $C(N + 2B_N)e^{-((1-p)\varepsilon N)^c}$, which tends to zero.

The fourth term. Fix $\delta' \in (0, \alpha_*)$. Since $c_* t \geq 1$, Lemma 2.1(ii) gives $T_+(y_i) \leq Ct^{-\alpha_+ + \delta'} T_+(B_N)$. So the term is at most $CNT_+(y_i)/t \leq CNT_+(B_N)t^{-1-\alpha_+ + \delta'}$, and $NT_+(B_N) \leq NT_+(b_{+,N}) \to s$ by Lemma 2.1(v). These sum to a constant.

For $i > R$ the same four steps use (2.4) for the minus sign, $x = B_N/b_{+,N}$, and $NT_-(B_N) \leq NT_-(b_{-,N}) \to s$. So the blocks with $|i| > R$ contribute at most $C_\varepsilon \overline{W}_N$ to $\Psi_N(k)$. Hence $\sup_{k \in Z_N^\varepsilon} \Psi_N(k) \leq C_{\varepsilon,R} \overline{W}_N = o(T_+(N))$. □

The gap between the two interior conditions is real. If a fixed law has $\nu_+(n_i) = T_+(n_i)$ on a sparse sequence and $\nu_+ = 0$ near it, then $W_n = O(T_+(n)/B_n)$, so under (LR) for both signs the initial run is invisible on every interior zone, by the window criterion (the sufficiency just proved). It is visible at the point $N - 1$ for $N = n_i + 1$, where $P_N(N-1) \geq T_+(N-1)$, and $\nu_+(n) \neq o(T_+(n))$. So the interior of the big-jump zone sees only window averages of the initial run, while the edge of the zone sees single atoms.

### 5.6 The stationary initial run

**Corollary 5.4** (stationary initial run)**.** *Assume (RV) and (SPAN), and let the initial run be stationary, $\nu_\sigma = T_\sigma/s$. Then the bulk criterion holds, and*

*(a) the atoms are $Q_\sigma(N) = s^{-1} \sum_{d \geq N} T_\sigma(d) \sim NT_\sigma(N)/(s(\alpha_\sigma - 1))$.*

*(b) if (LR) holds for the plus sign, then, with $\lambda_N$ as in* (4.5)*,*

$$\lim_{K \to \infty} \limsup_{N \to \infty} \sup_{pN + K\lambda_N B_N \leq k < N} \left| \frac{P_N(k)}{G_N^{\mathrm{st}}(k)} - 1 \right| = 0,$$

*where $G_N^{\mathrm{st}}(k) = (N - L_k) f_+(\lfloor L_k \rfloor)/\mu_- + T_+(\lfloor L_k \rfloor)\, \kappa_+^{\mathrm{st}}(N-k)$, and for fixed $v \in (p, 1)$, with $\ell = (v-p)/(1-p)$, $P_N(\lfloor vN \rfloor)/T_+(N) \to I_+(v) + \ell^{-\alpha_+}/\mu_-$. So in the interior of the big-jump zone the kernel gains the term $T_+(\lfloor L_k \rfloor)/\mu_-$, which is the complete initial plus run.*

*(c) the boundary kernel at the plus end is*

$$\frac{P_N(N-j)}{T_+(N)} \to \kappa_+^{\mathrm{st}}(j) = \frac{2}{s}\big(1 + \mu_+ u_-(j)\big) \qquad (j \geq 1),$$

*while $P_N(N)/T_+(N) \to \infty$. Moreover $\kappa_+^{\mathrm{st}}(j) \to 2/\mu_-$ as $j \to \infty$, which is the value of $I_+(v) + \ell^{-\alpha_+}/\mu_-$ at $v = 1$.*

*Proof.* The bulk criterion holds because $B_n T_\sigma(n) \to 0$ by Lemma 2.1(v). Part (a) is Lemma 2.1(iv).

We first compute $\kappa_+$. Here $\pi_+ = \sum_d T_+(d)/s = \mu_+/s = p$, and $\sum_{d=1}^{j} T_-(d) u_-(j-d) = 1$ for $j \geq 1$ by (2.9). So (5.4) gives $\kappa_+(j) = pu_-(j) + 1/s$ for $j \geq 1$.

(c) The function $\nu_+ = T_+/s$ is regularly varying with index $-\alpha_+ < 0$, so it satisfies (R$\nu$). Theorem C(d) gives, for $j \geq 1$,

$$\frac{P_N(N-j)}{T_+(N)} \to pu_-(j) + \frac{1}{s} + \frac{1}{s}\big(1 + \mu_+ u_-(j)\big) = \frac{2}{s} + \frac{2\mu_+}{s} u_-(j),$$

since $p = \mu_+/s$. Also $P_N(N) = Q_+(N)$, and $Q_+(N)/T_+(N) \to \infty$ by (a). By (2.10), $\kappa_+^{\mathrm{st}}(j) \to (2/s)(1 + \mu_+/\mu_-) = 2/\mu_-$.

(b) By the regular case of Theorem C(c), uniformly in the zone with $k < N$ and in the double limit,

$$P_N(k) = G_N^{\nu}(k) + \frac{T_+(\lfloor L_k \rfloor)}{s}\big(1 + \mu_+ u_-(N-k)\big) + o\big(\widehat{G}_N(k)\big),$$

where we used $T_+(\lfloor L_k \rfloor) \le \widehat{G}_N(k)$. By the computation of $\kappa_+$ the right side is $G_N^{\mathrm{st}}(k) + o(\widehat{G}_N(k))$. For the ratio form we need $G_N^{\mathrm{st}}(k) \ge c\widehat{G}_N(k)$. Since $\kappa_+^{\mathrm{st}} \ge 2/s$, this holds when $L_k > N/2$. When $L_k \le N/2$, (LR) gives $(N - L_k) f_+(\lfloor L_k \rfloor)/\mu_- \ge c(N/L_k) T_+(\lfloor L_k \rfloor)$ for large $N$. Now fix $v \in (p, 1)$ and let $k = \lfloor vN \rfloor$. Then $k$ lies in the zone for every $K$ when $N$ is large, because $\lambda_N B_N = o(N)$. Also $L_k = \ell N + O(1)$ and $N - k \to \infty$. By (LR) and (RV), $(N - L_k) f_+(\lfloor L_k \rfloor)/T_+(N) \to (1 - \ell)\alpha_+ \ell^{-\alpha_+ - 1}$ and $T_+(\lfloor L_k \rfloor)/T_+(N) \to \ell^{-\alpha_+}$, and $\kappa_+^{\mathrm{st}}(N - k) \to 2/\mu_-$. This gives the limit $I_+(v) + \ell^{-\alpha_+}/\mu_-$, with $I_+$ as in Corollary 4.9. At $v = 1$ it equals $2/\mu_-$. □

In $\kappa_+^{\mathrm{st}}(j)$ the term $pu_-(j)$ comes from a short first plus run, the term $1/s$ from a short first minus run, and the term $(1 + \mu_+ u_-(j))/s$ from a first plus run that ends within $O(1)$ steps of the horizon. Section 9 reports a check of part (c) against the exact law. Wang, Vezzani, Burioni and Barkai [74, Eqs. (28), (46) and (47)] compared the equilibrated and the ordinary start formally, for a biased continuous-time random walk. The equilibrated start adds the term $\xi^{-\alpha}$ to their infinite density, where $\xi$ is the fraction of the time taken by the longest trap, and an atom at the origin of order $t^{1-\alpha}$. They correspond to the term $\ell^{-\alpha_+}/\mu_-$ of (b) and to the atom of (a).

The same index comparisons place other initial runs. If $Q_\sigma \in \mathrm{RV}_{-\rho}$ with $\rho > 0$ and $\nu_\sigma(n) \sim \rho Q_\sigma(n)/n$, the bulk criterion holds, the criterion for the whole range holds for $\rho > 1/\alpha_*$ and fails for $\rho < 1/\alpha_*$, and since $\nu_+/T_+ \in \mathrm{RV}_{\alpha_+ - 1 - \rho}$ the initial run is invisible in the boundary layer and, under (LR) for the plus sign, in the big-jump zone for $\rho > \alpha_+ - 1$ and visible for $\rho < \alpha_+ - 1$. The stationary initial run sits on the line $\rho = \alpha_\sigma - 1$ with $\nu_\sigma/T_\sigma = 1/s$, so it is visible there.

### 5.7 No universal trim

Part (b) of Theorem C holds on every macroscopic interior with no condition. No sublinear trim of the range does the same.

**Corollary 5.5** (no universal trim)**.** *Assume (RV) and (SPAN). For every sequence of nonnegative integers $r_N = o(N)$ there is an initial run with a fixed deterministic sign for which the bulk approximation fails on $r_N \le k \le N - r_N$ along a subsequence.*

*Proof.* Put $R(n) = \max_{1 \le i \le 2n} r_i$ and

$$m(n) = \left\lceil \max\left\{ \sqrt{n},\ \frac{4(R(n) + 1)}{1 - p} \right\} \right\rceil.$$

Since $r_i = o(i)$, for every $\eta > 0$ we have $r_i \le \eta i$ for $i \ge i_\eta$, so $R(n) \le \max_{i < i_\eta} r_i + 2\eta n$ and $R(n) = o(n)$. Hence $m(n) \to \infty$ and $m(n) = o(n)$. Since $B$ is regularly varying with the positive index $1/\alpha_*$, Lemma 2.1(ii) gives $B_{m(n)}/B_n \to 0$. Choose $n_1 < n_2 < \cdots$ so that $n_{i+1} > 2n_i$, $m(n_i) \le n_i$ and $\sum_i B_{m(n_i)}/B_{n_i} \le \frac{1}{2}$. Put $m_i = m(n_i)$, $N_i = n_i + m_i \le 2n_i$ and $v_i = B_{m_i}/B_{n_i}$. The initial run has $\Sigma = +$ almost surely, $\nu_+(n_i) = v_i$ for every $i$, and the remaining mass at $d = 1$. Then $D$ is finite almost surely.

Fix $i$ and let $k_i' = \lfloor p m_i \rfloor$. By Theorem A(c) and Lemma 2.2(d) there is $c > 0$ with $B_{m_i} F_{m_i}^-(k_i') \ge c$ for large $i$. Let $k_i = n_i + k_i'$. Proposition 2.4, keeping only the term $d = n_i$, gives

$$B_{N_i} P_{N_i}(k_i) \ge B_{N_i}\, v_i\, F_{m_i}^-(k_i') \ge c\, \frac{B_{N_i}}{B_{n_i}} \ge c,$$

because $B$ is nondecreasing. On the other hand $k_i - pN_i = (1 - p)n_i + (k_i' - pm_i) \ge (1 - p)n_i - 1$, so $x_{N_i, k_i} \ge ((1 - p)n_i - 1)/B_{2n_i} \to \infty$, and $h_{N_i}(x_{N_i, k_i}) \to 0$ by Lemma 2.2(b). So $|B_{N_i} P_{N_i}(k_i) - h_{N_i}(x_{N_i, k_i})| \ge c/2$ for large $i$.

It remains to check that $k_i$ is retained. Since $N_i \le 2n_i$, we have $r_{N_i} \le R(n_i) = o(n_i)$, so $k_i \ge n_i \ge r_{N_i}$ for large $i$. Also $N_i - k_i = m_i - k_i' \ge (1 - p)m_i \ge 4(R(n_i) + 1) > r_{N_i}$. Hence $r_{N_i} \le k_i \le N_i - r_{N_i}$, and the approximation fails along $N_i$. □

The initial run violates the bulk criterion, since $B_{n_i}\nu_+(n_i) = B_{m_i} \to \infty$. Finally, the boundary layer is a renewal mass function, which determines its interarrival law, so the ends identify both run laws. The arXiv version of this paper makes this precise in a remark at the end of this section, for fresh and prepared starts, and shows by a geometric thinning of the runs that the ratios at the ends alone do not.

### 5.8 Aging: the law observed from time $t_a$

So far the walk starts at step 1. Now let it start fresh at step 1 and observe it from step $t_a + 1$ on, for an aging time $t_a = t_a(N) \geq 0$. Let $A_N^{(t_a)}$ be the number of plus steps among the steps $t_a + 1, \dots, t_a + N$, and let $P_N^{(t_a)}(k) = \mathbb{P}(A_N^{(t_a)} = k)$. In the physics literature this is the aged occupation time, and the stationary initial run of Section 5.6 is its limit as $t_a \to \infty$ with $N$ fixed. Akimoto, Sera, Yamato and Yano [1] proved the aged form of the arcsine law in distribution for $0 < \alpha < 1$, and Akimoto [4] computed the moments of the aged occupation time for $1 < \alpha < 2$. Here we give the aged law at every point. The result is that the bulk does not age, while the atom, the boundary layer and the big-jump zone do, and each aged piece interpolates between the fresh start at $\theta = 0$ and Corollary 5.4 at $\theta = \infty$, where $\theta = \lim t_a/N$.

The run in progress at step $t_a + 1$ is the initial run of the observed walk. Let $\Sigma$ be its sign and $D$ the number of its steps from step $t_a + 1$ on. Given $(\Sigma, D)$, the later runs are fresh and alternate. So $P_N^{(t_a)}$ is the law $P_N$ of Section 5.1 with the initial run

$$\nu_+^{(t_a)}(d) = \sum_{t=0}^{t_a} a_t\, f_+(t_a - t + d), \qquad \nu_-^{(t_a)}(d) = \sum_{t=0}^{t_a} b_t\, f_-(t_a - t + d), \tag{5.14}$$

where $a_t$ and $b_t$ are the probabilities that a plus run, respectively a minus run, starts at step $t+1$ of the fresh walk. Indeed, a plus run that starts at step $t+1 \leq t_a+1$ has $d$ steps from step $t_a + 1$ on exactly when its length is $t_a - t + d$. For a plus start the plus runs start at the renewal epochs of the cycles of the walk, whose law is $f_+ * f_-$ with mean $s$, so $a_t$ is the renewal sequence of that law and $b_t = \sum_{t'<t} a_{t'} f_+(t - t')$. For a minus start the two sequences are the delayed versions, with the first minus run as the delay. In both cases $a_t, b_t \leq 1$, and the proofs use only this bound and the limits below. Under (SPAN) for one sign the support of $X_+ + X_-$ lies in no progression $b + h\mathbb{Z}$ with $h \geq 2$, so by the renewal theorem [26, Section XIII.3]

$$a_t \to \frac{1}{s}, \qquad b_t \to \frac{1}{s} \qquad (t \to \infty). \tag{5.15}$$

This initial run depends on $N$ through $t_a$, which is why Theorem C is stated for families. We first collect what the family (5.14) satisfies.

**Lemma 5.6** (the aged initial run)**.** *Assume (RV) and (SPAN), and let $t_a = t_a(N) \geq 0$ be arbitrary.*

*(a) $\nu_\sigma^{(t_a)}(d) \leq T_\sigma(d)$ for every $d \geq 1$, and $Q_\sigma^{(t_a)}(N) \leq \sum_{i \geq N} T_\sigma(i)$. So the family satisfies* (5.1) *and $\bar\nu_\sigma(n) \leq T_\sigma(n)$.*

*(b) Uniformly in $t_a \geq 0$, as $d \to \infty$,*

$$\nu_+^{(t_a)}(d) = \frac{1}{s}\big[T_+(d) - T_+(d + t_a + 1)\big] + o(T_+(d)),$$

*and $\nu_+^{(t_a)}$ satisfies (R$\nu$) uniformly in $t_a$.*

*(c) If $t_a \to \infty$, then $\nu_\sigma^{(t_a)}(d) \to T_\sigma(d)/s$ for every fixed $d \geq 1$, $\pi_+^{(t_a)} = \mathbb{P}(\Sigma = +) \to p$, and $Q_+^{(t_a)}(N) = (1 + o(1))\, s^{-1} \sum_{i=N}^{N+t_a} T_+(i)$.*

*Proof.* (a) Since $a_t \le 1$, (5.14) gives $\nu_+^{(t_a)}(d) \le \sum_{t\le t_a} f_+(t_a - t + d) \le T_+(d)$, and summing over $d \ge N$ gives the bound on the atom. Then $\mathbb{P}(D > J) \le \sum_{i>J}(T_+(i) + T_-(i))$ for every $t_a$, which tends to zero as $J \to \infty$ because the means are finite.

(b) Write $a_t = 1/s + \eta_t$, so $\eta_t \to 0$ by (5.15). The main part of (5.14) is $s^{-1}\sum_{i=d}^{t_a+d} f_+(i) = s^{-1}[T_+(d) - T_+(d + t_a + 1)]$. Let $M = M(d) \to \infty$ with $M = o(d)$. The terms with $t \ge M$ contribute an error of at most $\sup_{t\ge M} |\eta_t|\, T_+(d)$. The terms with $t < M$ contribute at most $\sum_{t<M} f_+(t_a - t + d) = T_+(a) - T_+(a + M)$ with $a = t_a + d - M + 1 \ge d - M$, and by Lemma 2.1(i) this is $o(T_+(a)) = o(T_+(d))$, since $M/a \to 0$ and $a \ge d - M \ge d/2$. Both errors are uniform in $t_a$. For (R$\nu$), the first part holds with $C_\nu = C$ of Lemma 2.1(ii), since $\nu_+^{(t_a)}(d) \le T_+(d) \le CT_+(n)$ for $d \ge n/2$. For the second part let $|d - n| \le \delta n$. By the expansion just proved, $|\nu_+^{(t_a)}(d) - \nu_+^{(t_a)}(n)|$ is at most $s^{-1}|T_+(d) - T_+(n)| + s^{-1}|T_+(d + t_a + 1) - T_+(n + t_a + 1)| + o(T_+(n))$. The two arguments of each difference differ by at most $\delta n \le 2\delta \min(d, n)$ for $\delta \le \frac{1}{2}$, so by Lemma 2.1(i) each difference is at most $\epsilon(\delta, n)T_+(n)$ with $\lim_\delta \limsup_n \epsilon(\delta, n) = 0$, where we used $T_+(n + t_a + 1) \le T_+(n)$. The bounds do not depend on $t_a$.

(c) In (5.14) put $m = t_a - t$. Then $\nu_+^{(t_a)}(d) = \sum_{m=0}^{t_a} a_{t_a-m} f_+(m + d)$, the coefficients are at most 1, each tends to $1/s$, and $\sum_m f_+(m + d) = T_+(d) < \infty$. Dominated convergence gives $T_+(d)/s$, and the minus sign is the same with $b_t$. In the same way $\pi_+^{(t_a)} = \sum_{m=0}^{t_a} a_{t_a-m} T_+(m + 1) \to s^{-1}\sum_{m\ge 0} T_+(m + 1) = \mu_+/s = p$. For the atom, $Q_+^{(t_a)}(N) = \sum_{i=N}^{N+t_a} a_{t_a+N-i} T_+(i)$. Put $\Sigma_N = \sum_{i=N}^{N+t_a} T_+(i)$. Since $T_+$ is non-increasing, Lemma 2.1(ii) gives $\Sigma_N \ge \min(t_a, N)\, T_+(2N) \ge c \min(t_a, N)\, T_+(N)$. Let $M \to \infty$ with $M = o(\min(t_a, N))$. The $M$ terms with $t_a + N - i < M$ contribute at most $MT_+(N) = o(\Sigma_N)$, and in the other terms the coefficient is $1/s + o(1)$ by (5.15). Hence $Q_+^{(t_a)}(N) = (1/s + o(1))\Sigma_N$. □

By Lemma 2.1(i) and (iv), as $N \to \infty$ with $t_a/N \to \theta$,

$$\sum_{i=N}^{N+t_a} T_+(i) \sim \begin{cases} t_a T_+(N) & \text{if } \theta = 0 \text{ and } t_a \to \infty, \\ NT_+(N)\dfrac{1 - (1+\theta)^{1-\alpha_+}}{\alpha_+ - 1} & \text{if } 0 < \theta < \infty, \\ NT_+(N)\dfrac{1}{\alpha_+ - 1} & \text{if } \theta = \infty, \end{cases} \tag{5.16}$$

and $T_+(N + t_a)/T_+(N) \to (1 + \theta)^{-\alpha_+}$, with $(1 + \theta)^{-\alpha_+}$ read as 0 at $\theta = \infty$. Write $\kappa_+^{(t_a)}$ for the kernel (5.4) of the aged initial run, $\kappa_+^{(t_a)}(j) = \pi_+^{(t_a)} u_-(j) + \sum_{d=1}^{j} \nu_-^{(t_a)}(d) u_-(j - d)$ for $j \ge 1$, and put

$$\begin{aligned} G_N^{(t_a)}(k) = {} & \frac{N - L_k}{\mu_-} f_+(\lfloor L_k \rfloor) + T_+(\lfloor L_k \rfloor)\, \kappa_+^{(t_a)}(N - k) \\ & + \frac{T_+(\lfloor L_k \rfloor) - T_+(\lfloor L_k \rfloor + t_a)}{s}\big(1 + \mu_+ u_-(N - k)\big), \end{aligned} \tag{5.17}$$

the kernel (5.5) of the aged initial run plus one term.

**Theorem D** (aging)**.** *Let the walk start fresh with either sign at step* 1*, and let* $t_a = t_a(N) \ge 0$*.*

*(a) (The bulk does not age.) Assume (RV) and (SPAN). For every choice of* $t_a(N)$*,*

$$\sup_{1\le k<N} \big| B_N P_N^{(t_a)}(k) - h_N(x_{N,k}) \big| \to 0.$$

*(b) (The atom.) Assume (RV) and (SPAN), and let* $t_a \to \infty$ *with* $t_a/N \to \theta \in [0, \infty]$*. Then*

$P_N^{(t_a)}(N) = Q_+^{(t_a)}(N) \sim s^{-1}\sum_{i=N}^{N+t_a} T_+(i)$, *which* (5.16) *evaluates. For* $0<\theta<\infty$,

$$P_N^{(t_a)}(N) \sim \frac{NT_+(N)}{s(\alpha_+-1)}\big(1-(1+\theta)^{1-\alpha_+}\big).$$

*(c) (The big-jump zone and the boundary layer.) Assume (RV) and (SPAN) for both signs and (LR) for the plus sign. For every choice of* $t_a(N)$,

$$\lim_{K\to\infty}\limsup_{N\to\infty}\sup_{pN+K\lambda_N B_N\le k<N}\frac{\big|P_N^{(t_a)}(k)-G_N^{(t_a)}(k)\big|}{\widehat G_N(k)}=0.$$

*The last term of* (5.17) *is the probability that the run in progress at time* $t_a$ *is the giant run. If* $t_a\to\infty$ *with* $t_a/N\to\theta\in[0,\infty]$, *this gives the following limits.*

*(i) For every fixed* $j\ge 1$,

$$\frac{P_N^{(t_a)}(N-j)}{T_+(N)}\to\kappa_+^{(\theta)}(j)=p\,u_-(j)+\frac1s+\frac1s\big(1-(1+\theta)^{-\alpha_+}\big)\big(1+\mu_+u_-(j)\big).$$

*(ii) For every* $\varepsilon\in(0,\frac12)$, *uniformly over* $k$ *with* $\varepsilon N\le L_k\le(1-\varepsilon)N$,

$$P_N^{(t_a)}(k)=\frac{N-L_k}{\mu_-}\,f_+(\lfloor L_k\rfloor)+\frac{2T_+(\lfloor L_k\rfloor)-T_+(\lfloor L_k\rfloor+t_a)}{\mu_-}+o\big(T_+(N)\big),$$

*and for fixed* $v\in(p,1)$, *with* $\ell=(v-p)/(1-p)$, $P_N^{(t_a)}(\lfloor vN\rfloor)/T_+(N)\to I_+(v)+[\ell^{-\alpha_+}-(\ell+\theta)^{-\alpha_+}]/\mu_-$, *with* $I_+$ *as in Corollary 4.9 and* $(\ell+\theta)^{-\alpha_+}=0$ *at* $\theta=\infty$.

*The same holds at the minus end with the signs exchanged.*

At $\theta=0$ the layer and the interior kernel are those of the stationary initial run with its complete initial run removed, that is $\kappa_+^{(0)}=pu_-+1/s$ and the kernel of Theorem B(b), while the atom $t_aT_+(N)/s$ grows with $t_a$. At $\theta=\infty$ all parts are those of Corollary 5.4. For $0<\theta<\infty$ the straddling run reaches the horizon with probability $\nu_+^{(t_a)}(L_k)\approx s^{-1}[T_+(L_k)-T_+(L_k+t_a)]$, and this is the term that ages. The bulk never does.

*Proof.* Throughout, $\nu=\nu^{(t_a)}$ is the initial run (5.14), and Lemma 5.6(a) says that the family satisfies (5.1), so Theorem C applies to it.

(a) By Lemma 5.6(a), $B_N\bar\nu(N/2)\le B_N(T_+(N/2)+T_-(N/2))\to 0$, by Lemma 2.1(ii) and (v). Theorem C(b) concludes.

(b) This is Lemma 5.6(c) and (5.16), together with $P_N(N)=Q_+(N)$ from Proposition 2.4.

(c) By Lemma 5.6(b), $\nu_+$ satisfies (R$\nu$) uniformly in $t_a$, so the regular case of Theorem C(c) gives, on the zone with $k<N$ and in the double limit, $P_N(k)=G_N^{\nu}(k)+\nu_+(\lfloor L_k\rfloor)(1+\mu_+u_-(N-k))+o(\widehat G_N(k)+\nu_+(\lfloor L_k\rfloor))$. Here $\nu_+(\lfloor L_k\rfloor)\le T_+(\lfloor L_k\rfloor)\le\widehat G_N(k)$ by Lemma 5.6(a), and $L_k\to\infty$ uniformly on the zone, so Lemma 5.6(b) gives $\nu_+(\lfloor L_k\rfloor)=s^{-1}[T_+(\lfloor L_k\rfloor)-T_+(\lfloor L_k\rfloor+t_a+1)]+o(\widehat G_N(k))$, and $T_+(\lfloor L_k\rfloor+t_a)-T_+(\lfloor L_k\rfloor+t_a+1)=f_+(\lfloor L_k\rfloor+t_a)=o(T_+(\lfloor L_k\rfloor))$ under (RV). This is the display.

(i) Fix $j\ge1$ and let $k=N-j$, which lies in the zone for every $K$ once $N$ is large. Then $L_k=N-js/\mu_-$, so $\widehat G_N(k)\sim T_+(N)$ and $T_+(\lfloor L_k\rfloor)/T_+(N)\to1$ by Lemma 2.1(i), and $(N-L_k)f_+(\lfloor L_k\rfloor)\le CjT_+(N)/N=o(T_+(N))$ by (LR). By Lemma 5.6(c) and the covering identity (2.9), $\kappa_+^{(t_a)}(j)\to pu_-(j)+s^{-1}\sum_{d=1}^{j}T_-(d)u_-(j-d)=pu_-(j)+1/s$, and $T_+(\lfloor L_k\rfloor+t_a)/T_+(N)\to(1+\theta)^{-\alpha_+}$ by (5.16). Dividing the display by $T_+(N)$ gives $\kappa_+^{(\theta)}(j)$.

(ii) Let $Z_N^\varepsilon=\{k:\varepsilon N\le L_k\le(1-\varepsilon)N\}$. On $Z_N^\varepsilon$ we have $j=N-k=(1-p)(N-L_k)\ge(1-p)\varepsilon N$ and $T_+(N)\le\widehat G_N(k)\le C_\varepsilon T_+(N)$, as in the proof of Theorem C(e), and $Z_N^\varepsilon$

lies in the zone for every $K$ once $N$ is large, because $\lambda_N B_N = o(N)$. We claim that $\kappa_+^{(t_a)}(j) \to 1/\mu_-$ as $j \to \infty$, uniformly in $t_a$. Fix $J$ and split the sum in $\kappa_+^{(t_a)}(j)$ at $J$. The part $d > J$ is at most $\varepsilon_J$. In the part $d \le J$ each $u_-(j-d)$ tends to $1/\mu_-$ as $j \to \infty$ by (2.10), and $\sum_{d \le J} \nu_-(d)$ differs from $\pi_-$ by at most $\varepsilon_J$. Hence $|\kappa_+^{(t_a)}(j) - \pi_+ u_-(j) - \pi_-/\mu_-| \le 2\varepsilon_J + o(1)$, and since $\pi_+ + \pi_- = 1$ and $u_-(j) \to 1/\mu_-$ the claim follows. Also $1+\mu_+ u_-(j) \to 1+\mu_+/\mu_- = s/\mu_-$. Since $T_+(\lfloor L_k \rfloor) \le C_\varepsilon T_+(N)$ on $Z_N^\varepsilon$, the display becomes the stated expansion. For $k = \lfloor vN \rfloor$ with $v \in (p,1)$ fixed we have $L_k = \ell N + O(1)$, and as in the proof of Corollary 4.9, (LR) and (RV) give $(N - L_k) f_+(\lfloor L_k \rfloor)/T_+(N) \to \alpha_+(1-\ell)\ell^{-\alpha_+ - 1}$ and $T_+(\lfloor L_k \rfloor)/T_+(N) \to \ell^{-\alpha_+}$, while $T_+(\lfloor L_k \rfloor + t_a)/T_+(N) \to (\ell+\theta)^{-\alpha_+}$ when $t_a/N \to \theta$. The minus end is the mirror image. □

The golden ratio threshold of Theorem E(d) survives aging. In its proof, parts (a) and (b) of Theorem E use the initial run only through (6.2), which is Theorem D(a) for the aged law.

**Corollary 5.7** (the golden ratio under aging)**.** *Assume (RV) and (SPAN), a fresh start at step* 1 *and* $t_a/N \to \theta \in (0, \infty]$*. The endpoint of sign* $\sigma$ *beats every interior value of* $P_N^{(t_a)}$ *for all large* $N$ *when* $\alpha_\sigma - 1 < 1/\alpha_*$*, and loses to the largest interior value when* $\alpha_\sigma - 1 > 1/\alpha_*$*. With equal indices* $\alpha_+ = \alpha_- = \alpha$ *the global mode is an endpoint for all large* $N$ *when* $\alpha < \varphi$ *and is interior for all large* $N$ *when* $\alpha > \varphi$*. For* $T_+(n) = T_-(n) = n^{-\alpha}$ *and* $0 < \theta < \infty$*,* $B_N P_N^{(t_a)}(N)/H_N \sim (1 - (1+\theta)^{1-\alpha})\, \mathcal{C}(\alpha) N^{1-\alpha+1/\alpha}$*, so at* $\alpha = \varphi$ *the mode is interior for every* $\theta$*.*

*Proof.* By Theorem D(a), (6.2) holds for the aged law, so parts (a) and (b) of Theorem E hold for it with the same proof. By Theorem D(b) and (5.16), $P_N^{(t_a)}(N)$ is asymptotic to a constant multiple of $NT_+(N)$, which is regularly varying with index $1 - \alpha_+$, and $B_N \in \mathrm{RV}_{1/\alpha_*}$ by Lemma 2.1(v). So $B_N P_N^{(t_a)}(N)/H_N$ tends to infinity or to zero according to the sign of $1/\alpha_* - (\alpha_+ - 1)$, and parts (a) and (b) of Theorem E conclude. The minus end is the same. With equal indices the condition $\alpha - 1 < 1/\alpha$ is $\alpha < \varphi$. For the pure power laws the computation in the proof of Theorem E(d) applies with the atom multiplied by $1 - (1+\theta)^{1-\alpha} < 1$, and $\mathcal{C}(\varphi) < 1$. □

For $t_a = o(N)$ the atom is $t_a T_+(N)/s$ by Theorem D(b), and the comparison with the bulk depends on the rate at which $t_a$ grows. For the fresh start itself the mode is interior by Theorem E(d). Section 9 reports a check of parts (b) and (c) against the exact aged law.

## 6 The shape of the law

This section reads three features of the shape off Theorems A to C. The first is the global mode, where the golden ratio enters, the second is the interior minimum between the bulk and the plus end, and the third is the local extrema near the ends.

### 6.1 The global mode

Let $P_N^{\max} = \max_{1 \le k < N} P_N(k)$ be the largest interior value and $H_N = \sup_x h_N(x)$ the height of the profile. By Proposition 2.4 the two endpoint atoms are $P_N(N) = Q_+(N)$ and $P_N(0) = Q_-(N)$.

**Theorem E** (the global mode)**.** *Assume (RV) and (SPAN), and let the initial run satisfy the bulk criterion* $B_n \mathbb{P}(D = n) \to 0$*.*

*(a)* $P_N^{\max} = (H_N + o(1))/B_N$*, every interior maximizer lies in* $pN + O(B_N)$*, and along any subsequence on which* $h_N$ *converges uniformly the scaled maximizers* $(k - pN)/B_N$ *converge to the set of maximizers of the limit profile.*

*(b) If* $\liminf_N B_N Q_\sigma(N)/H_N > 1$ *for some sign* $\sigma$*, the endpoint atom of sign* $\sigma$ *exceeds every interior value for all large* $N$*, so the global mode of* $P_N$ *is an endpoint. If* $\limsup_N B_N Q_\sigma(N)/H_N < 1$ *for both signs, the global mode is interior for all large* $N$*.*

*(c) If* $Q_\sigma \in \mathrm{RV}_{-\rho_\sigma}$*, the endpoint of sign* $\sigma$ *beats every interior value for all large* $N$ *when* $\rho_\sigma < 1/\alpha_*$*, and loses to the largest interior value when* $\rho_\sigma > 1/\alpha_*$*.*

*(d) (The golden ratio.) For the stationary initial run,* $\rho_\sigma = \alpha_\sigma - 1$*. With equal indices* $\alpha_+ = \alpha_- = \alpha$*, the global mode is an endpoint for all large* $N$ *when* $\alpha < \varphi = (1+\sqrt{5})/2$ *and is interior for all large* $N$ *when* $\alpha > \varphi$*. For two equal laws* $T_+(n) = T_-(n) = n^{-\alpha}$ *and the stationary initial run,* $B_N Q_\sigma(N)/H_N \sim \mathcal{C}(\alpha) N^{1-\alpha+1/\alpha}$ *for each sign, where*

$$\mathcal{C}(\alpha) = \frac{\pi\big(K_\alpha |\cos\tfrac{\pi\alpha}{2}|\big)^{1/\alpha}}{4(\alpha-1)\,\Gamma(1+\tfrac{1}{\alpha})\,\zeta(\alpha)^{1+1/\alpha}}, \tag{6.1}$$

*and at* $\alpha = \varphi$ *the ratio tends to* $\mathcal{C}(\varphi) = 0.7761317\ldots < 1$*, so the global mode is interior. For a fresh start of sign* $\sigma_0$*,* $Q_{\sigma_0} = T_{\sigma_0}$*,* $Q_{\bar\sigma_0} = 0$ *and* $\rho_{\sigma_0} = \alpha_{\sigma_0} > 1 > 1/\alpha_*$*, so the global mode is interior for all large* $N$ *and every pair of indices.*

*Proof.* (a) Put $h_* = \inf_N H_N$, which is positive by Lemma 2.2(d). By Theorem C(b) the bulk criterion gives

$$\varepsilon_N = \sup_{1\le k<N} \big|B_N P_N(k) - h_N(x_{N,k})\big| \to 0. \tag{6.2}$$

Hence $B_N P_N^{\max} = \max_{1\le k<N} h_N(x_{N,k}) + O(\varepsilon_N)$, and $\max_k h_N(x_{N,k}) \le H_N$. For the lower bound, Lemma 2.2(b) gives $K$ with $\sup_{|x|>K} h_N(x) < h_*/2$ for all $N$, so $H_N$ is the supremum of $h_N$ over $[-K,K]$. For large $N$ the points $x_{N,k}$ with $1 \le k < N$ cover $[-K,K]$ with mesh $1/B_N$, since $pN/B_N \to \infty$ and $(1-p)N/B_N \to \infty$. The common Lipschitz constant of Lemma 2.2(a) gives $\max_k h_N(x_{N,k}) \ge H_N - C_2/B_N$. Hence $B_N P_N^{\max} = H_N + o(1)$.

Let $k_N$ be an interior maximizer. Then $h_N(x_{N,k_N}) \ge B_N P_N^{\max} - \varepsilon_N = H_N - o(1) > h_*/2$ for large $N$, so $|x_{N,k_N}| \le K$, which is the statement $k_N \in pN + O(B_N)$. Along a subsequence with $h_N \to h$ uniformly we have $H_N \to \sup h$ and $h(x_{N,k_N}) \ge H_N - o(1)$. Every limit point $x$ of the bounded sequence $x_{N,k_N}$ satisfies $h(x) = \sup h$ by continuity. So the distance from $x_{N,k_N}$ to the set of maximizers of $h$ tends to zero.

(b) If $B_N Q_\sigma(N) \ge (1+\eta) H_N$ for all large $N$, then by (a) $B_N Q_\sigma(N) - B_N P_N^{\max} \ge \eta H_N - o(1) \ge \eta h_* - o(1) > 0$, so the endpoint atom of sign $\sigma$ exceeds every interior value. If $B_N Q_\sigma(N) \le (1-\eta) H_N$ for both signs and all large $N$, then (a) gives $B_N P_N^{\max} - B_N Q_\sigma(N) \ge \eta h_* - o(1) > 0$ for both signs.

(c) By Lemma 2.1(v), $B_N \in \mathrm{RV}_{1/\alpha_*}$. Hence $B_N Q_\sigma(N) \in \mathrm{RV}_{1/\alpha_* - \rho_\sigma}$. A regularly varying function with a positive index tends to infinity and one with a negative index tends to zero [9, Proposition 1.5.1]. Since $H_N$ is bounded above and below by Lemma 2.2(a) and (d), part (b) applies.

(d) By Corollary 5.4(a) the stationary initial run satisfies the bulk criterion and $Q_\sigma(N) \sim N T_\sigma(N)/(s(\alpha_\sigma - 1))$, so $\rho_\sigma = \alpha_\sigma - 1$. With equal indices $\alpha_* = \alpha$, and $\alpha - 1 < 1/\alpha$ if and only if $\alpha^2 - \alpha - 1 < 0$, that is, $\alpha < \varphi$. Part (c) then shows that both endpoints beat the interior when $\alpha < \varphi$ and that both lose when $\alpha > \varphi$. For unequal indices part (c) shows that the endpoint of sign $\sigma$ beats the interior if and only if $\alpha_\sigma - 1 < 1/\alpha_*$, away from equality.

Now let $T_+(n) = T_-(n) = n^{-\alpha}$. Then $\mu_\sigma = \sum_{n\ge1} n^{-\alpha} = \zeta(\alpha)$, $s = 2\zeta(\alpha)$, $p = 1/2$, $\theta_{+,N} = \theta_{-,N} = 1$, and we may take $a_\sigma(t) = t^{1/\alpha}$, so $B_N = (N/s)^{1/\alpha}$. The profile does not

depend on $N$. It is the density $h$ of $\frac{1}{2}(Z_+ - Z_-)$ with $Z_+$ and $Z_-$ independent copies of $Z_\alpha$. Its characteristic function is

$$\exp\bigl(K_\alpha(-it/2)^\alpha + K_\alpha(it/2)^\alpha\bigr) = \exp\bigl(-c_\alpha|t|^\alpha\bigr), \qquad c_\alpha = 2^{1-\alpha}K_\alpha\bigl|\cos\tfrac{\pi\alpha}{2}\bigr|,$$

because $(-iy)^\alpha + (iy)^\alpha = 2y^\alpha\cos\frac{\pi\alpha}{2}$ for $y>0$ and $\cos\frac{\pi\alpha}{2}<0$. This function is positive and integrable, so by Fourier inversion $h(x)\le h(0)$ for every $x$, and

$$H_N = h(0) = \frac{1}{\pi}\int_0^\infty e^{-c_\alpha t^\alpha}\,dt = \frac{\Gamma(1+\frac{1}{\alpha})}{\pi\, c_\alpha^{1/\alpha}}.$$

The stationary atom is $Q_\sigma(N) = s^{-1}\sum_{d\ge N} d^{-\alpha} \sim N^{1-\alpha}/(2(\alpha-1)\zeta(\alpha))$. Multiplying,

$$\frac{B_N Q_\sigma(N)}{H_N} \sim \frac{N^{1/\alpha+1-\alpha}}{(2\zeta(\alpha))^{1/\alpha}\cdot 2(\alpha-1)\zeta(\alpha)}\cdot\frac{\pi\,2^{(1-\alpha)/\alpha}\bigl(K_\alpha|\cos\frac{\pi\alpha}{2}|\bigr)^{1/\alpha}}{\Gamma(1+\frac{1}{\alpha})} = \mathcal{C}(\alpha)\,N^{1-\alpha+1/\alpha},$$

since $2^{(1-\alpha)/\alpha}2^{-1/\alpha}2^{-1} = 2^{-2}$. At $\alpha=\varphi$ the exponent vanishes, and $\alpha-1 = 1/\varphi$, $1+1/\alpha=\varphi$ and $2-\alpha=\varphi^{-2}$, so $K_\varphi = \varphi\,\Gamma(\varphi^{-2})$ and

$$\mathcal{C}(\varphi) = \frac{\pi\varphi\bigl(\varphi\,\Gamma(\varphi^{-2})|\cos\frac{\pi\varphi}{2}|\bigr)^{1/\varphi}}{4\,\Gamma(\varphi)\,\zeta(\varphi)^\varphi} = 0.7761317205\ldots.$$

That $\mathcal{C}(\varphi)<1$ needs only four elementary bounds. Since $x^{-\varphi}$ is convex, the trapezoid rule gives $\zeta(\varphi)\ge\frac{1}{2}+1/(\varphi-1)=\varphi+\frac{1}{2}$. The minimum of $\Gamma$ on $(0,\infty)$ exceeds $0.8856$ [54, Section 5.4(iii)], so $\Gamma(\varphi)\ge 0.8856$. Since $\Gamma\le 1$ on $[1,2]$, $\Gamma(\varphi^{-2})=\varphi^2\Gamma(1+\varphi^{-2})\le\varphi^2$. Finally $\frac{\pi}{2}<\frac{\pi\varphi}{2}<\frac{5\pi}{6}$, so $|\cos\frac{\pi\varphi}{2}|\le\frac{\sqrt{3}}{2}$. Together these give $\mathcal{C}(\varphi)\le\pi\varphi(\sqrt{3}\varphi^3/2)^{1/\varphi}/(4\cdot 0.8856\,(\varphi+\frac{1}{2})^\varphi) = 0.9513\ldots<1$, and (b) shows that the global mode is interior.

For a fresh start of sign $\sigma_0$ we have $\mathbb{P}(D=n)=f_{\sigma_0}(n)\le T_{\sigma_0}(n)$, and $B_nT_{\sigma_0}(n)\to 0$ by Lemma 2.1(v), so the bulk criterion holds. Moreover $Q_{\sigma_0}=T_{\sigma_0}\in\mathrm{RV}_{-\alpha_{\sigma_0}}$ with $\alpha_{\sigma_0}>1>1/\alpha_*$, and the other atom vanishes. Part (c) concludes. □

For two equal laws $H_N=h(0)$, so $\mathcal{C}(\alpha)$ also compares the endpoint with the center point. On the line $\rho_\sigma=1/\alpha_*$ the ratio $B_NQ_\sigma(N)/H_N$ is slowly varying, the comparison depends on the slowly varying factors, and we leave it open.

### 6.2 The boundary layer at second order

By Theorem B the law follows the renewal mass function $u_-$ in the boundary layer, and $u_-(j)\to 1/\mu_-$ by (2.10). If $u_-$ stays above its limit, the layer lies above the level $T_+(N)/\mu_-$ at which it meets the big-jump zone, while the big-jump kernel increases as $k$ decreases from $N$. To locate the interior minimum between the two we need the boundary layer at second order, with an exact coefficient.

**Theorem 6.1** (second order in the boundary layer)**.** *Assume (RV) and (LR) for the plus sign, and let the start be fresh with sign $+$. Put*

$$\begin{aligned} C(j) &= \alpha_+\Bigl[j\,u_-(j) + 2\mu_+v_-(j) + \sum_{i=0}^{j-1}u_-(i)\Bigr] \\ &= \alpha_+\Bigl[j\,u_-(j) + \mu_+v_-(j) + \sum_{i=0}^{j-1}u_-(i)\bigl(1+\mu_+u_-(j-i)\bigr)\Bigr]. \end{aligned} \tag{6.3}$$

*(a) For every fixed $j\ge 0$,*

$$\frac{P_N(N-j)}{T_+(N)} = u_-(j) + \frac{C(j)}{N} + o\Bigl(\frac{1}{N}\Bigr).$$

*(b) For every* $\eta \in (0,1)$*,*

$$\sup_{1\le j\le N^{1-\eta}} \frac{N}{j}\Big|\frac{P_N(N-j)}{T_+(N)} - u_-(j) - \frac{C(j)}{N}\Big| \to 0.$$

*If in addition (RV) and (SPAN) hold for the minus sign, then* $C(j)/j \to 2\alpha_+ s/\mu_-^2$ *as* $j\to\infty$*.*

In the second form of $C(j)$ the terms $j\,u_-(j)+\mu_+ v_-(j)$ come from a giant run in progress at time $N$ that starts after $a$ steps, through $T_+(N-a)/T_+(N) = 1+\alpha_+ a/N + o(a/N)$, and the sum comes from a complete giant run followed by a fresh minus walk, through Lemma 2.5. We prove the first form. The convergence of $C(j)/j$ is slow, of the order of $jT_-(j)$ by (6.7) below. For the zeta law of Section 9 the limit is $3.0811$, while $C(j)/j$ is $4.968$, $3.762$ and $3.323$ at $j=5$, $40$ and $320$.

The proof uses the final-run identity (2.14) and two renewal identities besides (2.9). Write $u=u_-$ and $v=v_-$. For $j\ge 1$,

$$\sum_{m=1}^{j} T_-(m)\,v(j-m) = \sum_{i=1}^{j-1} u(i), \tag{6.4}$$

$$\sum_{i=0}^{j-1} u(i)\,u(j-i) = v(j). \tag{6.5}$$

For (6.4), $v(j-m)T_-(m) = \sum_r r\mathbb{P}(S_r^- = j-m,\, X_{r+1}^- \ge m)$, so the sum is the expectation of the index $r$ of the run that covers level $j$, minus one. That index minus one equals the number of $r\ge 1$ with $S_r^-\le j-1$, whose expectation is $\sum_{i=1}^{j-1}u(i)$. For (6.5), $\sum_{i=0}^{j} u(i)u(j-i) = \sum_{a,b\ge 0} u_{a+b}^-(j) = \sum_n (n+1)u_n^-(j) = v(j)+u(j)$, and the term $i=j$ equals $u(j)$. By (6.5) the two forms of (6.3) agree.

We need two facts about sums of plus runs. Write $f=f_+$, $T=T_+$, $\alpha=\alpha_+$, $\mu=\mu_+$ and $S_r = S_r^+$ in Lemmas 6.2 and 6.3. Under (LR) the sequence $f$ is regularly varying with index $-\alpha-1$, and Lemma 2.1(i) gives, for every $c>0$ and large $n$,

$$\sup_{m\ge cn}\frac{f(m)}{f(n)} \le C_c, \qquad \sup_{0\le w\le w_n}\Big|\frac{f(n-w)}{f(n)}-1\Big|\to 0 \quad \text{whenever } w_n = o(n). \tag{6.6}$$

By Lemma 2.1(iv), $\mathbb{E}[X_+^2; X_+\le y] \le \sum_{n\le y}(2n-1)T(n) \le Cy^2T(y)$ and $\mathbb{E}[X_+; X_+>y]\to 0$. Finally $nT(n)\to 0$, since $nT(n)\in \mathrm{RV}_{1-\alpha}$.

**Lemma 6.2** (uniform local big jump)**.** *Assume (RV) and (LR) for the plus sign, and fix* $\eta\in(0,1)$*. Then*

$$\sup_{1\le r\le n^{1-\eta}}\Big|\frac{f^{*r}(n)}{r\,f(n)}-1\Big|\to 0 \qquad (n\to\infty).$$

*Proof.* For a fixed $r$ this is the local subexponentiality of $f$. Indeed, for $r\ge 2$ split $f^{*r}(n) = \sum_m f^{*(r-1)}(m)f(n-m)$ at $m=n/2$. By (6.6) and dominated convergence the part $m\le n/2$ is $f(n)(1+o(1))$, and by induction on $r$ the part $m>n/2$, which is $\sum_{i<n/2} f(i)f^{*(r-1)}(n-i)$ up to one term of order $f(n/2)^2 = o(f(n))$, is $(r-1)f(n)(1+o(1))$. For growing $r$ we use the local big-jump theorem of Denisov, Dieker and Shneer [16, Theorem 9.1] for the centered step $\xi = X_+ - \mu$ and intervals of length $T=1$. Its hypotheses, those of [16, Proposition 9.1], hold here. By Karamata's theorem $\mathbb{E}[\xi^2; |\xi|\le x] \sim \frac{\alpha}{2-\alpha}x^2T(x)$, which gives their (25), the left tail of $\xi$ is bounded, which gives (26), and (27) follows from (25), $\mathbb{E}\xi = 0$ and the index $-\alpha<-1$ of the tail [16, Lemma 9.1]. The function $x\mapsto \mathbb{P}(x<\xi\le x+1) = f(\lfloor x+\mu\rfloor+1)$ is regularly varying with index $-\alpha-1<-1$, so it is intermediate regularly varying. The theorem then gives, for every $\varepsilon>0$, an $r_0$ such that $|\mathbb{P}(S_r - r\mu\in(x,x+1])/(r\mathbb{P}(x<\xi\le x+1))-1|\le\varepsilon$ for

all $r \geq r_0$ and all $x \geq x_r = \frac{1}{2} r^{1/(1-\eta)}$, because $rT(x_r) \to 0$ by Lemma 2.1(ii). Take $x = n - r\mu - \frac{1}{2}$. For $r \leq n^{1-\eta}$ and large $n$ we have $x \geq n/2 \geq x_r$, the event is $\{S_r = n\}$, and $\mathbb{P}(x < \xi \leq x+1) = f(n')$ with $|n' - n| \leq r\mu + 1 = o(n)$, so $f(n') = f(n)(1+o(1))$ by (6.6). Together with the finitely many $r < r_0$ this proves the lemma. □

**Lemma 6.3** (sums of plus runs)**.** *Assume (RV) for the plus sign. There is a constant $C$ such that, for all $r \geq 1$ and real $x > 0$,*

$$\mathbb{E}[S_r; S_r > x] \leq r\Big(\mathbb{E}[X_+; X_+ > x/2] + \frac{2r\mu^2}{x}\Big),$$

*and, if in addition $x \geq 4r\mu$,*

$$\mathbb{P}(S_r \geq x) \leq C\, rT(x).$$

*Proof.* If $S_r \geq x$, either some run is at least $x/2$, or the sum of the runs truncated at $x/2$ is at least $x$. The truncated sum has mean at most $r\mu \leq x/4$ and variance at most $r\,\mathbb{E}[X_+^2; X_+ < x/2] \leq Crx^2T(x)$. By Chebyshev's inequality the second event has probability at most $(4/(3x))^2 Crx^2T(x)$, and the first at most $rT(x/2) \leq CrT(x)$ by Lemma 2.1(ii), since $x \geq 4r\mu \geq 4$. For the other bound, which uses no condition on $x$, on $\{S_r > x\}$ either $X_i^+ > x/2$ or $S_r - X_i^+ > x/2$. Hence $\mathbb{E}[X_i^+; S_r > x] \leq \mathbb{E}[X_+; X_+ > x/2] + \mu\mathbb{P}(S_r - X_i^+ > x/2)$, and the last probability is at most $2r\mu/x$ by Markov's inequality. Sum over $i$. □

*Proof of Theorem 6.1.* We write $u = u_-$, $v = v_-$ and keep the notation of the two lemmas. By (LR), $Nf(N)/T(N) \to \alpha$. We use $v(j) \leq j\, u(j) \leq j$, which holds because $u_r^-(j) = 0$ for $r > j$.

(b) Fix $\eta \in (0,1)$, put $J_N = N^{1-\eta}$ and $x_N = N^{1-\eta/2}$, and let $N$ be so large that $2J_N \leq x_N$, $4J_N\mu \leq N/2$ and $J_N \leq (N/2)^{1-\eta/2}$. Every $o(\cdot)$ below is uniform in $1 \leq r \leq j \leq J_N$. In (2.14) the first sum runs over $1 \leq r \leq j$, since $u_0^-(j) = 0$ for $j \geq 1$, and the second over $1 \leq m \leq j$ and $0 \leq r \leq j - m$.

The final plus run. Split the expectation in the first sum of (2.14) at $S_r = N/2$. For integers $0 \leq x \leq 3N/4$ put

$$\mathcal{T}_N(x) = \frac{N}{T(N)}\big(T(N-x) - T(N)\big) = \frac{N}{T(N)} \sum_{n=N-x}^{N-1} f(n).$$

By (6.6) and $Nf(N)/T(N) \to \alpha$, $\mathcal{T}_N(x) = \alpha x(1+o(1))$ uniformly in $x \leq x_N$, and $\mathcal{T}_N(x) \leq Cx$ for $x \leq 3N/4$, since $f(n) \leq C_{1/4} f(N)$ for $n \geq N/4$. Since $j \leq N/4$,

$$\mathbb{E}\big[T(N-j-S_r); S_r \leq N/2\big] = T(N)\Big(\mathbb{P}(S_r \leq N/2) + \frac{1}{N}\mathbb{E}\big[\mathcal{T}_N(j+S_r); S_r \leq N/2\big]\Big).$$

On $\{S_r \leq x_N/2\}$ we have $j + S_r \leq x_N$. With $\mathbb{E}[j+S_r] = j + r\mu$ this gives

$$\big|\mathbb{E}[\mathcal{T}_N(j+S_r); S_r \leq N/2] - \alpha(j+r\mu)\big| \leq o(j+r) + C\,\mathbb{E}[j+S_r; S_r > x_N/2].$$

By Markov's inequality and Lemma 6.3, $\mathbb{E}[j+S_r; S_r > x_N/2] \leq 2jr\mu/x_N + r\big(\mathbb{E}[X_+; X_+ > x_N/4] + 4r\mu^2/x_N\big) = o(j)$, since $r \leq j \leq J_N = o(x_N)$. By Lemma 6.3 and Lemma 2.1(ii), $\mathbb{P}(S_r > N/2) \leq CrT(N)$. On the other part,

$$\mathbb{E}\big[T(N-j-S_r); N/2 < S_r \leq N-j-1\big] = \sum_{1 \leq l < N/2-j} f^{*r}(N-j-l)\, T(l).$$

Here $N - j - l \geq N/2$ and $r \leq J_N \leq (N-j-l)^{1-\eta/2}$. So Lemma 6.2 with $\eta/2$ in place of $\eta$ and (6.6) give $f^{*r}(N-j-l) = rf(N)(1+o(1))$ for $l \leq x_N$, since then $j + l = o(N)$,

and $f^{*r}(N-j-l) \le CrT(N)/N$ for all $l$ in the range. Since $\sum_{l\le x_N} T(l) \to \mu$ and $\sum_{l>x_N} T(l) \to 0$, the sum equals $r\alpha\mu T(N)N^{-1}(1+o(1))$. Together,

$$\mathbb{E}\big[T(N-j-S_r); S_r \le N-j-1\big] = T(N)\Big(1 - O\big(rT(N)\big) + \frac{\alpha(j+2r\mu)}{N} + o\Big(\frac{j}{N}\Big)\Big).$$

Multiply by $u_r^-(j)$ and sum over $r$. Since $\sum_r u_r^-(j) = u(j) \le 1$ and $\sum_r r u_r^-(j) = v(j)$, the first sum of (2.14) divided by $T(N)$ equals

$$u(j) - O\big(v(j)T(N)\big) + \frac{\alpha}{N}\big(j\,u(j) + 2\mu v(j)\big) + o\Big(\frac{j}{N}\Big).$$

Since $v(j)T(N) \le jT(N) = (j/N)\,NT(N)$ and $NT(N) \to 0$, the second term is $o(j/N)$.

The final minus run. Since $r+1 \le j \le J_N \le (N-J_N)^{1-\eta/2}$, Lemma 6.2 with $\eta/2$ in place of $\eta$ and (6.6) give $u_{r+1}^+(N-j) = f^{*(r+1)}(N-j) = (r+1)\alpha T(N)N^{-1}(1+o(1))$. So the second sum of (2.14) equals

$$\begin{aligned}&\frac{\alpha T(N)}{N}(1+o(1))\sum_{m=1}^{j} T_-(m)\sum_r (r+1)u_r^-(j-m)\\ &\quad= \frac{\alpha T(N)}{N}(1+o(1))\sum_{m=1}^{j} T_-(m)\big(u(j-m)+v(j-m)\big),\end{aligned}$$

with $v(0)=0$. By (2.9) and (6.4) the last sum equals $1+\sum_{i=1}^{j-1} u(i) = \sum_{i=0}^{j-1} u(i)$, which is at most $j$. Adding the two contributions gives $u(j) + C(j)/N + o(j/N)$ with the first form of (6.3), uniformly in $1 \le j \le J_N$. This is (b).

(a) For $j=0$, (2.14) gives $P_N(N) = T(N)$ exactly, and $C(0)=0$. For fixed $j \ge 1$, part (b) gives an error $o(j/N) = o(1/N)$.

The limit. By (2.10) for the minus sign, $u(j) \to 1/\mu_-$. By (6.5) and Cesàro's theorem, $v(j)/j = j^{-1}\sum_{i<j} u(i)u(j-i) \to \mu_-^{-2}$, and $j^{-1}\sum_{i<j} u(i) \to 1/\mu_-$. Hence $C(j)/j \to \alpha_+(2/\mu_- + 2\mu_+/\mu_-^2) = 2\alpha_+ s/\mu_-^2$. □

### 6.3 The antimode

Two classical facts about the renewal mass function enter. The first is the second-order renewal theorem. Under (RV) and (SPAN) for the minus sign,

$$u_-(j) - \frac{1}{\mu_-} \sim \frac{j\,T_-(j)}{\mu_-^2(\alpha_- - 1)} \qquad (j \to \infty). \tag{6.7}$$

This is Lemma 4 of Frenk [28], stated there for an aperiodic recurrent event whose tail $\mathbb{P}(X > n) \sim T(n)$ is regularly varying with index $-\alpha$, $\alpha > 1$ (see also [60] and [23, Theorem 1]). The second fact gives a simple condition for the hypothesis of the next theorem.

**Lemma 6.4** (log-convex run laws)**.** *Assume (RV) and (SPAN) for the minus sign. If $f_-(n) > 0$ for all $n \ge 1$ and $f_-(n-1)f_-(n+1) > f_-(n)^2$ for all $n \ge 2$, then $u_-$ is strictly decreasing and $u_-(j) > 1/\mu_-$ for every $j \ge 0$.*

*Proof.* By Theorem 1 of de Bruijn and Erdős [14], the renewal sequence of a strictly log-convex law is strictly log-convex, that is, $u_-(m-1)u_-(m+1) > u_-(m)^2$ for $m \ge 1$. Hence the ratios $u_-(m+1)/u_-(m)$ increase strictly in $m$. If one of them were at least 1, all later ones would exceed some $q > 1$, and $u_-$ would grow geometrically, which is impossible since $u_- \le 1$. So every ratio is smaller than 1, $u_-$ is strictly decreasing, and it decreases to its limit $1/\mu_-$ given by (2.10). □

**Theorem F** (the antimode)**.** *Assume (RV) and (SPAN) for both signs and (LR) for the plus sign, and let the start be fresh with sign $+$. Let $j_N^*$ be a minimizer of $j \mapsto P_N(N-j)$ over $0 \le j \le (1-p)N$.*

*(a) If $u_-(j) > 1/\mu_-$ for every $j \ge 1$, then $j_N^* \to \infty$, $j_N^* = o(N)$, and*

$$T_-(j_N^*) \sim \frac{2\alpha_+ s}{N}.$$

*Moreover $j_N^*/i_N \to 1$ for every choice of minimizers $i_N$ of $j \mapsto u_-(j) + C(j)/N$ over $j \ge 0$. So $j_N^*$ is of the order of $b_{-,N}$, the stable scale of the minus runs, and $j_N^* \asymp N^{1/\alpha_-}$ up to a slowly varying factor.*

*(b) If $\inf_{j\ge 1} u_-(j) < 1/\mu_-$, then $j_N^*$ is bounded and every value that $j_N^*$ takes infinitely often is a minimizer of $u_-$.*

The borderline case, in which $u_- \ge 1/\mu_-$ with equality at some $j \ge 1$, is not covered. By Lemma 6.4 part (a) applies to every strictly log-convex law of the minus runs, for instance the zeta law $f(n) = n^{-\alpha-1}/\zeta(\alpha+1)$.

*Proof.* Write $R_N(j) = P_N(N-j)/T_+(N)$, $u = u_-$ and $c_{\mathrm{a}} = 2\alpha_+ s/\mu_-^2$. In this proof $\mu_-$ and $\mu_+$ are always written with their signs. We first prove two lower bounds on $R_N$ and then treat (a) and (b).

Lower bound in the layer. Let $1 \le j \le N/(8\mu_+)$. In the first sum of (2.14), $T_+(N-j-S_r^+) \ge T_+(N-j)$, and $\mathbb{P}(S_r^+ \le N-j-1) \ge 1 - \mathbb{P}(S_r^+ \ge N/2) \ge 1 - CrT_+(N)$ by Lemma 6.3, since $4r\mu_+ \le N/2$. Dropping the second sum,

$$R_N(j) \ge \frac{T_+(N-j)}{T_+(N)}\big(u(j) - C\,v(j)\,T_+(N)\big) \ge \frac{T_+(N-j)}{T_+(N)}\big(u(j) - C\,j\,T_+(N)\big). \tag{6.8}$$

By (LR), $f_+(n) \ge \frac{3}{4}\alpha_+ T_+(n)/n \ge \frac{3}{4}\alpha_+ T_+(N)/N$ for $N/2 \le n \le N$ and large $N$, because $T_+(n)/n$ is nonincreasing. Hence $T_+(N-j)/T_+(N) \ge 1 + \frac{3}{4}\alpha_+ j/N$ for $j \le N/2$. We use (6.8) below only for $j \le \delta N$ with $\delta \le 1/(8\mu_+)$ and on ranges where $u(j)$ is bounded below by a positive constant. There $CjT_+(N) \le C\delta N T_+(N) \to 0$, so the second factor is positive for large $N$. Since the first factor is at least $1$, we may then drop it or expand the product.

Lower bound far from the layer. Fix $\delta \in (0, 1-p)$. We claim that

$$\liminf_N \min_{\delta N \le j \le (1-p)N} R_N(j) \ge \frac{(1-\delta)^{-\alpha_+}}{\mu_-}. \tag{6.9}$$

Let $\varepsilon > 0$. By Corollary 4.8 there is $K$ such that $P_N(k) \ge (1-\varepsilon)G_N^{(+)}(k)$ for $pN + KB_N \le k \le N$ and large $N$. For $k = N-j$ with $j \ge \delta N$ we have $L_k = N - j/(1-p) \le (1-\delta)N$, so $G_N^{(+)}(k) \ge T_+(\lfloor L_k \rfloor)u(j) \ge T_+(\lfloor (1-\delta)N \rfloor)u(j)$, and $T_+(\lfloor (1-\delta)N \rfloor)/T_+(N) \to (1-\delta)^{-\alpha_+}$ by (RV). For $pN \le k < pN + KB_N$, Theorem A(c) and Lemma 2.2(d) give $B_N P_N(k) \ge \frac{1}{2}\inf_N \inf_{[-K,K]} h_N > 0$ for large $N$, so $R_N(N-k) \ge c_K/(B_N T_+(N)) \to \infty$, since $B_N T_+(N) \to 0$ by Lemma 2.1(v). As $\varepsilon$ is arbitrary and $u(j) \to 1/\mu_-$, the claim follows.

(b) Let $m_* = \min_j u(j) < 1/\mu_-$, which is attained on a finite set because $u(j) \to 1/\mu_-$, and let $\varepsilon_* = (1/\mu_- - m_*)/3$. Choose $J$ with $u(j) \ge 1/\mu_- - \varepsilon_* > 0$ for $j \ge J$, and let $\delta = \min\{1/(8\mu_+), (1-p)/2\}$. For $J \le j \le \delta N$, (6.8) gives $R_N(j) \ge u(j) - CjT_+(N) \ge 1/\mu_- - \varepsilon_* - C\delta N T_+(N)$, which exceeds $m_* + \varepsilon_*$ for large $N$, since $NT_+(N) \to 0$. For $j \ge \delta N$, (6.9) gives $R_N(j) \ge m_* + \varepsilon_*$ for large $N$. For $j < J$, Theorem 6.1(a) gives $R_N(j) \to u(j)$. So for large $N$ every minimizer is smaller than $J$, and a value $j'$ taken

infinitely often with $u(j') > m_*$ would give $R_N(j') > R_N(j_0)$ for large $N$, where $u(j_0) = m_*$.

(a) Let $\Lambda_N = a_-(N)$, so that $NT_-(\Lambda_N) \to 1$ and $\Lambda_N \in \mathrm{RV}_{1/\alpha_-}$. Put $w(j) = u(j) - 1/\mu_-$ and $c_\mathrm{r} = 1/(\mu_-^2(\alpha_- - 1))$, so $w(j) \sim c_\mathrm{r} j T_-(j)$ by (6.7), and $w(j) > 0$ for $j \geq 1$. Let

$$\Upsilon(x) = c_\mathrm{r} x^{1-\alpha_-} + c_\mathrm{a} x, \qquad x > 0.$$

The function $\Upsilon$ is strictly convex with a unique minimizer $x_*$, given by $c_\mathrm{r}(\alpha_- - 1)x_*^{-\alpha_-} = c_\mathrm{a}$, that is, $x_*^{-\alpha_-} = c_\mathrm{a}\mu_-^2 = 2\alpha_+ s$.

We fix the constants first. They depend only on $c_\mathrm{r}$, $c_\mathrm{a}$, $\alpha_\pm$ and $\mu_\pm$. Choose $\varepsilon \in (0, x_*)$ so small that $\frac{c_\mathrm{r}}{5}\varepsilon^{-(\alpha_- - 1)/2} > \Upsilon(x_*) + 2$, and choose $K > x_*$ so large that $\alpha_+ K/(2\mu_-) > \Upsilon(x_*) + 1$. So $\varepsilon < x_* < K$.

Step 1 (the window). By (6.7) and Lemma 2.1(i), $(N/\Lambda_N)\, w(j) = c_\mathrm{r}(j/\Lambda_N)^{1-\alpha_-}(1 + o(1))$ uniformly in $\varepsilon\Lambda_N \leq j \leq K\Lambda_N$, because $NT_-(j) = NT_-(\Lambda_N) \cdot T_-(j)/T_-(\Lambda_N)$. By Theorem 6.1, $C(j) = c_\mathrm{a} j(1 + o(1))$ as $j \to \infty$. Choose $\eta > 0$ with $1 - \eta > 1/\alpha_-$, so that $K\Lambda_N \leq N^{1-\eta}$ for large $N$. Theorem 6.1(b) then gives

$$\frac{N}{\Lambda_N}\Big(R_N(j) - \frac{1}{\mu_-}\Big) = \Upsilon\Big(\frac{j}{\Lambda_N}\Big) + o(1) \quad \text{and} \quad \frac{N}{\Lambda_N}\Big(u(j) + \frac{C(j)}{N} - \frac{1}{\mu_-}\Big) = \Upsilon\Big(\frac{j}{\Lambda_N}\Big) + o(1), \tag{6.10}$$

uniformly in $\varepsilon\Lambda_N \leq j \leq K\Lambda_N$. Since $\varepsilon < x_* < K$, in particular the minimum of $R_N$ is at most $1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + o(1))$.

Step 2 (small $j$). We have $R_N(0) = 1$, and $u(0) + C(0)/N = 1$ for the comparison function. Since $\mu_- > 1$ and $\Lambda_N/N \to 0$, both exceed $1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + 1)$ for large $N$. Let $1 \leq j \leq \varepsilon\Lambda_N$. Then $u(j) > 1/\mu_-$, and (6.8) gives $R_N(j) \geq u(j) - C\varepsilon\Lambda_N T_+(N) = 1/\mu_- + w(j) - o(\Lambda_N/N)$. Choose $j_0$ with $w(j) \geq \frac{c_\mathrm{r}}{2} j T_-(j)$ for $j \geq j_0$. By Lemma 2.1(ii) applied to $jT_-(j) \in \mathrm{RV}_{1-\alpha_-}$, for $j_0 \leq j \leq \varepsilon\Lambda_N$ and $j_0$ large,

$$\frac{jT_-(j)}{\Lambda_N T_-(\Lambda_N)} \geq \frac{1}{2}\Big(\frac{j}{\Lambda_N}\Big)^{-(\alpha_- - 1)/2} \geq \frac{1}{2}\varepsilon^{-(\alpha_- - 1)/2}.$$

Since $\Lambda_N T_-(\Lambda_N) \sim \Lambda_N/N$, we get $w(j) \geq \frac{c_\mathrm{r}}{5}\varepsilon^{-(\alpha_- - 1)/2}\Lambda_N/N$ for large $N$. For $1 \leq j < j_0$, $w(j) \geq \min_{i<j_0} w(i) > 0$, which is much larger than $\Lambda_N/N$. By the choice of $\varepsilon$, $R_N(j) > 1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + 1)$ for $1 \leq j \leq \varepsilon\Lambda_N$ and large $N$. The same bound holds for $u(j) + C(j)/N \geq 1/\mu_- + w(j)$.

Step 3 (intermediate $j$). Let $\delta = \min\{1/(8\mu_+), (1-p)/2\}$ and $K\Lambda_N \leq j \leq \delta N$. By (6.8) and $u(j) > 1/\mu_-$,

$$R_N(j) \geq \Big(1 + \frac{3\alpha_+ j}{4N}\Big)\Big(\frac{1}{\mu_-} - CjT_+(N)\Big) \geq \frac{1}{\mu_-} + \frac{\alpha_+ j}{2\mu_- N} \geq \frac{1}{\mu_-} + \frac{\alpha_+ K}{2\mu_-} \cdot \frac{\Lambda_N}{N}$$

for large $N$, since $NT_+(N) \to 0$. By the choice of $K$ this exceeds $1/\mu_- + (\Lambda_N/N)(\Upsilon(x_*) + 1)$. For the comparison function, $C(j) \geq \alpha_+ j\, u(j) \geq \alpha_+ j/\mu_-$, so $u(j) + C(j)/N \geq 1/\mu_- + \alpha_+ j/(\mu_- N)$ for every $j$, and the same bound holds for all $j \geq K\Lambda_N$.

Step 4 (large $j$). By (6.9), $R_N(j) \geq (1 + \varepsilon_\delta)/\mu_-$ for $\delta N \leq j \leq (1-p)N$ and large $N$, with $\varepsilon_\delta = ((1-\delta)^{-\alpha_+} - 1)/2 > 0$, while $\Lambda_N/N \to 0$.

Conclusion. By Steps 2 to 4 every minimizer of $R_N$, and every minimizer of $u(j) + C(j)/N$, lies in $[\varepsilon\Lambda_N, K\Lambda_N]$ for large $N$. Fix $\eta' > 0$. Since $\Upsilon$ is continuous with a unique minimizer, $\Upsilon \geq \Upsilon(x_*) + \varepsilon'$ on $[\varepsilon, K] \setminus (x_* - \eta', x_* + \eta')$ for some $\varepsilon' > 0$, and (6.10) shows that both minimizers lie in $((x_* - \eta')\Lambda_N, (x_* + \eta')\Lambda_N)$ for large $N$. Hence $j_N^*/\Lambda_N \to x_*$ and $i_N/\Lambda_N \to x_*$, so $j_N^*/i_N \to 1$. By (RV), $NT_-(j_N^*) = NT_-(\Lambda_N) \cdot T_-(j_N^*)/T_-(\Lambda_N) \to x_*^{-\alpha_-} = 2\alpha_+ s$. Finally $\Lambda_N = a_-(N)$ and $b_{-,N} = a_-(N/s)$ differ by a bounded factor, and $a_- \in \mathrm{RV}_{1/\alpha_-}$. □

The leading law $T_-(j_N^*) \sim 2\alpha_+ s/N$ is approached slowly, because $C(j)/j$ converges slowly, and the minimizer of $u_-(j) + C(j)/N$ is the better prediction at moderate $N$, as Table 1 shows.

| $N$ | observed $j_N^*$ | refined $i_N$ | leading | $NT_-(j_N^*)/(2\alpha_+ s)$ |
|---|---|---|---|---|
| 600 | 7 | 7 | 10 | 1.534 |
| 1200 | 11 | 12 | 15 | 1.498 |
| 2400 | 19 | 20 | 23 | 1.282 |
| 4000 | 27 | 28 | 32 | 1.247 |
| 8000 | 44 | 45 | 50 | 1.186 |

Table 1: The antimode for the zeta law $f_\sigma(n) = n^{-5/2}/\zeta(5/2)$ and a fresh plus start. The observed $j_N^*$ minimizes $P_N(N-j)$ over $0 \le j \le N/2$. The refined $i_N$ minimizes $u_-(j) + C(j)/N$, and the leading prediction is the smallest $j$ with $T_-(j) < 2\alpha_+ s/N$.

### 6.4 Local extrema near the ends

Theorems E and F compare values of $P_N$ in different zones. Here we compare neighbouring values. A *local maximum* of a finite or infinite sequence $(y_i)$ is a maximal block $\{a, \dots, b\}$ of consecutive indices on which $y$ is constant, such that $y_{a-1} < y_a$ unless $a$ is the first index and $y_{b+1} < y_b$ unless $b$ is the last. Local minima are defined in the same way, and a *mode* of $P_N$ is a local maximum. So an endpoint is a mode when it beats its neighbour, and a plateau counts once.

Let the initial run satisfy $\nu_+(n) = o(T_+(n))$, and let $\widetilde{\kappa}_+(0) = \lim_N Q_+(N)/T_+(N)$ exist. The *end profile* is $\widetilde{\kappa}_+ = (\widetilde{\kappa}_+(0), \kappa_+(1), \kappa_+(2), \dots)$, with $\kappa_+$ as in (5.4). By Proposition 2.4 and Theorem C(d), under (RV) for the plus sign,

$$P_N(N-j)/T_+(N) \to \widetilde{\kappa}_+(j) \qquad \text{for every fixed } j \ge 0. \tag{6.11}$$

For a fresh plus start, a fresh minus start and the fair start, which is fresh with each sign with probability $\frac{1}{2}$, the profile is $(1, u_-(1), u_-(2), \dots)$, $(0, u_-(1), \dots)$ and $(\frac{1}{2}, u_-(1), \dots)$. The profile $\widetilde{\kappa}_-$ at the minus end is defined with the signs exchanged. We write $D_N(j) = P_N(N-j-1) - P_N(N-j)$ and $\Delta y(j) = y(j+1) - y(j)$.

**Lemma 6.5** (difference renewal theorem)**.** *Assume (RV) and (SPAN) for the minus sign, and put $d_n = u_-(n) - u_-(n-1)$ for $n \ge 0$, with $u_-(-1) = 0$. Then $\sum_n |d_n| < \infty$ and $d_n = -(1+o(1))T_-(n)/\mu_-^2$. In particular $u_-$ is eventually strictly decreasing.*

Summing over $n > j$ and using Lemma 2.1(iv) gives back (6.7), so the lemma is a local form of Frenk's theorem. It also shows that if $u_-$ is nonincreasing on $j \ge 1$, then $u_-(j) > 1/\mu_-$ for every $j \ge 1$, so Theorem F(a) applies, and the borderline case after Theorem F has a nonmonotone $u_-$.

*Proof.* Write $u = u_-$, $\mu = \mu_-$, $t_n = T_-(n+1)$ for $n \ge 0$, and $\hat{a}(z) = \sum_n a_n z^n$. Since $t_0 = 1$ and $t_{n-1} - t_n = f_-(n)$, we have $(1-z)\hat{t} = 1 - \hat{f}_-$, and $\hat{u} = 1/(1-\hat{f}_-)$, so $\hat{d} = (1-z)\hat{u} = 1/\hat{t}$. On $|z| \le 1$ the function $\hat{t}$ has no zero. At $z = 1$ it equals $\mu$, and elsewhere a zero would give $\hat{f}_-(z) = 1$, which (SPAN) excludes. By Wiener's lemma [75] the function $1/\hat{t}$ on the unit circle has absolutely summable Fourier coefficients. They vanish at negative indices since $1/\hat{t}$ is continuous on the closed disk and analytic inside. So they are the $d_n$, and $d \in \ell^1$.

For sequences on $\mathbb{Z}_{\ge 0}$ put $\|a\|_1 = \sum_n |a_n|$ and $\|a\|_t = \sup_n |a_n|/t_n$. The sequence $t$ is positive, nonincreasing and regularly varying, so $c_t = \sup_n t_{\lfloor n/2 \rfloor}/t_n < \infty$. Splitting a convolution at $n/2$ gives

$$\|a * b\|_t \le c_t\big(\|a\|_1 \|b\|_t + \|a\|_t \|b\|_1\big), \tag{6.12}$$

and the following product rule. If $a, b \in \ell^1$, $a_n/t_n \to \chi_a$ and $b_n/t_n \to \chi_b$, then

$$(a * b)_n/t_n \to \hat{a}(1)\chi_b + \hat{b}(1)\chi_a. \tag{6.13}$$

Indeed, in $\sum_{k \le n/2} a_k b_{n-k}/t_n$ each term tends to $a_k\chi_b$, since $t_{n-k}/t_n \to 1$, and the terms are dominated by $c_t\|b\|_t|a_k|$. The other half is the same.

Split $t = t' + t''$ with $t'_n = t_n \mathbf{1}_{\{n \le M\}}$, and take $M$ so large that $2c_t\|d\|_1\|t''\|_1 \le \frac{1}{4}$. Then $1 - \hat{t}''\hat{d}$ has the inverse $\sum_{k\ge 0}(\hat{t}''\hat{d})^k$ with summable coefficients, and $\hat{t}' = \hat{t}(1 - \hat{t}''\hat{d})$ has no zero on $|z| \le 1$. So the coefficients $e$ of $1/\hat{t}' = \hat{d}\sum_k(\hat{t}''\hat{d})^k$ satisfy $\|e\|_1 \le 2\|d\|_1$. As $\hat{t}'$ is a polynomial without zeros in the closed disk, $e_n$ decays geometrically, and $e_n = o(t_n)$. Put $g = e * t''$. Since $t''_n = t_n$ for $n > M$, (6.13) gives $g_n/t_n \to \hat{e}(1)$, and $c_t\|g\|_1 \le \frac{1}{4}$. By (6.12), $x_k = \|g^{*k}\|_t$ satisfies $x_k \le \frac{1}{4}x_{k-1} + c_t\|g\|_t 4^{1-k}$, so $x_k \le Ck4^{-k}$. By induction with (6.13), $(g^{*k})_n/t_n \to k\hat{g}(1)^{k-1}\hat{e}(1)$. By dominated convergence over $k$, $h = \sum_{k\ge 1}(-g)^{*k}$ satisfies $h_n/t_n \to -\hat{e}(1)/(1+\hat{g}(1))^2$. Finally $\hat{t} = \hat{t}'(1+\hat{g})$ gives $\hat{d} = \hat{e}(1+\hat{h})$, so $d = e + e * h$ and $d_n/t_n \to -\hat{e}(1)^2/(1+\hat{g}(1))^2$. Here $\hat{e}(1) = 1/\hat{t}'(1)$ and $1 + \hat{g}(1) = (\hat{t}'(1) + \hat{t}''(1))/\hat{t}'(1) = \mu/\hat{t}'(1)$, so the limit is $-1/\mu^2$. Since $t_n \sim T_-(n)$, the lemma follows. □

**Lemma 6.6** (the prepared profile)**.** *Assume (RV) and (SPAN) for the minus sign, and let $\nu_-(d) = (c + o(1))T_-(d)$ for some $c \ge 0$. Then $\kappa_+(j) - \kappa_+(j-1) = \big((c\mu_- - 1)/\mu_-^2 + o(1)\big)T_-(j)$.*

*Proof.* The generating function of $\kappa_+$ is $\pi_+(\hat{u}_- - 1) + \hat{\nu}_-\hat{u}_-$, so $\kappa_+(n) - \kappa_+(n-1) = \pi_+ d_n + (\nu_- * d)_n$ for $n \ge 2$. Apply (6.13) with $a = \nu_-$, $\hat{a}(1) = \pi_-$, $\chi_a = c$, and $b = d$, $\hat{b}(1) = \lim_n u_-(n) = 1/\mu_-$, $\chi_b = -1/\mu_-^2$, and add $\pi_+ d_n$. □

So $\kappa_+$ is eventually strictly decreasing when $c < 1/\mu_-$, for instance when $\nu_- = o(T_-)$ or for the stationary initial run ($c = 1/s$), and eventually strictly increasing when $c > 1/\mu_-$. If $\nu_- = T_-/\mu_-$ exactly, then $\kappa_+ \equiv 1/\mu_-$ on $j \ge 1$ by (2.9).

**Proposition 6.7** (copies of the profile)**.** *Assume (RV) for the plus sign, let $\nu_+(n) = o(T_+(n))$, and let $\widetilde{\kappa}_+(0)$ exist.*

*(a) If $\widetilde{\kappa}_+(j+1) \ne \widetilde{\kappa}_+(j)$ for $0 \le j < J$, then for large $N$ the sign of $D_N(j)$ is that of $\widetilde{\kappa}_+(j+1) - \widetilde{\kappa}_+(j)$ for every $0 \le j < J$.*

*(b) For every local maximum $\{a, \dots, b\}$ of $\widetilde{\kappa}_+$ and all large $N$, $P_N$ has a local maximum inside $\{N-b, \dots, N-a\}$. The same holds for local minima.*

*Proof.* (a) follows from (6.11), since $D_N(j) = T_+(N)(\Delta R_N)(j)$ with $R_N(j) = P_N(N - j)/T_+(N)$. For (b), let $\widetilde{\kappa}_+ = y$ on $\{a, \dots, b\}$, and let $3\varepsilon$ be the gap between $y$ and the larger of $\widetilde{\kappa}_+(a-1)$ (absent if $a = 0$) and $\widetilde{\kappa}_+(b+1)$. For large $N$, (6.11) puts $R_N(i)$ within $\varepsilon$ of $\widetilde{\kappa}_+(i)$ for $a - 1 \le i \le b + 1$. So the largest value of $P_N$ on $\{N-b, \dots, N-a\}$ exceeds its values at $N-b-1$ and $N-a+1$, and the maximal block on which $P_N$ takes this value and that meets an argmax lies inside $\{N-b, \dots, N-a\}$. It is a local maximum of $P_N$. Minima are the same. □

**Theorem 6.8** (a lower bound for the number of modes)**.** *Assume (RV) and (SPAN) for both signs, let $\nu_\sigma(n) = o(T_\sigma(n))$ for both signs, and let $\widetilde{\kappa}_+(0)$ and $\widetilde{\kappa}_-(0)$ exist. Then $\widetilde{\kappa}_+$ has finitely many local maxima, $M_+ \ge 1$ of them, and $\widetilde{\kappa}_-$ has $M_- \ge 1$. For all large $N$ the law $P_N$ has at least $M_+ + M_- + 1$ modes. In particular, if one end profile has two local maxima, then $P_N$ has at least four modes for all large $N$.*

*Proof.* By Lemma 6.6 with $c = 0$, $\widetilde{\kappa}_+$ is strictly decreasing on some $[J, \infty)$, with the positive limit $1/\mu_-$. So its local maxima lie in $[0, J]$, and there is at least one, the block of a global maximizer. By Proposition 6.7(b) each of them gives a mode of $P_N$ in $\{N - J, \dots, N\}$, and distinct local maxima give modes in disjoint blocks. The same holds at the minus end. The initial run satisfies the bulk criterion, since $B_n T_\sigma(n) \to 0$. By Theorem E(a) the interior maximizers lie in $pN + O(B_N)$, and the block of interior maximizers that contains one of them is a mode of $P_N$ at distance of order $N$ from both ends. □

For a fresh plus start $\widetilde{\kappa}_+ = (1, u_-(1), \dots)$ has exactly one local maximum if and only if $u_-$ is nonincreasing on $j \geq 1$, by Lemma 6.5. The preparation can add a mode, already for a log-convex law.

**Corollary 6.9** (a prepared start with four modes)**.** *Assume (RV) and (SPAN) for both signs, and let $f_-$ be strictly log-convex as in Lemma 6.4. Let the initial run be a fresh plus run with probability $\frac{1}{2}$ and a minus run of a fixed length $d_0 \geq 2$ with probability $\frac{1}{2}$. Then $P_N$ has at least four modes for all large $N$, while for a fresh plus start $\widetilde{\kappa}_+$ is strictly decreasing.*

*Proof.* Here $\nu_+ = f_+/2 = o(T_+)$, $\nu_-$ is a point mass, $Q_+(N) = T_+(N)/2$ and $Q_-(N) = 0$ for $N > d_0$. So $\widetilde{\kappa}_+(0) = \frac{1}{2}$ and, by (5.4), $\widetilde{\kappa}_+(j) = \frac{1}{2}(u_-(j) + u_-(j - d_0))$ for $j \geq 1$, with $u_- = 0$ at negative arguments. Since $u_-(1) = f_-(1) < 1$, $\{0\}$ is a local maximum. By Lemma 6.4, $u_-(d_0) > 1/\mu_-$, so $\widetilde{\kappa}_+(d_0) = \frac{1}{2}(1 + u_-(d_0))$ exceeds both $\widetilde{\kappa}_+(1) = \frac{1}{2}u_-(1)$ and $1/\mu_- = \lim_j \widetilde{\kappa}_+(j)$. So the largest value of $\widetilde{\kappa}_+$ on $j \geq 1$ is attained, and only at points $j \geq 2$, and its block is a second local maximum. Theorem 6.8 gives at least $2 + 1 + 1$ modes. For a fresh plus start Lemma 6.4 makes $(1, u_-(1), \dots)$ strictly decreasing. □

To go further we need differences of the law itself. Theorem 6.1(b) gives $R_N(j) = u_-(j) + C(j)/N + o(j/N)$, but near $j \asymp b_{-,N}$ both $N\Delta u_-(j)$ and $\Delta C(j)$ are of order one, so the error $o(j/N)$ says nothing about $\Delta R_N$. We take the difference inside the exact identities, and (LD) controls the one term that is a genuine difference of a convolution power.

**Lemma 6.10** (plus runs at the level of differences)**.** *Assume (RV), (LR) and (LD) for the plus sign. Write $f = f_+$, $T = T_+$, $\mu = \mu_+$, $V_m = f^{*m}$ with $V_0 = \delta_0$, $\delta V_m = V_{m+1} - V_m$ and $q_m = q_m^+$, and fix $\eta \in (0, 1)$. As $n \to \infty$, uniformly in $0 \leq m \leq n^{1-\eta}$, the following hold.*

*(i)* $q_{m+1}(n) \leq CT(n)$.

*(ii)* $|V_m(n-1) - V_m(n)| \leq Cmf(n)/n$.

*(iii)* $\delta V_m(n) = f(n)(1 + o(1))$.

*(iv)* $q_{m+2}(n) - q_{m+1}(n) = 2\mu f(n)(1 + o(1))$.

*Proof.* We use Lemma 6.2 at arguments of order $n$, with $\eta/2$ in place of $\eta$, and (6.6). So $V_m(x) \leq Cmf(n)$ for $m \geq 1$ and $x \geq n/2$, and $mT(n) \to 0$, $mf(n) \leq C/n$ uniformly. Below we also apply (ii) and (iii) at arguments $n' \geq n/2$. For large $n$ we have $n^{1-\eta} \leq (n/2)^{1-\eta/2} \leq (n')^{1-\eta/2}$, so there we use them with $\eta/2$ in place of $\eta$.

(i) In $q_{m+1}(n) = \sum_{x<n} V_m(x)T(n-x)$ the part $x \leq n/2$ is at most $T(n/2) \leq CT(n)$, and the part $x > n/2$ is at most $Cmf(n)\mu \leq CT(n)$.

(ii) Let $m \geq 1$, fix $K_0$ with $(K_0/4 - 1)(\alpha_+ - 1)/2 > \alpha_+ + 3$, and put $w = \lceil n/K_0 \rceil$. On the event that no summand reaches $w$, Lemma 4.3 with $z = n/2 - 1$ bounds $\mathbb{P}(S_m^+ = n')$ for $n' \in \{n-1, n\}$ by $C(mT(w))^{K_0/4-1} \leq Cn^{-\alpha_+-3} = o(f(n)/n)$, since $mT(w) \leq Cn^{-(\alpha_+-1)/2}$ by Lemma 2.1(ii). Otherwise let $I$ be the first index with $X_I^+ \geq w$, and let $h_i(z)$ be the

probability that $X_l^+ < w$ for $l < i$ and that the sum of the $X_l^+$ with $l \neq i$ equals $z$. Then $\mathbb{P}(S_m^+ = n', I = i) = \sum_{y \geq w} f(y) h_i(n' - y)$, and the difference at $n-1$ and $n$ is

$$\sum_{z \leq n-1-w} h_i(z)\big[f(n-1-z) - f(n-z)\big] - h_i(n-w) f(w).$$

In the first sum $n - 1 - z \geq n/K_0$, so (LD) and (6.6) bound the bracket by $Cf(n)/n$, and $\sum_z h_i(z) \leq 1$. In the second term $h_i(n-w) \leq V_{m-1}(n-w) \leq Cmf(n)$, and $mf(n)f(w) \leq Cf(n)/n$. Summing over $i \leq m$ gives (ii).

(iii) For $m = 0$ this is $\delta V_0(n) = f(n)$. Let $m \geq 1$. Then $\delta V_m(n) = \sum_{x \geq 1} f(x)[V_m(n-x) - V_m(n)]$. For $x \leq n/2$, (ii) at the arguments $n-x, \dots, n$ bounds the bracket by $Cmxf(n)/n$, so this part is at most $C\mu m f(n)/n = o(f(n))$. The part $x > n/2$ equals $\sum_{1 \leq y < n/2} f(n-y) V_m(y) - V_m(n) \sum_{x > n/2} f(x)$. The second term is at most $Cmf(n)T(n/2) = o(f(n))$. In the first, $f(n-y)/f(n)$ is bounded and tends to one uniformly in $y \leq \varepsilon n$, while $\mathbb{P}(S_m^+ < n/2) \to 1$ and $\mathbb{P}(S_m^+ > \varepsilon n) \leq m\mu/(\varepsilon n) \to 0$. So it is $f(n)(1 + o(1))$.

(iv) Putting one more run in front gives $q_{m+2}(n) = \sum_{x \geq 1} f(x) q_{m+1}(n-x)$, with $q_{m+1} = 0$ at nonpositive arguments. Comparing the events in (2.7) gives $q_{m+1}(l-1) - q_{m+1}(l) = \delta V_m(l-1)$ for $l \geq 1$. So $q_{m+2}(n) - q_{m+1}(n) = \sum_x f(x)[q_{m+1}(n-x) - q_{m+1}(n)]$. For $x \leq n/2$ the bracket is $\sum_{i=1}^{x} \delta V_m(n-i)$, which is at most $Cxf(n)$ and is $xf(n)(1 + o(1))$ for fixed $x$, by (iii). Since $\sum_x x f(x) = \mu$, this part is $\mu f(n)(1 + o(1))$. The part $x > n/2$ equals $\sum_{1 \leq y < n/2} f(n-y) q_{m+1}(y) - q_{m+1}(n) \sum_{x > n/2} f(x)$. The second term is $O(T(n)^2) = o(f(n))$ by (i). In the first, $\sum_y q_{m+1}(y) = \mathbb{E} X_{m+1}^+ = \mu$, and $\sum_{y \geq \varepsilon n} q_{m+1}(y) \leq \mathbb{E}[X_+; X_+ \geq \varepsilon n/2] + \mu \mathbb{P}(S_m^+ \geq \varepsilon n/2) \to 0$. So it is $\mu f(n)(1 + o(1))$. □

**Theorem 6.11** (first differences in the layer)**.** *Assume (RV) and (SPAN) for the minus sign and (LR) and (LD) for the plus sign, and fix $\eta \in (0,1)$. Let the start be fresh, with sign $+$ with probability $w_+$ and sign $-$ with probability $w_- = 1 - w_+$. Put $R_N(j) = P_N(N-j)/T_+(N)$ and $C_w(j) = C(j) - 2\alpha_+ \mu_+ w_- u_-(j)$, with $C$ as in* (6.3)*. Then, uniformly in $1 \leq j \leq N^{1-\eta}$,*

$$N\,\Delta R_N(j) = N\,\Delta u_-(j) + \Delta C_w(j) + o(1),$$

*and $\Delta C_w(j) \to c_{\mathrm{a}} = 2\alpha_+ s/\mu_-^2$ as $j \to \infty$.*

*Proof.* By linearity it suffices to treat the two fresh starts. Write $u = u_-$, $v = v_-$, $a_m = u_m^-$, $V_m = u_m^+$ and $n = N - j$. By Proposition 2.3 and its minus form, $P_N(N-j) = \sum_{m \geq 1} [a_m(j) Q_m(n) + V_m(n) b_m(j)]$, with $Q_m = q_{m+1}^+$ and $b_m = q_m^-$ for a plus start, and $Q_m = q_m^+$ and $b_m = q_{m+1}^-$ for a minus start. Taking the difference, $D_N(j) = \mathrm{I} + \mathrm{II} + \mathrm{III} + \mathrm{IV}$ with

$$\mathrm{I} = \sum_m \Delta a_m(j)\, Q_m(n-1), \qquad \mathrm{II} = \sum_m a_m(j)\big[Q_m(n-1) - Q_m(n)\big],$$
$$\mathrm{III} = \sum_m V_m(n-1)\, \Delta b_m(j), \qquad \mathrm{IV} = \sum_m \big[V_m(n-1) - V_m(n)\big] b_m(j).$$

Only $m \leq j+1$ occurs, by (K2), so Lemma 6.10 applies at $n - 1 \geq N/2$ with $\eta/2$ in place of $\eta$. We use six facts on the minus side.

(1) $\sum_m a_m(j) = u(j)$ and $\sum_m m a_m(j) = v(j)$.

(2) The sum $\bar{A}_i(j) = \sum_{m > i} a_m(j)$ equals $\sum_{x < j} u_i^-(x) u(j-x)$. Hence

$$\Delta \bar{A}_i(j) = \sum_{x < j} u_i^-(x)\, d_{j+1-x} + u_i^-(j)\, u(1),$$

and $\sum_{i \geq 1} |\Delta \bar{A}_i(j)| \leq 1 + \|d\|_1$, which is finite by Lemma 6.5.

(3) $(j+1)|\Delta u(j)| \le C$, by Lemma 6.5 and $jT_-(j) \to 0$.

(4) By (6.5), $v + u = u * u$, so $\Delta(v+u)(j) = \sum_{i \le j+1} d_i u(j+1-i)$ is the coefficient of $z^{j+1}$ in $\hat{u}\hat{d}$, and $|\Delta v(j)| \le 1 + \|d\|_1$.

(5) $\bar{B}_i(j) = \sum_{m \ge i} b_m(j)$ equals $\mathbb{P}(S_i^- < j)$ for a plus start and $\mathbb{P}(S_{i+1}^- < j)$ for a minus start. So $\Delta\bar{B}_i(j)$ is $u_i^-(j)$, respectively $u_{i+1}^-(j)$, and $\sum_m \Delta b_m(j)$ is $0$, respectively $f_-(j)$.

(6) $\sum_m m b_m(j) \le j$, since the index of the run that covers level $j$ is at most $j$.

By Lemma 6.10 and (6.6), every $\delta V_i(n-1)$ below is $f_+(N)(1+o(1))$ and every $Q_{i+1}(n-1) - Q_i(n-1)$ is $2\mu_+ f_+(N)(1+o(1))$, uniformly in $i \le j+1$. Also $T_+(N-j-1) = T_+(N) + (j+1)f_+(N)(1+o(1))$.

II. Here $Q_m(n-1) - Q_m(n)$ is $\delta V_m(n-1)$, respectively $\delta V_{m-1}(n-1)$, so $\mathrm{II} = f_+(N)u(j) + o(f_+(N))$ by (1).

III. Write $V_m = V_1 + \sum_{i<m} \delta V_i$ and sum by parts. For a plus start $\mathrm{III} = \sum_{i \ge 1} \delta V_i(n-1)u_i^-(j) = f_+(N)u(j) + o(f_+(N))$ by (5). For a minus start $\mathrm{III} = V_1(n-1)f_-(j) + \sum_{i \ge 1} \delta V_i(n-1)u_{i+1}^-(j) = f_+(N)[f_-(j) + u(j) - u_1^-(j)] + o(f_+(N))$, which is the same since $u_1^- = f_-$.

IV. By Lemma 6.10(ii) and (6), $|\mathrm{IV}| \le C f_+(N) j/N = o(f_+(N))$.

I. Write $Q_m = Q_1 + \sum_{i<m}(Q_{i+1} - Q_i)$ and sum by parts. Then $\mathrm{I} = Q_1(n-1)\Delta u(j) + \sum_{i \ge 1}(Q_{i+1} - Q_i)(n-1)\,\Delta\bar{A}_i(j)$. Since $\sum_{i \ge 1} \bar{A}_i(j) = \sum_m (m-1)a_m(j) = v(j) - u(j)$, fact (2) makes the sum $2\mu_+ f_+(N)(\Delta v - \Delta u)(j) + o(f_+(N))$. For a plus start $Q_1(n-1) = q_2^+(N-j-1) = T_+(N) + (j+1+2\mu_+)f_+(N) + o((j+1)f_+(N))$ by Lemma 6.10(iv) with $m=0$, and for a minus start $Q_1(n-1) = T_+(N-j-1) = T_+(N) + (j+1)f_+(N) + o((j+1)f_+(N))$. By (3) the errors times $\Delta u(j)$ are $o(f_+(N))$.

Adding the four parts,

$$D_N(j) = T_+(N)\Delta u(j) + f_+(N)\big[(j+1)\Delta u + 2\mu_+\Delta v + 2u - 2\mu_+ w_-\Delta u\big](j) + o(f_+(N)),$$

with $w_- = 0$ for a plus start and $w_- = 1$ for a minus start. The bracket is bounded by (3) and (4), and $N f_+(N)/T_+(N) \to \alpha_+$ by (LR). Since $\Delta C(j) = \alpha_+[(j+1)\Delta u + 2\mu_+\Delta v + 2u](j)$ by (6.3), multiplying by $N/T_+(N)$ gives the claim. For the limit, $(j+1)\Delta u(j) \to 0$, $u(j) \to 1/\mu_-$, and $\Delta(v+u)(j) \to \hat{d}(1)/\mu_- = 1/\mu_-^2$ by (4) and dominated convergence, so $\Delta C_w(j) \to \alpha_+(2\mu_+/\mu_-^2 + 2/\mu_-) = c_{\mathrm{a}}$. □

**Theorem 6.12** (the extrema up to distance $N^{1-\eta}$)**.** *Assume (RV), (SPAN), (LR) and (LD) for both signs, and let the start be fresh with either sign or the fair start. Let $J_+ \ge 1$ be such that $u_-$ is strictly decreasing on $[J_+, \infty)$, and assume $\widetilde{\kappa}_+(j+1) \ne \widetilde{\kappa}_+(j)$ for $0 \le j < J_+$. Put $\Lambda_N = a_-(N)$ and $x_* = (2\alpha_+ s)^{-1/\alpha_-}$, and fix $\eta \in (0, 1 - 1/\alpha_*)$ and $\eta' \in (0, x_*)$. For all large $N$ the following hold.*

*(a) $D_N(j)$ has the sign of $\widetilde{\kappa}_+(j+1) - \widetilde{\kappa}_+(j)$ for $0 \le j < J_+$.*

*(b) $D_N(j) < 0$ for $J_+ \le j \le (x_* - \eta')\Lambda_N$.*

*(c) $D_N(j) > 0$ for $(x_* + \eta')\Lambda_N \le j \le N^{1-\eta}$.*

*(d) On $(x_* - \eta')\Lambda_N < j < (x_* + \eta')\Lambda_N$ the sign of $D_N(j)$ changes an odd number of times, after the zeros of $D_N$ are deleted.*

*The closed window $[(x_* - \eta')\Lambda_N, (x_* + \eta')\Lambda_N]$ contains every minimizer of $P_N(N-j)$ over $J_+ \le j \le N^{1-\eta}$. For a fresh plus start with $u_-(j) > 1/\mu_-$ for all $j \ge 1$ it contains the minimizer $j_N^*$ of Theorem F. The same holds at the minus end with the signs exchanged.*

*Proof.* (a) and (b) on $J_+ \le j < J'$, for any fixed $J'$, follow from Proposition 6.7(a), since $\widetilde{\kappa}_+ = u_-$ on $j \ge 1$ for these starts. By Theorem 6.11 and Lemma 6.5,

$$\frac{N\,D_N(j)}{T_+(N)} = -\frac{NT_-(j+2)}{\mu_-^2}(1+\epsilon_j) + \Delta C_w(j) + o(1)$$

uniformly in $1 \le j \le N^{1-\eta}$, with $\epsilon_j \to 0$ as $j \to \infty$. Here $NT_-(x\Lambda_N) \to x^{-\alpha_-}$ for $x > 0$, and $x_*^{-\alpha_-} = c_{\mathrm{a}}\mu_-^2$. For (b) on $J' \le j \le (x_* - \eta')\Lambda_N$, the first term is at most $-(1-\epsilon)((x_*-\eta')^{-\alpha_-} + o(1))/\mu_-^2$ and $\Delta C_w(j) \le c_{\mathrm{a}} + \epsilon$ once $J'$ is large, so the right side is negative for $\epsilon$ small. For (c), the first term is at least $-(1+\epsilon)((x_*+\eta')^{-\alpha_-} + o(1))/\mu_-^2$ and $\Delta C_w(j) \ge c_{\mathrm{a}} - \epsilon$, so it is positive. The range of (c) is not empty, since $\Lambda_N \in \mathrm{RV}_{1/\alpha_-}$ and $1/\alpha_- \le 1/\alpha_* < 1 - \eta$.

For (d) and the minimizers, (b) and (c) at $\eta'$ are not enough, since they leave $D_N(j)$ open for the integers $j$ in the open window nearest to its ends. We apply them with $\eta'/2$ in place of $\eta'$. Put $L = (x_* - \eta')\Lambda_N$ and $U = (x_* + \eta')\Lambda_N$, and let $j_L = \lfloor L \rfloor + 1$ and $j_U = \lceil U \rceil - 1$ be the first and last integers in $(L, U)$. Since $\eta'\Lambda_N/2 \to \infty$, for large $N$ we have $j_L + 1 \le (x_* - \eta'/2)\Lambda_N$ and $j_U - 1 \ge (x_* + \eta'/2)\Lambda_N$. So $D_N(j_L) < 0$ and $D_N(j_U) > 0$, and the sign of $D_N$ changes an odd number of times on $[j_L, j_U]$, which is (d). Write $g(j) = P_N(N - j)$, so $D_N(j) = g(j+1) - g(j)$. By (b) at $\eta'/2$, $g$ decreases strictly on $J_+ \le j \le j_L + 1$, so no minimizer is below $j_L + 1$. By (c) at $\eta'/2$, $g$ increases strictly on $j_U - 1 \le j \le N^{1-\eta}$, so no minimizer is above $j_U - 1$. Hence every minimizer lies in the open window, and so in the closed one. For the fresh plus start, $j_N^*/\Lambda_N \to x_*$ by Theorem F(a) and its proof. At the minus end the same argument uses (LR) and (LD) for the minus sign. □

So within distance $N^{1-\eta}$ of the plus end the local extrema of $P_N$ are the copies of those of $\widetilde{\kappa}_+$ and, inside the window of (d), alternating minima and maxima with one more minimum than maximum, plateaus counted once.

### 6.5 The exact number of modes

Fresh and fair starts satisfy the hypotheses of Theorem 6.8, which gives at least $M_+ + M_- + 1$ modes. Here we prove the matching upper bound. Three steps are needed beyond Theorem 6.12. The window of Theorem 6.12(d) must hold a single minimum. The law must be monotone on the big-jump zone, where the number of runs is of order $N$ and Lemma 6.2 no longer applies. Finally $\Delta P_N$ must change sign only once in the bulk. Each step needs one more order of smoothness, which is (LD2). The last two also need the two scales to be comparable, which is (EQ). We keep $\Lambda_N = a_-(N)$ and $x_* = (2\alpha_+ s)^{-1/\alpha_-}$ from Theorem 6.12, and we write $x_\theta$ for the mode of the profile $h_\theta$, which is unique by Lemma B.12.

**Theorem G** (the exact number of modes)**.** *Assume (SPAN), (LR) and (LD2) for both signs, and (EQ). Let the start be fresh with either sign or the fair start, and assume the no-tie condition of Theorem 6.12 at both ends. Fix $\eta \in (0, 1 - 1/\alpha_*)$ and $\eta' \in (0, x_*)$. There is $K_0$, which depends only on the laws of the runs and on $\eta$ and $\eta'$, such that for all large $N$ the following hold.*

- *(i) $D_N(j)$ has the sign of $\widetilde{\kappa}_+(j+1) - \widetilde{\kappa}_+(j)$ for $0 \le j < J_+$.*
- *(ii) $D_N(j) < 0$ for $J_+ \le j \le (x_* - \eta')\Lambda_N$.*
- *(iii) $D_N$ is strictly increasing on $(x_* - \eta')\Lambda_N < j < (x_* + \eta')\Lambda_N$. So its sign changes exactly once there, and it vanishes at most once.*
- *(iv) $D_N(j) > 0$ for $(x_* + \eta')\Lambda_N \le j \le (1-p)N - K_0 B_N$.*

*(v) On $|x_{N,k}| \le K_0 + 1$ the sign of $\Delta P_N(k) = P_N(k+1) - P_N(k)$ changes exactly once, from positive to negative, at a point $k_N$ with $x_{N,k_N} - x_{\theta_N} \to 0$, and $\Delta P_N$ is strictly decreasing near $k_N$.*

*The same holds at the minus end with the signs exchanged. Consequently $P_N$ has exactly $M_+ + M_- + 1$ modes for all large $N$. There are $M_+$ of them within distance $J_+$ of $N$, one in the bulk at $pN + (x_{\theta_N} + o(1))B_N$, and $M_-$ within distance $J_-$ of $0$. Between the bulk mode and the modes near each end lies exactly one local minimum, the antimode, in the window of Theorem 6.12(d) or its mirror.*

Parts (i), (ii) and the existence of a change of sign in (iii) are Theorem 6.12. Appendix B proves the rest. The strict increase in (iii) comes from a second-order form of Theorem 6.11 (Theorem B.4). Part (iv) comes from a difference form of Theorem B(b) (Theorem B.8), which rests on Lemma 6.18 below. Part (v) comes from a local limit theorem for the first and second differences of $P_N$ in the bulk (Theorem B.13) together with the bell shape of stable densities. Parts (i) to (iii) do not use (EQ).

*Proof of the count.* We take parts (i) to (v) and their mirrors as given. Section B.7 checks that their ranges overlap and cover $0 \le j < N$. Write $g(j) = P_N(N-j)$, so $D_N(j) = g(j+1) - g(j)$, and let $S$ be the sequence of the nonzero signs among $D_N(0), \dots, D_N(N-1)$. The modes of $P_N$ are the local maxima of $g$. Each local maximum corresponds to exactly one of the following, an adjacent pair $(+,-)$ in $S$, a first sign $-$, or a last sign $+$. Let $\sigma_+$ be the sequence of signs of $\Delta\widetilde{\kappa}_+$ on $[0, J_+)$, all nonzero by the no-tie condition, and define $\sigma_-$ at the minus end in the same way. At the minus end $D_N(j) = -(P_N(i+1) - P_N(i))$ with $i = N - j - 1$, and $i$ decreases as $j$ increases. So (i) to (v) and their mirrors give

$$S = \sigma_+\,(-)^a\,(+)^b\,(-)^c\,(+)^d\,\bar{\sigma}_-, \qquad a, b, c, d \ge 1,$$

where $\bar{\sigma}_-$ is $\sigma_-$ read backwards with every sign reversed. The two changes from $-$ to $+$ are the antimodes and the change from $+$ to $-$ is the bulk mode. Since $\widetilde{\kappa}_+ = u_-$ on $j \ge 1$ is strictly decreasing on $[J_+, \infty)$, the full sign sequence of $\Delta\widetilde{\kappa}_+$ is $\sigma_+$ followed by minus signs. So a first sign $-$ and the pairs $(+,-)$ in $\sigma_+(-)$ count the local maxima of $\widetilde{\kappa}_+$, which are $M_+$. The middle block $(-)^a(+)^b(-)^c(+)^d$ holds one pair $(+,-)$. Reading a sequence backwards and reversing its signs keeps the pairs $(+,-)$ and exchanges a first sign $-$ with a last sign $+$. So the pairs in $(+)\bar{\sigma}_-$ and a last sign $+$ count the local maxima of $\widetilde{\kappa}_-$, which are $M_-$. The total is $M_+ + 1 + M_-$. □

**Corollary 6.13** (three modes)**.** *Under the hypotheses of Theorem G, $P_N$ has exactly three modes for all large $N$ if and only if each end profile has exactly one local maximum.*

*Proof.* By Theorem G the number of modes is $M_+ + M_- + 1$ for all large $N$, and $M_+, M_- \ge 1$. □

The "only if" half needs only Theorem 6.8. The next lemma says when an end profile of a fresh or fair start has one local maximum.

**Lemma 6.14** (one local maximum)**.** *Assume (RV) and (SPAN) for the minus sign, let $c_0 \in [0,1]$, and let $y = (c_0, u_-(1), u_-(2), \dots)$.*

*(a) If $u_-$ is nonincreasing on $j \ge 1$, then $y$ has exactly one local maximum.*

*(b) If $c_0 > u_-(1)$, then $y$ has exactly one local maximum if and only if $u_-$ is nonincreasing on $j \ge 1$.*

*(c) If $c_0 < u_-(1)$, then the local maxima of $y$ are those of the sequence $(u_-(j))_{j \ge 1}$.*

*Proof.* Write $u = u_-$. By Lemma 6.5 $u$ is eventually strictly decreasing, with the positive limit $1/\mu_-$, so every maximal block of equal values of $y$ is finite. (a) A block $\{a, \dots, b\}$ with $a \geq 2$ is not a local maximum, since $y_{a-1} \geq y_a$. If $c_0 > u(1)$, the block $\{0\}$ is the only local maximum. Otherwise the block that contains $1$ is a local maximum, since it is followed by a strict decrease, and it is the only one. (b) The "if" part is (a). Let $u(i+1) > u(i)$ for some $i \geq 1$. The values of $u$ on $[i+1, \infty)$ lie above $1/\mu_-$ from some point on and tend to it, so their maximum is attained. The maximal block of a maximizer starts after $i$, since $u(i) < u(i+1)$, and it is a local maximum of $y$. So is $\{0\}$, since $c_0 > u(1)$. (c) Since $c_0 < u(1)$, no local maximum contains $0$, and a block that starts at $1$ meets the condition at its left end. □

For a fresh start of sign $\sigma_0$ the profile at the end of sign $\sigma_0$ has $c_0 = 1 > u(1)$, so (b) applies, and this is the criterion stated after Theorem 6.8. Part (b) makes monotonicity necessary at every end whose profile starts above $u(1)$. This includes the fair start, with $c_0 = \frac{1}{2}$, when $f(1) < \frac{1}{2}$. The profile at the other end of a fresh start has $c_0 = 0$, and (c) applies when $f(1) > 0$. If $f(1) = 0$ the profile starts with the tie $0 = u(1)$, and neither (b) nor (c) applies. Part (c) also applies to the fair start when $f(1) > \frac{1}{2}$. In case (c) monotonicity of $u$ is sufficient by (a). It is not necessary in the example with $f(1) = 0.6$ below, but there $u$ is checked only on a finite range. For the fair start and the zeta law of Section 9, $\widetilde{\kappa}_+ = (\frac{1}{2}, u_-(1), u_-(2), \dots)$ with $u_-(1) = 1/\zeta(5/2) = 0.745\dots$. The profile is not monotone and has one local maximum, and Corollary 6.15 below gives exactly three modes for all large $N$. Now let $f(1) = 0.6$, $f(2) = 0.3$ and $f(n) = cn^{-5/2}$ for $n \geq 3$ with $\sum_{n \geq 3} f(n) = 0.1$, for both signs. This law satisfies (SPAN), (LR), (LD2) and (EQ). Here $u(1) = 0.6 < u(2) = 0.66$, so $u$ is not monotone, while the computed $u$ decreases strictly on $2 \leq j \leq 10^4$. The computed law of the fair start has exactly three modes, at $k = 2$, $N/2$ and $N - 2$, for $N = 1000$, $1500$ and $2000$. For the fresh plus start the profile $(1, 0.6, 0.66, \dots)$ has at least two local maxima by (b), so $P_N$ has at least four modes for all large $N$ by Theorem 6.8, and the computed law has four.

**Corollary 6.15** (log-convex run laws)**.** *Assume (SPAN), (LR) and (LD2) for both signs and (EQ), and let $f_+$ and $f_-$ be strictly log-convex as in Lemma 6.4. For a fresh start of either sign, and for the fair start when $f_+(1) \neq \frac{1}{2}$ and $f_-(1) \neq \frac{1}{2}$, $P_N$ has exactly three modes for all large $N$.*

*Proof.* By Lemma 6.4, $u_-$ is strictly decreasing, so $J_+ = 1$ is admissible. The no-tie condition asks $u_-(1) = f_-(1) \neq \widetilde{\kappa}_+(0)$. This holds since $0 < f_-(1) < 1$ and $\widetilde{\kappa}_+(0)$ is $1$, $0$ or $\frac{1}{2}$. By Lemma 6.14(a) the profile has one local maximum. The minus end is the same, and Corollary 6.13 concludes. □

The zeta law $f(n) = n^{-\alpha-1}/\zeta(\alpha+1)$ with the same $\alpha$ for both signs satisfies these hypotheses, and $f(1) = 1/\zeta(\alpha+1) > \frac{1}{2}$ for $\alpha > 1$.

(EQ) enters the proof only at the scale $N$. So the theorem also holds along any sequence of times at which the two scales are comparable.

**Corollary 6.16** (comparable scales along a sequence)**.** *Assume the hypotheses of Theorem G with (EQ) replaced by $\alpha_+ = \alpha_-$, and fix $\theta_0 > 0$. There is $N_0$ such that for every $N \geq N_0$ with $\min_\sigma \theta_{\sigma,N} \geq \theta_0$ parts (i) to (v) hold and $P_N$ has exactly $M_+ + M_- + 1$ modes.*

Under (EQ) every large $N$ qualifies for some $\theta_0$ (Section B.7). The corollary also covers equal indices with a ratio $T_+/T_-$ that oscillates between $0$ and $\infty$, at the times when the two scales are comparable.

In continuous time the propagator of a Lévy walk is pictured as a central peak between two side peaks at the fronts [78]. Corollary 6.13 makes this picture exact on the lattice and gives its criterion. It also gives a family of laws where the picture fails.

**Example 6.17** (runs of length at least two)**.** If $f_+(1) = f_-(1) = 0$, then for the fair start $\widetilde{\kappa}_\pm = (\frac{1}{2}, 0, u_\mp(2), \dots)$, and Lemma 6.14(b) gives at least two local maxima at each end. So $P_N$ has at least five modes for all large $N$, by Theorem 6.8. Now let $f_\sigma(n) = n^{-5/2}/(\zeta(5/2)-1)$ for $n \ge 2$ for both signs. This law satisfies (SPAN), (LR), (LD2) and (EQ). A finite computation gives $u(2i) > \max\{u(2i-1), u(2i+1)\}$ for $1 \le i \le 6$, so each profile has at least seven local maxima, at $j = 0, 2, \dots, 12$, and $P_N$ has at least $15$ modes for all large $N$. The computed $u$ decreases strictly on $12 \le j \le 10^4$. If it decreases strictly on $[12, \infty)$, the no-tie condition holds with $J_\pm = 12$, each profile has exactly seven local maxima, and Theorem G gives exactly $15$ modes for all large $N$. The computed law has exactly $15$ modes for every $N = 2500, 2600, \dots, 4000$, and $17$, $19$ and $17$ modes at $N = 1000$, $1500$ and $2000$. The number $15$ belongs to this law and this start. For $f_\sigma(n) = n^{-12/5}/(\zeta(12/5)-1)$, $n \ge 2$, of index $7/5$, the computed $u$ has peaks at $j = 2, 4, \dots, 10$ only, so the proved lower bound is $13$, and the computed law has $13$ modes at $N = 3000$ and $4000$. For the fresh plus start with the law of index $3/2$ above, the minus-end profile $(0, 0, u(2), \dots)$ has a tie at $j = 0$ and six local maxima, so the lower bound is $14$.

The main new tool of the proof is a difference form of the local big-jump theorem for sums of $m$ runs, uniform in $m$. We state it here because it may be of use elsewhere.

**Lemma 6.18** (difference form of the local big jump)**.** *Assume (SPAN), (LR) and (LD2) for the plus sign. Write $f = f_+$, $T = T_+$, $\mu = \mu_+$, $\alpha = \alpha_+$, $a = a_+$, $V_m = f^{*m}$, $\nabla V_m(n) = V_m(n-1) - V_m(n)$, $\delta V_m = V_{m+1} - V_m$ and $q_m = q_m^+$, and read $f(x)$ as $f(\lfloor x \rfloor)$ for real $x \ge 1$. There is $C$ such that for every $\varepsilon > 0$ there is $K$ with the following property. For all $m \ge 1$ and all integers $n$ with $L = n - m\mu \ge Ka(m)$,*

$$\Big|\nabla V_m(n) - (\alpha+1)\, m\, \frac{f(L)}{L}\Big| \le \varepsilon\, m\, \frac{f(L)}{L}, \qquad V_m(n) \le C\, m\, f(L),$$
$$\Big|\delta V_m(n) - f(L) - \mu(\alpha+1)\, m\, \frac{f(L)}{L}\Big| \le \varepsilon\Big(f(L) + m\frac{f(L)}{L}\Big),$$
$$\Big|q_{m+2}(n) - q_{m+1}(n) - 2\mu f(L) - \mu^2(\alpha+1)\, m\, \frac{f(L)}{L}\Big| \le \varepsilon\Big(f(L) + m\frac{f(L)}{L}\Big),$$

*and $q_{m+1}(n) \le C(T(L) + mf(L))$. The bound for $q_{m+2} - q_{m+1}$ also holds for $m = 0$ and $n \ge K$, with $L = n$.*

The first bound is the derivative of the local big-jump formula $f^{*m}(n) \approx mf(n - m\mu)$. For a fixed $m$ it follows from the asymptotic form $f(n) - f(n+1) \sim (\alpha+1)f(n)/n$ of Lemma B.1. The point is the uniformity in all $m \ge 1$ with $L \ge Ka(m)$, with no upper bound on $m$, since the big-jump zone needs $m$ of order $N$. The local big-jump theorems for sums, due to Doney [20] for lattice walks with a regularly varying mass function, to Denisov, Dieker and Shneer [16, Theorem 9.1] and to Berger [6, Theorem 2.4], give the values $f^{*m}(n)$ up to a factor $1 + o(1)$. Such a factor says nothing about the difference of two neighbouring values, which is smaller by a factor of order $1/L$. We take the difference inside the decomposition by the first giant summand, where it falls on a single summand, and we bound the sums without a giant summand by a Fourier estimate for truncated sums. For a fixed number of summands the difference form is within reach of the asymptotic expansions of Barbe and McCormick [3] for convolutions of laws with regularly varying tails. What is new is the uniformity in $m \ge 1$ on $L \ge Ka(m)$, in the lattice setting.

**Remark 6.19** (what remains open)**.** Two questions remain. The first is unequal scales. If $\theta_{\sigma,N} \to 0$ for one sign, the kernels of the smaller scale have differences much larger than $B_N^{-1-r}$, and our proof of (v) fails. The window $KB_N \le k - pN \le K\lambda_N B_N$ of Theorem B is then open already for the values of $P_N$ (Theorem 4.14). Parts (i) to (iii) hold without (EQ).

The second is a bound for every $N$. Theorem G is eventual, and its threshold depends on the laws. For the law of Example 6.17 and the fair start the computed count is $17$ or $19$ for $N \le 2000$ and is $15$ at every computed $N$ from $2500$ to $4000$, once the antimode at about $x_*\Lambda_N$ has moved past the oscillations of $u_-$. We know no bound on the number of modes that holds for every $N$.

## 7 Exposure in experiments that switch at regenerations

A switchback experiment assigns one of two treatments, the arms, to a whole system for a period of time, then the other, and alternates. The quantity under the experimenter's control is the exposure, the number of time slots spent in the treatment arm. The designs in the literature switch at fixed times or at the hits of a recurrent state [8, 35, 30, 45, 38, 34], and each of them assumes that the carryover from one period to the next decays geometrically, through a Markov state on a finite set, a bounded queue, or a fixed memory. Then the regeneration cycles have exponential moments and the exposure count is Gaussian at scale $\sqrt{N}$. When the system is a queue with heavy-tailed service the cycles have infinite variance, and none of these results applies. This section works out the exposure law for a switchback that switches arm at the regenerations of a slotted queue, where the count is exactly the occupation count $A_N$ of this paper and the cycles are the runs. The results are the theorems of Sections 3 to 6 read in the language of the design, once the cycle law is shown to satisfy the hypotheses. The second step is the content of Proposition 7.1. Its first part, the local tail of the cycle, is known. The difference bounds (LD) and (LD2) for the cycle are new, and they are what the mode count needs.

### 7.1 The queue, the design and the reduction

Time is slotted. In each slot a customer arrives with probability $\lambda \in (0,1)$, a constant unrelated to the factor $\lambda_N$ of (4.5), and brings $Y$ slots of work, where the $Y$ are independent with values in $\{1,2,\dots\}$. So the work $W$ that arrives in a slot has $\mathbb{P}(W=0) = 1-\lambda$ and $\mathbb{P}(W=n) = \lambda f_Y(n)$ for $n \ge 1$, and its mean is $\rho = \lambda \mathbb{E}Y$, which we assume to be less than $1$. The server does one unit of work per slot, so the workload at the end of slot $t$ is $V_t = \max(V_{t-1} - 1, 0) + W_t$. A regeneration is a slot that ends with $V_t = 0$, and the cycle $C$ is the number of slots between consecutive regenerations. An idle slot is a cycle of length $1$, so $\mathbb{P}(C=1) = 1-\lambda$. Since the fraction of busy slots is $\rho$ and the regenerations are the idle slots, $\mathbb{E}C = 1/(1-\rho)$.

The 2 arms set their own arrival probabilities $\lambda_+$ and $\lambda_-$ and share the law of $Y$, so their loads $\rho_\pm = \lambda_\pm \mathbb{E}Y$ differ. The experiment switches arm at every regeneration. At a switch the system is empty, so the cycles are independent, each cycle has the cycle law of its own arm, and consecutive cycles belong to opposite arms. Hence the number of slots spent in the treatment arm among the first $N$ is the count $A_N$ of Section 2.1, with the plus runs the cycles under treatment and the minus runs the cycles under control. In this section $f_\pm$ and $T_\pm$ are the cycle laws of the two arms, $\mu_\pm = 1/(1-\rho_\pm)$, $s = \mu_+ + \mu_-$ and $p = \mu_+/s$ is the long-run treated share. Three starts occur. A fresh start of either arm is an experiment that starts on an empty system. A random-time start is an experiment that starts at a typical time of a long-running switchback, and its initial run is the stationary initial run $\nu_\sigma = T_\sigma/s$ of Section 5.6. An aged start is an experiment that starts $t_a$ slots after a switchback started on an empty system, and its law is the aged law of Section 5.8.

### 7.2 The cycle law

Let $U_n = W_1 + \cdots + W_n$. Starting from an empty system, $V_1 = W_1$ and $V_t = V_{t-1} - 1 + W_t$ as long as $V_{t-1} \geq 1$, so $V_t = U_t - t + 1$ up to the end of the cycle, and $C = \inf\{t \geq 1 : U_t - t = -1\}$ is the first passage of the downward skip-free walk $U_t - t$ to $-1$. The hitting time theorem of Kemperman [40] (see van der Hofstad and Keane [68] for a short proof) gives the cyclic lemma

$$\mathbb{P}(C = n) = \frac{1}{n}\,\mathbb{P}(U_n = n - 1) \qquad (n \geq 1). \tag{7.1}$$

Equivalently, $C$ is the total progeny of a Galton–Watson tree with offspring law $W$ [24]. Let $W' = W + 1$, a law on $\{1, 2, \dots\}$ with mass function $f_{W'}(n) = f_W(n-1)$ and mean $\mu' = 1 + \rho$, and let $V_m = f_{W'}^{*m}$. Then $U_n = U'_n - n$ with $U'_n$ a sum of $n$ copies of $W'$, so (7.1) reads

$$\varphi(n) := n\, f_C(n) = V_n(2n - 1), \tag{7.2}$$

a convolution power evaluated at the deviation $L = (2n-1) - n\mu' = (1-\rho)n - 1$ above its mean. This puts the cycle law inside the big-jump range of the lemmas of Section 6 and Appendix B, with the number of summands $m = n$ of the order of the deviation.

**Proposition 7.1** (the cycle law)**.** *Let $Y$ satisfy (LR) with index $\alpha \in (1,2)$, that is $f_Y(n) \sim \alpha T_Y(n)/n$ with $T_Y \in \mathrm{RV}_{-\alpha}$.*

*(a) $f_C(n) \sim f_W\big((1-\rho)n\big) \sim \lambda(1-\rho)^{-\alpha-1} f_Y(n)$ and $T_C(n) \sim \lambda(1-\rho)^{-\alpha-1} T_Y(n)$. So $C$ satisfies (LR) with the index $\alpha$, and (SPAN).*

*(b) If $Y$ also satisfies (LD2), then $C$ satisfies (LD) and (LD2), and $f_C(n+1) - f_C(n) = -(\alpha + 1 + o(1)) f_C(n)/n$.*

*(c) If 2 arms share $Y$ and have loads $\rho_\pm < 1$, their cycle laws satisfy (EQ), and $T_+(n)/T_-(n) \to (\lambda_+/\lambda_-)\big((1-\rho_-)/(1-\rho_+)\big)^{\alpha+1}$.*

Part (a) is known. For the total progeny of a Galton–Watson tree it is Kortchemski [42, Eq. (14)] and Janson [37, Remark 18.34], and for the busy period of the M/G/1 queue in continuous time it is Denisov and Shneer [17, Corollary 2.2], whose $f_\tau(t) \sim f_B((1-\rho)t)$ is the analogue of the first relation in (a). The global tail of the busy period, with the factor $\mathbb{E}\nu = 1/(1-\rho)$ for the mean number served, is due to De Meyer and Teugels [15] and Zwart [76]. We include the short proof of (a) because (b) continues it. We did not find (b) in the literature.

*Proof.* (a) The law $W'$ satisfies (LR) with index $\alpha$, since $f_{W'}(n) = \lambda f_Y(n-1)$ for $n \geq 2$, and (SPAN), since $f_{W'}(1) = 1 - \lambda > 0$ and $f_{W'}(n) > 0$ for all large $n$. The local big-jump theorem of Denisov, Dieker and Shneer [16, Theorem 9.1], whose hypotheses were verified for a law with (LR) in the proof of Lemma 6.2, gives for every $\varepsilon > 0$ an $r_0$ with $|\mathbb{P}(U'_r - r\mu' \in (x, x+1])/(r\mathbb{P}(x < W' - \mu' \leq x+1)) - 1| \leq \varepsilon$ for all $r \geq r_0$ and $x \geq x_r = (1-\rho)r/2$, because $rT_{W'}(x_r) \to 0$. Take $r = n$ and $x = 2n - 1 - n\mu' - \frac{1}{2}$. The event is $\{U'_n = 2n-1\}$ and $\mathbb{P}(x < W' - \mu' \leq x+1) = f_{W'}(n')$ with $n' = (1-\rho)n + O(1)$. By (6.6) for the law $W'$, $f_{W'}(n') = f_{W'}((1-\rho)n)(1+o(1))$, and $f_{W'}((1-\rho)n) = f_W((1-\rho)n - 1) \sim f_W((1-\rho)n)$ by the same lemma. Hence $\varphi(n) = n f_W((1-\rho)n)(1+o(1))$, which is the first relation, and the second is the regular variation of $f_W = \lambda f_Y$ on $n \geq 1$. The tail follows by Karamata's theorem, and (LR) for $C$ follows from (LR) for $Y$. Finally $f_C(1) = 1 - \lambda > 0$ and $f_C(n) > 0$ for large $n$, which gives (SPAN).

(b) The law $W'$ satisfies (LD2) because $Y$ does, and (LD) by Lemma B.1. Write $f = f_{W'}$, $a = a_{W'}$, and $L_x = x - n\mu'$ for the deviation of the point $x$ at $m = n$ summands,

so that $L_{2n-1} = L$ and $L_{2n+i} = L + i + 1$. For large $n$, $L \geq Ka(n)$ for any fixed $K$, since $a(n) = o(n)$. By (7.2),

$$\varphi(n+1) - \varphi(n) = V_{n+1}(2n+1) - V_n(2n-1) = \delta V_n(2n+1) - \nabla V_n(2n+1) - \nabla V_n(2n),$$

in the notation of Lemma 6.18. That lemma, applied to the law $W'$, gives, with $o(1)$ as $n \to \infty$,

$$\begin{aligned} \delta V_n(2n+1) &= f(L) + \mu'(\alpha+1)\frac{nf(L)}{L} + o\Big(f(L) + \frac{nf(L)}{L}\Big), \\ \nabla V_n(2n+i) &= (\alpha+1)\frac{nf(L)}{L}\big(1+o(1)\big) \qquad (i = 0, 1), \end{aligned}$$

where we used $f(L+i) = f(L)(1 + O(1/L))$ for $|i| \leq 2$, by (LD). Since $n/L \to 1/(1-\rho)$ and $\mu' - 2 = -(1-\rho)$,

$$\varphi(n+1) - \varphi(n) = f(L)\Big[1 + (\alpha+1)\frac{n}{L}(\mu'-2) + o(1)\Big] = -\alpha f(L)(1+o(1)).$$

By (a), $\varphi(n) = nf(L)(1+o(1))$. Hence

$$f_C(n+1) - f_C(n) = \frac{\varphi(n+1) - \varphi(n)}{n+1} - \frac{\varphi(n)}{n(n+1)} = -\frac{(\alpha + 1 + o(1))f(L)}{n},$$

and since $f_C(n) \sim f(L)$ by (a), this is (LD) for $C$ with the stated constant. For (LD2) we use the second-difference bounds of Lemma B.10 with $c = (1-\rho)/2$, which apply at $m = n$ and at the points $2n - 1 \leq x \leq 2n + 3$ for large $n$, since there $L \geq (1-\rho)n/2 \geq \max(K_1 a(n), L_0, cn)$. Put $D(n) = \varphi(n+1) - \varphi(n)$. Then

$$\begin{aligned} D(n+1) - D(n) &= \big[\delta V_{n+1}(2n+3) - \delta V_n(2n+1)\big] - \big[\nabla V_{n+1}(2n+3) - \nabla V_n(2n+1)\big] \\ &\quad - \big[\nabla V_{n+1}(2n+2) - \nabla V_n(2n)\big] \\ &= \delta^2 V_n(2n+3) - \nabla\delta V_n(2n+3) - \nabla\delta V_n(2n+2) \\ &\quad - \nabla\delta V_n(2n+3) + \nabla^2 V_n(2n+3) + \nabla^2 V_n(2n+2) \\ &\quad - \nabla\delta V_n(2n+2) + \nabla^2 V_n(2n+2) + \nabla^2 V_n(2n+1), \end{aligned}$$

where each bracket was split into a difference in $m$ at the larger point and a difference in $x$ at $m = n$, and $X(x+2) - X(x) = -\nabla X(x+2) - \nabla X(x+1)$. By Lemma B.10 each of the nine terms is at most $Cf(L)/L$ in absolute value, since $n \leq L/c$. So $|\varphi(n+2) - 2\varphi(n+1) + \varphi(n)| \leq Cf(L)/n$. With $g(n) = 1/n$ the product rule gives

$$\Delta^2(g\varphi)(n) = g(n+2)\Delta^2\varphi(n) + \Delta\varphi(n)\big[\Delta g(n) + \Delta g(n+1)\big] + \varphi(n+1)\Delta^2 g(n),$$

and the three terms are $O(f(L)/n^2)$, $O(f(L)) \cdot O(n^{-2})$ and $O(nf(L)) \cdot O(n^{-3})$. Since $f(L) \asymp f_C(n)$ by (a), this is (LD2) for $C$.

(c) By (a), $T_\pm(n) \sim \lambda_\pm(1-\rho_\pm)^{-\alpha-1} T_Y(n)$, and the ratio of the two constants is the stated limit. □

### 7.3 The exposure law

Throughout this subsection $Y$ satisfies (LR) with index $\alpha \in (1, 2)$, so both cycle laws satisfy (RV), (LR), (SPAN) and (EQ) by Proposition 7.1, and the exposure count $A_N$ is the count of this paper with these two run laws. By Proposition 7.1(c) and Lemma 2.1, the ratio $\theta_{\sigma,N} = b_{\sigma,N}/B_N$ of the two scales converges, to $\theta_\sigma = \min\{1, (\kappa_\sigma/\kappa_{\bar\sigma})^{1/\alpha}\}$ with $\kappa_\sigma = \lambda_\sigma(1-\rho_\sigma)^{-\alpha-1}$, so the moving profile $h_N$ converges uniformly to the fixed density $h_\theta$ of (2.5).

**Corollary 7.2** (the exposure imbalance)**.** *For a fresh start of either arm,*

$$\sup_{0\le k\le N}\Big|B_N\mathbb{P}(A_N=k)-h_\theta\Big(\frac{k-pN}{B_N}\Big)\Big|\to 0,\qquad \sup_{N,k}B_N\mathbb{P}(A_N=k)<\infty.$$

*For a random-time start the same holds over* $1\le k<N$*. It holds over* $0\le k\le N$ *when* $\alpha>\varphi$*, and when* $\alpha<\varphi$ *it fails at the endpoints, where* $B_N\mathbb{P}(A_N\in\{0,N\})\to\infty$*. In all cases* $(A_N-pN)/B_N$ *converges in law to* $W_\theta$*, so the treated share* $A_N/N$ *misses* $p$ *by a random error of exact order* $N^{1/\alpha-1}$ *with a skewed stable law, in addition to the deterministic imbalance* $(2p-1)N$ *of the counts when* $\rho_+\neq\rho_-$*.*

*Proof.* The fresh start is Theorem A(a) and (b) with $h_N\to h_\theta$. The random-time start satisfies the bulk criterion by Corollary 5.4, so Theorem C(b) gives the statement over $1\le k<N$. Over the whole range it holds if and only if $B_NQ_\sigma(N)\to 0$ for both arms, and $B_NQ_\sigma(N)\in\mathrm{RV}_{1/\alpha+1-\alpha}$ by Corollary 5.4(a), which tends to zero for $\alpha>\varphi$ and to infinity for $\alpha<\varphi$. The convergence in law follows from the local statement over $1\le k<N$ and $Q_\sigma(N)\to 0$. □

The limit in law is classical. For the sojourn time of an alternating renewal process it is due to Takács [65], and Godrèche and Luck [31] give it for one law. What the corollary adds is the local form at every $k$, uniformly, and the bound $C/B_N$ at every point, together with the exceptions at the endpoints.

**Corollary 7.3** (severe imbalance and no switch)**.** *Fix* $\varepsilon\in(0,1-p)$ *and put* $\ell_\varepsilon=1-\varepsilon/(1-p)$*.*

*(a) For a fresh start of either arm,*

$$\mathbb{P}(N-A_N<\varepsilon N)\sim NT_+(N)\,\frac{\varepsilon}{\mu_-}\,\ell_\varepsilon^{-\alpha},$$

*and for a random-time start*

$$\mathbb{P}(N-A_N<\varepsilon N)\sim NT_+(N)\Big[\frac{\varepsilon}{\mu_-}\,\ell_\varepsilon^{-\alpha}+\frac{\ell_\varepsilon^{1-\alpha}}{s(\alpha-1)}\Big].$$

*The mirror statements hold for* $\mathbb{P}(A_N<\varepsilon N)$ *with the signs exchanged.*

*(b) (No switch.) The event* $\{A_N\in\{0,N\}\}$ *is the event that the first cycle covers all* $N$ *slots. For a fresh start in the treatment arm its probability is* $T_+(N)$ *exactly. For a random-time start it is* $Q_+(N)+Q_-(N)$ *exactly, with* $Q_\sigma(N)=s^{-1}\sum_{d\ge N}T_\sigma(d)\sim NT_\sigma(N)/(s(\alpha-1))$*. Hence*

$$\frac{\mathbb{P}_{\mathrm{random}}(\textit{no switch})}{\mathbb{P}_{\mathrm{fresh}\,+}(\textit{no switch})}\sim\frac{N}{s(\alpha-1)}\Big(1+\frac{T_-(N)}{T_+(N)}\Big).$$

*For an aged start with* $t_a/N\to\theta\in(0,\infty)$ *both atoms carry the factor* $1-(1+\theta)^{1-\alpha}$*, by Theorem D(b), and so does the ratio.*

*Proof.* (b) is Proposition 2.3, Proposition 2.4 and Corollary 5.4(a). For (a), let $\delta\in(0,\varepsilon)$. The points $(1-\varepsilon)N\le k\le(1-\delta)N$ have $L_k\ge(\ell_\varepsilon N)(1+o(1))$, so they lie in the zone of Theorem B(b) for every $K$ once $N$ is large, because $\lambda_NB_N=o(N)$, and there $\widehat{G}_N(k)\le C_\varepsilon T_+(N)$. By the additive form of Theorem B(b), the sum of $P_N(k)$ over these points is $\sum_k G_N(k)+o(NT_+(N))$. By the computation in the proof of Corollary 4.9, together with the uniform convergence of Lemma 2.1(i) and (4.4), $G_N(\lfloor vN\rfloor)/T_+(N)\to I_+(v)$ uniformly in $v\in[1-\varepsilon,1-\delta]$, and $I_+$ is continuous there. So the Riemann sum gives $\sum_k G_N(k)\sim NT_+(N)\int_{1-\varepsilon}^{1-\delta}I_+(v)\,dv$. The points $(1-\delta)N<k<N$ have $L_k\ge N/2$ for small $\delta$, so $G_N(k)\le C\widehat{G}_N(k)\le CT_+(N)$ by (4.14), and by the additive form again they

contribute at most $C\delta NT_+(N)$. The atom at $k=N$ is $T_+(N)=o(NT_+(N))$. Let $\delta\to 0$. With $\ell=(v-p)/(1-p)$, $dv=(1-p)\,d\ell$ and $(1-p)/\mu_-=1/s$,

$$\int_{1-\varepsilon}^{1} I_+(v)\,dv=\frac{1}{s}\int_{\ell_\varepsilon}^{1}\big[\alpha(1-\ell)\ell^{-\alpha-1}+\ell^{-\alpha}\big]d\ell=\frac{1}{s}\Big[-(1-\ell)\ell^{-\alpha}\Big]_{\ell_\varepsilon}^{1}=\frac{(1-\ell_\varepsilon)\ell_\varepsilon^{-\alpha}}{s}=\frac{\varepsilon}{\mu_-}\,\ell_\varepsilon^{-\alpha},$$

since $(1-\ell)\ell^{-\alpha}$ has derivative $-\alpha(1-\ell)\ell^{-\alpha-1}-\ell^{-\alpha}$ and $1-\ell_\varepsilon=\varepsilon/(1-p)$. For the random-time start, Corollary 5.4(b) gives the ratio form $P_N(k)\sim G_N^{\mathrm{st}}(k)$ uniformly on the zone with $k<N$, with $G_N^{\mathrm{st}}(\lfloor vN\rfloor)/T_+(N)\to I_+(v)+\ell^{-\alpha}/\mu_-$ uniformly on compacts, and $G_N^{\mathrm{st}}(k)\le CT_+(N)$ on $L_k\ge N/2$. The same argument adds $\int_{1-\varepsilon}^{1}\ell^{-\alpha}\mu_-^{-1}\,dv=(\ell_\varepsilon^{1-\alpha}-1)/(s(\alpha-1))$ to the integral and the atom $Q_+(N)\sim NT_+(N)/(s(\alpha-1))$ to the sum, which together give the second term. □

Part (b) is the design rule. Starting at a regeneration, that is on an empty system, makes the no-switch probability smaller by a factor of order $N$ than starting at an arbitrary time, and by Corollary 5.4(b) it also removes the complete initial cycle from the big-jump kernel, which is the second term in (a). The plus and minus statements differ only by the two loads, so the heavier arm, the one with the larger load, is the one whose cycle is more likely to swallow the experiment.

**Corollary 7.4** (the most likely exposure)**.** *(a) For a fresh start of either arm, the most likely value of $A_N$ is interior for all large $N$, within $O(B_N)$ of $pN$.*

*(b) For a random-time start, and for an aged start with $t_a/N\to\theta\in(0,\infty]$, the single most likely value of $A_N$ is an endpoint, that is no switch, for all large $N$ when $\alpha<\varphi=(1+\sqrt{5})/2$, and is interior for all large $N$ when $\alpha>\varphi$. The same holds for an experiment started at a typical time of a system run in control only, with the cycle in progress kept in control.*

*(c) If $Y$ also satisfies (LD2), the start is fresh or fair, the no-tie condition of Theorem 6.12 holds at both ends and $u_\mp$ is nonincreasing on $j\ge 1$, then $A_N$ has exactly 3 modes for all large $N$, the bulk mode and one within distance $1$ of each end. For a fresh treatment start they are at $A_N=1$, within $O(B_N)$ of $pN$, and at $A_N=N$. The monotonicity holds when $f_\pm$ is strictly log-convex.*

*Proof.* (a) and the random-time and aged cases of (b) are Theorem E(d) and Corollary 5.7, with $\alpha_+=\alpha_-=\alpha$. For the control-only start the initial run is $\nu_-=T_-/\mu_-$ and $\nu_+=0$, so $B_n\mathbb{P}(D=n)=B_nT_-(n)/\mu_-\to 0$ and $Q_-(N)=\mu_-^{-1}\sum_{d\ge N}T_-(d)\in \mathrm{RV}_{1-\alpha}$, and Theorem E(c) gives the same threshold. (c) By Proposition 7.1, the cycle laws satisfy (SPAN), (LR), (LD2) and (EQ), so Theorem G and Corollary 6.13 apply. A nonincreasing $u_\mp$ gives each end profile one local maximum by Lemma 6.14, the positions follow from Theorem G and Proposition 6.7, and Corollary 6.15 covers the log-convex case. □

Four points of (b) deserve care. The threshold is not an equivalence at $\alpha=\varphi$, where slowly varying factors decide. For the cycle laws of the queue the tails are pure powers up to the factor of $T_Y$, and the ratio $B_NQ_\sigma(N)/H_N$ tends to a constant that we have not computed. The theorem compares each endpoint atom with the largest interior value, not the sum of the two atoms with it. The statement holds for all large $N$, not for every $N$. With different service laws in the two arms the indices may differ, and the criterion is then $\alpha_\sigma-1<1/\alpha_*$ for each arm separately, by Theorem E(c). For (c), strict log-convexity of the total-progeny law is checked only numerically, for the zeta service law, and we do not know a proof. Section 9 reports the checks of this section against the exact law, computed from (7.1) and the recursion (9.1), and a simulation of the queue itself.

## 8 Open problems

We list what the paper leaves open, in the order of the sections.

(1) (The start of the big-jump zone.) When the plus scale is the smaller one, the window $KB_N \le k - pN \le K\lambda_N B_N$ lies between Theorems A and B. Theorem 4.14 locates the start of the zone up to a factor $1 \pm \delta$ for fresh starts and describes $P_N$ below it on the logarithmic scale. A relative-error statement on the whole window, the strip $(1 \pm \delta)\xi_N$ around the start, and the sign of the first differences on the window (Remark 6.19) are open. Since (EQ) excludes this case, the exact mode count is open whenever the two scales are not comparable.

(2) (Every $N$.) Theorem G holds for all large $N$, and its threshold depends on the laws. The count for the law of Example 6.17 is 17 or 19 at $N \le 2000$ and 15 from $N = 2500$ on. We know no bound on the number of modes that holds for every $N$, even for the fair start of a log-convex law.

(3) (Total-progeny laws.) Corollary 7.4(c) needs each end profile to have one local maximum, which holds for strictly log-convex cycle laws. For the cycle law of the slotted queue with zeta service we have checked strict log-convexity numerically up to $n = 2^{14}$. A proof, or a direct proof that $u_C$ is decreasing, would make the 3 modes unconditional.

(4) (The line $\alpha = \varphi$.) On the line $\rho_\sigma = 1/\alpha_*$ of Theorem E(c) the comparison of the endpoint with the bulk depends on slowly varying factors. For pure power laws and the stationary or aged start the constant is explicit and below 1. For the cycle laws of Section 7 it is not computed.

(5) (The boundary indices.) At $\alpha = 1$ the runs may or may not have a finite mean, and the scale $B_N$ is $N$ up to a slowly varying factor. The local tools of Berger [6] for the Cauchy domain should give the bulk and the big-jump zone, with the centering $pN$ replaced by a truncated mean. At $\alpha = 2$ the law has a Gaussian bulk with a slowly varying variance and a big-jump zone that starts at a logarithmic multiple of the scale. We have not worked out either line. The case $0 < \alpha < 1$ of infinite means, where the bulk is the Lamperti law and the scale is $N$, is the subject of [57].

## 9 Computations

No proof uses a computed number. The computations check the exact identities, produce Figure 1 and Table 1, and test the limit theorems at moderate $N$. Except for the identities, two examples of Section 6.5 and the queue of Section 7 they use the zeta law $f_\sigma(n) = n^{-5/2}/\zeta(5/2)$ for both signs, so $\alpha_\pm = 3/2$, $p = 1/2$ and $s = 3.894745$, with a fresh plus start unless we say otherwise.

For a fresh plus start let $a_t(k)$ and $b_t(k)$ be the probabilities that a plus run, respectively a minus run, starts at step $t+1$ and the first $t$ steps contain exactly $k$ plus steps. So $a_t(k) = \rho^{(+)}_{t-k}(k)$, $a_0(0) = 1$ and $b_0(0) = 0$. Since the runs alternate, with $a_t(k) = b_t(k) = 0$ for $k < 0$,

$$b_t(k) = \sum_{\ell=1}^{t} a_{t-\ell}(k-\ell)\, f_+(\ell), \qquad a_t(k) = \sum_{\ell=1}^{t} b_{t-\ell}(k)\, f_-(\ell) \qquad (t \ge 1), \tag{9.1}$$

and by (2.13) the identity (2.14) reads

$$P_N(N-j) = \sum_{t=j}^{N-1} a_t(t-j)\, T_+(N-t) + \sum_{m=1}^{j} b_{N-m}(N-j)\, T_-(m). \tag{9.2}$$

This gives the whole law in $O(N^3)$ operations, and an initial run enters by adding its weights to $a_t$ and $b_t$. Every entry is a dot product with nonnegative terms, so in double precision every computed value has relative error at most $(1+\gamma_{N+2})^{N+2}-1$ with $\gamma_n = nu/(1-nu)$ and $u = 2^{-53}$, which is $2.8 \cdot 10^{-9}$ at $N = 5000$. At $N = 600$ a run in ball arithmetic confirms the double-precision law.

*The fresh law.* In Figure 1, with $B_N = 74$, the largest relative error of the bulk on $|k - pN| \le 3B_N$ is $0.071$, that of the big-jump kernel where $|k - pN| \ge 4B_N$ and $\min(k, N-k) \ge 2B_N$ is $0.38$, attained at the junction with the bulk, and that of the boundary kernel on $1 \le \min(k, N-k) \le B_N$ is $0.094$. At $N = 1200$ and $3000$ the ratio $P_N(k)/G_N(k)$ at $k = \lceil pN + KB_N \rceil$ is $0.354$, $1.482$, $1.036$ and $0.337$, $1.538$, $1.059$ for $K = 1$, $4$, $16$, against the limits $0.31$, $1.68$, $1.09$ of Remark 4.10. The minimizers in Table 1 are certified against the rounding bound. For the stationary initial run the ratio of $P_N(N-j)/T_+(N)$ to $\kappa_+^{\mathrm{st}}(j)$ is $1.0007$, $1.0031$ and $1.0098$ at $j = 1$, $2$ and $5$ for $N = 2400$, about half its deviation at $N = 1200$. For Section 6.4, the ratio $d_j \mu_-^2 / T_-(j)$ of Lemma 6.5 is $-0.963$, $-0.996$ and $-0.9996$ at $j = 10$, $100$ and $1000$. The error term of Theorem 6.11 for a fresh plus start at $j = 20$ is $0.73$ at $N = 1000$ and $0.23$ at $N = 3000$, so at this $j$ it decays like $1/N$. For the fair start the law has exactly three modes, at $k = 1$, $N/2$ and $N-1$, for $N = 400$ and $1000$. For the start of Corollary 6.9 with $d_0 = 5$ it has exactly four modes at $N = 1000$, at $k = 1$, $499$, $995$ and $1000$. These modes are certified, since every relative difference of neighbouring values exceeds twice the rounding bound. For Section 6.5, the zeta law has exactly three modes for the fresh plus and the fair start at $N = 2500$, $3000$, $3500$ and $4000$. For the law $f(n) \propto n^{-5/2}$, $n \ge 2$, of Example 6.17, $u(2i) > \max\{u(2i-1), u(2i+1)\}$ for $1 \le i \le 6$, $u$ decreases strictly on $12 \le j \le 10^4$, and the fair start has exactly $15$ modes, at $j$ and $N-j$ for $j = 0, 2, \dots, 12$ and at $N/2$, for every $N = 2500, 2600, \dots, 4000$, and $17$, $19$ and $17$ modes at $N = 1000$, $1500$ and $2000$. For the law $f(n) \propto n^{-12/5}$, $n \ge 2$, of index $7/5$ in the same example, the fair start has exactly $13$ modes at $N = 3000$ and $4000$. For the law with $f(1) = 0.6$ and $f(2) = 0.3$ after Lemma 6.14, $u$ decreases strictly on $2 \le j \le 10^4$, and the fair start has exactly three modes, at $k = 2$, $N/2$ and $N-2$, while the fresh plus start has four, at $N = 1000$, $1500$ and $2000$. The renewal mass functions are sums of nonnegative terms with the same rounding bound, and all these signs are certified in the same way.

*The aged law.* For Theorem D with $t_a = N$, so $\theta = 1$, the aged law at $N = 500$, $1000$ and $2000$ is computed from the first-run decomposition with the initial run (5.14), after a check of the method against a sum over all run sequences for $t_a + N \le 12$. The largest relative difference between the aged and the fresh law on $|k - N/2| \le B_N$ is $0.054$, $0.043$ and $0.035$. At $N = 2000$ the atom is $1.020$ times the limit in part (b), the ratios $P_N(N-j)/(T_+(N)\kappa_+^{(1)}(j))$ of part (c)(i) are $1.015$ to $1.028$ for $j = 1, \dots, 5$, and the aging increment $P_N^{(t_a)}(k) - P_N^{(0)}(k)$ at $L_k = \ell N$ is $0.945$, $0.998$ and $1.002$ times the two terms that the kernel of part (c) adds to the fresh kernel, for $\ell = 0.3$, $0.5$ and $0.7$. The aged kernel inherits the slow convergence of the fresh kernel near the bulk, where $P_N/G_N$ is $1.30$ at $\ell = 0.3$ for the fresh start and $1.24$ for the aged one. For Corollary 5.7 the ratio of the larger endpoint atom to the largest interior value at $\theta = 1$ is $0.258$, $0.292$ and $0.330$ for $\alpha = 3/2$, growing by the factor $1.13$ per doubling of $N$ against the asymptotic factor $2^{1-\alpha+1/\alpha} = 1.122$, while the stationary start has already crossed, with the ratio $1.117$ at $N = 2000$. For $\alpha = 9/5$ the aged ratio is $0.024$, $0.022$ and $0.019$, against the factor $0.844$. For the zeta law the factor $1 - 2^{1-\alpha} = 0.29$ in the aged atom moves the crossover at $\alpha = 3/2$ beyond $N = 10^6$.

*The queue.* For Section 7 the cycle law is computed from (7.1) by FFT convolution up to $n = 2^{14}$, with the zeta service law of index $\alpha = 3/2$ and the loads $\rho_+ = 0.6$ and $\rho_- = 0.4$. It agrees with a dynamic program on the workload itself to a relative $10^{-15}$ for $n \le 300$, and with a simulation of $4 \cdot 10^5$ cycles. At $n = 2^{14}$ the ratio $f_C(n)/(\lambda(1-\rho)^{-\alpha-1} f_Y(n))$ of

Proposition 7.1(a) is $0.9996$ for $\rho = 0.4$ and $0.9972$ for $\rho = 0.6$, and at $n = 2^{13}$ the quantities $n\Delta f_C(n)/f_C(n)$ and $n^2\Delta^2 f_C(n)/f_C(n)$ are $-2.499$ and $8.745$ for $\rho = 0.4$, against the limits $-5/2$ and $35/4$ of part (b). The computed $f_C$ is strictly log-convex on $2 \le n < 2^{14}$ for both loads. The exposure law is computed by the recursion (9.1) with the initial run injected, and a simulation of $3 \cdot 10^5$ switching queues at $N = 200$ agrees with it ($\chi^2 = 10.8$ on $11$ degrees of freedom). At $N = 1000$, $2000$ and $4000$, $B_N P_N(\lfloor pN \rfloor)$ is $0.341$, $0.328$ and $0.318$ against $h_\theta(0) = 0.282$. For Corollary 7.3(a) the ratio of $\mathbb{P}(N - A_N < \varepsilon N)$ to its prediction at $N = 4000$ is $1.076$, $1.063$ and $1.066$ for the fresh start and $1.009$, $1.012$ and $1.022$ for the random-time start, for $\varepsilon = 0.05$, $0.1$ and $0.2$, down from $1.15$, $1.12$ and $1.09$ at $N = 1000$ for the fresh start. For part (b) the ratio of the two no-switch probabilities is $607$, $1204$ and $2396$ against the asymptotic values $599$, $1195$ and $2387$. For Corollary 7.4 the ratio of the larger atom to the largest interior value for the random-time start is $3.28$, $3.81$ and $4.39$ at $\alpha = 3/2$ and $0.291$, $0.253$ and $0.220$ at $\alpha = 9/5$, and the fresh law has exactly 3 modes at the three values of $N$.

The exact identities of Section 2 and the renewal identities of Section 6.2 are checked in exact rational arithmetic against a sum over all sign strings for $N \le 12$. The identities of Section 2.5, except the bound in Lemma 2.7, are also proved in the Lean library PaperF, whose file MAP.md maps the labels to the Lean names. The scripts, their outputs with the hash of each script, and the Lean files are in the folder kagey-problems/problem131 of https://github.com/apemm/Kagey-Problems. In paperF the command `python -I verify_all.py` runs the checks, and README.md lists the software versions and what each script tests.

## A Classical inputs

This appendix gives the proofs of Lemmas 2.1 and 2.2, four classical facts on sums of runs, and the bounds on the kernels of one sign that enter the proof of Theorem A.

### A.1 Regular variation and the profile

*Proof of Lemma 2.1.* Parts (i) to (iii) are the cited theorems, and $T_\sigma(x/c) \le C_c T_\sigma(x)$ follows from (i) for large $x$ and from positivity for the others. For (iv), $nT_\sigma(n)$ has index $1 - \alpha_\sigma > -1$ and $T_\sigma$ has index $-\alpha_\sigma < -1$, so the cited propositions give the asymptotics, and positivity gives the bounds for all $w \ge 1$. For (v), $B_N$ is nondecreasing since each $a_\sigma$ is increasing. If $\alpha_+ \ne \alpha_-$, then $B_N = b_{\sigma,N}$ for the sign with the smaller index and all large $N$. If the indices are equal, $B_{cN}/B_N$ lies between the two ratios $b_{\sigma,cN}/b_{\sigma,N}$, which tend to $c^{1/\alpha_*}$. So $B_N T_\sigma(N)$ has the negative index $1/\alpha_* - \alpha_\sigma$ and tends to zero [9, Proposition 1.5.1], the definition of $a_\sigma$ gives $NT_\sigma(b_{\sigma,N}) \to s$, and $B_N \log N = o(N)$ since $1/\alpha_* < 1$. For (vi), take $\delta = (\alpha_\sigma - 1)/2$ in (ii) and use $s/2 \le NT_\sigma(b_{\sigma,N}) \le 2s$ for large $N$. □

*Proof of Lemma 2.2.* Under (RV), $0 < p < 1$. For $1 < \alpha < 2$ the real parts of $(-it)^\alpha$ and $(it)^\alpha$ equal $|t|^\alpha \cos\frac{\pi\alpha}{2} < 0$, and $(-ict)^\alpha = c^\alpha(-it)^\alpha$ for $c \ge 0$. So $W_\theta$ has the characteristic function

$$\widehat{h}_\theta(t) = \exp\Big(K_{\alpha_+}\big((1-p)\theta_+\big)^{\alpha_+}(-it)^{\alpha_+} + K_{\alpha_-}\big(p\,\theta_-\big)^{\alpha_-}(it)^{\alpha_-}\Big),$$

continuous in $\theta$. One coordinate of $\theta \in \Theta$ equals 1, so $|\widehat{h}_\theta| \le \Phi$ with $\Phi(t) = 1$ for $|t| < 1$ and $\Phi(t) = e^{-c|t|^{\alpha_*}}$ otherwise, where $c = \min_\sigma K_{\alpha_\sigma}|\cos\frac{\pi\alpha_\sigma}{2}|\min(p, 1-p)^{\alpha_\sigma} > 0$. Fourier inversion gives (a) with $C_1 = \frac{1}{2\pi}\int \Phi$ and $C_2 = \frac{1}{2\pi}\int |t|\Phi(t)\,dt$, and dominated convergence gives the continuity in (c). If $N'/N \to 1$, then $\theta_{N'} - \theta_N \to 0$ by Lemma 2.1(i), and uniform continuity on the compact $\Theta$ gives the rest of (c).

For (b), the set $\{h_\theta : \theta \in \Theta\}$ is compact in the uniform norm and each $h_\theta$ vanishes at infinity, so a finite net gives $\varepsilon_h(R) \to 0$. If $\varepsilon_h'(R) \not\to 0$, there are $\varepsilon > 0$, $R_i \to \infty$ and $N_i$

with $\int_{|x|>R_i} h_{N_i} \geq \varepsilon$ and, by compactness, $\theta_{N_i} \to \theta$. Then $h_{N_i} \to h_\theta$ in $L^1$ by Scheffé's lemma, which is a contradiction.

For (d), fix a sign and write $Z = Z_\sigma$, $g = g_\sigma$, $\alpha = \alpha_\sigma$. A sum of $n$ independent copies of $Z$ has the law of $n^{1/\alpha}Z$. If $Z \geq 0$ almost surely, then the strong law of large numbers applied to $\min(Z_i, 1)$ makes $n^{-1/\alpha}(Z_1 + \cdots + Z_n) \to \infty$, which contradicts tightness, and the same holds for $-Z$. So the open set $U = \{g > 0\}$ meets both half lines. The density $g^{*n}$ is continuous, positive at $y_1 + \cdots + y_n$ when every $y_i \in U$, and equal to $n^{-1/\alpha} g(n^{-1/\alpha}\,\cdot)$. Hence $(a - \varepsilon, a + \varepsilon) \subset U$ with $0 < \varepsilon < a$ gives $c_n(a - \varepsilon, a + \varepsilon) \subset U$ with $c_n = n^{1-1/\alpha}$, and since $c_{n+1}/c_n \to 1$ these intervals cover a half line. The same holds on the other side, and the case $n = 2$ gives $U = \mathbb{R}$. If $\theta_\sigma = 1$, then $W_\theta = aZ_\sigma + Y$ with $a \neq 0$ and $Y$ independent, so $h_\theta(x) = \mathbb{E}[|a|^{-1} g_\sigma((x - Y)/a)] > 0$. The two minima in (d) are minima of positive continuous functions on $\Theta$.

For (e) with $\tau = +$, condition on $Z_- = y$ and use $1 - p = \mu_-/s$ and $p/(1-p) = \mu_+/\mu_-$. For $\tau = -$ condition on $Z_+$. □

### A.2 Sums of runs

We write $\bar{T}_\sigma(x) = \mathbb{P}(X_\sigma > x) \sim T_\sigma(x)$.

**Lemma A.1** (stable and local limits)**.** *Assume (RV) and (SPAN) for the sign $\sigma$, and write $\alpha = \alpha_\sigma$, $\mu = \mu_\sigma$, $a = a_\sigma$, $g = g_\sigma$, $T = T_\sigma$ and $S_m = S_m^\sigma$. Parts (i), (iii) and (iv) need only (RV).*

*(i) $Z_m = (S_m - m\mu)/a(m)$ converges in law to $Z_\sigma$, whose density $g$ is bounded, uniformly continuous, and tends to zero at infinity. Moreover $m\big(\mathbb{E}e^{-(X_\sigma - \mu)/a(m)} - 1\big) \to K_\alpha$.*

*(ii) (Gnedenko's local limit theorem) $\sup_{k \in \mathbb{Z}} \big| a(m)\mathbb{P}(S_m = k) - g\big((k - m\mu)/a(m)\big)\big| \to 0$ as $m \to \infty$, and $\bar{u}_\sigma = \sup_{m \geq 1} \sup_k a(m)\mathbb{P}(S_m = k) < \infty$.*

*(iii) (local large deviations) There is a constant $C_{\mathrm{LD}} = C_{\mathrm{LD}}(\sigma)$ such that $\mathbb{P}(S_m = j) \leq C_{\mathrm{LD}}\, m\, T(j - m\mu)/a(m)$ for all $m \geq 1$ and all integers $j$ with $j - m\mu \geq 1$.*

*(iv) (left deviations) The constant $C_- = C_-(\sigma) = \sup_{m \geq 1} m \log \mathbb{E}e^{-(X_\sigma - \mu)/a(m)}$ is finite, and $\mathbb{P}(S_m \leq m\mu - y) \leq e^{C_-}\, e^{-y/a(m)}$ for all $m \geq 1$ and $y \geq 0$.*

*Proof.* (i) The domain of attraction is classical [27, Section XVII.5], and we check the constant $K_\alpha$. Let $X = X_\sigma$ and $\Lambda(\xi) = \mathbb{E}[e^{-\xi X} - 1 + \xi X]$ for $\operatorname{Re}\xi \geq 0$. Fubini's theorem gives, for $\xi = te^{i\vartheta}$ with $t > 0$ and $|\vartheta| \leq \pi/2$,

$$\Lambda(\xi) = \int_0^\infty \xi\big(1 - e^{-\xi y}\big)\bar{T}(y)\, dy = \int_0^\infty e^{i\vartheta}\big(1 - e^{-e^{i\vartheta} v}\big)\bar{T}(v/t)\, dv.$$

The integrand is at most $\min(2, v)\bar{T}(v/t)$ in modulus, and $\bar{T}(v/t)/\bar{T}(1/t) \to v^{-\alpha}$ as $t \to 0$, dominated by Lemma 2.1(ii) for $v \geq tx_\delta$ with $1 < \alpha \pm \delta < 2$, while $v < tx_\delta$ contributes $o(\bar{T}(1/t))$ since $\alpha < 2$. So, uniformly in $\vartheta$, $\Lambda(te^{i\vartheta}) = \bar{T}(1/t)(F(e^{i\vartheta}) + o(1))$ with $F(\xi) = \int_0^\infty \xi(1 - e^{-\xi v})v^{-\alpha} dv$. The function $F$ is continuous on $\operatorname{Re}\xi \geq 0$, analytic inside and homogeneous of degree $\alpha$, and an integration by parts gives $F(1) = \Gamma(2-\alpha)/(\alpha - 1) = K_\alpha$, so $F(\xi) = K_\alpha \xi^\alpha$. With $\mathbb{E}e^{-\xi(X-\mu)} - 1 = \Lambda(\xi) + O(|\xi|^2)$, $\xi = -i\theta/a(m)$, $m\bar{T}(a(m)/|\theta|) \to |\theta|^\alpha$ and $m/a(m)^2 \to 0$ we get $m(\mathbb{E}e^{i\theta(X-\mu)/a(m)} - 1) \to K_\alpha(-i\theta)^\alpha$, and Lévy's continuity theorem gives $Z_m \to Z_\sigma$. The limit characteristic function is integrable, which gives the properties of $g$. With $\xi = 1/a(m)$ the same expansion gives the last claim.

(ii) By (SPAN) the maximal span of $X_\sigma$ is 1, and Gnedenko's theorem [36, Theorem 4.2.1] (see also [9, Theorem 8.4.1]) gives the limit, with any norming $a_m \sim a(m)$. The bound $\bar{u}_\sigma$ follows.

(iii) This is [10, Theorem 1.1, bound (1.6)] for the centered variable $X_\sigma - \mu$, with $A = 1/\bar{T}$, the interval $(-1, 0]$ and $x = j - m\mu$. Its right tail is $\bar{T}(x)(1 + o(1))$ and its left tail vanishes for $x > \mu$. The bound needs no lattice assumption, its norming sequence is asymptotic to $a(m)$ [9, Theorem 1.5.12], and $\bar{T} \le T$.

(iv) By (i), $m(\mathbb{E}e^{-(X_\sigma-\mu)/a(m)} - 1)$ is bounded, and $\log(1 + z) \le z$ makes $C_-$ finite. Markov's inequality for $e^{-(S_m-m\mu)/a(m)}$ gives the bound. □

### A.3 Kernels of one sign

For each sign $\sigma$ the fresh identities use the kernels $u^\sigma_m$, $q^\sigma_m$ and $q^\sigma_{m+1}$ ($m \ge 1$), with weights $c_{\mathsf{k}} = 1$, $\mu_\sigma$ and $\mu_\sigma$. A kernel of sign $\sigma$ is one of these sequences, written $\mathsf{k}_m$. We use four facts.

(K1) $\sum_{m\ge1} u^\sigma_m(j) \le 1$, $\sum_{m\ge1} q^\sigma_m(j) = 1$ and $\sum_{m\ge1} q^\sigma_{m+1}(j) \le 1$ for $j \ge 1$, by (2.8).

(K2) $\mathsf{k}_m(j) = 0$ for $m > j$, because $S^\sigma_m \ge m$.

(K3) Let $J_\sigma(j) = \min\{m \ge 1 : S^\sigma_m \ge j\}$. Then $\mathbb{P}(J_\sigma(j) = m) = q^\sigma_m(j)$ by (2.7), and $u^\sigma_m(j) \le q^\sigma_m(j)$, since $\{S^\sigma_m = j\} \subset \{J_\sigma(j) = m\}$.

(K4) $\sum_j q^\sigma_m(j) = \mu_\sigma$ and $\sum_j u^\sigma_m(j) = 1$.

**Lemma A.2** (kernel local limits)**.** *Assume (RV) and (SPAN). For each sign $\sigma$ and each kernel $\mathsf{k}$ of sign $\sigma$,*

$$\delta_{\mathsf{k}}(m) = \sup_{j\in\mathbb{Z}} \big|a_\sigma(m)\mathsf{k}_m(j) - c_{\mathsf{k}}\, g_\sigma\big((j - m\mu_\sigma)/a_\sigma(m)\big)\big| \to 0 \qquad (m \to \infty),$$

*and $\mathsf{k}_m(j) \le C_3/a_\sigma(m)$ for all $m \ge 1$ and $j$, where $C_3 = C_3(\sigma) = \max\big(a_\sigma(1),\, \mu_\sigma r_\sigma \bar{u}_\sigma\big)$ and $r_\sigma = \sup_{m\ge2} a_\sigma(m)/a_\sigma(m-1) < \infty$.*

*Proof.* For $u^\sigma_m$ this is Lemma A.1(ii). The kernel $q^\sigma_m$ is a sum of shifts of $u^\sigma_{m-1}$ by $v \ge 1$ with weights $T_\sigma(v)$ of total mass $\mu_\sigma$, and $q^\sigma_{m+1}$ is the same sum built on $u^\sigma_m$. This gives the bounds, with $q^\sigma_1 = T_\sigma \le 1$. For the limit of $q^\sigma_m$ write $a = a_\sigma$, $g = g_\sigma$ and $z = (j - m\mu)/a(m)$, and fix $v_0$ and $Z$. The shifts $v > v_0$ contribute at most $(\bar{u}_\sigma + \|g\|_\infty)\sum_{v>v_0} T(v)$ to $|a(m-1)q_m(j) - \mu g(z)|$. For $v \le v_0$ the scaled point moves by at most $Z|a(m)/a(m-1) - 1| + (v_0 + \mu)/a(m-1)$ when $|z| \le Z$, and stays beyond $Z/2$ in modulus for large $m$ when $|z| > Z$. So the uniform continuity of $g$, its decay and Lemma A.1(ii) give the limit as $m \to \infty$, then $v_0, Z \to \infty$. The kernel $q^\sigma_{m+1}$ is the same. □

**Lemma A.3** (small counts)**.** *Assume (RV). For each sign $\sigma$ there is $C_4 = C_4(\sigma)$ such that every kernel of sign $\sigma$ satisfies*

$$\mathsf{k}_m(j) \le C_4\big(1 + m/a_\sigma(m)\big)T_\sigma(j) \qquad \textit{whenever } 8m\mu_\sigma \le j.$$

*Consequently, for $0 < \beta \le 1$ there is $C_5 = C_5(\sigma, \beta)$ with*

$$\mathsf{k}_m(j) \le C_5\, e_{\sigma,N}, \qquad e_{\sigma,N} = \big(1 + N/a_\sigma(N)\big)T_\sigma(N), \tag{A.1}$$

*for all $N \ge 1$, all $\beta N \le j \le N$ and all $m$ with $8m\mu_\sigma \le \beta N$. Moreover $B_N e_{\sigma,N} \to 0$.*

*Proof.* Write $T = T_\sigma$, $a = a_\sigma$, $\mu = \mu_\sigma$, and let $T(x/2) \le C_T T(x)$. If $4m\mu \le i$, then $i - m\mu \ge i/2 \ge 1$, and Lemma A.1(iii) gives $u_m(i) \le C_{\mathrm{LD}} C_T\, m\, T(i)/a(m)$, which is the claim for $u_m$. Let $8m\mu \le j$ and $m \ge 2$, and split $q_m(j) = \sum_v T(v)u_{m-1}(j - v)$ at $v = j/2$. For $v \le j/2$ the bound for $u_{m-1}$ applies at $j - v \ge 4(m-1)\mu$, and the weights add up to at most $\mu$. For $v > j/2$ we have $T(v) \le C_T T(j)$, and the values $u_{m-1}(j - v)$ add up to at most 1. This gives the claim with $C_4 = C_T + \mu C_{\mathrm{LD}} C_T^2 r_\sigma$, and $q_{m+1}$ is the same. For (A.1), $m/a(m) \le C_P N/a(N)$ for $m \le N$ by Lemma 2.1(iii), and $T(j) \le C_{T,\beta} T(N)$. Finally $B_N N T_\sigma(N)/a_\sigma(N)$ has the index $1/\alpha_* - 1/\alpha_\sigma + 1 - \alpha_\sigma \le 1 - \alpha_* < 0$, since $1/\alpha_* - 1/\alpha_\sigma \le \alpha_\sigma - \alpha_*$, and $B_N T_\sigma(N) \to 0$. □

## B Proofs for the exact number of modes

This appendix proves parts (iii) to (v) of Theorem G, Lemma 6.18 and Corollary 6.16, and checks the junctions of the five zones. Parts (i) and (ii) are Theorem 6.12.

*Notation.* For a sequence $X$ on $\mathbb{Z}$ we write $\Delta X(j) = X(j+1) - X(j)$ and $\nabla X(n) = X(n-1) - X(n)$, and $\Delta^2$, $\nabla^2$ for the iterates. For a family $X_m$ indexed by $m$ we write $\delta X_m = X_{m+1} - X_m$ and $\delta^2 X_m = \delta X_{m+1} - \delta X_m$. For real $x \geq 1$ we read $f_\sigma(x)$ as $f_\sigma(\lfloor x \rfloor)$. By Lemma 2.1(ii) and $mT_\sigma(a_\sigma(m)) \to 1$, for every $\varepsilon' > 0$ there are $c, C > 0$ with

$$c\, y^{-\alpha_\sigma - \varepsilon'} \leq m\, T_\sigma\big(y\, a_\sigma(m)\big) \leq C\, y^{-\alpha_\sigma + \varepsilon'} \qquad (m \geq 1,\ y \geq 1). \tag{B.1}$$

For bounded $m$ this follows from the bounds for $T_\sigma$ alone, since $a_\sigma(m)$ is then bounded above and below. Below $y = 1$ the exponents exchange. With $x_0$ the threshold of Potter's bounds in Lemma 2.1(ii), for every $\varepsilon' > 0$ there are $c, C > 0$ with

$$c\, y^{-\alpha_\sigma + \varepsilon'} \leq m\, T_\sigma\big(y\, a_\sigma(m)\big) \leq C\, y^{-\alpha_\sigma - \varepsilon'} \qquad (m \geq 1,\ 0 < y \leq 1,\ y\, a_\sigma(m) \geq x_0), \tag{B.2}$$

since then both $a_\sigma(m)$ and $y a_\sigma(m)$ are at least $x_0$ and $mT_\sigma(a_\sigma(m))$ is bounded above and below.

### B.1 Smoothness and the renewal sequence

**Lemma B.1** (consequences of (LD2))**.** *Let $f$ be a positive sequence that is regularly varying with index $-\alpha - 1$, $\alpha > 0$, and satisfies $|f(n+1) - 2f(n) + f(n-1)| \leq C f(n)/n^2$. Then $|f(n+1) - f(n)| \leq C' f(n)/n$ for large $n$, and $f(n) - f(n+1) = (\alpha + 1 + o(1)) f(n)/n$.*

So under (LR), (LD2) implies (LD), and $\nabla f_\sigma(y) = (\alpha_\sigma + 1 + o(1)) f_\sigma(y)/y$ as $y \to \infty$.

*Proof.* Since $f(n) \to 0$, $\Delta f(n) \to 0$, and $\Delta f(n) = -\sum_{i \geq n} \Delta^2 f(i)$. Here $|\Delta^2 f(i)| \leq C f(i+1)/(i+1)^2 \leq C' f(i)/i^2$, and $\sum_{i \geq n} f(i)/i^2 \leq C'' f(n)/n$ by Karamata's theorem, since $f(i)/i^2$ has index $-\alpha - 3$. This is the first claim. For the second put $r(n) = n\Delta f(n)/f(n)$, which is bounded. Fix $\lambda > 1$ and let $k = \lfloor (\lambda - 1) n \rfloor$. For $0 \leq i < k$ the bound on $\Delta^2 f$ gives $|\Delta f(n+i) - \Delta f(n)| \leq C_\lambda i f(n)/n^2$, with $C_\lambda$ bounded for $\lambda \leq 2$. Summing over $i < k$,

$$\frac{f(n+k)}{f(n)} - 1 = \frac{k}{n}\, r(n) + O\big((\lambda - 1)^2\big).$$

The left side tends to $\lambda^{-\alpha - 1} - 1$ by Lemma 2.1(i), and $k/n \to \lambda - 1$. Hence $\limsup_n |r(n) - (\lambda^{-\alpha-1} - 1)/(\lambda - 1)| \leq C(\lambda - 1)$. The quotient tends to $-(\alpha + 1)$ as $\lambda \downarrow 1$, so $r(n) \to -(\alpha + 1)$. □

**Lemma B.2** (second-difference renewal theorem)**.** *Assume (RV), (SPAN) and (LR) for the minus sign, and write $u = u_-$, $v = v_-$, $t_n = T_-(n+1)$ and $d_n = u(n) - u(n-1)$ as in Lemma 6.5. Then $n(d_n - d_{n-1})/t_n \to \alpha_-/\mu_-^2$. Equivalently*

$$\Delta^2 u(j) = u(j+2) - 2u(j+1) + u(j) = \big(1 + o(1)\big) \frac{f_-(j)}{\mu_-^2}.$$

*Moreover $|\Delta^2 u(j)| \leq C T_-(j)/(j+1)$ for $j \geq 0$, $\Delta^2 v(j) = O(T_-(j))$, and the sequence $\tilde{v}$ equal to $v$ on $\mathbb{Z}_{\geq 0}$ and to $0$ elsewhere has $\Delta^2 \tilde{v} \in \ell^1$.*

*Proof.* We repeat the proof of Lemma 6.5 with one more seminorm. For sequences $a$ on $\mathbb{Z}_{\geq 0}$ put $\nabla a_n = a_n - a_{n-1}$ with $a_{-1} = 0$, and $\|a\|_{t'} = \sup_{n \geq 1} n|\nabla a_n|/t_n$. For $n \geq 2$ and $M = \lfloor n/2 \rfloor$ a direct computation gives

$$\nabla(a * b)_n = \sum_{k \leq M} a_k \nabla b_{n-k} + \sum_{l \leq n-M-2} b_l \nabla a_{n-l} + a_{M+1} b_{n-M-1}.$$

In the two sums the index of the differenced factor is at least $n/2$, and the last term is at most $\|a\|_t\|b\|_t c_t'^2 t_n^2$, where $c_t' = \sup_n t_{\lfloor n/2\rfloor-1}/t_n$. Since $nt_n \to 0$,

$$\|a*b\|_{t'} \le C\big(\|a\|_1\|b\|_{t'} + \|a\|_{t'}\|b\|_1 + \|a\|_t\|b\|_t\big), \tag{B.3}$$

where the constant of the first two terms is $2c_t$. If moreover $n\nabla a_n/t_n \to \chi_a'$ and $n\nabla b_n/t_n \to \chi_b'$, then dominated convergence in the two sums, as for (6.13), gives $n\nabla(a*b)_n/t_n \to \hat a(1)\chi_b' + \hat b(1)\chi_a'$.

By (LR), $n\nabla t_n/t_n = -nf_-(n)/T_-(n+1) \to -\alpha_-$, and the same holds for the truncation $t''$ of the proof of Lemma 6.5. Take $M$ there so large that also $2c_t\|g\|_1 \le \frac14$. This is possible because $\|e\|_1 \le 2\|d\|_1$ for every $M$. The coefficients $e$ decay geometrically, so $n\nabla e_n/t_n \to 0$, and $g = e*t''$ has $n\nabla g_n/t_n \to -\alpha_-\hat e(1)$. By (B.3) and the bound $\|g^{*k}\|_t \le Ck4^{-k}$ of that proof, $\|g^{*k}\|_{t'} \le Ck^2 4^{-k}$. By induction $n\nabla(g^{*k})_n/t_n \to k\hat g(1)^{k-1}(-\alpha_-\hat e(1))$, and dominated convergence over $k$ gives, for $h = \sum_{k\ge1}(-g)^{*k}$, the limit $\alpha_-\hat e(1)/(1+\hat g(1))^2$ of $n\nabla h_n/t_n$. Since $d = e + e*h$, the limit of $n\nabla d_n/t_n$ is $\hat e(1)\alpha_-\hat e(1)/(1+\hat g(1))^2 = \alpha_-/\mu_-^2$, with $\hat e(1)/(1+\hat g(1)) = 1/\mu_-$ as in that proof. By (LR) this is the stated form of $\Delta^2 u(j) = \nabla d_{j+2}$, and $|\nabla d_n| \le Ct_n/n$ gives the bound on $\Delta^2 u$.

The generating function of $d*d$ is $(1-z)^2\hat u^2$, so $d*d = \nabla^2(u*u)$, and $(d*d)_n/t_n \to 2\hat d(1)(-1/\mu_-^2)$ by (6.13). By (6.5), $u*u = v+u$ on $\mathbb{Z}_{\ge0}$. Hence $\nabla^2 v_n = (d*d)_n - \nabla^2 u_n = O(t_n)$, that is $\Delta^2 v(j) = O(T_-(j))$. Since $T_-$ is summable, $\Delta^2\tilde v \in \ell^1$. □

### B.2 One antimode

Write $f = f_+$, $T = T_+$, $\mu = \mu_+$, $V_m = u_m^+$ with $V_0 = \delta_0$ and $q_m = q_m^+$, as in Lemma 6.10.

**Lemma B.3** (plus runs at second order)**.** *Assume (LR) and (LD2) for the plus sign, and fix $\eta \in (0,1)$. As $n\to\infty$, uniformly in $0 \le m \le n^{1-\eta}$,*

*(i)* $|\nabla^2 V_m(n)| \le Cmf(n)/n^2$,

*(ii)* $|\nabla\delta V_m(n)| \le Cf(n)/n$,

*(iii)* $|\delta^2 V_m(n)| \le Cf(n)/n$,

*(iv)* $|q_{m+3}(n) - 2q_{m+2}(n) + q_{m+1}(n)| \le Cf(n)/n$.

*Proof.* Let $m \ge 1$. For $m \le n^{1-\eta}$ and large $n$ the deviation $L = n - m\mu$ satisfies $n/2 \le L \le n$ and $L \ge \max(K_1 a(m), L_0, m)$, since $a(m) \le a(n^{1-\eta}) = o(n)$ by Lemma 2.1(ii). So Lemma B.10 with $c = 1$, whose proof uses only Lemmas B.1, 4.3, B.6 and 6.18, gives (i) to (iii) with $f(L)$ and $L$ in place of $f(n)$ and $n$, and $f(L) \le Cf(n)$ by (6.6). The same holds at every argument $n' \ge n/2$, since $m \le n^{1-\eta} \le (n')^{1-\eta/2}$ for large $n$. For $m = 0$, (i) is trivial, $\nabla\delta V_0(n) = \nabla f(n)$ for $n \ge 2$, which is (LD), and $\delta^2 V_0(n) = f*f(n) - 2f(n)$ for $n \ge 1$, with

$$|f*f(n) - 2f(n)| \le 2\sum_{x<n/2} f(x)\,|f(n-x) - f(n)| + 2f(n)T(n/2) + f(n/2)^2 \le C\frac{f(n)}{n}$$

by (LD), $f(n/2) \le Cf(n)$ and $nT(n) \to 0$. By Lemma B.1, (LD) holds for the plus sign, and by Lemma 6.10 and (6.6), $V_m(x) \le Cmf(n)$ for $m \ge 1$ and $x \ge n/2$, and $mnf(n) \le Cn^{1-\eta}T(n) \to 0$.

(iv) Write $\delta q_{m+1} = q_{m+2} - q_{m+1}$. As in the proof of Lemma 6.10(iv), $\delta q_{m+1}(n) = \sum_x f(x)[q_{m+1}(n-x) - q_{m+1}(n)]$ and $q_{m+1}(l-1) - q_{m+1}(l) = \delta V_m(l-1)$. Hence $\nabla\delta q_{m+1}(l) = \delta^2 V_m(l-1)$, and the quantity in (iv) is $\sum_x f(x)[\delta q_{m+1}(n-x) - \delta q_{m+1}(n)]$. We split the sum at $x = n/2$. The part $x \le n/2$ is at most $\sum_{x\le n/2} f(x)\, x \sup_{l\ge n/2}|\nabla\delta q_{m+1}(l)| \le C\mu f(n)/n$ by (iii), and the term $-\delta q_{m+1}(n)\sum_{x>n/2} f(x)$ is $O(f(n)T(n))$ by Lemma 6.10(iv). The rest,

$\sum_{y<n/2} f(n-y)\delta q_{m+1}(y)$, equals $f(n)\sum_{y<n/2}\delta q_{m+1}(y) + \sum_y \gamma(y)\delta q_{m+1}(y)$ with $\gamma(y) = (f(n-y)-f(n))\mathbf{1}_{\{0\le y<n/2\}}$. Since $\delta q_{m+1} = f * q_{m+1} - q_{m+1}$,

$$\sum_y \gamma(y)\delta q_{m+1}(y) = \sum_x f(x)\sum_{y'} q_{m+1}(y')\big[\gamma(y'+x)-\gamma(y')\big].$$

The terms with $y'+x < n/2$ add up to at most $C\mu\sum_{y'} q_{m+1}(y')f(n)/n = C\mu^2 f(n)/n$. The terms with $y' < n/2 \le y'+x$ are at most $Cf(n)\sum_{y'<n/2} q_{m+1}(y')T(\lceil n/2\rceil - y')$, and this sum is at most $C\mu T(n)$. Indeed, for $y' \le n/4$ the factor $T$ is at most $T(n/4)$ and the weights add up to $\mu$, and for $y' > n/4$ the weight $q_{m+1}(y')$ is at most $CT(n)$ by Lemma 6.10(i) while the factors $T$ add up to at most $\mu$. Finally the full sum of $\delta q_{m+1}$ vanishes, and for every $L'$

$$\Big|\sum_{y\ge L'}\delta q_{m+1}(y)\Big| = \sum_x f(x)\sum_{L'-x\le y<L'} q_{m+1}(y) \le (C+1)\mu\, T(L'/2),$$

by the same split at $x = L'/2$. With $L' = \lceil n/2\rceil$ this part is $f(n)O(T(n))$. □

**Theorem B.4** (second differences in the layer)**.** *Assume (RV), (SPAN) and (LR) for the minus sign and (LR) and (LD2) for the plus sign, and fix $\eta \in (0,1)$. Let the start be fresh with either sign, or a mixture of the two fresh starts. Then, uniformly in $1 \le j \le N^{1-\eta}$,*

$$D_N(j+1) - D_N(j) = T_+(N)\,\Delta^2 u_-(j) + O\Big(f_+(N)\Big(T_-(j) + \frac{1}{N}\Big)\Big).$$

*Proof.* By linearity we may treat the two fresh starts. We keep the notation of the proof of Theorem 6.11, with $n = N - j$, $a_m = u_m^-$, and $Q_m$, $b_m$ as there, and we use its six facts. Differencing $D_N = \mathrm{I} + \mathrm{II} + \mathrm{III} + \mathrm{IV}$ once more in $j$, with $n$ replaced by $n-1$, gives the exact identity

$$D_N(j+1) - D_N(j) = A_1 + 2A_2 + A_3 + 2A_4 + A_5 + A_6$$

for a plus start, where

$$A_1 = \sum_m \Delta^2 a_m(j)\, Q_m(n-2), \qquad A_2 = \sum_m \Delta a_m(j)\,\delta V_m(n-2),$$
$$A_3 = \sum_m a_m(j)\,\nabla\delta V_m(n-1), \qquad A_4 = \sum_m \nabla V_m(n-1)\,\Delta b_m(j),$$
$$A_5 = \sum_m \nabla^2 V_m(n)\, b_m(j), \qquad A_6 = \sum_m V_m(n-2)\,\Delta^2 b_m(j).$$

Here we used $\nabla Q_m(l) = \delta V_m(l-1)$. Only $m \le j+2$ contributes, so Lemma B.3 applies at the arguments $n-2 \ge N/2$ with $\eta/2$ in place of $\eta$, and $f(n') \le Cf(N)$ there.

By Lemma B.3(i) and fact (6), $|A_5| \le Cf_+(N)j/N^2$. By (ii) and fact (1), $|A_3| \le Cf_+(N)/N$. Writing $\nabla V_m = \nabla V_1 + \sum_{i<m}\nabla\delta V_i$ and using $\sum_m \Delta b_m(j) = 0$ and fact (5), $A_4 = \sum_{i\ge1}\nabla\delta V_i(n-1)u_i^-(j)$, which is at most $Cf_+(N)/N$. In the same way $A_6 = \sum_{i\ge1}\delta V_i(n-2)\Delta a_i(j)$, which has the form of $A_2$. Summation by parts with $\delta V_m = \delta V_1 + \sum_{i<m}\delta^2 V_i$ gives $A_2 = \delta V_1(n-2)\Delta u(j) + \sum_{i\ge1}\delta^2 V_i(n-2)\Delta\bar{A}_i(j)$. By Lemma 6.10(iii), Lemma 6.5, Lemma B.3(iii) and fact (2), $A_2 = O(f_+(N)T_-(j)) + O(f_+(N)/N)$.

For $A_1$ write $Q_m = Q_1 + (m-1)\delta Q_1 + \sum_{i=1}^{m-2}(m-1-i)\delta^2 Q_i$ and sum by parts twice. With $\sum_m a_m = u$, $\sum_m (m-1)a_m = v - u$ and

$$W_i(j) = \sum_{m\ge i+2}(m-1-i)\,a_m(j) = \sum_x u_{i+1}^-(x)\,\tilde{v}(j-x),$$

this gives

$$A_1 = Q_1(n-2)\Delta^2 u(j) + \delta Q_1(n-2)\Delta^2(v-u)(j) + \sum_{i\ge 1}\delta^2 Q_i(n-2)\Delta^2 W_i(j).$$

Here $Q_1(n-2) = q_2(N-j-2) = T_+(N) + O((j+1)f_+(N))$ by Lemma 6.10(iv), and $(j+1)|\Delta^2 u(j)| \le CT_-(j)$ by Lemma B.2. So the first term is $T_+(N)\Delta^2 u(j) + O(f_+(N)T_-(j))$. In the second, $\delta Q_1 = O(f_+(N))$ by Lemma 6.10(iv) and $\Delta^2(v-u)(j) = O(T_-(j))$ by Lemma B.2. In the third, $|\delta^2 Q_i(n-2)| \le Cf_+(N)/N$ by Lemma B.3(iv), and $\sum_i |\Delta^2 W_i(j)| \le \sum_x u(x)|\Delta^2\tilde v(j-x)| \le \|\Delta^2\tilde v\|_1$. Adding up gives the theorem for a plus start.

For a minus start $Q_m = q^+_m$ and $b_m = q^-_{m+1}$. The index of every $\delta V$ and $\delta^2 Q$ above drops by one, which Lemmas 6.10 and B.3 allow, since they include $m = 0$. Now $Q_1 = T_+$, $\sum_m \Delta b_m(j) = f_-(j)$ and $\sum_m \Delta^2 b_m(j) = \Delta f_-(j)$. This adds the terms $\nabla V_1(n-1)f_-(j) = O(f_+(N)/N)$ to $A_4$ and $V_1(n-2)\Delta f_-(j) = O(f_+(N)T_-(j))$ to $A_6$. The rest is unchanged. □

**Corollary B.5** (convexity near the plus end)**.** *Assume the hypotheses of Theorem B.4 with $\eta < 1 - 1/\alpha_-$. Let $J_0$ be such that $f_- > 0$ and $\Delta^2 u_- > 0$ on $[J_0,\infty)$, which exists by (LR) and Lemma B.2. For every $K$ and all large $N$, $D_N$ is strictly increasing on $J_0 \le j \le K\Lambda_N$.*

*Proof.* By Lemma B.2 there is $c > 0$ with $\Delta^2 u_-(j) \ge cf_-(j)$ for $j \ge J_0$. Since $f_+(N) \le CT_+(N)/N$, it suffices that $Nf_-(j)/T_-(j) \to \infty$ and $N^2 f_-(j) \to \infty$ uniformly on $[J_0, K\Lambda_N]$, which lies in $[1, N^{1-\eta}]$ for large $N$. The first ratio is at least $cN/j \ge cN/(K\Lambda_N)$, by (LR). For the second, Lemma 2.1(ii) gives $f_-(j) \ge cf_-(K\Lambda_N)$ on the range, and $N^2 f_-(K\Lambda_N) \ge cN\cdot NT_-(K\Lambda_N)/(K\Lambda_N) \ge cK^{-\alpha_- -1}N/\Lambda_N$ for large $N$. Both tend to infinity since $\Lambda_N = o(N)$. □

*Proof of Theorem G(iii).* For large $N$ the window lies in $[J_0, (x_*+1)\Lambda_N]$. Apply Corollary B.5 with $K = x_* + 1$. By Theorem 6.12(d) $D_N$ changes sign in the window, and a strictly increasing sequence changes sign at most once and vanishes at most once. (EQ) was not used. □

### B.3 Fourier bounds and truncated sums

In this subsection we fix a sign and drop it, and we assume (RV) and (SPAN). So $f$, $T$, $\mu$, $\alpha$, $a$, $S_m$ and $u_m = f^{*m}$ refer to that sign, and $\varphi(t) = \mathbb{E}e^{itX}$. We use three elementary bounds. First, for all $m \ge 1$ and $y > 0$,

$$\mathbb{P}(S_m - m\mu \ge y) \le C\, mT(y). \tag{B.4}$$

If $mT(y/8) \ge 1$ this holds since $T(y/8) \le C_8 T(y)$. Otherwise either some summand is at least $y/8$, which has probability at most $mT(y/8)$, or none is, and Lemma 4.3 with $w = y/8$ bounds this case by $e^{C_K}(mT(y/8))^4$. Second, for $m \ge 1$ and $y \ge 0$,

$$u_m(m\mu - y) \le \frac{C}{a(m)}\, e^{-y/(2a(m))}. \tag{B.5}$$

For $m = 1$ the left side vanishes unless $y < \mu$. For $m \ge 2$ write $u_m = u_{m_1} * u_{m_2}$ with $m_1 = \lfloor m/2\rfloor$. One of the two blocks lies at least $y/2$ below its mean, so the left side is at most $\sup u_{m_2}\mathbb{P}(S_{m_1} \le m_1\mu - y/2) + \sup u_{m_1}\mathbb{P}(S_{m_2} \le m_2\mu - y/2)$, and Lemma A.1(ii) and (iv) conclude, since $a(m_i) \le a(m) \le Ca(m_i)$. Third, $\mathbb{E}|S_m - m\mu| \le Ca(m)$, since by (B.4) and (2.3) $\mathbb{E}(S_m - m\mu)_+ \le a(m) + CmC_K a(m)T(a(m))$, and the negative part has the same mean.

We also need $1-|\varphi(t)| \ge cT(1/|t|)$ for $|t| \le \pi$. For small $|t|$ this follows from $1-|\varphi(t)|^2 = \mathbb{E}[1-\cos t(X-X')] \ge \frac{1}{4}t^2\mathbb{E}[(X-X')^2; |X-X'| \le 1/|t|]$, with $X'$ an independent copy, and from $\mathbb{P}(X-X' \in (y/2, y]) \ge f(n_0)\mathbb{P}(X - n_0 \in (y/2, y]) \ge cT(y)$ for any $n_0$ with $f(n_0) > 0$. For the other $t$ it follows from (SPAN), since $|\varphi| < 1$ on $0 < |t| \le \pi$.

**Lemma B.6** (truncated sums)**.** *For $w \ge 1$ let $u_m^{(w)}(x) = \mathbb{P}(S_m = x,\ \max_{i\le m} X_i < w)$.*

*(a) For all $m \ge 1$, $w \ge 1$ and integers $x$ with $x - m\mu \ge y \ge 2w$, $u_m^{(w)}(x) \le Ca(m)^{-1}(mT(w))^{y/(4w)}$ whenever $mT(w) \le 1$.*

*(b) If $mT(w) \le 1$, then $\sup_x |\nabla u_m^{(w)}(x)| \le Ca(m)^{-2}$.*

*Proof.* (a) For $m = 1$ the left side vanishes, since $x \ge y + \mu > w$. For $m \ge 2$ split $m = m_1 + m_2$ as above. One of the two blocks exceeds its mean by at least $y/2$, so $u_m^{(w)}(x) \le \sum_{i=1,2} \sup u_{m_{3-i}} \cdot \mathbb{P}(S_{m_i} - m_i\mu \ge y/2,\ \max < w)$, and Lemma A.1(ii) and Lemma 4.3 give the bound.

(b) The generating function of $u_m^{(w)}$ is $\varphi_w^m$ with $\varphi_w(t) = \sum_{n<w} f(n)e^{itn}$, and $|\varphi_w| \le |\varphi| + T(w)$. Hence $|\varphi_w(t)|^m \le e^{mT(w)}e^{-cmT(1/|t|)} \le e \cdot e^{-cmT(1/|t|)}$ and

$$|\nabla u_m^{(w)}(x)| \le \frac{e}{2\pi}\int_{-\pi}^{\pi} |t|\, e^{-cmT(1/|t|)}\, dt.$$

For $m \ge m_0$ substitute $t = s/a(m)$. By (B.2) with $y = 1/s$, $mT(a(m)/s) \ge cs^{\alpha-\varepsilon'}$ for $1 \le s \le a(m)/x_0$, and for $|t| \ge 1/x_0$ the integrand is at most $e^{-cmT(x_0)}$. So the integral is at most $Ca(m)^{-2}$. For $m < m_0$ the integral is bounded and $a(m)$ is bounded. □

**Lemma B.7** (local limits for differences)**.** *Let* k *be a kernel of the fixed sign, that is $u_m$, $q_m$ or $q_{m+1}$, with weight $c_{\mathsf{k}}$ as in Appendix A.3, and let $r \in \{1, 2\}$. Then*

$$\sup_{j\in\mathbb{Z}}\Big|a(m)^{1+r}\Delta^r \mathsf{k}_m(j) - c_{\mathsf{k}}\, g^{(r)}\Big(\frac{j-m\mu}{a(m)}\Big)\Big| \to 0 \quad (m \to \infty),$$

*and $\sup_{m\ge1}\sup_j a(m)^{1+r}|\Delta^r \mathsf{k}_m(j)| < \infty$.*

*Proof.* This is Gnedenko's proof [36, Theorem 4.2.1] with one more factor. We have $\Delta^r u_m(j) = (2\pi)^{-1}\int_{-\pi}^{\pi}\varphi(t)^m(e^{-it}-1)^r e^{-itj}\,dt$. On $\varepsilon \le |t| \le \pi$, (SPAN) gives $|\varphi| \le e^{-c_\varepsilon}$, so this range contributes $O(e^{-c_\varepsilon m})$. On $|t| < \varepsilon$ put $t = \theta/a(m)$ and $y = (j - m\mu)/a(m)$. The integrand becomes $a(m)^{-1-r}(e^{-i\theta\mu/a(m)}\varphi(\theta/a(m)))^m(a(m)(e^{-i\theta/a(m)}-1))^r e^{-i\theta y}$. The first factor tends to $\exp(K_\alpha(-i\theta)^\alpha)$ by Lemma A.1(i) and the second to $(-i\theta)^r$, with $|a(m)(e^{-i\theta/a(m)}-1)| \le |\theta|$. As in Lemma B.6(b), the modulus of the first factor is at most $\exp(-c|\theta|^{\alpha-\varepsilon'})$ for $1 \le |\theta| < \varepsilon a(m)$. Dominated convergence, uniform in $y$ since $|e^{-i\theta y}| = 1$, gives the limit $(2\pi)^{-1}\int e^{K_\alpha(-i\theta)^\alpha}(-i\theta)^r e^{-i\theta y}d\theta = g^{(r)}(y)$. For $q_m$ with $m \ge 2$, $q_m(j) = \sum_{v\ge1} T(v)u_{m-1}(j-v)$, so the integrand carries the extra factor $\hat\tau(t) = \sum_{v\ge1} T(v)e^{itv}$, which is continuous, bounded by $\mu$ and equal to $\mu$ at 0. Since $a(m-1)/a(m) \to 1$ and $g^{(r)}$ is uniformly continuous, the same proof applies. The kernel $q_{m+1}$ is the same. The uniform bound follows from the domination for large $m$ and from $|\Delta^r \mathsf{k}_m| \le 2^r \max(1, \mu)$ for the others. □

### B.4 Proof of Lemma 6.18

We write $f$, $T$, $\mu$, $\alpha$, $a$, $V_m$ for the plus sign as in the lemma, and $u_m^{(w)}$ as in Lemma B.6. By (LR), $f(L) \ge cT(L)/L$ for large $L$, so by (B.1), with $r = L/a(m)$,

$$m\,\frac{f(L)}{L} \ge c\,r^{-\alpha-2-\varepsilon'}a(m)^{-2}, \qquad mT(\delta L) \le C_\delta r^{-\alpha+\varepsilon'}. \tag{B.6}$$

*The first difference.* For $m = 1$ the claim is Lemma B.1, since $L = n - \mu$ and $f(n)/n \sim f(L)/L$. Let $m \geq 2$. Fix $\delta \in (0, 1/16]$ so small that $(\alpha - \varepsilon')/(32\delta) > \alpha + 2 + \varepsilon'$, and put $w = \lceil \delta L \rceil$, so $w \leq 2\delta L \leq L/8$ for large $L$.

Step 1, no summand of size at least $w$. Put $m_1 = \lfloor m/2 \rfloor$, $m_2 = m - m_1$, $g_i = u_{m_i}^{(w)}$ and $Y = \lceil m_1\mu + L/2 \rceil - 1$. Summation by parts in the sum over $y \leq Y$ gives the exact identity

$$\nabla u_m^{(w)}(n) = \sum_{y \leq Y} \nabla g_1(y)\, g_2(n-y) + g_1(Y)\, g_2(n-1-Y) + \sum_{y > Y} g_1(y)\, \nabla g_2(n-y).$$

In the first sum $n - y - m_2\mu \geq L/2$, and in the last $y - m_1\mu \geq L/2$. By Lemma B.6(b) and Lemma 4.3 these two sums are at most $Ca(m)^{-2}(mT(w))^{L/(4w)}$. In the middle term $Y - m_1\mu \geq L/4 \geq 2w$, so by Lemma B.6(a) and Lemma A.1(ii) it is at most $Ca(m)^{-2}(mT(w))^{L/(16w)}$. Since $L/(16w) \geq 1/(32\delta)$ and $mT(w) \to 0$ as $r \to \infty$ by (B.6), the whole is at most $Ca(m)^{-2}(C_\delta r^{-\alpha+\varepsilon'})^{1/(32\delta)}$, which is $o(mf(L)/L)$ as $r \to \infty$ by (B.6) and the choice of $\delta$.

Step 2, a giant summand. Let $I$ be the first index with $X_I^+ \geq w$, and let $h_i(z)$ be the probability that $X_l^+ < w$ for $l < i$ and that the sum of the $X_l^+$ with $l \leq m$, $l \neq i$, equals $z$. Then

$$\nabla \mathbb{P}(S_m^+ = \cdot,\, I = i)(n) = \sum_{z \leq n-1-w} h_i(z) \nabla f(n-z) - h_i(n-w) f(w).$$

Fix $\varepsilon_1 > 0$. On the main range $|z - (m-1)\mu| \leq \varepsilon_1 L$ we have $n - z = L + \mu - (z - (m-1)\mu)$, so Lemma B.1 and Lemma 2.1(i) give $\nabla f(n-z) = (\alpha+1)(f(L)/L)(1+\theta_1)$ with $|\theta_1| \leq \omega(\varepsilon_1) + o(1)$ as $L \to \infty$, where $\omega(\varepsilon_1) \to 0$. The mass of the main range lies between $(1 - T(w))^{i-1} - p_m$ and $(1 - T(w))^{i-1}$, with $p_m = \mathbb{P}(|S_{m-1}^+ - (m-1)\mu| > \varepsilon_1 L)$. Off the main range $n - z \geq w \geq \delta L$, so $|\nabla f(n-z)| \leq C_\delta f(L)/L$ by (LD) and Lemma 2.1(ii), and the mass is at most $p_m$. By (B.4) and Lemma A.1(iv), $p_m \leq CmT(\varepsilon_1 L) + e^{C_-} e^{-\varepsilon_1 L/a(m)}$, which tends to 0 as $r \to \infty$, uniformly in $m$, by (B.1). Next, $\sum_{i \leq m} (1 - T(w))^{i-1} = m(1 + O(mT(w)))$. Finally the deviation of $n - w$ from $(m-1)\mu$ is at least $L/2$, so $h_i(n - w) \leq u_{m-1}^+(n-w) \leq CmT(L)/a(m)$ by Lemma A.1(iii), and $f(w) \leq C_\delta f(L)$. Summed over $i \leq m$ these terms are at most $Cm^2T(L)f(L)/a(m)$, which relative to $mf(L)/L$ is $CmT(L)L/a(m) \leq Cr^{1-\alpha+\varepsilon'}$. Adding Steps 1 and 2, $\nabla V_m(n) = (\alpha+1)mf(L)/L\,(1+\theta_2)$ with $|\theta_2| \leq C\omega(\varepsilon_1) + o(1)$ as $r \to \infty$, uniformly in $m$. Choose $\varepsilon_1$ and then $K$.

*The bound on $V_m(n)$.* The same decomposition without differences gives

$$V_m(n) \leq u_m^{(w)}(n) + m \sup_{y \geq w} f(y).$$

By Lemma B.6(a) the first term is at most $Ca(m)^{-1}(mT(w))^{L/(4w)}$, which is $o(mf(L))$ by the same comparison. The second is at most $C_\delta mf(L)$.

*The difference in the index.* We have $\delta V_m(n) = \sum_x f(x)[V_m(n-x) - V_m(n)]$. Fix $\varepsilon_1, \varepsilon_2 > 0$. For $x \leq \varepsilon_1 L$ the bracket is $\sum_{i<x} \nabla V_m(n-i)$, and the first bound at the deviations $L - i \geq (1 - \varepsilon_1)L$, with Lemma 2.1(i), makes it $x(\alpha+1)mf(L)/L$ up to a factor $1 + \theta$ with $|\theta| \leq \varepsilon + \omega(\varepsilon_1)$. Since $\sum_{x \leq \varepsilon_1 L} x f(x) \to \mu$, this part is $\mu(\alpha+1)mf(L)/L$ up to the same factor and $o(1)$. The term $-V_m(n) \sum_{x > \varepsilon_1 L} f(x)$ is at most $Cmf(L)T(\varepsilon_1 L) = o(mf(L)/L)$. The rest is $\mathbb{E}[f(n - S_m^+);\, n - S_m^+ > \varepsilon_1 L]$. On $|S_m^+ - m\mu| \leq \varepsilon_2 L$ the integrand is $f(L)(1 + O(\omega(\varepsilon_2)))$, and this event has probability $1 - o(1)$ as $r \to \infty$. On $S_m^+ - m\mu > \varepsilon_2 L$ the integrand is at most $C_{\varepsilon_1} f(L)$, with probability at most $CmT(\varepsilon_2 L)$. On $S_m^+ - m\mu < -\varepsilon_2 L$ it is at most $Cf(L)$, with probability at most $e^{C_-} e^{-\varepsilon_2 r}$. So the rest is $f(L)(1 + o(1))$ up to $O(\omega(\varepsilon_2))f(L)$.

*The difference of the $q$ kernels.* We have $q_{m+2}(n) - q_{m+1}(n) = \sum_x f(x)[q_{m+1}(n - x) - q_{m+1}(n)]$, with $q_{m+1} = 0$ at nonpositive arguments. For $x \leq \varepsilon_1 L$ the bracket is $\sum_{l=1}^{x} \delta V_m(n-l)$, and the previous step makes this part $\mu[f(L) + \mu(\alpha+1)mf(L)/L]$

up to small relative errors. For $m = 0$ use $\delta V_0(y) = f(y)$ instead. The term $-q_{m+1}(n)\sum_{x>\varepsilon_1 L} f(x)$ is at most $C(T(L)+mf(L))T(\varepsilon_1 L)$, which is $o(f(L)+mf(L)/L)$. The rest equals $\mathbb{E}\sum_{y=S^+_{m+1}+1}^{S^+_{m+1}} f(n-y)\mathbf{1}_{\{n-y>\varepsilon_1 L\}}$. On the event that $|S^+_m - m\mu| \le \varepsilon_2 L$ and $X^+_{m+1} \le \varepsilon_2 L$ the sum is $X^+_{m+1} f(L)(1+O(\omega(2\varepsilon_2)))$, and the expectation of $X^+_{m+1}$ on this event tends to $\mu$, by independence. On the complement the sum is at most $C_{\varepsilon_1} f(L) X^+_{m+1}$, whose expectation there is at most $C_{\varepsilon_1} f(L)(\mathbb{E}[X_+; X_+ > \varepsilon_2 L] + \mu\mathbb{P}(|S^+_m - m\mu| > \varepsilon_2 L)) = o(f(L))$. Adding the parts gives the claim.

*The bound on $q_{m+1}$.* Split $q_{m+1}(n) = \sum_x V_m(x)T(n-x)$ at $x = n - L/2$. The lower part is at most $T(L/2) \le CT(L)$. In the upper part the deviation of $x$ is at least $L/2$, so the bound on $V_m$ gives $V_m(x) \le Cmf(L)$, and the weights $T(n-x)$ add up to at most $\mu$. □

### B.5 The big-jump zone

From here on we assume $\alpha_+ = \alpha_- = \alpha$, fix $\theta_0 > 0$, and consider only the $N$ with $\min_\sigma \theta_{\sigma,N} \ge \theta_0$. "For all large $N$" means for all large such $N$, and constants may depend on $\theta_0$. Under these conditions $a_+(m)$ and $a_-(m)$ are comparable, uniformly in $m \in [\varepsilon N, 2N]$ for every fixed $\varepsilon > 0$. Indeed $a_+/a_-$ is slowly varying, its value at $N/s$ lies in $[\theta_0, 1/\theta_0]$, and Lemma 2.1(i) transfers this to $m \in [\varepsilon N, 2N]$.

Put $L = L_{N-j} = N - js/\mu_-$, $i_0 = j/\mu_-$ and

$$\mathcal{D}_N(j) = \frac{2s}{\mu_-^2} f_+(L) + \frac{js^2}{\mu_-^3}(\alpha+1)\frac{f_+(L)}{L}.$$

For a plus start this is $(1+o(1))$ times the derivative in $j$ of the kernel (4.3), by $\gamma(j) \sim js/\mu_-^2$ and $u_-(j) \to 1/\mu_-$.

**Theorem B.8** (a difference form of Theorem B(b))**.** *Assume (SPAN) and (LR) for both signs and (LD2) for the plus sign, and the conditions above. Let the start be fresh with either sign or fair, and fix $\eta \in (0, 1-1/\alpha)$. Then*

$$\lim_{K\to\infty} \limsup_{N\to\infty} \sup_{N^{1-\eta} \le j \le (1-p)N - KB_N} \Big| \frac{D_N(j)}{\mathcal{D}_N(j)} - 1 \Big| = 0.$$

*Proof.* Take the plus start and write $n = N - j$. In the range of the theorem $L \ge KB_N$. The summations by parts in the proof of Theorem 6.11 are exact for every $j \ge 1$, and give $D_N = \mathrm{I} + \mathrm{II} + \mathrm{III} + \mathrm{IV}$ with

$$\mathrm{I} = Q_1(n-1)\Delta u(j) + \sum_{i\ge 1} \delta Q_i(n-1)\Delta \bar{A}_i(j), \qquad \mathrm{II} = \sum_m u^-_m(j)\,\delta V_m(n-1),$$
$$\mathrm{III} = \sum_{i\ge 1} \delta V_i(n-1)\,u^-_i(j), \qquad \mathrm{IV} = \sum_m \nabla V_m(n)\,q^-_m(j),$$

where $u = u_-$, $Q_m = q_{m+1}$ and $\delta Q_i = q_{i+2} - q_{i+1}$. Only $m \le j+1$ occurs. Put $L_m = n - 1 - m\mu_+$, so $L_{i_0} = L - 1$. Fix $M \ge 2$, then $K$, then $N$, and write $I = I^-_j(M)$ for the index window of Lemma 3.1.

The window. For $m \in I$, $|m - i_0| \le Ma_-(j/\mu_-)/\mu_- \le CMB_N$, so $L_m = L(1+O(M/K))$. Also $m = i_0(1+o(1))$, since $a_-(j) = o(j)$ and $j \ge N^{1-\eta}$. Since $a_+(m) \le a_+(N) \le CB_N$, we have $L_m \ge Ka_+(m)/C$. So Lemma 6.18 and Lemma 2.1(i) give, uniformly in $m \in I$,

$$\nabla V_m(n) = (\alpha+1)m\frac{f(L)}{L}(1+o(1)), \quad \delta V_m(n-1) = \Big[f(L) + \mu_+(\alpha+1)i_0\frac{f(L)}{L}\Big](1+o(1)),$$

and $\delta Q_m(n-1) = [2\mu_+ f(L) + \mu_+^2(\alpha+1)i_0 f(L)/L](1+o(1))$, where $f = f_+$ and $o(1)$ tends to 0 as $N \to \infty$ and then $K \to \infty$.

The other indices with $m \le m_* = i_0 + L/(2\mu_+)$. For them $L_m \ge L/3$, so Lemma 6.18 gives $|\nabla V_m(n)| \le Cmf(L)/L$ and $|\delta V_m(n-1)|$, $|\delta Q_m(n-1)| \le C(f(L) + i_0 f(L)/L)$, using $m \le j+1 \le Ci_0$. By Lemma 3.1(a) the masses $\sum_{m\notin I} u_m^-(j)$ and $\sum_{m\notin I} q_m^-(j)$ are at most $\eta_-(M) + o(1)$. Moreover $\sum_{m\notin I} m q_m^-(j) \le i_0(2\eta_-(M) + o(1))$. For $m \le 2i_0$ this follows from the masses, and $\sum_{m>2i_0} m q_m^-(j) \le (j+1)\mathbb{P}(J_-(j) > 2i_0)$ is at most $(j+1)e^{C_-}e^{-cj/a_-(j)} = o(1)$ by Lemma A.1(iv).

The bad indices $m_* < m \le j+1$. For them $m\mu_- - j \ge \mu_- L/(2\mu_+) = c_* L$, a left deviation of the minus runs. By (B.5), $u_m^-(x) \le Ca_-(m)^{-1}e^{-c_* L/(2a_-(m))}$ for $x \le j$, and $q_m^-(j) = \sum_v T_-(v) u_{m-1}^-(j-v)$ obeys the same bound with $c_* L/2$ in place of $c_* L$. For the plus factors we use the crude bounds of Lemma B.7, $|\nabla V_m| \le Ca_+(m)^{-2}$, $|\delta V_m| \le \mu_+ \sup|\nabla V_m|$ and $|\delta Q_m| \le \mu_+ \sup|\nabla q_{m+1}| \le Ca_+(m)^{-2}$. Bad indices exist only if $m_* < j+1$, which forces $L \le Cj$ and so $j \ge cN$. Then $m \in [cN, N]$, and $a_+(m)$, $a_-(m)$ are comparable to $A = a_-(j)$. Summing the exponential bound over $m$ gives at most $Ce^{-cL/A}$ for the minus masses, so every bad contribution is at most $CA^{-2}e^{-cL/A}$. With $\rho = L/A \ge cK$ we have $\mathcal{D}_N(j) \ge ci_0 f(L)/L \ge c\rho^{-\alpha-2-\varepsilon'}A^{-2}$ by (B.1), since $i_0 T_+(A)$ is bounded below. So the bad contributions are at most $C\rho^{\alpha+2+\varepsilon'}e^{-c\rho}\mathcal{D}_N(j)$, which is small as $K \to \infty$.

The four parts. By the window, the masses and $\sum_m m q_m^-(j) = \mathbb{E}J_-(j) = \sum_{x<j} u(x) = i_0(1+o(1))$,

$$\mathrm{IV} = (\alpha+1)\, i_0\, \frac{f(L)}{L}\big(1 + O(\eta_-(M)) + o(1)\big),$$

$$\mathrm{II},\ \mathrm{III} = \frac{1}{\mu_-}\Big[f(L) + \mu_+(\alpha+1)i_0\frac{f(L)}{L}\Big]\big(1 + O(\eta_-(M)) + o(1)\big),$$

where we used $\sum_{m\in I} u_m^-(j) = u(j) + O(\eta_-(M)) + o(1)$ and $u(j) \to 1/\mu_-$. In I, $Q_1(n-1) = q_2(n-1) \le CT_+(n) \le CT_+(L)$ and $|\Delta u(j)| \le CT_-(j)$, so the first term is at most $CLT_-(j)f(L) \le CNT_-(N^{1-\eta})f(L) = o(f(L))$, since $(1-\eta)\alpha > 1$. For the sum let $i_* = \lfloor i_0 \rfloor \in I$ and write it as $\delta Q_{i_*}(n-1)\Delta(v-u)(j) + \sum_i (\delta Q_i - \delta Q_{i_*})(n-1)\Delta\bar{A}_i(j)$, using $\sum_i \bar{A}_i = v - u$. Here $\Delta(v-u)(j) \to 1/\mu_-^2$ by the proof of Theorem 6.11. By fact (2) of that proof, $\Delta\bar{A}_i(j) = \sum_{x<j} u_i^-(x) d_{j+1-x} + u_i^-(j)u(1)$. Fix $R'$. The terms with $j - x > R'$ contribute at most $(\sum_{r>R'}|d_r|)\sup_{x\le j}\sum_i u_i^-(x)(|\delta Q_i| + |\delta Q_{i_*}|)$, and the index classes at the level $x$ bound the last sum by $C(f(L) + i_0 f(L)/L) + o(\mathcal{D}_N(j))$. For the terms with $j - x \le R'$, the indices $i \in I_x^-(M)$ satisfy $|\delta Q_i - \delta Q_{i_*}| = o(f(L) + i_0 f(L)/L)$ by the window, since $|x - j| \le R'$, and the other indices carry mass at most $\eta_-(M) + o(1)$. Hence

$$\mathrm{I} = \frac{1}{\mu_-^2}\Big[2\mu_+ f(L) + \mu_+^2(\alpha+1)i_0\frac{f(L)}{L}\Big] + O\Big(\eta_-(M) + \sum_{r>R'}|d_r| + o(1)\Big)\Big(f(L) + i_0\frac{f(L)}{L}\Big).$$

Adding the four parts, the coefficient of $f(L)$ is $2\mu_+/\mu_-^2 + 2/\mu_- = 2s/\mu_-^2$ and that of $(\alpha+1)i_0 f(L)/L$ is $(\mu_+ + \mu_-)^2/\mu_-^2 = s^2/\mu_-^2$. So $D_N(j) = \mathcal{D}_N(j)(1 + O(\eta_-(M) + \sum_{r>R'}|d_r|) + o(1))$, since $\mathcal{D}_N(j) \ge c(f(L) + i_0 f(L)/L)$. Let $N \to \infty$, then $K \to \infty$, then $M, R' \to \infty$.

For a minus start the indices shift by one. The extra term $V_1(n-1)f_-(j)$ in III is $O(f_+(L)f_-(j)) = o(f(L))$, and $\sum_m m q_{m+1}^-(j) = \mathbb{E}J_-(j) - 1$. The leading terms are the same, and the fair start follows by linearity. □

*Proof of Theorem G(iv).* Choose $K_0$ such that $|D_N/\mathcal{D}_N - 1| \le \frac{1}{2}$ on $N^{1-\eta} \le j \le (1-p)N - K_0 B_N$ for all large $N$. Then $D_N > 0$ there. On $(x_* + \eta')\Lambda_N \le j \le N^{1-\eta}$ we have $D_N > 0$ by Theorem 6.12(c). At $j = N^{1-\eta}$ both statements hold, and $\mathcal{D}_N \sim c_{\mathrm{a}} T_+(N)/N$ matches the limit in Theorem 6.11. □

The proof uses the comparability of the scales only for the bad indices, which occur only for $j \ge cN$. Without it the same bound needs $e^{-cL/a_-(j)}$ to beat $(a_-(j)/a_+(j))^2$, which fails near the window of Theorem 4.14.

### B.6 The bulk

We keep the conditions of Section B.5. Fix a sign and drop it, as in Section B.3, put $A = a(j/\mu)$ and let $I_j(K)$ be the index window of Lemma 3.1.

**Lemma B.9** (a pointwise bound)**.** *Assume (RV) and (SPAN), and let $r \in \{1,2\}$. For $m \geq 1$ and integers $x$ put $z = x - m\mu$. Then*

$$|\Delta^r u_m(x)| \leq Ca(m)^{-1-r} \min\{1, mT(z/4)\} \qquad (z \geq 0),$$
$$|\Delta^r u_m(x)| \leq Ca(m)^{-1-r} e^{-|z|/(8a(m))} \qquad (z < 0),$$

*and $\sum_x |\Delta u_m(x)| \leq C/a(m)$.*

*Proof.* The bound with $\min\{1,\cdot\} = 1$ is Lemma B.7. So for $z \geq 0$ we may assume $mT(z/4) \leq 1$ and $z \geq 24$, and for $z < 0$ we may assume $|z| \geq 24\max(1, a(m))$. For $m \leq 2$ the claims are direct, since $|\Delta^r u_1(x)| \leq 4T(x)$, $u_2(y) \leq 2T(y/2)$, and $x \geq m$ gives $|z| \leq 2\mu$ when $z < 0$.

Let $r = 1$, $m \geq 2$ and $z \geq 24$. Write $u_m = u_{(1)} * u_{(2)}$ with block sizes $m_1 = \lfloor m/2 \rfloor$ and $m_2 = m - m_1$, means $\nu_i = m_i\mu$, and put $c = \lceil \nu_1 + z/2 \rceil$. Then $\Delta u_m(x) = \sum_{x_1} u_{(1)}(x_1)\Delta u_{(2)}(x - x_1)$. The part $x_1 \geq c - 1$ is at most $\sup|\Delta u_{(2)}|\mathbb{P}(S_{m_1} - \nu_1 \geq z/2 - 1) \leq Ca(m)^{-2}mT(z/3)$, by Lemma B.7 and (B.4). On the part $x_1 \leq c-2$ summation by parts gives

$$-\sum_{x_1 \leq c-2} \Delta u_{(1)}(x_1 - 1)\, u_{(2)}(x - x_1) + u_{(1)}(c-2)\, u_{(2)}(x - c + 1).$$

The sum is at most $\sup|\Delta u_{(1)}|\mathbb{P}(S_{m_2} - \nu_2 \geq z/2)$, and in the last term $u_{(1)}(c-2) \leq Cm_1T(z/3)/a(m)$ by Lemma A.1(iii) and $u_{(2)} \leq C/a(m)$.

Let $r = 2$, $m \geq 3$ and $z \geq 24$. Use three blocks $u_{(1)}, u_{(2)}, u_{(3)}$ of sizes $\lfloor m/3 \rfloor$ or $\lceil m/3 \rceil$, with means $\nu_i$, and cuts $c_i = \lceil \nu_i + z/3 \rceil$ for $i = 1, 2$. Split $u_m(x)$ into the parts with $x_1 \geq c_1$, with $x_1 < c_1$ and $x_2 \geq c_2$, and with $x_1 < c_1$ and $x_2 < c_2$, where $x_i$ is the value of block $i$. The second difference of the first part is at most $\mathbb{P}(S_{(1)} - \nu_1 \geq z/3)\sup|\Delta^2 u_{(2)} * u_{(3)}| \leq CmT(z/4)a(m)^{-3}$, and the second part is the same. In the third part block 3 lies above $\nu_3 + z/3 - 2$. Put $H(y) = \sum_{x_2 < c_2} u_{(2)}(x_2)u_{(3)}(y - x_2)$ and $F(x) = \sum_{x_1 < c_1} u_{(1)}(x_1)H(x - x_1)$. Summation by parts as for $r = 1$ gives

$$\Delta^2 F(x) = -\sum_{x_1 < c_1} \Delta u_{(1)}(x_1 - 1)\, \Delta H(x - x_1) + u_{(1)}(c_1 - 1)\, \Delta H(x - c_1),$$

$$\Delta H(y) = -\sum_{x_2 < c_2} \Delta u_{(2)}(x_2 - 1)\, u_{(3)}(y - x_2) + u_{(2)}(c_2 - 1)\, u_{(3)}(y - c_2).$$

Expanding, each term is a product of factors of three kinds. Either it contains $\|\Delta u_{(1)}\|_1 \sup|\Delta u_{(2)}|$ and a tail probability of block 3 at deviation at least $z/3$, or it contains a local probability $u_{(i)}(c_i - 1)$ at deviation at least $z/3 - 1$, together with $\|\Delta u_{(1)}\|_1$ or $\|\Delta u_{(2)}\|_1$ or $u_{(1)}(c_1 - 1)$, and $\sup u_{(3)}$. By (B.4), Lemma A.1(ii) and (iii) and the bound $\|\Delta u_{(i)}\|_1 \leq C/a(m)$ proved below from the case $r = 1$, each is at most $Ca(m)^{-3}mT(z/4)$.

For $z < 0$ mirror the argument. The cuts become $\lfloor \nu_i + z/2 \rfloor$ or $\lfloor \nu_i + z/3 \rfloor$, the summations by parts run over the sums restricted from below, and (B.5) and Lemma A.1(iv) replace Lemma A.1(iii) and (B.4). The deviations are at least $|z|/4$, which gives the exponent.

For the total variation, the two-block argument for $r = 1$, mirrored when $z < 0$, gives for every $x$

$$|\Delta u_m(x)| \leq Ca(m)^{-2} \sum_{i=1,2} \mathbb{P}\big(|S_{m_i} - \nu_i| \geq |z|/2 - 3\big) + u_{(1)}(c_x)\, u_{(2)}(x - c_x + e_x),$$

with $|c_x - \nu_1 - z/2| \le 3$ and $e_x \in \{-1, 1\}$. Summing over $x$, the probabilities add up to at most $C\mathbb{E}|S_{m_i} - \nu_i| + C \le Ca(m)$. The point $c_x$ takes each value for at most three $x$, so the products add up to at most $3 \sup u_{(2)} \le C/a(m)$. □

**Lemma B.10** (second differences in the big-jump range)**.** *Assume (SPAN), (LR) and (LD2), and fix $c > 0$. There are $K_1$, $L_0$ and $C$, depending on the law and on $c$, such that for all $m \ge 1$ and all integers $n$ with $L = n - m\mu \ge \max(K_1 a(m), L_0, cm)$,*

$$|\nabla u_m(n)| \le Cm\frac{f(L)}{L}, \qquad |\nabla^2 u_m(n)| \le Cm\frac{f(L)}{L^2},$$
$$|\nabla\delta u_m(n)| \le C\frac{f(L)}{L}, \qquad |\delta^2 u_m(n)| \le C\frac{f(L)}{L}.$$

The case $c = \mu$ covers the deviations beyond the mean that the bulk needs below, the case $c = 1$ gives Lemma B.3(i) to (iii), and Section 7 uses deviations of the order of the number of summands.

*Proof.* Throughout, $L \ge cm$ gives $m \le L/c$, $mf(L) \le CT(L)$ by (LR) and $mT(\delta L) \le C_\delta LT(L)$ by Lemma 2.1(ii), while $L \ge K_1 a(m)$ gives $mT(L) \le C$ and $mT(L)L/a(m) \le C$ by (B.1), and Lemma 2.1(ii) gives $f(L') \le C_b f(L)$ for $L' \ge bL$ and $L$ large. The first bound is Lemma 6.18, which also gives $u_m(x) \le Cmf(L_x)$ and $|\delta u_m(x)| \le C(f(L_x) + mf(L_x)/L_x)$ at every point $x$ with $L_x = x - m\mu \ge Ka(m)$. We enlarge $K_1$ and $L_0$ so that every deviation used below lies in the range of these bounds, and in the range of the bounds already proved at the constant $c/2$ where a deviation of at least $L/2$ occurs. All constants may depend on $c$.

*The second bound.* For $m = 1$ it is (LD2). Let $m \ge 2$, fix $\delta > 0$ small and put $w = \lceil \delta L \rceil$. Then $mT(w) \le C_\delta LT(L) \le C_\delta L^{1-\alpha+\varepsilon'} \le 1$ for $L \ge L_0$.

No summand of size at least $w$. With $g_i$ as in the proof of Lemma 6.18, $\nabla^2 u_m^{(w)} = (\nabla g_1) * (\nabla g_2)$ exactly. For each $y$ one of the blocks exceeds its mean by at least $L/2$. Since $\sum_{y \ge Y} |\nabla g(y)| \le 2\mathbb{P}(S \ge Y - 1)$ for the law $g$ of a sum $S$, Lemma B.6(b) and Lemma 4.3 bound this part by $Ca(m)^{-2}(mT(w))^{L/(8w)} \le CL^{-(\alpha-1-\varepsilon')/(16\delta)}$. This is $o(f(L)/L^2)$ for $\delta$ small, since $f(L)/L^2 \ge L^{-\alpha-3-\varepsilon'}$ for large $L$.

A giant summand. With $I$ and $h_i$ as in that proof, $\mathbb{P}(S_m = n', I = i) = \sum_z h_i(z) f(n' - z)\mathbf{1}_{\{n'-z \ge w\}}$, and its second difference at $n$ is

$$\sum_{z \le n-2-w} h_i(z)\nabla^2 f(n-z) + h_i(n-1-w)\big(f(w+1) - f(w)\big) + f(w)\big(h_i(n-w) - h_i(n-1-w)\big).$$

The main sum is at most $C_\delta f(L)/L^2$ by (LD2), since $n - z \ge \delta L$. In the first boundary term $h_i(n-1-w) \le u_{m-1}(n-1-w) \le CmT(L)/a(m)$ by Lemma A.1(iii) and $|f(w+1) - f(w)| \le C_\delta f(L)/L$. Summed over $i$ this is at most $Cmf(L)/L^2$ times $CmT(L)L/a(m) \le C$. For the second boundary term we apply the proof of the first bound of Lemma 6.18, as an upper bound, to the law $h_i$ at the level $n - w$, with the threshold $w$ itself. The deviation is at least $L/2$, a truncated variable is never a giant summand, and the no-giant part is bounded as above. This gives $|h_i(n-w) - h_i(n-1-w)| \le Cmf(L)/L$, so the term is at most $C_\delta f(L)\, mf(L)/L$. Summed over $i$ it is $Cmf(L)/L^2$ times $CmLf(L) \le CmT(L) \le C$.

*The third bound.* Since $\delta u_m = f * u_m - u_m$ and $\nabla$ commutes with convolution,

$$\nabla\delta u_m(n) = \sum_{x \le L/2} f(x)\big[\nabla u_m(n-x) - \nabla u_m(n)\big] - \nabla u_m(n)\, T(L/2) + \sum_{y < n - L/2} f(n-y)\, \nabla u_m(y).$$

For $x \le L/2$ and $1 \le i \le x$ the point $n - i + 1$ has deviation at least $L/2$, so the second bound at the constant $c/2$ gives $|\nabla u_m(n-x) - \nabla u_m(n)| \le \sum_{i=1}^{x} |\nabla^2 u_m(n - i + 1)| \le Cxmf(L)/L^2$, and the first sum is at most $C\mu mf(L)/L^2 \le Cf(L)/L$. The middle

term is at most $Cmf(L)T(L/2)/L \le C(mT(L))f(L)/L \le Cf(L)/L$. In the last sum put $Y = \lceil n - L/2 \rceil - 1$ and sum by parts,

$$\sum_{y \le Y} f(n-y)\big[u_m(y-1) - u_m(y)\big] = \sum_{y \le Y-1} \big[f(n-y-1) - f(n-y)\big]u_m(y) - f(n-Y)u_m(Y).$$

For $y \le Y$ we have $n - y \ge L/2$, so $f(n-y) \le Cf(L)$ and, by (LD), $|f(n-y-1) - f(n-y)| \le Cf(L)/L$. Since $\sum_y u_m(y) \le 1$, the sum is at most $Cf(L)/L$. The boundary term has $Y - m\mu \ge L/2 - 1$, so $u_m(Y) \le Cmf(L)$ and the term is at most $Cmf(L)^2 \le C(mT(L)/L)f(L) \le Cf(L)/L$.

*The fourth bound.* Since $\delta^2 u_m = f * \delta u_m - \delta u_m$,

$$\delta^2 u_m(n) = \sum_{x \le L/2} f(x)\big[\delta u_m(n-x) - \delta u_m(n)\big] - \delta u_m(n)\, T(L/2) + \sum_{y < n - L/2} f(n-y)\, \delta u_m(y).$$

By the third bound at the constant $c/2$, the first sum is at most $\sum_x f(x)\, x\, Cf(L)/L \le C\mu f(L)/L$. The middle term is at most $C(f(L) + mf(L)/L)T(L) \le Cf(L)T(L) = o(f(L)/L)$, since $LT(L) \to 0$. For the last sum put $\gamma(y) = f(n-y)\mathbf{1}_{\{y < n - L/2\}}$. With $S_m$ of law $u_m$ and $X$ an independent copy of the law $f$, $\delta u_m(y) = \mathbb{P}(S_m + X = y) - \mathbb{P}(S_m = y)$, so the last sum is $\mathbb{E}[\gamma(S_m + X) - \gamma(S_m)]$. If $S_m + X < n - L/2$, then both arguments lie below $n - L/2$, and (LD) applied $X$ times gives $|\gamma(S_m + X) - \gamma(S_m)| \le CXf(L)/L$, with expectation at most $C\mu f(L)/L$. If $S_m < n - L/2 \le S_m + X$, the difference is $\gamma(S_m) \le Cf(L)$, and this event has probability $\sum_{k \ge 1} T(k)u_m(n' - k)$ with $n' = \lceil n - L/2 \rceil$. For $k \le L/4$ the point $n' - k$ has deviation at least $L/4 - 1$, so $u_m(n' - k) \le Cmf(L)$, and these terms add up to at most $C\mu m f(L) \le CT(L)$. For $k > L/4$, $T(k) \le T(L/4) \le CT(L)$, and the $u_m$ add up to at most 1. So the event has probability at most $CT(L)$, and its contribution is at most $Cf(L)T(L) = o(f(L)/L)$. If $S_m \ge n - L/2$, both values of $\gamma$ vanish. □

**Lemma B.11** (tails of the kernel differences)**.** *Assume (SPAN), (LR), (LD2), let* $\mathsf{k}$ *be one of* $u_m$, $q_m$, $q_{m+1}$, *and let* $r \in \{1, 2\}$.

*(a) For* $K \ge 2$ *and* $j \to \infty$, $\sum_{m \notin I_j(K)} |\Delta^r \mathsf{k}_m(j)| \le C(K^{1-\alpha+\delta} + e^{-cK})A^{-r} + o(A^{-r})$, *and* $\sum_m |\Delta^r \mathsf{k}_m(i)| \le Ca(i)^{-r}$ *for all* $i \ge 1$.

*(b) Fix* $\varepsilon > 0$ *and put* $m_\varepsilon = \varepsilon N/(8\mu_{\max})$. *Uniformly in levels* $x \in [\varepsilon N, N]$, $|\Delta^r \mathsf{k}_m(x)| \le Ca(N)^{-1-r}$ *for* $m \ge m_\varepsilon$. *For* $m < m_\varepsilon$, $|\Delta^r u_m(x)| \le Cmf(x)x^{-r}$, *and* $|\Delta^r q_m(x)|$ *and* $|\Delta^r q_{m+1}(x)|$ *are at most* $Cf(x)x^{1-r}(1 + m/x)$.

*(c) Let* $\mathsf{k}'$ *be a kernel of the other sign and let* $x' \in [\varepsilon N, N]$. *Then*

$$\sum_{m < m_\varepsilon} |\Delta^r \mathsf{k}_m(x)|\, \mathsf{k}'_m(x') = o(B_N^{-1-r}),$$

*and* $\sum_{m < m_\varepsilon} |\Delta \mathsf{k}_m(x)|\, |\Delta \mathsf{k}'_m(x')| = o(B_N^{-3})$ *when one of* $\mathsf{k}, \mathsf{k}'$ *is a* $u$ *kernel and the other a* $q$ *kernel.*

*Proof.* (a) for $u$. Split $m \notin I_j(K)$ into three ranges. For $m \le j/(2\mu)$ we have $L = j - m\mu \ge j/2 \ge \max(K_1 a(m), L_0, m\mu)$ for large $j$, so Lemma B.10 with $c = \mu$ gives $|\Delta^r u_m(j)| \le Cmf(j)j^{-r}$, and the range contributes at most $Cj^{2-r}f(j) \le CT(j)j^{1-r}$. This is $o(A^{-r})$, since $T(j)a(j) \to 0$ and $T(j)a(j)^2/j \to 0$ for $\alpha > 1$. For $j/(2\mu) < m < (j - KA)/\mu$ we have $a(m) \asymp A$, and Lemma B.9 bounds the terms by $CA^{-1-r}jT(z/4)$ with $z = j - m\mu \ge KA$ running through a progression of step $\mu$. By Karamata's theorem the sum is at most $CA^{-1-r}j\, KAT(KA) \le CK^{1-\alpha+\delta}A^{-r}$, since $jT(A) \to \mu$. For $m > (j + KA)/\mu$ only $m \le j + 2$ contributes, and Lemma B.9 gives at most $CA^{-1-r}\sum_{l \ge 0} e^{-(KA + l\mu - 2)/(CA)} \le CA^{-r}e^{-cK}$. The window $I_j(2)$ contributes at most $CA^{-r}$, which gives the second claim.

(a) for $q$. For $m=1$, $\Delta^r q_1(j)=\Delta^r T(j)=O(f(j))=o(A^{-r})$. For $m\ge 2$, $\Delta^r q_m(j)=\sum_{v\ge 1}T(v)\Delta^r u_{m-1}(j-v)$. Fix $v_0$. For $v\le v_0$ and large $j$, $m\notin I_j(K)$ implies $m-1\notin I_{j-v}(K/2)$, so these shifts contribute at most $\mu$ times the bound for $u$ with $K/2$. The shifts $v_0<v\le j/2$ contribute at most $(\sum_{v>v_0}T(v))CA^{-r}$, by the second claim for $u$. For the shifts $v>j/2$ we do not take absolute values inside the sum over $v$. For each $m'=m-1$ we sum by parts $r$ times in $i=j-v<j/2$, which moves the differences onto $T(j-i)$. By (LD), $|\Delta_i^r T(j-i)|\le Cf(j)j^{1-r}$ there, and $\sum_{m'}u_{m'}(i)\le 1$. The boundary terms are $T(j/2+O(1))|\Delta^{r-1}u_{m'}(\lfloor j/2\rfloor+O(1))|$ and, for $r=2$, a term $f(j/2+O(1))u_{m'}(\lfloor j/2\rfloor+O(1))$. Summed over all $m'$, by the second claim for $u$, these shifts contribute at most $CT(j)j^{1-r}+CT(j)a(j)^{1-r}+Cf(j)=o(A^{-r})$. Let $v_0\to\infty$. The kernel $q_{m+1}=u_m*\tau$ is the same.

(b) For $m\ge m_\varepsilon$ this is Lemma B.7, since $a(m)\ge a(m_\varepsilon)\ge ca(N)$, and $|\Delta^r q_m|\le \mu\sup|\Delta^r u_{m-1}|$. For $m<m_\varepsilon$ and $u$ we have $x-m\mu\ge x/2\ge\max(K_1a(m),L_0,m\mu)$ for large $N$, and Lemma B.10 with $c=\mu$ applies. For $q_m$ split $\sum_v T(v)\Delta^r u_{m-1}(x-v)$ at $v=x/2$. The part $v\le x/2$ is at most $C\mu m f(x)x^{-r}$ by the bound for $u$. For the part $v>x/2$ the summation by parts above, for the single index $m-1$, gives at most $Cf(x)x^{1-r}$ plus boundary terms. These are $T(x/2)u_{m-1}(x/2)\le CT(x)mf(x)\le Cf(x)$ for $r=1$, and $T(x/2)|\Delta u_{m-1}(x/2)|+f(x/2)u_{m-1}(x/2)\le Cf(x)x^{-1}(1+m/x)$ for $r=2$, using $mT(x)\le xT(x)\to 0$ and $xT(x)\le 1$.

(c) Write $T'$, $f'$ and $a'$ for the other sign. By Lemma A.3, $\mathsf{k}'_m(x')\le C(1+m/a'(m))T'(N)$ for $m<m_\varepsilon$. By (b) and $f(x)\le Cf(N)$, the first sum is at most

$$C\frac{f(N)}{N^r}T'(N)\sum_{m<\varepsilon N}m\Big(1+\frac{m}{a'(m)}\Big)+Cf(N)N^{1-r}T'(N)\sum_{m<\varepsilon N}\Big(1+\frac{m}{a'(m)}\Big)$$
$$\le C\frac{N^{2-r}T(N)T'(N)}{a'(N)}.$$

Both signs have the index $\alpha$, so the right side is regularly varying with the index $2-r-2\alpha-1/\alpha$, and $B_N^{-1-r}$ with the index $-(1+r)/\alpha$. The claim follows from $1-2\alpha-1/\alpha<-2/\alpha$ and $-2\alpha-1/\alpha<-3/\alpha$, which hold for $\alpha>1$. For the mixed sum, (b) bounds the product of a $u$ difference and a $q$ difference by $Cmf(N)N^{-1}\cdot Cf'(N)$, so the sum is at most $CNf(N)f'(N)\le CT(N)T'(N)/N$, whose index $-2\alpha-1$ is below $-3/\alpha$. □

**Lemma B.12** (the bell shape)**.** *Let $\alpha_+=\alpha_-=\alpha$ and $\theta\in\Theta$ with $\min_\sigma\theta_\sigma>0$. Then $h_\theta$ extends to an entire function, $h'_\theta$ has exactly one zero $x_\theta$, and $h'_\theta>0$ on $(-\infty,x_\theta)$, $h'_\theta<0$ on $(x_\theta,\infty)$ and $h''_\theta(x_\theta)<0$. On $\Theta_{\theta_0}=\{\theta\in\Theta:\min_\sigma\theta_\sigma\ge\theta_0\}$ the map $\theta\mapsto x_\theta$ is continuous, and there are $\varepsilon_0,c_0>0$ with $h''_\theta\le -c_0$ on $|x-x_\theta|\le\varepsilon_0$. For every $R$ and $\varepsilon>0$ there is $c>0$ with $|h'_\theta(x)|\ge c$ for $|x|\le R$, $|x-x_\theta|\ge\varepsilon$ and $\theta\in\Theta_{\theta_0}$.*

*Proof.* By the proof of Lemma 2.2, $W_\theta$ is the sum of two independent strictly $\alpha$-stable variables, so it is strictly $\alpha$-stable, and it is nondegenerate. Its characteristic function satisfies $|\widehat{h}_\theta(t)|=e^{-c_\theta|t|^\alpha}$ with $c_\theta>0$, so $\int e^{|t||\operatorname{Im}z|}|\widehat{h}_\theta(t)|\,dt<\infty$ for every $z\in\mathbb{C}$, and $h_\theta$ is entire. By Kwaśnicki [43, Corollary 1.3], every stable density on $\mathbb{R}$ is bell-shaped in the sense of his Definition 2.1, that is, for every $n\ge 0$ the derivative $h_\theta^{(n)}$ tends to 0 at $\pm\infty$ and changes sign exactly $n$ times.

Let $g=h_\theta^{(n)}$. It is analytic and not identically zero, so its zeros are isolated and of finite order. If $g$ has $D$ consecutive zeros, then $g'$ changes sign in each of the $D-1$ gaps between them, since on a gap $|g|$ rises from 0 and returns to 0. As $g'$ changes sign exactly $n+1$ times, $g$ has finitely many zeros, say $D$ of them. Of these, $S$ have odd order and $T_0$ even order, and $S=n$ since $g$ changes sign exactly $n$ times. The derivative $g'$ changes sign once in each gap, once in each of the two unbounded intervals, where $g\ne 0$ and $g\to 0$ at infinity, and once at each zero of even order, where $g'$ has a zero of odd order.

These are distinct places, so $n+1 \geq D+1+T_0 = n+1+2T_0$, and $T_0 = 0$. This holds for every $n \geq 1$. If a zero of $g$ had order $k \geq 3$, it would be a zero of $g'$ of even order. So all zeros of every derivative are simple. For $n = 1$, $h'_\theta$ has one simple zero, $h_\theta > 0$ tends to $0$ at $\pm\infty$, so the zero is the maximum, $h'_\theta$ has the stated signs and $h''_\theta(x_\theta) < 0$.

The map $\theta \mapsto h_\theta$ is continuous from the compact set $\Theta_{\theta_0}$ into $C^2$ with the uniform norms, by dominated convergence in the Fourier integrals with the weights $1$, $|t|$ and $t^2$, and $c_\theta$ is bounded below there. Since $x_\theta$ is the unique zero of $h'_\theta$ and $h''_\theta(x_\theta) < 0$, the implicit function theorem makes $x_\theta$ continuous. The last two claims follow from the continuity of $(\theta, x) \mapsto h''_\theta(x)$ and of $|h'_\theta(x)|$, which is positive on the compact set of the statement. □

Yamazato's unimodality of stable laws does not suffice here, since it gives neither $h'_\theta \neq 0$ away from the mode nor $h''_\theta < 0$ at the mode.

**Theorem B.13** (a local limit theorem for differences in the bulk)**.** *Assume (SPAN), (LR) and (LD2) for both signs and the conditions of Section B.5, and let the start be fresh with either sign or fair. For every $R$, uniformly in $|x_{N,k}| \leq R$,*

$$B_N^2 \Delta P_N(k) = h'_N(x_{N,k}) + o(1), \qquad B_N^3 \Delta^2 P_N(k) = h''_N(x_{N,k}) + o(1).$$

*Proof.* By linearity we take a fresh start of sign $\sigma_0$, so $P_N = F_N^{\sigma_0}$. We follow Step 5 of the proof of Theorem A(a) and keep its notation. For large $N$ the points with $|x_{N,k}| \leq R$ satisfy $\varepsilon N \leq k \leq (1-\varepsilon)N$. For one product $\phi(k) = A(k)B(N-k)$ in (3.1), with $A$ the plus kernel and $B$ the minus kernel,

$$\begin{aligned}
\Delta\phi(k) &= \Delta A(k)\, B(N-k-1) - A(k)\, \Delta B(N-k-1),\\
\Delta^2\phi(k) &= \Delta^2 A(k)\, B(N-k-2) - 2\Delta A(k)\, \Delta B(N-k-2) + A(k)\, \Delta^2 B(N-k-2).
\end{aligned}$$

So a difference on the factor of sign $\sigma$ carries the sign $\varsigma_\sigma$. Fix $K \geq 2$ and let $I = I_{k_{\bar\tau}}^{\bar\tau}(K)$.

Indices in $I$. Here $a_\tau(m) = B_N(1+o(1))$ and $a_{\bar\tau}(m) = b_{\bar\tau,N}(1+o(1))$, with $b_{\bar\tau,N} = \theta_{\bar\tau,N} B_N \geq \theta_0 B_N$. By Lemma B.7 and the uniform continuity of $g_\sigma^{(r)}$, uniformly in $m \in I$ and in shifts of the levels by at most $2$,

$$\Delta^{r_1} \mathsf{k}_m^\tau(k_\tau) = \frac{c\, g_\tau^{(r_1)}(\zeta_{N,x}(y_m)) + o(1)}{B_N^{1+r_1}}, \qquad \Delta^{r_2} \mathsf{k}_m^{\bar\tau}(k_{\bar\tau}) = \frac{c'\, g_{\bar\tau}^{(r_2)}(y_m) + o(1)}{b_{\bar\tau,N}^{1+r_2}},$$

with $c, c'$ the weights of the two kernels. The points $y_m$ form a progression of step $\mu_{\bar\tau}/b_{\bar\tau,N} \cdot (1+o(1))$ in $[-K,K]$, with at most $Cb_{\bar\tau,N}$ points. So the sum over $m \in I$ of the product is a Riemann sum,

$$\sum_{m\in I} \Delta^{r_1}\mathsf{k}_m^\tau\, \Delta^{r_2}\mathsf{k}_m^{\bar\tau} = \frac{cc'}{\mu_{\bar\tau} B_N^{1+r_1} b_{\bar\tau,N}^{r_2}} \int_{-K}^{K} g_\tau^{(r_1)}(\zeta_{N,x}(y))\, g_{\bar\tau}^{(r_2)}(y)\, dy + o(B_N^{-1-r_1-r_2}),$$

where we used $b_{\bar\tau,N} \geq \theta_0 B_N$ for the error. Extending the integral to $\mathbb{R}$ costs at most $C\int_{|y|>K} |g_{\bar\tau}^{(r_2)}|$. Put $\Phi(z,y) = g_\tau(z) g_{\bar\tau}(y)$ and $\theta = \theta_{\bar\tau,N}$. Summing over the products and over $i$ with (3.2), the main term of $B_N^{1+r}\Delta^r P_N(k)$ is

$$\frac{s}{\mu_{\bar\tau}} \int_{\mathbb{R}} \Big[\Big(\varsigma_\tau \partial_1 + \frac{\varsigma_{\bar\tau}}{\theta}\partial_2\Big)^r \Phi\Big](\zeta_{N,x}(y), y)\, dy.$$

Since $\zeta_{N,x}$ has the slope $\zeta' = \mu_\tau\theta/\mu_{\bar\tau}$, every integrable $\Psi$ with integrable derivatives satisfies $\int \partial_2\Psi(\zeta_{N,x}(y),y)dy = -\zeta' \int \partial_1\Psi(\zeta_{N,x}(y),y)dy$. Applied once or twice, this turns the operator into $(\varsigma_\tau - \varsigma_{\bar\tau}\zeta'/\theta)^r \partial_1^r = (\varsigma_\tau s/\mu_{\bar\tau})^r \partial_1^r$, since $\varsigma_{\bar\tau} = -\varsigma_\tau$. By (2.6) the result is $h_N^{(r)}(x)$.

Indices outside $I$. If the differences fall only on the $\tau$ factor, then for $m \geq m_\varepsilon$ Lemma B.11(b) and Lemma 3.1(a) give $O(\eta(K))B_N^{-1-r}$, and for $m < m_\varepsilon$ Lemma B.11(c) gives $o(B_N^{-1-r})$. If they fall only on the $\bar{\tau}$ factor, then $\mathsf{k}_m^\tau \leq C/B_N$ by (3.3), and Lemma B.11(a) at the level $k_{\bar{\tau}}$, with $a_{\bar{\tau}}(k_{\bar{\tau}}/\mu_{\bar{\tau}}) \geq c\theta_0 B_N$, gives $O(K^{1-\alpha+\delta} + e^{-cK})B_N^{-1-r}$. In the mixed term for $r = 2$ we combine Lemma B.11(b) for the $\tau$ factor with Lemma B.11(a) for $r = 1$ when $m \geq m_\varepsilon$, and use Lemma B.11(c) when $m < m_\varepsilon$. Every product in (3.1) pairs a $u$ kernel with a $q$ kernel. Letting $N \to \infty$ and then $K \to \infty$ proves the theorem. □

**Corollary B.14** (one change of sign in the bulk)**.** *Under the hypotheses of Theorem B.13, let $R_0 = \sup_{\Theta_{\theta_0}} |x_\theta|$ and $R \geq R_0 + 1$. For all large $N$ the sign of $\Delta P_N(k)$ on $|x_{N,k}| \leq R$ changes exactly once, from positive to negative, at a point $k_N$ with $x_{N,k_N} - x_{\theta_N} \to 0$, and $\Delta P_N$ is strictly decreasing on $|x_{N,k} - x_{\theta_N}| \leq \varepsilon_0/2$.*

*Proof.* Let $0 < \varepsilon \leq \varepsilon_0/2$. By Theorem B.13 with $R + 1$ and Lemma B.12, for large $N$, $\Delta P_N(k) > 0$ when $x_{N,k} \leq x_{\theta_N} - \varepsilon$ and $\Delta P_N(k) < 0$ when $x_{N,k} \geq x_{\theta_N} + \varepsilon$, and $\Delta^2 P_N(k) < 0$ when $|x_{N,k} - x_{\theta_N}| \leq \varepsilon_0/2$, since $x_{N,k+2} - x_{N,k} = 2/B_N \to 0$. So on the window around $x_{\theta_N}$ the sequence $\Delta P_N$ decreases strictly from a positive to a negative value, and its sign changes exactly once, inside $|x - x_{\theta_N}| < \varepsilon$. Since $\varepsilon$ is arbitrary, $x_{N,k_N} - x_{\theta_N} \to 0$. □

### B.7 Junctions and the proof of Theorem G

*(EQ) and comparable scales.* Under (EQ) the ratio $T_+/T_-$ is regularly varying with index $\alpha_- - \alpha_+$ and bounded above and below, so $\alpha_+ = \alpha_-$. If $a_-(t) \geq Ca_+(t)$ with $C$ large, then by Lemma 2.1(ii) $1 \sim tT_-(a_-(t)) \leq tT_-(Ca_+(t)) \leq 2C^{-\alpha/2}tT_-(a_+(t)) \leq c'C^{-\alpha/2}$, which is impossible. So $a_+ \asymp a_-$, and $\min_\sigma \theta_{\sigma,N} \geq \theta_0$ for all large $N$ and some $\theta_0 > 0$. Hence Theorem G is the case of Corollary 6.16 in which every large $N$ qualifies, and it suffices to prove the corollary. Its hypotheses are those under which Sections B.5 and B.6 were proved.

*The zones.* Parts (i) and (ii) are Theorem 6.12(a) and (b), which need (RV), (SPAN), (LR) and (LD) for both signs. (RV) follows from (LR) and (LD) from Lemma B.1. Part (iii) was proved in Section B.2 and part (iv) in Section B.5. For (v) take $K_0$ as in the proof of (iv), enlarged if necessary so that $K_0 \geq R_0$, and apply Corollary B.14 with $R = K_0 + 1$. The mirror statements at the minus end follow from the same proofs with the signs exchanged, using (LD2) for the minus sign.

*The junctions.* The ranges overlap as follows.

- Parts (i) and (ii) meet at $J_+$.
- Part (ii) ends at $(x_* - \eta')\Lambda_N$ and part (iv) starts at $(x_* + \eta')\Lambda_N$, and the window of (iii) is the open interval between them. By the proof of Theorem 6.12(d), $D_N < 0$ at the first integer of the window and $D_N > 0$ at its last integer.
- Part (iv) is the union of Theorem 6.12(c) up to $N^{1-\eta}$ and Theorem B.8 from $N^{1-\eta}$ on, with the same $\eta < 1 - 1/\alpha$.
- Part (iv) covers the points with $x_{N,N-j} \geq K_0$, since $j \leq (1-p)N - K_0B_N$ is $N - j \geq pN + K_0B_N$. Part (v) covers $|x_{N,k}| \leq K_0 + 1$. So the two overlap.
- The minus end is the mirror image, and covers $x_{N,k} \leq -K_0$.

The constants are chosen in the order $\eta, \eta'$, then $K_0$, which therefore depends on the laws and on $\eta$ and $\eta'$, then $N$. Each zone holds for all large $N$, so all hold together for all large $N$, and the proof of the count in Section 6.5 applies. This proves Corollary 6.16 and Theorem G. □

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